Mathematics Atlas

2023–21 October 2026

A domain synthesis, complete treatment catalogue, and ten interactive worked models

1,051 retained treatments · 16 domains · 71 concept families · 10 working models

Definitions. Theorems. Proofs. Exact examples. Limits. Cross-domain bridges.

This atlas unifies the four completed retrospective mathematics volumes: 114 treatments for 2023, 237 for 2024, 351 for 2025 and 349 for 2026, for a total of 1,051 main treatments. Fourteen additional 2023 synthesis treatments are retained separately, giving 1,065 retained treatment bodies including supplements. These are treatments, not a count of unique conversations or distinct mathematical ideas. The source boundary is the four retrieved annual bundles, not a fresh complete export of every account conversation. The 2026 boundary is 21 October 2026; no later month is inferred. Historical dates identify the dated topics; revised proofs and examples are retrospective exposition and must not be read as discoveries made on those historical dates.

The self-contained interactive companion includes the full source-treatment text, search and filters, and parameterized mathematical labs. The portable bundle retains all four annual source volumes and replayable checks.

Contents and reading map

Contents and reading map 2

Scope, method and central argument 3

The domain map 4

D01 · Arithmetic and number fields 5

D02 · Algebraic and arithmetic geometry 22

D03 · Algebra and representation theory: structures, actions and exact certificates 33

D04 · Categories and homological methods 48

D05 · Topology, cohomology and gluing 59

D06 · Differential and symplectic geometry 70

D07 · Analysis, transforms and special functions 81

D08 · Differential equations and asymptotics 91

D09 · Dynamics, mechanics and mathematical modeling 101

D10 · Probability and stochastic processes 111

D11 · Information, statistics and inference 122

D12 · Optimization and decision theory 133

D13 · Combinatorics and graph mathematics 144

D14 · Logic, computation and exact certification 154

D15 · Control, observability and inverse problems 164

D16 · Quantum, deformation and integrable structures: finite models with explicit boundaries 174

Six cross-domain bridges 189

Interactive capacity contracts 195

Complete treatment index 200

Verification and reproducibility 249

Scope, method and central argument

Source boundary

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Editorial method

The chronological volumes remain intact. Above them, this report groups related material into domain chapters and canonical concept families. A family gives precise objects, a scoped theorem, a proof, a worked example, limits and concrete next questions. Its source identifiers allow the reader to return to every original treatment. Domain assignment is an editorial navigation aid, with secondary links for real cross-domain content; it is not an assertion that mathematical subjects have disjoint boundaries. Repeated annual treatments may support one family, and one treatment may support several families.

The central argument

The most productive organizing principle is a sequence of obligations: specify the object and admissible maps, identify the target, state hypotheses, derive a result, compute a checkable example, and expose what remains unresolved. Quotients ask which distinctions matter; cohomology asks whether local data glue; spectra quantify suppression and amplification; counting turns finite configuration spaces into exact laws and certificates; deformation tests whether a leading-order structure survives higher compatibility conditions; arithmetic reduction turns some infinite families into finite representative checks. These are connected through explicit constructions, not through a universal theory claimed to contain them all.

Interactive capacities

The interactive companion has two different functions. Its catalogue provides search, domain/year/capability filters and the complete retained text of all 1,051 source treatments. Its ten labs implement small, explicitly specified mathematical models. A capability tag on a chapter means the mathematics supports that kind of operation; it does not mean a general solver for that domain has been implemented. Each lab identifies its inputs, arithmetic, output and limits. Exact rational or integer results are distinguished from floating evaluations, analytic upper bounds and conceptual research directions.

How to read the atlas

Read the domain overview to orient the objects and questions, then a concept family for its theorem and proof. Follow a source identifier to inspect the original revised question and treatment. Use a lab to vary parameters inside a stated model. Use a cross-domain bridge only with its displayed hypotheses. The report is deliberately not another concatenation of the annual volumes: the full source bodies are retained in the interactive catalogue and portable bundle, while this document develops the shared architecture and the strongest concept families in depth.

What has been checked

The four original annual verification suites were replayed successfully from fresh copies. They contain 61 verifier programs and 1,173 scoped records; records are heterogeneous and include a proof-only record, so these counts are not counts of formally proved theorems. New domain scripts check selected worked examples, and the lab engine is tested against independent exact fixtures and boundary cases. A passing finite check supports its named model and does not automatically certify an entire exposition or an unrecovered historical instance.

Source year	Treatments	Original verifier programs	Scoped records
2023	114	16	174
2024	237	15	272
2025	351	14	360
2026	349	16	367

The domain map

Domain	Primary treatments	Concept families
D01 · Arithmetic and number fields	88	7
D02 · Algebraic and arithmetic geometry	57	4
D03 · Algebra and representation theory	88	6
D04 · Categories and homological methods	48	4
D05 · Topology, cohomology and gluing	54	4
D06 · Differential and symplectic geometry	55	4
D07 · Analysis, transforms and special functions	126	4
D08 · Differential equations and asymptotics	61	4
D09 · Dynamics, mechanics and mathematical modeling	110	4
D10 · Probability and stochastic processes	37	4
D11 · Information, statistics and inference	59	4
D12 · Optimization and decision theory	40	4
D13 · Combinatorics and graph mathematics	39	4
D14 · Logic, computation and exact certification	55	4
D15 · Control, observability and inverse problems	51	4
D16 · Quantum, deformation and integrable structures	83	6

Primary counts are editorial placements after a complete routing review. Secondary domains and family membership preserve overlap. A retained treatment may have no expanded family of its own; it remains fully available in the catalogue.

D01 · Arithmetic and number fields

Organizing question

How can an exact arithmetic construction retain its scalar domain, local data and completeness certificate as it is transported into a larger mathematical framework?

The strongest arithmetic thread across the recovered treatments is the replacement of suggestive numerical association with a declared object and a certificate. A number such as 384 becomes useful after it is assigned to the degree of a cyclotomic field. A recurrence becomes a classification after descent proves every solution lies in its seed orbits. A polynomial becomes an arithmetic field description after irreducibility, discriminant and ramification are checked. A residue becomes a root certificate after compatibility, a unit derivative and completeness supply the infinite conclusion. These transitions are the organizing structure of this chapter.

The entry point is scalar type. The 2023 distinction between Q, Q_p and characteristic-p series rings prevents invalid maps before any calculation begins. The same issue returns in finite-precision Frobenius tests and formal-series invariants: zero modulo p^N can hide a nonzero obstruction, and an unknown tail can hide the first unit coefficient. The p-adic family therefore couples exact digits with a proof about the completed object. Valuation arithmetic then gives the complementary integer picture. Prime exponents turn monomial constraints into affine integer lattices, divisor order into a product of chains, and factorials into a p-power part plus a stripped unit. The gcd-matrix factorization demonstrates how such arithmetic can legitimately create an energy form through an explicit incidence map.

Two families support implemented finite interactive models. The conductor-1680 model enumerates 384 units, reduces them to eight primitive order-30 classes and verifies 48 points in every fibre. The tower formula then gives exact trace transport. The Pell model generates positive and negative norm orbits, displays inverse steps and computes periodic congruence acceptance. Its written seed reduction is essential: enumeration explains a finite modular period, while descent proves global coverage. Both labs are worked models with exact inputs and outputs, rather than general arithmetic-solving services.

Number fields provide the main obstruction language. The S_3 closure of the discriminant-minus-23 cubic forbids cyclotomic containment. The $Q(\sqrt{10})$ examples distinguish a genuine unramified class-field candidate from an extension of the right degree with the wrong ramification. Cubic norm formulas preserve the lattice defining the original binary equation, showing exactly why a unit action can leave the solution plane. The completely proved Mordell example $y^2=x^3-2$ illustrates the stronger standard: a norm-Euclidean ring and coprimality argument can deliver a global finite answer, whereas an ambient unit identity alone cannot.

The final two families connect this arithmetic to analytic and modular structure. Modular forms require a group, weight and normalization before dimensions and Hecke relations make sense. Euler products require a convergence half-plane before coefficient algebra becomes analysis. Pointwise $\mu\Lambda$, convolution $\mu*\log$, prime counting and prime-power counting remain different objects. These distinctions make cross-domain bridges more productive because each bridge names its actual map. The chapter does not force a chronological storyline where sources are absent: the conductor thread is supported in 2025–2026, the explicit Pell thread in 2024 and 2026, and the modular thread primarily in 2024–2025. The common framework is a new synthesis of those supported mathematical treatments.

Core objects

Integer valuation vectors and divisor incidence

p-adic fields, residue systems and integral series

Quadratic units and complete Pell orbits

Cyclotomic fields and finite Galois fibres

Ideal classes, norm lattices and ramification

Modular coefficients, local factors and convergent Dirichlet series

Open frontier

The most valuable extensions would certify complete unit bases and lattice-return conditions for the norm examples, extend seed proofs to additional Pell cores, and provide explicit integral intertwiners for the proposed conductor-to-lattice bridge. Each is a precise missing theorem or certificate rather than an unspecified generalization.

D01-F01 · Arithmetic types, completions and certified p-adic precision

Which arithmetic statements survive a change of scalar domain, and what can a finite p-adic calculation actually certify?

An arithmetic expression includes its scalar ring, topology and precision. The rational field, a p-adic completion, a residue ring and a formal power-series ring support different operations. A unital homomorphism between characteristic zero and characteristic p would send the nonzero integer p both to an invertible element and to zero, an immediate obstruction when the source is a field. Completion instead preserves rational identities while changing which sequences converge. These distinctions turn the early request to separate meanings of Q into a practical specification for every later lifting algorithm. The useful output is a compatible sequence of exact congruences accompanied by derivative and residual valuations. Such a sequence proves an existence statement only with a convergence argument; a zero residue at one precision cannot certify an exact zero. Perfectoid and prismatic language extends this discipline, but carries additional structural hypotheses rather than merely renaming small errors.

Definitions

For nonzero rational x, $v_p(x)$ is its prime-p exponent and $|x|_p=p^{(-v_p(x))}$; Q_p is the metric completion and Z_p its unit ball.

A simple root modulo p is an integer a with $f(a)\equiv 0 \bmod p$ and $f'(a)$ not divisible by p. A precision-N representative specifies a coset modulo p^N.

For a nonzero integral series $F=\Sigma a_n T^n$, $\mu(F)=\min v_p(a_n)$, and $\lambda(F)$ is the first index attaining that minimum. Infinite tails must be controlled before certifying these minima.

Theorem and hypotheses

If $f\in Z_p[X]$ and a_n is a root modulo p^n with unit derivative, there is exactly one $a_{n+1}=a_n+t\,p^n$ modulo $p^{(n+1)}$ reducing to a_n. The compatible sequence converges to the unique root of f in a_1+pZ_p.

Proof

Taylor expansion modulo p^(n+1) gives f(a_n+t p^n)≡f(a_n)+t p^n f′(a_n), since all terms of order two are divisible by p^(2n). Dividing by p^n gives one linear equation for t modulo p. The derivative is a unit, so t is unique. Successive representatives differ by an element of p^nZ_p and hence form a Cauchy sequence. Completeness supplies a limit a; continuity of polynomial evaluation gives f(a)=0. If b is a second root in the same residue class, factor f(a)−f(b)=(a−b)g(a,b). The reduction of g is the nonzero derivative at a_1, so g is a unit and a=b. Newton's update a−f(a)/f′(a) accelerates the same argument: an initial residual valuation at least n becomes at least 2n when the derivative is a unit. An exact vanishing claim requires this theorem or a separate identity, whereas finite arithmetic alone establishes only congruences.

Worked example

Lift the residue 3 of X^2-2 modulo 7, then compare a filtered Frobenius residual at two precisions.

At modulus 7, $3^2-2=7$ and $f'(3)=6$ is a unit. Writing $a_2=3+7t$ gives $1+6t\equiv 0 \bmod 7$, hence $t=1$ and $a_2=10$. Then $f(10)=98=2\cdot 49$; $2+20t\equiv 0 \bmod 7$ gives $t=2$, so $a_3=108$ and $108^2-2=34\cdot 343$. Separately, for $\varphi=\mathrm{diag}(1,5)$, $\mathrm{Fil}^1=\mathrm{span}(e_2)$, the integral quotient residual $R(x,y)=\pi((5\varphi-I)(x,y))$ is $4x$. At $(25,1)$ it is 100: invisible modulo 25, nonzero modulo 125. This second calculation certifies failure of exact membership once sufficient precision is supplied.

The root computation supplies certified digits of a genuine 7-adic algebraic number; the residual example explains why a temporary zero display cannot replace a proof.

Continuity across the source years

2023 — The 2023 treatments supply the scalar distinctions and simple-root lifting foundation; this synthesis uses their explicit hypotheses rather than attributing a later precision calculus to the original exchanges. Sources: 2023:M001, 2023:M002

2024 — Perfectoid fields and prisms introduce completeness, Frobenius and structured pairs. Their appearance broadens the objects, while the elementary lifting theorem remains a separate local result. Sources: 2024:Y24-10-023, 2024:Y24-10-024

2025 — Filtered Frobenius and power-series invariants expose the distinction between a finite residue and an exact infinite object. Newton lifting gives a positive certificate when its derivative assumptions hold. Sources: 2025:Y25-08-907, 2025:Y25-09-010, 2025:Y25-10-003

2026 — The later completion and quintic examples return to concrete compatible lifts. Their exact residue-and-derivative checks support a reusable verification contract, without claiming global factorization or rational roots. Sources: 2026:Y26-07-031, 2026:Y26-09-013

Connections

D14 — Exact certification. A verifier checks f(a_N)≡0 mod p^N and invertibility of f′(a_N), while the written convergence proof upgrades the certificate to existence. Finite checking and the infinite theorem have distinct responsibilities.

D02 — Comparison and lifting data. A crystalline or filtered calculation must name its base, lattice and filtration before a p-adic residual is meaningful. The elementary residual example is not a proof of a general p-adic Hodge comparison theorem.

Capacity and interactive scope

Lab ids: []

Inputs: Prime p, integer polynomial f, initial residue, requested precision N; optional exact filtered matrices.

Outputs: A table of compatible roots, derivative-unit checks and residual valuations; a precision-aware membership verdict.

Validation: Replay each modular root and reduction step exactly; distinguish zero modulo p^N from an identically zero obstruction.

Status: Mathematical recipe; no general solver implemented

Limits

Simple-root lifting does not cover a vanishing derivative without stronger hypotheses. Truncated series cannot reveal unseen unit coefficients. Perfectoid and prismatic constructions require their full structural definitions; a numerical resemblance to a weight or height supplies none of them.

Next questions

For a singular residue, what extra valuation inequalities guarantee a lift, and can an exact counterexample show why the simple-root contract fails?

Can a proposed filtered calculation provide a certified lower bound on input precision sufficient to distinguish nonvanishing from an unresolved zero residue?

Source anchors: 2023:M001, 2023:M002, 2025:Y25-08-907, 2025:Y25-09-010, 2025:Y25-10-003, 2026:Y26-07-031, 2026:Y26-09-013

NIST DLMF q calculus — https://dlmf.nist.gov/17.2
Keith Conrad p-adic course materials — https://kconrad.math.uconn.edu/math3094f16/
Keith Conrad Hensel’s Lemma — https://kconrad.math.uconn.edu/blurbs/gradnumthy/hensel.pdf
Milne, Class Field Theory, local fields and valuations — https://www.jmilne.org/math/CourseNotes/cft.html
Coates Schneider Sujatha Modules over Iwasawa algebras — https://arxiv.org/abs/math/0110342
Milne Algebraic Number Theory — https://www.jmilne.org/math/CourseNotes/ANT.pdf

D01-F02 · Divisor incidence, valuations and constructive integer parametrization

How can multiplicative arithmetic become a finite lattice calculation with a complete certificate?

Prime factorization converts multiplication into addition of exponent vectors. Divisibility becomes a coordinate order, gcd and lcm become coordinate minimum and maximum, and a monomial equation becomes an integer linear system at every prime. This does more than simplify computation: it states exactly which sign cases, valuation bounds and prime supports a complete answer must cover. Divisor incidence then transports scalar arithmetic identities into matrix factorizations. The same identity $n=\Sigma_{(d|n)}\varphi(d)$ controls a positive gcd kernel, its determinant, and a sieve implementation. Thus arithmetic, combinatorics and finite-dimensional energy geometry meet through a declared transform, not merely because they share the word prime. The family includes factorial valuations and prime-stripped block products: one computes the p-power removed, the other computes the remaining unit, and their recombination reconstructs the original factorial exactly.

Definitions

For $n=\prod p_i^{a_i}$, a divisor corresponds to exponents $0\le b_i\le a_i$. Its divisor lattice is the product of chains $C_{(a_i+1)}$.

Dirichlet convolution is $(f*g)(n)=\Sigma_{(d|n)}f(d)g(n/d)$. The incidence matrix E has $E_{(i,d)}=1$ when d divides i, and zero otherwise.

Legendre’s formula gives v_p(n!)=Σ_(j≥1)⌊n/p^j⌋. The stripped unit U_p(n)=n!/p^v_p(n!) is an integer not divisible by p.

Theorem and hypotheses

For G_(i,j)=gcd(i,j), 1≤i,j≤N, one has G=E diag(φ(1),…,φ(N))Eᵀ. Consequently G is positive definite and det G=∏_(d=1)^Nφ(d). Moreover positive solutions of $x^2y^3=2^{20}3^{12}$ are exactly x=2^(10−3r)3^(6−3s), y=2^(2r)3^(2s), with 0≤r≤3 and 0≤s≤2.

Proof

The (i,j) entry of the displayed matrix product is Σ_(d|i,d|j)φ(d)=Σ_(d|gcd(i,j))φ(d)=gcd(i,j). The identity Σ_(d|m)φ(d)=m follows by grouping 1,…,m according to gcd with m. E is lower triangular with diagonal one; every φ(d)>0, giving positive definiteness and the determinant. For the monomial equation, a prime outside {2,3} has 2v_p(x)+3v_p(y)=0, forcing both exponents zero. At p=2, nonnegative integral solutions to 2a+3b=20 require b=2r and a=10−3r, with r=0,1,2,3. At p=3 the same argument gives b=2s and a=6−3s, with s=0,1,2. Independent choices prove both existence and completeness of the twelve solutions. A bounded search would show existence of examples but would not establish that no extra prime or exponent pattern occurs.

Worked example

Factor the 4×4 gcd matrix, count a divisor lattice, and recover a stripped factorial.

For N=4 the diagonal weights are (1,1,2,2), so det G=4. The row for i=4 in E is (1,1,0,1), explaining gcd(4,2)=φ(1)+φ(2)=2. For $n=72=2^3 3^2$, the rank polynomial is $(1+t+t^2+t^3)(1+t+t^2)$, whose coefficients are 1,2,3,3,2,1; its value at one is 12 and its degree is five. For 10!, v_3=3+1=4, so U_3(10)=3628800/81=44800. The recursion U_3(10)=(1·2·4·5·7·8·10)U_3(3), with U_3(3)=2, reproduces 44800.

A positive kernel, a complete Diophantine parametrization and an efficient arithmetic block algorithm arise from distinct but compatible uses of divisor structure.

Continuity across the source years

2023 — Exact exponent manipulation is the early foundation. It supplies arithmetic scale control, without implying that later divisor kernels were already formulated. Sources: 2023:M081

2024 — The exponent-tower treatment makes parse structure explicit. The same expression discipline is needed before converting a multiplicative constraint into an exponent equation. Sources: 2024:Y24-10-028

2025 — Divisor convolution and factorial valuations provide the algebraic and computational components separately, each with an exact domain and proof. Sources: 2025:Y25-05-020, 2025:Y25-06-004

2026 — Species, lattice order, kernel factorizations and full valuation parametrizations make the shared exponent structure particularly explicit in the later treatments. Sources: 2026:Y26-05-019, 2026:Y26-05-020, 2026:Y26-10-003, 2026:Y26-10-004, 2026:Y26-10-005, 2026:Y26-10-007

Connections

D13 — Incidence and lattice structure. The divisor poset is a product of chains; its zeta matrix is E and its Möbius inverse supplies arithmetic inversion. This finite poset statement does not identify arbitrary compatibility graphs with complete arithmetic solution sets.

D12 — Energy certificates. G=EWEᵀ with W positive gives an exact quadratic energy and strict convexity on the declared finite coordinates. Optimization follows from this factorization, not from prime support

alone.

D14 — Complete finite search. Nonnegative valuation bounds prove that enumeration of the exponent rectangle is exhaustive. Integer sign and zero cases require separate checks when the original equation permits them.

Capacity and interactive scope

Lab ids: []

Inputs: A factored integer, exponent matrix, target valuations and sign domain; or an integer kernel cutoff N.

Outputs: Exact exponent solutions, divisor-rank counts, incidence factors and determinant certificates.

Validation: Reconstruct each integer, substitute into the original equation, and prove prime-support and valuation bounds before enumerating.

Status: Mathematical recipe; no general solver implemented

Limits

Factorization is an input cost, and exact arithmetic operation counts are not bit-complexity bounds. Real or complex solutions of a monomial equation do not inherit the integer valuation classification. A positive matrix constructed from divisors does not automatically encode an arithmetic L-function.

Next questions

For a general nonnegative exponent matrix, which finite target supports produce a bounded polytope of integer valuation vectors?

Can the O(N log N) divisor-kernel action be paired with a bit-complexity bound and an exact sparse factorization certificate?

Source anchors: 2025:Y25-05-020, 2025:Y25-06-004, 2026:Y26-01-006, 2026:Y26-05-020, 2026:Y26-10-003, 2026:Y26-10-005, 2026:Y26-10-007

NIST DLMF Dirichlet series — https://dlmf.nist.gov/27.4
MIT number-theory lecture notes — https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/
MIT elementary number theory notes — https://math.mit.edu/~dav/finalCheatSheet.pdf
Henry J. S. Smith, On the Value of a Certain Arithmetical Determinant — https://doi.org/10.1112/plms/s1-7.1.208
J. S. Milne, Algebraic Number Theory — https://www.jmilne.org/math/CourseNotes/ANT.pdf
NIST DLMF: arithmetic functions — https://dlmf.nist.gov/27.2

D01-F03 · Cyclotomic degrees, Galois fibres and trace transport

What exact structure follows from conductor 1680, and which additional certificates are needed to connect it to another representation?

The number 384 becomes mathematically productive when assigned a specific role: it is the degree of $Q(\zeta_{1680})$, not the order of a primitive root, the largest Galois-element order, a Lie algebra dimension or a category rank. Distinguishing those meanings allows a much stronger construction. The subgroup corresponding to $Q(\zeta_{30})$ produces uniform fibres of size 48; trace transport is then a theorem of the field tower and can be checked through finite residue classes. This organizes a recurring numerical framework into an auditable arithmetic object. The same discipline applies to trace orthogonality: a coefficient p, rather than p+1, follows from which residue sums are actually being taken. Any proposed passage to an integral E_8 lattice must exhibit a basis matrix, its Gram form and an intertwining equation. Equal degree or a handful of coincident traces cannot supply those maps.

Definitions

For $N \geq 1$, $\zeta_N = \exp(2\pi i/N)$; $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$ identifies with $U_N = (\mathbb{Z}/N\mathbb{Z})^\times$ by $\sigma_a(\zeta_N) = \zeta_N^a$.

The Ramanujan sum $c_N(n) = \sum_{a \in U_N} \zeta_N^{an}$ is an integer and equals $\sum_{d \mid \gcd(N,n)} d\,\mu(N/d)$.

For $L \subseteq K$ and $x \in L$, $\mathrm{Tr}_{K/\mathbb{Q}}(x) = [K:L]\,\mathrm{Tr}_{L/\mathbb{Q}}(x)$. Relative trace is a linear map, not a choice of an integral basis.

Theorem and hypotheses

Let $K = \mathbb{Q}(\zeta_{1680})$, $L = \mathbb{Q}(\zeta_{30})$, and $\eta = \zeta_{1680}^{56}$. Then $[K:\mathbb{Q}] = 384$, $[L:\mathbb{Q}] = 8$, $[K:L] = 48$; reduction $U_{1680} \to U_{30}$ is onto with every fibre of size 48. Hence $\mathrm{Tr}_{K/\mathbb{Q}}(\eta^n) = 48\,c_{30}(n)$. The Galois group of K is $C_4^2 \times C_2^3 \times C_3$ and has exponent 12.

Proof

Since $1680 = 16 \cdot 3 \cdot 5 \cdot 7$, the totient product is $8 \cdot 2 \cdot 4 \cdot 6 = 384$; $\varphi(30) = 1 \cdot 2 \cdot 4 = 8$. The order of ζ_{1680}^{56} is $1680/\gcd(1680,56) = 30$, giving $L \subseteq K$ and the tower degree. A class coprime to 30 can be extended to a unit modulo 1680 by prescribing a compatible odd residue modulo 16 and a nonzero residue modulo 7; CRT proves surjectivity. A surjective finite-group homomorphism has fibres equal to kernel cosets, all of size $384/8$. Summing η^{an} across these fibres proves the trace formula. For the group structure CRT gives $U_{16} \times U_3 \times U_5 \times U_7 \cong (C_4 \times C_2) \times C_2 \times C_4 \times C_6$, and $C_6 \cong C_2 \times C_3$. The least common multiple of these cyclic orders is 12, so the group cannot be cyclic of order 384. These conclusions use classical cyclotomic irreducibility as a structural input; the residue enumeration independently verifies the finite arithmetic.

Worked example

Count the fibres and compute four exact traces without floating-point roots of unity.

Enumerating $0 \leq a < 1680$ with $\gcd(a,1680) = 1$ gives 384 residues. Reducing modulo 30 produces precisely $1,7,11,13,17,19,23,29$, each 48 times. Inclusion–exclusion gives $c_{30}(1) = -1$, $c_{30}(2) = 1$, $c_{30}(15) = -8$ and $c_{30}(30) = 8$. The associated K-traces are $-48, 48, -384, 384$. Complex conjugation is the nontrivial involution $a \mapsto -a$, so its fixed maximal real subfield has degree 192. This is distinct from the extension degree 48 and the number of target phases eight.

The arithmetic framework now supports a real interactive finite model: every displayed multiplicity is an exact residue count and every displayed trace has an integer formula.

Continuity across the source years

2025 — The explicit user conductor framework supports the degree and Galois computation; the fourth-power counterexample and corrected trace coefficient delimit what that arithmetic can imply. Sources: 2025:Y25-04-001, 2025:Y25-05-003, 2025:Y25-10-026

2026 — The later treatments correct the totient to 384 and derive the concrete conductor-30 field tower. The uniform phase fibres are the strongest exact transport statement supported by these entries. Sources: 2026:Y26-01-016, 2026:Y26-10-030

Connections

D03 — Representation comparison. A matrix C on one lattice and D on another are intertwined only after an explicit B satisfies $CB = BD$; then $C^n B = BD^n$ follows by induction. Trace coincidences give necessary diagnostics, not an integral lattice isomorphism.

D16 — Finite phase models. The primitive phase pushforward is an exact finite distribution with uniform multiplicity 48. It does not determine a fusion category, quantum dimension or Hamiltonian.

D14 — Certificate design. CRT enumeration, group-element orders and integer Ramanujan sums give independent finite checks. The field-degree interpretation still cites cyclotomic irreducibility.

Capacity and interactive scope

Lab ids: ['L07']

Inputs: Select a supported divisor m of 1680; the default target is m=30.

Outputs: All reduced unit classes, their fibre sizes, source and target counts, and a uniformity verdict. The trace identities are explained in the family and replayed by verify.py; the current lab has no trace-exponent control.

Validation: Enumerate every gcd-one residue modulo 1680; compare target counts with 384/φ(m). At m=30 assert eight fibres of size 48.

Status: Implemented residue-fibre worked model; trace controls are a specified extension

Limits

This is a worked conductor pair, not a universal cyclotomic-field or integral-representation solver. No missing E_8 matrices are reconstructed from dimensions. The sources do not supply arithmetic continuity back to 2023, so this family's chronology begins with the supported 2025 conductor treatment.

Next questions

Can the proposed integral map provide explicit integer matrices B,A,G,C,D with $B^TAB=G$ and $CB=BD$, and can its index be certified?

For a general divisor M of N, can the interface expose the CRT proof of surjectivity alongside the uniform reduction fibres, while retaining exceptional small-conductor conventions?

Source anchors: 2025:Y25-04-001, 2025:Y25-05-003, 2025:Y25-10-026, 2026:Y26-01-016, 2026:Y26-10-030

Milne, Algebraic Number Theory — https://www.jmilne.org/math/CourseNotes/ANT.pdf
Elkies, On $A^4+B^4+C^4=D^4$ — https://doi.org/10.1090/S0025-5718-1988-0930224-9
Elkies mathematical publications — https://people.math.harvard.edu/~elkies/math_pubs.html
Milne Fields and Galois Theory — https://www.jmilne.org/math/CourseNotes/FT.pdf
Lawrence C. Washington, Introduction to Cyclotomic Fields — https://doi.org/10.1007/978-1-4612-1934-7
NIST DLMF: Ramanujan sums and finite Fourier series — https://dlmf.nist.gov/27.10

D01-F04 · Quadratic units, complete Pell orbits and congruence populations

When does a recurrence describe every integer solution, and how do finite congruence filters change an infinite population?

A norm-preserving recurrence generates solutions immediately; proving that it generates all of them is the substantive step. The D=2 Pell equation has an elementary descent to the identity, while the negative equation $x^2-2y^2=-7$ requires two distinct canonical seeds. A compact reduction interval makes the seed count finite and proves uniqueness. This exposes an important difference between a finite list, an infinite parametrized family, and a complete classification. Once completeness is established, modular arithmetic is useful again: the unit recurrence is invertible modulo every q, so its state orbit is genuinely periodic. Counting accepted index classes gives an exact density within the orbit and a logarithmic coefficient in a height-bounded population. The same quadratic-unit mechanism explains Fibonacci matrix eigenvalues, but geometric length normalizations must still be declared separately.

Definitions

In Z[√2], conjugation sends x+y√2 to x−y√2 and the norm is x²−2y². The norm-one unit ε=3+2√2 has inverse 3−2√2.

Multiplication by ε sends (x,y) to (3x+4y,2x+3y), an integral matrix of determinant one.

A canonical seed is a norm solution reduced to a specified half-open interval under multiplication by ε. A modular filter is a predicate on the full residue state, not solely on a visually selected coordinate.

Theorem and hypotheses

Every solution x,y≥0 of x²−2y²=1 is ε^j for a unique j≥0. Every solution of x²−2y²=−7 with y>0 belongs uniquely to (1+2√2)ε^j or (5+4√2)ε^j, j∈Z. For the positive Pell orbit, a modulo-q predicate accepting w positions in its period P yields a population (w/P)log N/log ε+O(1) under y≤N.

Proof

For norm one and y>0, y≥2 and √2<x/y≤3/2. The inverse step gives x′=3x−4y>0 and 0≤y′=3y−2x<y; iteration reaches (1,0), proving coverage. For norm −7, α=x+y√2 is positive whenever y>0. Multiply by a unique power of ε to put β in [√7,ε√7). Since β=−7/β, its coordinates satisfy x_β=(β−7/β)/2≥0 and y_β=(β+7/β)/(2√2), which lies below 3√(7/2)<6. Thus y_β∈{2,3,4,5}; the required squares 2y_β²−7 are 1,11,25,43. Exactly two seeds remain. The interval is half-open, so seed and exponent are unique. For the positive orbit, conjugation gives y_j=(ε^j−ε^(−j))/(2√2), hence j≤log N/log ε+O(1). The determinant-one recurrence permutes a finite modular state set, so its orbit has no transient tail. A periodic acceptance pattern with w of P positions contributes w/P of the indices, up to a bounded endpoint error.

Worked example

Generate the two negative-norm seeds, then apply an even-index filter to the positive-norm orbit.

The first seed (1,2) maps to (11,8), and 11²−2·8²=−7. The second (5,4) maps to (31,22), also of norm −7. The inverse sends (11,8) back to (1,2). The positive norm-one sequence begins (3,2),(17,12),(99,70),(577,408). Modulo 3 its full states (x,y) are (1,0),(0,2),(2,0),(0,1), followed by (1,0); the period is four. The condition y≡0 mod 3 accepts precisely the even indices, so the filtered count has coefficient 1/(2log ε).

Completeness rests on descent and seed reduction; finite modular enumeration then safely determines the accepted fraction of that already complete family.

Continuity across the source years

2024 — The recovered catalog name is developed into an elementary complete D=2 Pell example, with a proof explicitly authored for the annual treatment rather than claimed as recovered historical wording. Sources: 2024:Y24-03-024

2026 — The later treatments connect quadratic units to matrix decay, periodic filters and logarithmic populations, then make the multiple-seed issue explicit through the negative-norm equation. No intervening 2025 Pell source is invented. Sources: 2026:Y26-05-005, 2026:Y26-09-029, 2026:Y26-10-006

Connections

D09 — Discrete exact dynamics. The recurrence is an invertible integer map preserving a quadratic invariant; its finite reductions are permutation dynamics. Modular periodicity does not by itself prove completeness over the integers.

D08 — Growth and counting. An exact conjugate formula turns exponential orbit growth into logarithmic population growth. The O(1) counting remainder is tied to this orbit model, not to arbitrary Diophantine

families.

D14 — Global certificates. A compact seed bound plus exact square testing proves that every unit orbit is represented. A finite search without a derived reduction bound is not a global theorem.

Capacity and interactive scope

Lab ids: ['L10']

Inputs: Choose negative-norm seed (1,2) or (5,4), then an integer unit power from −20 to 20.

Outputs: Exact transported coordinates, signed step and norm −7. Positive-Pell sequences and modular acceptance periods are proved in the family and replayed by verify.py; they are not current lab controls.

Validation: Compute with arbitrary-precision integers and assert $x^2-2y^2=-7$ after each displayed transport. The accompanying half-open seed interval proves completeness.

Status: Implemented two-seed negative-Pell worked model; positive-orbit and congruence-filter controls are specified extensions

Limits

The lab covers these D=2 equations. A different discriminant or norm can have more seeds, a different unit group and nontrivial sign conventions. Floating-point logarithms may display growth but are not used to certify integer membership. No unsupplied global shift-uniqueness theorem follows from the population calculation.

Next questions

For each additional quadratic core, can all seeds be bounded and enumerated with a proof of descent coverage before attaching any congruence filter?

Can an exact interval method decide a height cutoff without rounding error when ε^j lies close to the chosen bound?

Source anchors: 2024:Y24-03-024, 2026:Y26-05-005, 2026:Y26-09-029, 2026:Y26-10-006

MIT number theory notes: Euclidean algorithm and Brahmagupta–Pell — https://ocw.mit.edu/courses/18-781-theory-of-numbers-spring-2012/pages/lecture-notes/
J. S. Milne, Algebraic Number Theory — https://www.jmilne.org/math/CourseNotes/ANT.pdf
Conrad Pell equations, author notes — https://kconrad.math.uconn.edu/blurbs/ugradnumthy/pelleqn1.pdf
Lenstra: solving the Pell equation — https://www.ams.org/notices/200202/fea-lenstra.pdf

D01-F05 · Number fields, norm forms and arithmetic obstructions

Which field, ideal and lattice data are sufficient to turn a norm identity into a complete arithmetic statement?

A defining polynomial is the beginning of a number-field calculation. Irreducibility, the polynomial-order index, ramification, unit structure and the actual sublattice representing the original equation each impose additional requirements. The squarefree discriminants −23 and −283 give unusually clean maximal-order certificates, while nonabelian Galois closure supplies a decisive obstruction to cyclotomic containment. The class-field examples add a second kind of obstruction: an extension with the right degree can still ramify at finite or real places. Finally, the 2026 norm forms show why unit transport alone need not classify a two-variable equation. Multiplication preserves a cubic norm surface, but often leaves the binary plane that encoded the original problem. Retaining the embedding lattice makes this loss visible and turns a plausible reduction into a mathematically correct one.

Definitions

For an algebraic integer α, the discriminant of Z[α] equals the field discriminant times [O_K:Z[α]]². A squarefree polynomial discriminant therefore forces index one.

The field norm N_(K/Q)(x) is the determinant of multiplication by x on K. In a specified integral order, norm ±1 implies a unit of that order.

A binary cubic equation can embed in a norm equation only together with a rank-two lattice condition; the full norm surface has an extra coordinate. An exceptional unit is an algebraic integer z for which both z and 1−z are units.

Theorem and hypotheses

For $f(T)=T^3-T+1$, irreducibility and discriminant −23 imply S_3 Galois closure and exclude every root from every cyclotomic field. Separately, in the order Z[z] with $z^3=-4$, $N(a+bz+cz^2)=a^3-4b^3+16c^3+12abc$; norm-one transport need not preserve c=0.

Proof

The only possible rational roots of the monic cubic are ±1, neither of which is a root. Its discriminant is $-4(-1)^3-27=-23$, not a rational square. An irreducible cubic has transitive Galois group A_3 or S_3; the discriminant-square criterion excludes A_3. Every subfield of a finite abelian Galois extension is itself Galois because every subgroup is normal. A nonnormal cubic with S_3 closure therefore cannot be contained in a cyclotomic field. For the norm formula, multiplication by $a+bz+cz^2$ in basis $(1,z,z^2)$ has matrix $[[a,-4c,-4b],[b,a,-4c],[c,b,a]]$. Its determinant is the displayed form, giving an exact integral certificate without approximate roots. The unit $\varepsilon=1-z-z^2$ has norm one. Multiplying the binary element $-1-3z$ by ε gives $-13-2z+4z^2$. Both have norm 107, but the latter has nonzero third coordinate. Consequently unit-orbit completeness in the ambient order, even if independently proved, would still require an exact return-to-plane test to solve $u^3+4v^3=107$.

Worked example

Contrast a correct unramified class-field candidate with a wrong one, then retain a Thue lattice.

For K=Q(√10), the quadratic discriminant is 40 and the class number is two. The biquadratic L=Q(√2,√5) has discriminant $8\cdot5\cdot40=1600=40^2$, so its relative discriminant has norm one; L is totally real, giving no real-place ramification. The alternative K(i) has discriminant $40\cdot(-4)\cdot(-40)=6400$ and relative norm four, so it is ramified. For $H(u,v)=-3u^3+9uv^2-2v^3$ and $\beta^3-27\beta-18=0$, determinant expansion gives $N(-3u-\beta v)=9H(u,v)$. The relevant lattice is $-3Z+Z\beta$, not every norm element of the field.

Degree, discriminant, class-field identification and binary-lattice membership answer different questions and require distinct certificates.

Continuity across the source years

2024 — Quadratic splitting in Q(i) supplies a small exact local-to-global model and illustrates how local factors depend on prime ideals rather than solely rational primes. Sources: 2024:Y24-10-011, 2024:Y24-12-018

2025 — Ray classes, nonprincipal ideals, ramification tests and finite-field descent make several independent meanings of arithmetic obstruction concrete. Sources: 2025:Y25-07-022, 2025:Y25-07-023, 2025:Y25-08-903, 2025:Y25-08-902

2026 — The later sources expose nonabelian field obstructions, exceptional-unit symmetries and binary norm-plane restrictions. The Mordell equation adds a genuinely complete proof using a norm-Euclidean quadratic ring. Sources: 2026:Y26-06-009, 2026:Y26-06-016, 2026:Y26-06-017, 2026:Y26-09-030,

2026:Y26-10-009, 2026:Y26-10-036

Connections

D02 — Birational coordinates and divisors. The corner cubic is related by an explicit Möbius substitution, and its rational function has divisor 3[1]−2[0]−[∞]. Matching discriminants alone does not identify coordinates or embeddings.

D03 — Galois representation obstruction. The S_3 closure distinguishes a cubic field from subfields of abelian cyclotomic extensions. A common numerical degree gives no representation equivalence.

D14 — Completeness conditions. An exact multiplication matrix certifies a norm identity; unit bases and lattice-return constraints are additional obligations for exhaustive solving. No unreplayed global unit search is promoted to a theorem.

Capacity and interactive scope

Lab ids: []

Inputs: Defining polynomial, specified order basis, element coordinates, candidate unit and the original embedding lattice.

Outputs: Exact multiplication determinant, norm-preservation certificate, discriminant diagnostics and lattice-membership checks.

Validation: Expand determinants over integers; verify substitutions both ways; distinguish polynomial from field discriminants and unit transport from exhaustive orbit coverage.

Status: Mathematical recipe; no general solver implemented

Limits

A norm-one element is a unit of the chosen order, but a displayed unit pair need not generate the full unit group. Ramification and class-field assertions use stated classical inputs. No uniqueness of a field from a small discriminant or a polynomial list is claimed.

Next questions

Can a proven complete unit basis and finite residue classes be combined with an exact lattice-return argument for the recovered Thue forms?

Which explicitly declared exceptional-unit height bounds would turn a proposed field-uniqueness statement into a finite, complete classification?

Source anchors: 2025:Y25-07-022, 2025:Y25-07-023, 2026:Y26-06-009, 2026:Y26-06-016, 2026:Y26-09-030, 2026:Y26-10-009, 2026:Y26-10-036

Milne, Algebraic Number Theory — https://www.jmilne.org/math/CourseNotes/ANT.pdf
Milne, Class Field Theory — https://www.jmilne.org/math/CourseNotes/CFT.pdf
J. S. Milne, Fields and Galois Theory — https://www.jmilne.org/math/CourseNotes/FT.pdf
Mathlib: algebra norm as a determinant — https://leanprover-community.github.io/mathlib4_docs/Mathlib/RingTheory/Norm/Basic.html
MIT algebraic number theory course notes — https://math.mit.edu/~drew/18.785/
The Stacks Project, Kummer theory — https://stacks.math.columbia.edu/tag/03PK

D01-F06 · Modular forms, Hecke relations and arithmetic local factors

Which identities genuinely characterize modular arithmetic, rather than an arbitrary prime-indexed statistic?

A modular-form assertion includes the group, weight, character, normalization and transformation law. Once those are fixed, dimensions, Hecke recurrences and Euler factors become exact algebraic constraints. The number 24 enters the logarithmic derivative of Dedekind eta through a specific product normalization; it is not interchangeable with a root count or an abstract dimension. Similarly, a genuine Frobenius trace is tied to a point count and determinant p. Its symmetric square has three eigenvalue products, not a guessed two-factor denominator. This family places q-expansion calculations beside local arithmetic while retaining the distinction between an Eisenstein series, a cusp eigenform and an elliptic-curve L-function. Modular symbols supply a boundary algebra that can later support periods, but their elementary relations alone do not execute a modular-symbol algorithm or establish modularity.

Definitions

For the full modular group over C, the graded ring of holomorphic modular forms is $C[E_4,E_6]$, with weights 4 and 6. $\Delta=(E_4^3-E_6^2)/1728$ has weight 12.

For weight k, a normalized Hecke eigenform satisfies $a_{(p^2)}=a_p^2-p^{(k-1)}$, with suitable unramified and normalization hypotheses.

For an elliptic local factor, $\alpha+\beta=a_p$ and $\alpha\beta=p$. The symmetric square eigenvalues are α^2,p,β^2.

Theorem and hypotheses

The weight-38 full-level modular-form space has basis $E_4^8E_6$, $E_4^5E_6^3$, $E_4^2E_6^5$ and dimension three; its cusp subspace is ΔM_{26} of dimension two. For elliptic eigenvalues α,β, the symmetric-square denominator is $1-(a_p^2-p)t+p(a_p^2-p)t^2-p^3t^3$.

Proof

A monomial $E_4^{\wedge}aE_6^{\wedge}b$ has weight $4a+6b$. Solving $4a+6b=38$ in nonnegative integers gives $b=1,3,5$ and $a=8,5,2$. Polynomial independence in $C[E_4,E_6]$ proves that these monomials form a basis. Multiplication by the nonzero Δ identifies cusp forms of weight 38 with weight-26 forms, whose equation $4a+6b=26$ has solutions $(a,b)=(5,1),(2,3)$. The structural description of the modular-form ring and its cusp ideal are classical inputs, while the dimension calculation is elementary. For the symmetric square, $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=a_p^2-2p$. Adding the middle eigenvalue p gives sum a_p^2-p. The sum of pairwise products is $p(\alpha^2+\beta^2)+p^2=p(a_p^2-p)$, and the triple product is p^3. Expanding the three linear factors gives the stated polynomial. The determinant p is essential: substituting an unrelated prime-power ratio destroys this derivation.

Worked example

Check an Eisenstein Hecke relation and an elliptic symmetric-square factor.

The normalized nonconstant coefficients of E_4 are $\sigma_3(n)$. At $p=2$, $\sigma_3(2)=9$ and $\sigma_3(4)=73=9^2-2^3$. The constant coefficient of E_4 is also compatible with $T_2E_4=9E_4$ under the stated weight-four convention. For $p=2$ and $a_p=-2$, the symmetric-square denominator becomes $1-2t+4t^2-8t^3$. For eta, its defining product gives $q\,d(\log\eta)/dq=1/24-\Sigma_{(n\geq1)}\sigma_1(n)q^{\wedge}n=E_2/24$. The translation phase of an eta quotient must still be tested; a q-series with appealing coefficients is not automatically invariant under $\tau\mapsto\tau+1$.

Small exact coefficient tests make the hypotheses auditable, while the theorem of transformation or modularity remains a separate structural obligation.

Continuity across the source years

2024 — The exact Hecke formula, full-level weight-38 dimension and eta normalization make distinct modular invariants calculable. Sources: 2024:Y24-06-008, 2024:Y24-09-025, 2024:Y24-12-020

2025 — Transformation phases, modular-symbol boundaries and local eigenvalue products impose additional consistency tests. The later Hecke critique explains why an arbitrary prime-index statistic cannot be substituted for those invariants. Sources: 2025:Y25-03-033, 2025:Y25-07-001, 2025:Y25-09-007, 2025:Y25-11-012

Connections

D02 — Elliptic local arithmetic. A point-count trace fixes a_p and determinant p, supplying the exact local quadratic factor that feeds the symmetric square. A local factor alone does not prove modularity, analytic rank or BSD.

D04 — Boundary algebra. The modular-symbol relation {a,b}+{b,c}={a,c} is realized by [b]−[a], a degree-zero cusp boundary. The elementary relation is a precursor to relative homology and periods, not the whole theory.

D07 — Generating functions. Eta logarithmic derivatives and Eisenstein Dirichlet series connect products to coefficient convolutions in their convergence domains. Analytic continuation and modular transformation require their own theorems.

Capacity and interactive scope

Lab ids: []

Inputs: Declared group and weight, exact q-expansion coefficients, good prime p and genuine local trace a_p.

Outputs: Dimension and monomial checks, Hecke coefficient residuals, eta phases and local symmetric-square factors.

Validation: Check transformation conventions and coefficient normalization before applying recurrences; distinguish finite coefficient consistency from a universal modularity proof.

Status: Mathematical recipe; no general solver implemented

Limits

The sources do not support a full four-year continuous modular-form project. This family records the supported 2024–2025 treatments. Full-level ring formulas cannot be applied unchanged to arbitrary level or character. A finite q-expansion match requires a proved comparison bound before it identifies two forms.

Next questions

For a proposed eta quotient, do its translation and inversion laws produce a genuine modular form on a declared subgroup with a declared multiplier?

Can each proposed arithmetic statistic pass the exact p^2 Hecke recurrence and a finite certified comparison bound after all normalization choices are stated?

Source anchors: 2024:Y24-06-008, 2024:Y24-09-025, 2024:Y24-12-020, 2025:Y25-03-033, 2025:Y25-07-001, 2025:Y25-09-007, 2025:Y25-11-012

Milne Modular Functions and Modular Forms — https://www.jmilne.org/math/CourseNotes/MF.pdf
NIST DLMF eta definitions — https://dlmf.nist.gov/23.15
NIST DLMF modular transformations — https://dlmf.nist.gov/23.18
Milne, Elliptic Curves — https://www.jmilne.org/math/Books/EC2.pdf
Deligne Formes modulaires et représentations l-adiques — https://www.numdam.org/item/SB_1968-1969__11__139_0/
NIST DLMF divisor sums — https://dlmf.nist.gov/27.4

D01-F07 · Euler products, prime sums and the analytic burden of identification

How do multiplicative identities become analytic formulas without losing signs, prime powers, convergence or main terms?

The divisor algebra gives formal identities; absolute convergence permits their interpretation as analytic functions. This transition is where several recovered proposals needed repair. Pointwise μΛ is a negative prime indicator weighted by logarithm, while convolution μ*log equals the positive prime-power function Λ. A prime-ideal sum changes again when norms and splitting types enter. An Euler product in Re(s)>1 is nonvanishing there, but this fact does not settle zeros after analytic continuation. Likewise π(x) and x/log x are not interchangeable at square-root error scale: the omitted x/log²x term is much larger. The family organizes these issues around exact operation and domain. Spectral determinants, sampled zeta layouts and prime oscillatory tones can be valid mathematical objects while their proposed identification with arithmetic zeros remains an additional, explicitly missing theorem.

Definitions

The von Mangoldt function Λ(n) is log p when n=p^k, k≥1, and zero otherwise. The Möbius function μ vanishes on integers with a squared prime factor.

A Dirichlet series is absolutely convergent where Σ|a_n|n^(−σ) is finite; only in a justified domain may one freely multiply, rearrange or differentiate the relevant series.

The arithmetic prime sums θ(x)=Σ_(p≤x)log p and ψ(x)=Σ_(p^k≤x)log p differ by prime-power contributions. A spectral zeta requires its own operator, eigenvalue domain and summability.

Theorem and hypotheses

The exact convolution identity μ*log=Λ gives −ζ′(s)/ζ(s)=ΣΛ(n)n^(−s) for Re(s)>1. Also Σφ(n)n^(−s)=ζ(s−1)/ζ(s) for Re(s)>2, and for real σ>2 its tail after N is at most N^(2−σ)/(σ−2).

Proof

For n=∏p_i^a_i, log n=Σ_(d|n)Λ(d), since the prime powers dividing n contribute Σ_i a_i log p_i. Convolution with μ, using μ*1=ε, gives μ*log=Λ. Absolute convergence in σ>1 justifies differentiating the Euler product and summing its geometric prime-power expansions. By contrast μ(n)Λ(n) is −log p at n=p and zero elsewhere, because μ(p^k)=0 for k≥2; its sum is −θ(x), not ψ(x). For the totient identity, n=Σ_(d|n)φ(d) means id=1*φ. Multiplication of absolutely convergent Dirichlet series in σ>2 gives ζ(s−1)=ζ(s)Σφ(n)n^(−s); the Euler product ensures ζ(s)≠0 there. Since 0≤φ(n)≤n, the remaining tail is bounded by Σ_(n>N)n^(1−σ)≤∫_N^∞x^(1−σ)dx. This explicitly states both the stronger half-plane required by φ(n) and an error certificate for truncation.

Worked example

Compare the two operations at n=12 and bound a real totient series.

The divisors of 12 with nonzero μ are 1,2,3,6. Thus (μ*log)(12)=log 12−log 6−log 4+log 2=0, agreeing with Λ(12)=0. At n=8 the convolution is log 8−log 4=log 2, while μ(8)Λ(8)=0. At n=3 the latter product is −log 3. For σ=4 and N=100, the totient tail is at most 100^(−2)/2=1/20000. This is a deterministic truncation error, not an empirical convergence observation or a zero-location theorem.

A single symbol change between multiplication and convolution changes the arithmetic object; a single half-plane change can invalidate an analytic manipulation.

Continuity across the source years

2024 — The strongest supported early thread is operation and analytic-domain repair: signs, convolution, prime powers, Euler tails and the correct prime-counting main term. Sources: 2024:Y24-09-018, 2024:Y24-10-009, 2024:Y24-10-010, 2024:Y24-10-013, 2024:Y24-11-004

2025 — Valuation-shell integrals and divisor convolution give exact analytic models. Prime oscillatory sums retain valid finite calculus without gaining a zero correspondence merely from their construction. Sources: 2025:Y25-05-010, 2025:Y25-05-020, 2025:Y25-07-017, 2025:Y25-10-022

2026 — The later sources distinguish sampled layouts from invariants and give a totient-series tail on its correct domain. The cyclotomic Mellin kernel supplies an exact special identity, not a universal spectral realization. Sources: 2026:Y26-04-026, 2026:Y26-06-024, 2026:Y26-10-033

Connections

D07 — Transform and convergence. Mellin transforms, logarithmic derivatives and Euler products connect coefficient algebra to analytic functions once a common absolute-convergence domain is proved. Continuation outside that domain needs a separate theorem.

D14 — Arithmetic correspondence burden. A claimed spectral realization must state an operator, domain, determinant regularization and an equality to the normalized L-function. Finite spectral data or finitely many zero checks cannot supply that global equality.

D11 — Sampling and invariance. A bijective zeta sampling layout preserves the sampled multiset but may change adjacency statistics. A visualization statistic must specify its dependence on ordering and truncation.

Capacity and interactive scope

Lab ids: []

Inputs: Cutoff N, real σ in the declared half-plane, explicit arithmetic coefficient function and operation type.

Outputs: Finite sums, rigorous tail bounds, prime-power contributions and a table distinguishing pointwise from convolution operations.

Validation: Check exact divisor identities first; attach analytic domain and explicit tail before reporting a limit.

Status: Mathematical recipe; no general solver implemented

Limits

The worked integer checks certify arithmetic identities, while the convergence proof supplies infinite validity. No Riemann hypothesis, BSD conclusion or arithmetic-spectral identification is inferred. Floating-point samples on a line do not certify unsampled zeros or a global distribution.

Next questions

Can every proposed spectral equality state its exact normalization, analytic domain and continuation theorem, rather than only matching selected terms?

For a chosen arithmetic transform, what explicit tail bound is strong enough to support the claimed precision uniformly across its parameter region?

Source anchors: 2024:Y24-09-018, 2024:Y24-10-009, 2024:Y24-10-010, 2024:Y24-10-013, 2024:Y24-11-004, 2025:Y25-10-022, 2026:Y26-06-024

NIST DLMF, Euler products and Dirichlet series — https://dlmf.nist.gov/27.4
J. S. Milne, Algebraic Number Theory — https://www.jmilne.org/math/CourseNotes/ANT.pdf
NIST DLMF, Asymptotic formulas for primes — https://dlmf.nist.gov/27.12
NIST DLMF Euler products — https://dlmf.nist.gov/25.2
NIST DLMF Riemann zeta zeros — https://dlmf.nist.gov/25.10
Igusa, theory of local zeta functions — https://doi.org/10.1090/amsip/014

D02 · Algebraic and arithmetic geometry

Organizing question

Which geometric model, maps and comparison data make a claim about arithmetic, singularities or cohomology precise enough to prove?

This domain is organized by the object to which a statement applies. An affine equation, its singular curve, the normalized function-field model and the smooth compact curve answer different questions. A point count, a rational group, a divisor class and a cohomology realization likewise retain different information. The best revision of the earlier questions is to specify that object first, then ask for a map or invariant with hypotheses that can be checked. This replaces a broad request for additional geometric structure with a sequence of concrete mathematical tasks.

The curve family begins with maps. Rational parametrization includes an exceptional point and an inverse domain. The cusp map $t \mapsto (t^2, t^3)$ is injective but has zero derivative at the origin; its normalization ring and local intersection multiplicity expose what a set-theoretic description misses. The 2026 corner cubic adds an exact birational coordinate change, while the superelliptic family computes genus only after finite and infinite ramification are counted on the smooth compact normalization. These examples provide reusable questions for any proposed parameter map: where is it defined, where does rank drop, what happens at infinity, and which inverse identity is actually proved?

Divisors and positivity organize the 2023 arithmetic-quotient cluster. Neatness addresses stabilizers in the open quotient; the cusp coordinate fills a puncture; a regular fan addresses local boundary smoothness; the canonical or log-canonical divisor addresses growth of sections; a very ample line bundle addresses an embedding. The modular curves X(3) and X(7) make ordinary and logarithmic general type visibly different. Riemann–Roch and Serre duality then give a precise degree threshold for a curve embedding. Picard calculations on $P^1 \times P^1$ and invariant divisor subspaces provide finite exact models of the same language. Higher-dimensional claims remain bounded by the actual compactification and positivity data supplied.

Elliptic arithmetic is the strongest local-to-global example. The discriminant-37 curve recurs in the 2024 and 2026 treatments with exact finite counts and rational group computations. Two good reductions exclude every possible rational torsion prime; the nonzero point then proves a rank lower bound. Heights turn independence into a Gram certificate, but a subgroup regulator differs by an index square from the full regulator. Selmer groups add hidden torsors and therefore supply information different from an explicit generator list. The interactive capacity here is specified as a recipe for exact point and group checks; no general certified rank or integral-point engine is claimed.

The cohomological family supplies a general caution about forgetting structure. An Euler realization can erase nonzero groups or take negative values. Rational monodromy ranks can erase integral torsion. A nilpotent operator determines a weight filtration but does not by itself produce a limiting mixed Hodge structure. Étale, crystalline, de Rham and Betti comparisons require compatible bases, coefficients and geometric hypotheses; a false universal motivic–Betti isomorphism already fails at a point. These are constructive constraints: they tell a future interface which fields and degrees to require and which counterexamples to expose. Together the four families form an atlas of models, maps and information loss, with each purported cross-domain identification stated as a theorem still requiring its own certificate.

Core objects

Affine curves, normalizations and projective models

Divisors, line bundles and spaces of sections

Modular curves, cusps and toric boundary charts

Elliptic groups, Frobenius factors and height lattices

Cohomological realizations and monodromy filtrations

Open frontier

The next substantial work would supply explicit global compactification data in a fixed higher-dimensional example, certified descent and saturation for the elliptic subgroup, and correctly graded functorial comparison maps. A numerical invariant can assist those tasks but cannot replace the relevant geometry or cohomology.

D02-F01 · Algebraic maps, singularities and the choice of smooth model

Which geometric information is preserved by a parametrization, a birational coordinate change or a normalization?

An equation is not yet a complete specification of the geometric object being studied. An affine curve may be singular, its normalization smooth, and its projective closure may acquire additional points at infinity. A parametrization can be injective and still lose differential information. Conversely, an explicit rational change of coordinate can identify function fields even when the displayed equations look unrelated. This family builds a hierarchy of certificates: substitution proves that a map lands on the curve; an inverse on a dense open proves birationality; an integral-closure argument proves normalization; Jacobian or local-ring calculations detect singularities; ramification on the smooth compact model determines genus. Cross-ratio symmetry supplies a finite quotient of ordered configurations, but singular special values and ramification must be handled before treating a parameter as a moduli coordinate.

Definitions

A normalization replaces an integral curve by the curve whose coordinate ring is integrally closed in its function field. A birational map is an isomorphism of function fields, not necessarily an everywhere-defined isomorphism.

For a plane curve $F=0$ in characteristic zero, a point with $F=F_x=F_y=0$ is singular. Local intersection multiplicity measures the length of the corresponding local quotient when finite.

For squarefree P of degree m and $d\geq 2$, the superelliptic curve means the smooth projective normalization of $y^d=P(x)$, with $\delta=\gcd(d,m)$ points over infinity.

Theorem and hypotheses

The cusp $y^2=x^3$ is normalized by $t\mapsto(t^2,t^3)$, and its intersection multiplicity with $y=0$ at the origin is three. For squarefree P of degree m, the smooth projective model of $y^d=P(x)$ has genus $g=((d-1)(m-1)-\gcd(d,m)+1)/2$.

Proof

The ring $A=C[t^2,t^3]$ has fraction field $C(t)$, since $t=t^3/t^2$, and t is integral over A because it satisfies $T^2-t^2=0$. $C[t]$ is integrally closed and finite over A, so it is the normalization. The cusp is singular at the origin because the two partial derivatives of y^2-x^3 vanish there; the parametrization derivative is also zero there, explaining why injectivity does not imply an immersion. Quotienting the local equation by y gives $C[x]_(x)/(x^3)$, whose length is three. In the blowup chart $y=ux$, the total equation is $x^2(u^2-x)=0$ and the strict transform $u^2-x=0$ is smooth. For the superelliptic genus, each of the m simple finite zeros has ramification index d, contributing $d-1$. At infinity there are δ points, each of index d/δ, contributing $d-\delta$ in total. Riemann–Hurwitz gives $2g-2=-2d+m(d-1)+(d-\delta)$. Rearrangement yields the stated formula. Both the squarefree assumption and the use of the smooth compact normalization are essential.

Worked example

Compare a cusp, two compact superelliptic models and an exact corner-coordinate substitution.

For $y^2=x^3-x$, $d=2,m=3,\delta=1$ gives genus one. For $y^3=x^4-1$, $d=3,m=4,\delta=1$ gives genus three; its compact surface therefore has Euler characteristic -4. The rational function $R(z)=(1-z)^3/z^2$ on P^1 has divisor $3[1]-2[0]-[\infty]$. Setting $u=(z-1)/z$ transforms $R(z)=1$ into $u^3-u+1=0$, with inverse $z=1/(1-u)$. The equation excludes $z=0,1$ and $u=0,1$, so the denominators introduce no missing or extraneous solutions. A matching discriminant would not have proved this coordinate identification; the forward and inverse substitutions do.

Each model supports a different valid question: singular local structure, smooth genus, ordered branch data or arithmetic field identity. The chosen object must remain explicit throughout.

Continuity across the source years

2023 — Rational parametrization and the ordered cross-ratio provide early concrete maps with domains and exceptional points already visible. Sources: 2023:M098, 2023:M102

2024 — Rational-coordinate and cusp treatments separate set-theoretic parametrization, ambient dimension and differential regularity. Sources: 2024:Y24-06-004, 2024:Y24-10-019

2025 — Jacobian algebras quantify isolated hypersurface singularities, while the Legendre j-map makes finite cross-ratio collisions explicit. These are related geometric tools rather than a claimed single historical construction. Sources: 2025:Y25-04-009, 2025:Y25-09-011

2026 — The later cusp blowup, birational corner coordinate and superelliptic ramification calculations supply increasingly complete map-and-model certificates. Sources: 2026:Y26-05-031, 2026:Y26-06-010, 2026:Y26-10-008

Connections

D06 — Branch monodromy and curvature. A simple branch point has local coordinate $x-a=t^d$, and analytic continuation multiplies y by $\exp(2\pi i/d)$; Gauss–Bonnet converts the compact genus to total curvature. Local covering monodromy is distinct from an arbitrary connection holonomy.

D01 — Exceptional units and fields. The Möbius corner substitution identifies the cubic fields, while the divisor of R records zeros and poles exactly. Neither an approximate root match nor a common discriminant supplies the rational map.

D14 — Map verification. Substitution, inverse-domain checks, Jacobian rank and local quotient length form complementary certificates. A finite sample of mapped points proves none of the universal map properties.

Capacity and interactive scope

Lab ids: []

Inputs: An explicit polynomial curve, a rational map with denominators, and a declared affine, normalized or compact model.

Outputs: Exact substitution identities, excluded loci, singular points, local intersection length and genus under verified squarefree hypotheses.

Validation: Verify forward and inverse maps on their stated domains; count ramification at infinity as well as at finite roots.

Status: Mathematical recipe; no general solver implemented

Limits

The genus formula does not apply unchanged to repeated roots, positive characteristic dividing d or an unreduced affine equation. A normalization is not automatically an embedding, and a blowup chart does not by itself prove global resolution. General singularity classification requires more than a handful of invariants.

Next questions

For a proposed parameter family, where do repeated roots or denominator poles change the normalization and branch profile?

Can every claimed moduli identification exhibit its exact finite symmetry group and distinguish generic fibres from special ramified fibres?

Source anchors: 2023:M098, 2023:M102, 2024:Y24-10-019, 2025:Y25-04-009, 2025:Y25-09-011, 2026:Y26-05-031, 2026:Y26-06-010, 2026:Y26-10-008

NIST Digital Library of Mathematical Functions — https://dlmf.nist.gov/
Milnor, Singular Points of Complex Hypersurfaces — https://www.jstor.org/stable/j.ctt1bd6kvv
Milne Elliptic Curves, author text — https://www.jmilne.org/math/Books/EC2.html
The Stacks Project, computing intersection multiplicities — https://stacks.math.columbia.edu/tag/0B01
J. S. Milne, Fields and Galois Theory — https://www.jmilne.org/math/CourseNotes/FT.pdf
Don Zagier, The Dilogarithm Function — https://www.maths.dur.ac.uk/users/herbert.gangl/zagier_dilog.pdf

D02-F03 · Divisors, compactification and projective positivity

What separate certificates establish a smooth quotient, a well-behaved boundary, positivity and an actual projective embedding?

The early arithmetic-geometry material becomes coherent when organized around distinct geometric properties. Neatness removes finite stabilizer phenomena in an open arithmetic quotient. A compactification adds boundary strata. Regular cones make local toric charts smooth. A positive canonical or log-canonical divisor controls a birational growth property. A sufficiently positive line bundle supplies sections that define an embedding. None of these steps is interchangeable with the others. Modular curves make this hierarchy unusually explicit: X(3) is a smooth genus-zero compact curve, while its four-punctured interior is hyperbolic and of log-general type. Divisor intersection on $P^1 \times P^1$ then gives a second exact arena for positivity and invariant classes. The family's strongest result is a curve embedding criterion whose proof states exactly why enough sections separate both points and tangent directions.

Definitions

For a smooth compact curve X of genus g, deg K_X=2g−2; with c reduced boundary points D, deg(K_X+D)=2g−2+c.

A line bundle L is very ample when its complete linear system gives a closed projective embedding. On a smooth curve, separation of every length-two subscheme is the relevant section criterion.

A two-dimensional toric cone with primitive generators u,v is regular when |det(u,v)|=1. Smoothness is local lattice regularity; projectivity additionally requires a compatible support function.

Theorem and hypotheses

For a smooth projective curve of genus g and a line bundle L of degree d≥2g+1, $h^0(L)=d+1-g$ and the complete linear system embeds X into P^(d−g). On the principal modular curve X(3), deg K=−2 while deg(K+D)=2; at level seven these degrees are four and 28.

Proof

For any length-two subscheme Z, Serre duality gives $H^1(L(-Z))$ dual to $H^0(K_X\otimes L^{-1}(Z))$. Its degree is 2g−2−d+2≤−1, so it has no nonzero section. The restriction sequence therefore makes $H^0(L)\to H^0(L|_Z)$ surjective. Distinct points and doubled points give point separation and tangent separation; the length-one version also gives base-point freeness. The standard embedding criterion then proves very ampleness. Vanishing of $H^1(L)$, followed by Riemann–Roch, yields $h^0(L)=d+1-g$. For a torsion-free effective modular group of index μ and c cusps, Gauss–Bonnet gives 2g−2+c=μ/6. Principal level N has c=μ/N. At N=3, μ=12,c=4,g=0; at N=7, μ=168,c=24,g=3. The degree formulas then distinguish ordinary from logarithmic general type. These arguments use the declared smooth projective curve and classical modular-curve inputs; they do not algebraize an arbitrary analytic space.

Worked example

Compare a toric boundary chart, a divisor lattice and two projective embeddings.

The cone generated by (1,0),(1,2) has determinant two and a singular toric chart. Adding (1,1) gives determinant-one adjacent cones, producing a smooth local toric modification. On $P^1\times P^1$, classes (a,b) have intersection pairing (a,b)·(c,d)=ad+bc and self-intersection 2ab; for a,b≥0, $h^0(O(a,b))=(a+1)(b+1)$. Swapping the factors fixes the line a=b in the real divisor space. For g=1,d=3, the embedding theorem gives three sections and a plane cubic. For g=3,d=7, it gives five sections and an embedding into P^4.

Boundary regularity, divisor intersections and sections now have computable meanings rather than being treated as interchangeable claims of geometric richness.

Continuity across the source years

2023 — The foundational cluster already separates arithmetic quotient smoothness, compactification charts, canonical positivity, embedding sections and adjunction dimensions. The atlas links these explicit treatments without upgrading their higher-dimensional scope. Sources: 2023:M027, 2023:M028, 2023:M029, 2023:M030, 2023:M031, 2023:M096

2024 — Picard and Néron–Severi calculations, invariant divisor classes and a finite-index modular quotient give complementary algebraic descriptions of the same questions of class, quotient and positivity. Sources: 2024:Y24-03-001, 2024:Y24-03-002, 2024:Y24-08-006

2025 — The cusp area example makes finite volume distinct from compactness, while the product abelian-surface locus fixes a specific moduli discriminant. Neither is claimed as a general compactification theorem. Sources: 2025:Y25-02-025, 2025:Y25-11-009

Connections

D06 — Quotient geometry. Hyperbolic area and cusp widths determine genus in the modular-curve case; $q=\exp(2\pi i\tau/N)$ fills each puncture. Finite area does not imply compactness, and neatness does not choose a higher-dimensional smooth fan.

D03 — Invariant divisor spaces. Averaging a finite group action projects the real Picard class space to its invariants; factor swap leaves (1,1). Fixed-space linear algebra does not classify the whole Picard group or its torsion.

D13 — Fan combinatorics. Determinants of adjacent primitive rays certify regularity of a two-dimensional subdivision. Global projectivity and arithmetic admissibility remain additional conditions.

Capacity and interactive scope

Lab ids: []

Inputs: Genus g, degree d, boundary count c; or lattice rays and a specified divisor-class basis.

Outputs: Canonical and log-canonical degrees, section counts, projective target dimension, intersection values and cone determinants.

Validation: Check smoothness assumptions, integral lattice primitiveness, cohomology-vanishing hypotheses and exact effective quotient indices.

Status: Mathematical recipe; no general solver implemented

Limits

The degree criterion is sufficient, not necessary. A projective curve is an input to its proof, so the theorem cannot establish algebraization from scratch. A local toric subdivision is not a proof of a global arithmetic compactification. Higher-dimensional general type needs explicit boundary and ramification contributions.

Next questions

For a specified higher-dimensional arithmetic quotient, what canonical-divisor formula and admissible fan give simultaneous smoothness, projectivity and a proved positivity bound?

Can an intended analytic compactification exhibit a concrete ample bundle and enough sections to verify its claimed algebraization and embedding?

Source anchors: 2023:M027, 2023:M028, 2023:M029, 2023:M030, 2023:M031, 2023:M096, 2024:Y24-03-001, 2024:Y24-08-006

J S Milne Introduction to Shimura Varieties — https://www.jmilne.org/math/articles/2005aX.pdf
J S Milne Toroidal Compactification of Shimura Varieties — https://www.jmilne.org/math/articles/1990aT.pdf
Stacks project algebraic curves — https://stacks.math.columbia.edu/download/curves.pdf
Stacks project very ample line bundles on curves — https://stacks.math.columbia.edu/tag/0H2V
Ravi Vakil Foundations of Algebraic Geometry adjunction — https://math.stanford.edu/~vakil/216blog/FOAGjul2724public.pdf
Eisenbud and Harris 3264 and All That — https://dash.harvard.edu/bitstreams/7312037e-7c66-6bd4-e053-0100007fdf3b/download

D02-F02 · Elliptic arithmetic: local counts, global points and height lattices

How far can exact local and group-law certificates take us toward rational rank, and where do descent and global conjectures enter?

Elliptic arithmetic joins several structures whose numerical outputs must not be conflated. A point count over a finite field gives a Frobenius trace. A rational group-law calculation gives an exact point and its multiples. Good reduction can rule out rational torsion, producing a rank lower bound from one nonzero

point. A height pairing then certifies independence and measures the covolume of a chosen subgroup, while a Selmer computation can give an upper bound with an additional Tate–Shafarevich contribution. Each is valuable on its own. The most coherent cross-year example is $E: y^2+y=x^3-x$, of discriminant 37: exact finite counts in 2024 recur as group and rank certificates in 2026. That continuity supports a reusable arithmetic laboratory recipe, while the stronger assertions of exact rank, BSD or complete integral points need separate proofs.

Definitions

For a good prime p, $a_p=p+1-\#E(F_p)$ and the local polynomial is $1-a_p\,t+pt^2$. Good reduction means the discriminant is nonzero modulo p.

For $E: y^2+y=x^3-x$, negation is $(x,y)\mapsto(x,-y-1)$. Rational points form an abelian group with identity O.

The canonical height pairing is positive definite on $E(Q)$ modulo torsion. A Gram determinant for a subgroup of index I is I^2 times the full-lattice regulator.

Theorem and hypotheses

For $E: y^2+y=x^3-x$, the rational torsion subgroup is trivial and $P=(0,0)$ has infinite order, so rank $E(Q)\geq 1$. The exact point counts $\#E(F_5)=8$ and $\#E(F_7)=9$ suffice, provided the prime-to-p restriction on torsion reduction is respected.

Proof

The discriminant is 37, so both primes are good. Suppose there is a rational torsion point of prime order ℓ. If $\ell\neq 5$, injection of ℓ-torsion under reduction at 5 forces $\ell\mid 8$ and thus $\ell=2$. Reduction at 7 then forces $2\mid 9$, impossible. If $\ell=5$, reduction at 7 instead forces $5\mid 9$, also impossible. Every nontrivial finite abelian torsion group contains an element of prime order, so all rational torsion is excluded. The point $(0,0)$ is nonzero and lies on E, hence has infinite order. This proves a rank lower bound without computing an L-function. For heights, bilinearity gives $G'=M^{T}GM$ under a generator-coordinate matrix M and therefore $\det G'=(\det M)^2 \det G$. A positive Gram determinant proves independence of the displayed points but says nothing about saturation unless the index is certified. The exact Selmer sequence adds the separate quotient $Sha[n]$ to $E(Q)/nE(Q)$; a Selmer cardinality is therefore not automatically a count of visible rational generators.

Worked example

Count the two finite reductions and verify exact rational multiples.

Enumerating $y^2+y=x^3-x$ over F_5 gives seven affine points plus O, while over F_7 it gives eight affine points plus O. At $P=(0,0)$, the tangent slope is $(3x^2-1)/(2y+1)=-1$, yielding $2P=(1,0)$. The line through P and $2P$ has slope zero, yielding $3P=(-1,-1)$. Exact addition gives $7P=(-5/9,8/27)$, where both sides of the equation are $280/729$. Independently, for a short Weierstrass curve $y^2=x^3+ax+b$, clearing the denominator in the doubling formula gives a quartic condition on the x-coordinate of a half-point; every candidate must still satisfy the original curve and group equation.

The finite reductions establish the global non-torsion result through a stated theorem; the visible multiples verify the arithmetic implementation but are not themselves the proof of infinite order.

Continuity across the source years

2024 — Finite-field zeta factors, the discriminant-37 counts, a non-torsion proof and height-lattice interpretation establish four distinct arithmetic certificates. Sources: 2024:Y24-05-005, 2024:Y24-09-012, 2024:Y24-09-013, 2024:Y24-12-025

2025 — Weierstrass invariants, true local traces, independence and Selmer bounds sharpen the distinction between a displayed subgroup and the full arithmetic object. Sources: 2025:Y25-07-003, 2025:Y25-07-005, 2025:Y25-09-006, 2025:Y25-10-007, 2025:Y25-10-008

2026 — The later sources make hidden torsors, CM local traces, higher-genus Frobenius reconstruction and multiplication fibres explicit. The same discriminant-37 curve supplies a supported return to the earlier rank lower bound. Sources: 2026:Y26-01-031, 2026:Y26-01-032, 2026:Y26-02-028, 2026:Y26-07-025, 2026:Y26-10-037

Connections

D01 — Local arithmetic and Euler products. Finite counts determine a_p and the determinant p, supplying local factors and the input to Hecke and symmetric-square identities. A list of factors does not alone prove analytic continuation or BSD.

D05 — Torsors and descent. The exact Selmer sequence separates E(Q)/nE(Q) from Sha[n]. A cohomological upper bound is not automatically a constructive list of rational points.

D12 — Lattice geometry. A positive height Gram matrix certifies independence and supports metric optimization on the free quotient. Torsion is collapsed, and index or saturation must be certified before claiming the full regulator.

Capacity and interactive scope

Lab ids: []

Inputs: Exact Weierstrass coefficients, a good prime, rational points and optional certified height-pairing data.

Outputs: Point counts, Frobenius traces, exact group-law residuals, torsion exclusions and basis-dependent Gram determinants.

Validation: Count every finite-field point including infinity; check good reduction; respect prime-to-p injection; substitute every rational output exactly.

Status: Mathematical recipe; no general solver implemented

Limits

No general elliptic solver or certified height engine is implemented here. The rank-one lower bound is not an upper bound, and an analytic determinant analogy is not BSD. Halving quartics yield candidates requiring sign and exceptional-case checks. Genus-two counts require their own Frobenius degree and cannot use an elliptic quadratic factor.

Next questions

Can a documented descent give an independent upper bound matching the exhibited non-torsion subgroup, with saturation certified?

For multiplication fibres, can the field of definition, rational torsion kernel and all exceptional doubling cases be incorporated into a complete exact candidate procedure?

Source anchors: 2024:Y24-09-012, 2024:Y24-09-013, 2024:Y24-12-025, 2025:Y25-07-003, 2025:Y25-07-005, 2025:Y25-10-007, 2026:Y26-07-025, 2026:Y26-10-037

LMFDB 37.a1 record (access challenge encountered; calculations below are independent) — https://www.lmfdb.org/EllipticCurve/Q/37/a/1

J. S. Milne, Elliptic Curves — https://www.jmilne.org/math/Books/EC2.pdf

MIT elliptic curves notes — https://math.mit.edu/~drew/18.783/

Bernstein and Lange: explicit Weierstrass formulas — https://www.hyperelliptic.org/EFD/g1p/auto-shortw.html

J S Milne Lectures on Etale Cohomology — https://www.jmilne.org/math/CourseNotes/LEC210.pdf
Harris–Shepherd-Barron–Taylor, A family of Calabi–Yau varieties and potential automorphy — https://annals.math.princeton.edu/2010/171-2/p04

D02-F04 · Cohomological realizations, Hodge filtrations and comparison types

Which information survives a realization, and what additional structure is required before two cohomology theories can be compared?

Cohomology provides several forms of organized information: dimensions, graded pieces, mixed structures, integral torsion, Frobenius action and functorial maps. A numerical realization may preserve addition and multiplication while forgetting almost everything else. This is why an Euler characteristic can be zero for a nontrivial elliptic curve, and why multiplying it by Shannon entropy cannot automatically become a positive complexity index. The same caution governs comparison theorems: a characteristic-p variety has no complex analytic space until appropriate characteristic-zero data have been supplied, and a motivic chain index is not a singular cohomology degree. A finite monodromy model makes the loss of information especially concrete. Rational ranks can remain unchanged while integer coinvariants gain torsion; the nilpotent operator determines a weight filtration but not an entire limiting mixed Hodge structure.

Definitions

The Grothendieck ring of complex varieties imposes [X]=[Z]+[X minus Z] for closed Z and multiplication by Cartesian product. A motivic realization is a declared ring homomorphism out of that ring.

Compact-support Euler characteristic is $\chi_c(X)=\Sigma_i(-1)^i \dim H_c^i(X)$; it is additive but need not be positive or separating.

A monodromy weight filtration centered at one obeys $N(W_k)\subseteq W_{(k-2)}$ and $N^j:Gr_{(1+j)}^W\to Gr_{(1-j)}^W$ isomorphically. A Hodge filtration and polarization are extra data.

Theorem and hypotheses

For $V=Qe_1\oplus Qe_2$ and $N(e_2)=e_1, N(e_1)=0$, the centered weight filtration is $W_k=0$ for $k<0$, $W_0=W_1=Qe_1$ and $W_k=V$ for $k\geq 2$. For integral monodromy $M_n=I+nN$, $n>0$, the coinvariants are $Z\oplus Z/nZ$, although all these rational models have the same ranks.

Proof

The image and kernel of N are both Qe_1. The proposed filtration lowers by two under N, and its only nonzero graded pieces are Gr_0 and Gr_2, both one-dimensional. N maps the class of e_2 to the class of e_1, an isomorphism, while all higher required isomorphisms concern zero spaces. Thus the defining conditions are satisfied. On the integer lattice, $(M_n-I)Z^2=nZe_1$, so the quotient is generated freely by e_2 and by e_1 modulo n. Tensoring with Q removes the latter torsion. For realization loss, the scissor relation $A^1=\{0\}\sqcup G_m$ gives $[G_m]=L-1$. Compact-support cohomology of G_m has one copy of the coefficient field in degrees one and two, so its Euler characteristic is zero despite nonzero cohomology. A projective genus-g curve has Betti numbers (1,2g,1) and Euler characteristic $2-2g$. Therefore $\chi_c(X)H(p)$ can vanish or be negative for a positive entropy H(p). Additivity alone gives no positivity or faithful geometric encoding.

Worked example

Test comparison types on P^1 and exhibit a failure of an unrestricted motivic–Betti isomorphism.

P^1 over F_5 has six points and its standard smooth lift has cohomological ranks (1,0,1). Frobenius acts by 1 in degree zero and 5 on the hyperplane class in degree two, giving trace count 1+5=6; over F_{25} it gives 1+25=26. This is a useful consistency test, not a proof of the general comparison theorem. At Spec C, the weight-one motivic group H^{1,1}=C× is nonzero, whereas singular H^1 of a point vanishes. Even after tensoring with C the source is nonzero, since log|·| maps C× onto R and R⊗_Z C maps nontrivially to C. The universal isomorphism claim therefore fails at a point.

Small examples discriminate a valid comparison with explicit coefficients and hypotheses from a false formula that merely passes a rank coincidence.

Continuity across the source years

2023 — The early Hodge treatment supplies concrete Betti and Hodge-number constraints for curves and $P^1 \times P^1$, making geometric compatibility a prerequisite for proposed tables. Sources: 2023:M097

2024 — Motivic scalars, finite-field trace counts and prismatic structure introduce distinct realization targets. Their codomains and additional data must be kept visible. Sources: 2024:Y24-07-011, 2024:Y24-08-009, 2024:Y24-10-024

2025 — Monodromy, localization, a point counterexample and typed étale–crystalline statements sharpen the information-loss and comparison questions. No unsupported 2026 continuation is added. Sources: 2025:Y25-01-001, 2025:Y25-01-019, 2025:Y25-06-010, 2025:Y25-06-013, 2025:Y25-08-907

Connections

D04 — Exact and derived structures. Localization and compact-support cohomology explain a subtraction in a Grothendieck group without implying a direct-sum decomposition of actual objects. Equality of virtual classes is weaker than an isomorphism of complexes or mixed Hodge structures.

D01 — Integral and p-adic precision. The integer coinvariant torsion and filtered Frobenius residual retain information erased by rational ranks or insufficient digits. Rational comparison cannot by itself recover integral torsion.

D11 — Information functionals. A scalar geometric realization can be multiplied by entropy only after a codomain is fixed and desired positivity and sensitivity are tested. The resulting authored functional is not an intrinsic nonnegative complexity measure.

Capacity and interactive scope

Lab ids: []

Inputs: Exact nilpotent matrix and integer lattice; or a variety, cohomology theory, coefficient ring, base and proposed comparison degree.

Outputs: Weight graded dimensions, integer coinvariants, Euler realizations and explicit type checks for comparison formulas.

Validation: Verify N-action on each graded piece; preserve torsion before tensoring; test proposed universal comparisons on a point and projective line.

Status: Mathematical recipe; no general solver implemented

Limits

The finite matrix model is not a general degeneration or mixed-Hodge solver. A limiting Hodge filtration, polarization, geometric origin and admissibility remain extra requirements. Étale–crystalline comparison uses stated lifting or period-ring hypotheses; numerical agreement of a cellular test object is insufficient.

Next questions

Can a proposed comparison specify an actual functor and a correctly graded natural map before asking whether it is an isomorphism?

Which integral or filtered invariants must a proposed geometric complexity measure retain to distinguish objects with identical Euler characteristic?

Source anchors: 2023:M097, 2024:Y24-07-011, 2024:Y24-08-009, 2025:Y25-01-001, 2025:Y25-01-019, 2025:Y25-06-010, 2025:Y25-06-013

Stacks Project varieties and cohomology background — https://stacks.math.columbia.edu/
Deligne Théorie de Hodge II — https://www.numdam.org/item/PMIHES_1971__40__5_0/
Pierre Deligne Théorie de Hodge II — https://doi.org/10.1007/BF02684692
Stacks Project Grothendieck groups of varieties — https://stacks.math.columbia.edu
Mark Goresky Lecture notes on sheaves and perverse sheaves — https://arxiv.org/abs/2105.12045
Mazza, Voevodsky and Weibel, motivic cohomology — https://sites.math.rutgers.edu/~weibel/motiviclectures.html

D03 · Algebra and representation theory: structures, actions and exact certificates

Organizing question

What algebraic structure is specified, and which exact maps or identities justify the invariant being claimed?

Algebra provides the atlas with a language for separating a structure from its numerical shadow. A carrier with a multiplication law is more information than a cardinality; a representation is more information than a dimension; a spectral decomposition is more information than an eigenvalue list. Across the source years, the strongest developments begin when the multiplication, action, field and equivalence relation become explicit. The six families in this domain organise those developments around operations, linear certificates, symmetry, bracket structure, finite local models and operator states.

Typed operations come first because they determine which equations even make sense. Transporting a field through a bijection produces a valid isomorphic field, while merely changing notation or mixing transported and ordinary operations does not. Capped counters create finite monoids with an absorbing top layer and a minimal multiplication ideal; they do not automatically create a vector-space nullspace. Ring quotients, maximal ideals and squarefree branch decompositions then show how discarded information is certified by an actual homomorphism. The finite cases are useful because every product and every ideal condition can be replayed.

Linear structure supplies a second layer of certificates. Rank needs both upper and lower witnesses, singular values need declared metrics, and projectors need their idempotent and annihilation equations. A squarefree annihilating polynomial yields a decomposition only after checking the polynomial identity, and every advertised spectral point still needs nonzero support. These distinctions connect algebra to observation and stability throughout the atlas. They also prevent exact rational rank, integral torsion and floating effective rank from being treated as the same quantity.

Symmetry becomes productive through actual actions. Finite averaging gives invariant projectors; orbit counts give stabilizers; characters and branching specify which modules occur. The B_4 action over F_3 has a complete Hamming-weight classification, while a signed-permutation invariant quadratic form must be scalar. Neither result forces an empirical dataset rank or a unique frequency ratio on a multiplicity space. Exceptional symmetry is retained at its honest level: the E_8 root accounting and 120+128 decomposition are exact, but an exceptional bracket needs equivariant maps and Jacobi beyond a dimension count. Octonions sharpen this lesson by providing a nonassociative product with a sharp, convention-sensitive defect theorem.

Finite local and path algebras add the geometry of relations. The four-dimensional jet algebra needs cubic closure; its symmetric Hilbert function coexists with a two-dimensional socle. The rank-thirteen quiver pencil has a global path-bundle proof of flatness, while its full Hochschild profile remains another calculation. Finally, faithful finite operator states supply modular flow and KMS identities in a setting where all formulas can be derived directly. These finite models connect outward to categories, geometry, quantum theory and inference through explicit maps. The domain therefore aims at inspectable structure and proof, with larger classification and infinite analytic problems stated as precise frontiers rather than inferred from familiar names.

Core objects

Typed multiplication laws and quotients

Matrices, metrics and projectors

Group actions, characters and intertwiners

Octonion and graded brackets; exceptional roots

Finite local and quiver algebras

Faithful finite operator states

Open frontier

Recover the actual multiplication, representation and cochain matrices behind the larger historical objects; then test completeness of invariants, exceptional bracket identities and parameter-stratified derived invariants.

D03-F01 · Typed operations, quotients and finite algebraic structure

Which operations, equations and equivalences actually determine an algebra, and which information survives quotienting?

The algebraic backbone of the collection is the move from a named collection of symbols to a multiplication law whose identities can be proved. A field, a monoid, a ring and a partial resource algebra have different admissible operations. Their quotients therefore require different compatibility conditions. The useful revised question is not whether a construction looks algebraic, but whether its carrier, operations, units, domains of definition and notion of equality have all been specified. This viewpoint makes the number-system comparisons, transported operations, finite maximal ideals, capped trace monoids and squarefree branch decomposition parts of one coherent programme.

Transport of structure is the first constructive tool. Given a bijection, pulling all operations back creates an isomorphic object, so associativity and distributivity are inherited by proof. This gives a complete model of a coordinate deformation while explaining why it supplies no new intrinsic field invariant. Quotients then remove information in a controlled way: a ring congruence is determined by a two-sided ideal; a monoid quotient requires a multiplication-compatible equivalence; a partial algebra needs its composability relation retained. Approximate numerical closeness cannot silently serve as any of these equalities.

Finite models make the distinction executable. Enumerating ideals of a finite ring can produce maximal ideals and field quotients. Capping an additive counter produces an absorbing top layer, but does not turn its semigroup kernel into a linear nullspace. The latter distinction matters throughout the atlas: a minimal two-sided ideal, a kernel of a homomorphism and a vector-space kernel answer different questions. Algebra becomes useful for interactive reconstruction precisely when the requested invariant is named before its computation.

Definitions

For a unital ring R, a quotient R/I is defined by a two-sided ideal I; multiplication of classes is [a][b]=[ab].

For a finite monoid S, K(S) denotes its unique minimal nonempty two-sided ideal. It is not a matrix nullspace.

For $k \geq 1$, $L_k=\{0,\ldots,k\}$ has operation $a \oplus b=\min(a+b,k)$; k is absorbing.

For a bijection $\varphi:X\to Y$, transported multiplication is $x\star y=\varphi^{-1}(\varphi(x)\varphi(y))$, with all other required operations transported likewise.

Theorem and hypotheses

For a finite monoid S and positive integers $k_1,\dots,k_r$, $K(S\times\prod L_{k_i})=K(S)\times\{(k_1,\dots,k_r)\}$. Thus adding saturated counters preserves the cardinality of the minimal ideal, although it enlarges the monoid.

Proof

Let $P=S\times\prod L_{k_i}$. The displayed product J is a nonempty two-sided ideal: multiplication preserves $K(S)$, while every capped coordinate remains at its maximum. Let I be any nonempty two-sided ideal of P and choose $(s,a)\in I$. Multiplication by $(1,k_1,\dots,k_r)$ puts $(s,k_1,\dots,k_r)$ in I. The set of $x\in S$ for which $(x,k_1,\dots,k_r)\in I$ is a nonempty ideal of S. It contains $K(S)$, hence I contains J. Therefore J is the unique minimal ideal. For completeness, existence and uniqueness of $K(S)$ in a finite monoid follow by choosing a minimal nonempty ideal and observing that the product of any two nonempty ideals is nonempty and lies in their intersection. Minimality then forces the two chosen minimal ideals to agree. This proof concerns multiplication ideals only; no vector-space structure is used.

Worked example

Take $S=C_2=\{1,g\}$, $g^2=1$, and one counter L_2. Also compare the finite commutative ring $\mathbb{Z}/12\mathbb{Z}$.

The six monoid elements are (1,0),(g,0),(1,1),(g,1),(1,2),(g,2). Their product is (g^a,i)(g^b,j)=(g^{a+b mod2},min(i+j,2)). Exactly (1,0) and (1,2) are idempotent. The minimal ideal is $\{(1,2),(g,2)\}$, a copy of C_2. Exhaustive multiplication proves associativity on all 216 triples and finds no smaller nonempty ideal. In $\mathbb{Z}/12\mathbb{Z}$ the maximal ideals are $(2)=\{0,2,4,6,8,10\}$ and $(3)=\{0,3,6,9\}$; their quotients have two and three elements. These outputs distinguish minimal monoid ideals from maximal ring ideals rather than relying on the shared word ideal.

A small exact multiplication table can certify an invariant of the declared model. It cannot identify an unnamed historical transition object with that model without a multiplication-preserving map.

Continuity across the source years

2024 — Number systems and transported operations supply the operation-first theme; the historical topics do not themselves establish a new field. Sources: 2024:Y24-06-001, 2024:Y24-09-003, 2024:Y24-10-015, 2024:Y24-10-016

2025 — The capped monoid and partial-share models sharpen which products are total and which equalities are legitimate. Sources: 2025:Y25-12-020, 2025:Y25-05-014, 2025:Y25-09-013

2026 — Saturation, finite maximal ideals and quotient branch splitting give finite certificates of retained or discarded structure. Sources: 2026:Y26-01-001, 2026:Y26-05-007, 2026:Y26-10-011

Connections

D14 — Semantic quotients. A transition congruence must preserve all future outputs before a machine quotient represents behaviour. A multiplication table alone does not establish observational equivalence.

D01 — Chinese remainder decomposition. Coprime factors and Bézout identities construct orthogonal idempotents and branch inverses. Repeated factors require local or primary methods.

D04 — Universal quotient property. A homomorphism killing relations factors uniquely through their generated ideal. The category and type of ideal must be declared.

Capacity and interactive scope

Lab ids: []

Inputs: Finite multiplication table, distinguished identity, chosen capped-counter sizes or finite ring table.

Outputs: Associativity certificate, units, idempotents, relevant ideals and explicit quotient map.

Validation: Exhaustive finite products; ideal closure; no conflation of ring, semigroup and linear kernels.

Status: Worked mathematical recipe; no dedicated general algebra user interface is implemented.

Limits

No unspecified M29 table is reconstructed. Transport is an isomorphism construction. Enumeration scales exponentially, and finite maximal-ideal search proves no algorithmic result for arbitrary infinite rings.

Next questions

Given a recovered finite transition table, which multiplication congruence is also complete for its actual observations?

For a squarefree polynomial quotient, which sign or arithmetic admissibility conditions survive the algebraic product decomposition?

Source anchors: 2024:Y24-06-001, 2024:Y24-09-003, 2025:Y25-12-020, 2026:Y26-01-001, 2026:Y26-05-007, 2026:Y26-10-011

Etingof et al., Introduction to Representation Theory — https://math.mit.edu/~etingof/reprbook.pdf

D03-F02 · Exact matrix structure, metrics and spectral certificates

Which finite identities certify rank, spectrum and decomposition, and which metric is each certificate using?

Linear algebra is a common language across the source years, but a useful synthesis must preserve the question encoded by each matrix. Rank counts independent directions, singular values quantify amplification after metrics are fixed, Jordan structure measures nilpotent mixing, and a projector certifies a subspace only through its defining identities. A large Frobenius norm, a symmetry label and an eigenvalue list cannot replace one another. The revised framework starts with a domain, codomain and scalar field, then attaches the appropriate certificate.

The exact approach is particularly effective when a numerical claim can be reduced to polynomial identities. A verified factorization $A=UV$ supplies a rank upper bound; a nonzero minor supplies a lower bound. Reduction modulo a prime can certify that a rational minor is nonzero, while a rank drop modulo that prime can arise from arithmetic cancellation. A spectral polynomial $p(T)=0$ gives projectors only when its roots are distinct; actual nonzero support witnesses are still needed to certify that every advertised eigenvalue occurs. These are constructive, inspectable statements rather than a generic reliance on diagonalization software.

Metrics are part of this structure. A diagonalizable matrix with real spectrum can be self-adjoint in a suitable positive metric without being symmetric in the original coordinates. Conversely a nontrivial Jordan block cannot be made self-adjoint by a positive metric, because an isometric change to ordinary Euclidean coordinates would make it diagonalizable. The same rule governs anisotropy and whitening: coordinate changes preserve metric singular values only when both source and target metrics transform. This family supplies algebraic certificates which the inverse, control and statistical chapters can then use with their own observation and error models.

Definitions

A rank certificate pairs an exact rank-r factorization with a nonzero r×r minor.

A positive metric η defines ⟨x,y⟩_η=x*ηy; η-self-adjointness is A*η=ηA.

For commuting idempotents P_i, define E_ε=∏_i P_i^{ε_i}(I−P_i)^{1−ε_i}, ε∈{0,1}^m.

For a squarefree polynomial p(t)=∏_{λ∈S}(t−λ), Lagrange polynomials e_λ(t)=∏_{μ≠λ}(t−μ)/(λ−μ) define spectral projectors when p(T)=0.

Theorem and hypotheses

If P_1,...,P_m are commuting idempotents on a finite-dimensional vector space, the E_ε are pairwise annihilating idempotents summing to I, and im(E_ε) is the simultaneous eigenspace P_i v=ε_i v. Hence the nonzero blocks give an exact direct-sum decomposition.

Proof

All factors in E_ε commute, and each factor is idempotent, so E_ε²=E_ε. If ε≠δ, some index i has one product containing P_i and the other containing I−P_i. Their product is zero, proving E_εE_δ=0. Expanding ∏_i(P_i+(I−P_i)) gives Σ_εE_ε=I. Multiplication by P_i gives P_iE_ε=ε_iE_ε. Conversely, if v satisfies all simultaneous eigenvalue equations, every factor of E_ε acts as identity on v, so E_εv=v. Thus its image is exactly that eigenspace. Any v equals ΣE_εv; applying each projector to a supposed zero sum proves directness. The proof requires commutativity: arbitrary idempotents can have incompatible images and the product construction can fail even to be idempotent.

Worked example

Use P=diag(1,1,0,0), Q=diag(1,0,1,0), and T=2P−Q.

The Boolean blocks E_11,E_10,E_01,E_00 are the four coordinate projections. T has respective eigenvalues 1,2,−1,0, so p(T)=T(T−I)(T−2I)(T+I)=0. Every block is nonzero, witnessed by its coordinate basis vector. A separate rank example is A=[[1,2,3],[2,4,6],[0,1,1]]: its factorization into its first two independent columns times coefficient rows gives rank at most two, while the minor on rows 1,3 and columns 1,2 equals one. It remains nonzero modulo every prime. In contrast, [[2]] has rational rank one and rank zero modulo two; modular rank loss is not a rational vanishing certificate.

The examples provide independent upper and lower witnesses. The eigenvalue list is supported by actual vectors, and no conditioning statement is inferred from exact rank alone.

Continuity across the source years

2023 — Norm, block-energy and rank distinctions establish the finite matrix vocabulary. Sources: 2023:M059, 2023:M068, 2023:M083, 2023:M088

2024 — Jordan nullities separate generalized eigenspaces from a bare list of eigenvalues. Sources: 2024:Y24-12-005

2025 — Positive metrics, modular minors, Perron projections and nilpotent mixing identify the hypotheses behind spectral claims. Sources: 2025:Y25-01-006, 2025:Y25-09-019, 2025:Y25-08-005, 2025:Y25-10-002, 2025:Y25-11-014, 2025:Y25-11-024, 2025:Y25-12-001, 2025:Y25-12-005

2026 — Boolean blocks, annihilating polynomials and integer intertwiner kernels provide exact reconstruction certificates. Sources: 2026:Y26-03-016, 2026:Y26-07-001, 2026:Y26-09-009, 2026:Y26-03-019

Connections

D15 — Observability and inverse stability. A matrix kernel becomes an invisible state direction only after a specific observation map is declared. Exact rank gives no noise margin without singular values.

D11 — Whitening and covariance. $W^*\Sigma W=I$ is a metric normalization, distinct from an invariance equation $P^*\Sigma P=\Sigma$. A singular covariance can be whitened only on its range.

D13 — Graph spectra. A characteristic polynomial or equitable quotient can certify a conditional spectrum. A spectrum alone does not construct the graph.

Capacity and interactive scope

Lab ids: ['L01', 'L04']

Inputs: Exact matrix, field, input/output metrics and proposed projectors or factors.

Outputs: Rank bounds, exact residuals, Boolean blocks, spectral support and metric-aware singular directions.

Validation: Check products over rationals; use only certified prime reductions; report exact identity separately from numerical conditioning.

Status: Related worked-model labs expose rank and singular-value effects; the general matrix recipe is documented.

Limits

Finite exact identities do not settle infinite operator domains or asymptotic boundedness. Rational rank, integral torsion and floating-point effective rank remain different invariants.

Next questions

Which nonzero projector witnesses suffice to make a proposed finite spectral rigidity theorem independently replayable?

For a family of varying metrics, can the smallest retained singular value be bounded uniformly on the declared parameter domain?

Source anchors: 2023:M059, 2023:M068, 2023:M083, 2025:Y25-01-006, 2025:Y25-09-019, 2026:Y26-03-016, 2026:Y26-07-001, 2026:Y26-09-009

Etingof et al., Introduction to Representation Theory — https://math.mit.edu/~etingof/reprbook.pdf

D03-F03 · Symmetry as an action, representation and orbit calculation

What does a specified symmetry force, and which numerical features remain free after imposing it?

A symmetry label becomes mathematical information only after a group acts on a stated space. The collection repeatedly returns to Weyl groups, permutation characters, signed permutations, crystals and branching. These topics connect through invariant subspaces and orbit decompositions, not through shared large integers. The useful question is what the actual action fixes, which representations occur, and what independent parameters survive in the commuting algebra.

The elementary averaging projector is the organising construction. It turns a finite group action into an exact invariant subspace over a field where the group order is invertible. Character traces then measure multiplicities, and orbit splitting gives concrete restrictions and induced representations. Low-rank Weyl calculations provide a more structured version: a signed orbit sum becomes a polynomial identity after

fixing the root coordinates and normalization. The A_2 denominator can be expanded completely; a nonconstant Kazhdan–Lusztig polynomial requires its actual Bruhat interval and Hecke convention. Neither object is a universal score of computational complexity.

The signed-permutation examples are especially useful because they expose the limit of a symmetry inference. Over F_3, sign changes interchange the two nonzero entries, so the B_4 orbits are exactly the five Hamming-weight classes. Over the reals, invariance of a symmetric matrix under all independent sign changes kills off-diagonal entries, and permutation invariance makes its diagonal entries equal. This forces a scalar matrix on that irreducible coordinate representation, but two copies of a representation still admit independent scalar blocks and hence arbitrary positive frequency ratios. Symmetry narrows a model class; it need not pick a numerical constant, an empirical rank or a physical scale.

Definitions

A representation is a homomorphism $\rho:G\to GL(V)$; its invariants are $V^G=\{v:\rho(g)v=v \text{ for all } g\}$.

The character $\chi_V(g)=tr\rho(g)$ records the trace of the actual action, not just $\dim V=\chi_V(1)$.

The signed permutation group $B_n=C_2^n\rtimes S_n$ changes coordinate signs and permutes coordinates.

A Weyl denominator is an alternating group-algebra sum; its signs and root lattice are part of the input.

Theorem and hypotheses

If G is finite and $|G|$ is invertible in k, $R=|G|^{-1}\Sigma_g\rho(g)$ is an idempotent with image V^G. For the B_n action on F_3^n, the orbits are indexed by Hamming weight j and have sizes $2^j \binom{n}{j}$.

Proof

For $h\in G$, $\rho(h)R=|G|^{-1}\Sigma_g\rho(hg)=R$ because left multiplication permutes G. Hence every vector in $\mathrm{im}R$ is invariant. If v is invariant then $Rv=v$, so $\mathrm{im}R=V^G$ and $R^2=R$. The division hypothesis is essential; in characteristic dividing $|G|$ this formula is unavailable and invariant functors need not be exact. For the orbit claim, every group element preserves the number of nonzero coordinates. Conversely, two vectors of equal weight can first have their supports matched by a coordinate permutation, then have their entries matched by sign changes, since $F_3\setminus\{0\}=\{1,-1\}$. Thus weight is a complete orbit invariant. Choose the support in $\binom{n}{j}$ ways and independently choose each nonzero sign in 2^j ways. This proves both completeness and the sizes without enumerating all group elements.

Worked example

Enumerate B_4 orbits on F_3^4, and compare a symmetry constraint on a real 5×5 quadratic form.

The orbit sizes are $1,8,24,32,16$ and sum to 81. Their stabilizer sizes are $384,48,16,12,24$, giving orbit×stabilizer$=384$ in each case. For a symmetric real κ invariant under every coordinate sign flip, choose the flip of coordinate i; its conjugation reverses every off-diagonal entry in row or column i, forcing those entries to zero. Conjugation by coordinate permutations then forces all diagonal entries equal. Thus $\kappa=cI$. On $V\oplus V$, however, $K=\mathrm{diag}(a^2I,b^2I)$ commutes with $\rho(g)\oplus\rho(g)$ for every $a,b>0$. Its two oscillation frequencies have freely chosen ratio b/a, so no particular ratio follows from equivariance alone.

An orbit computation can be complete at a stated finite level while a proposed exceptional-group action remains unconstructed. Large familiar orders are consistency constraints, not replacements for matrices or relations.

Continuity across the source years

2023 — Weyl denominators, Hecke polynomials and compact-group cutoffs turn symmetry into finite calculations. Sources: 2023:M033, 2023:M036, 2023:M038, 2023:M045

2024 — Character tables, branching diagrams and invariant averaging add actual representation-theoretic maps. Sources: 2024:Y24-12-019, 2024:Y24-09-024, 2024:Y24-11-010, 2024:Y24-11-015, 2024:Y24-11-017, 2024:Y24-12-030, 2024:Y24-12-031

2025 — Signed-permutation orbits, Mackey splitting and rank-one crystals separate group actions from speculative numerical identifications. Sources: 2025:Y25-06-023, 2025:Y25-07-035, 2025:Y25-07-019, 2025:Y25-01-007, 2025:Y25-08-019, 2025:Y25-08-015, 2025:Y25-10-031

2026 — Intertwiner kernels and completeness of invariant maps frame exactly what an action can reconstruct. Sources: 2026:Y26-09-009, 2026:Y26-03-021

Connections

D13 — Orbit enumeration. Hamming-weight orbit counts and stabilizers are finite counting certificates. The count does not identify unrelated datasets.

D07 — Harmonic decomposition. Character orthogonality yields exact observable projection loss in compact-group examples. Noncentral functions need matrix coefficients, not characters alone.

D01 — Galois actions. A splitting-field group acts on roots and preserves field operations. An automorphism is not an evolution in physical time.

Capacity and interactive scope

Lab ids: ['L02']

Inputs: Generators with explicit matrices or permutations, scalar field, proposed invariant and representation space.

Outputs: Orbits, stabilizers, invariant projector, character traces and intertwiner kernels.

Validation: Verify generator relations and the actual action; compare independent orbit counts or kernel certificates.

Status: The finite-counting lab illustrates enumeration; the representation-specific calculations are worked recipes.

Limits

This family does not construct Monster moonshine, every KL polynomial, an E_8 crystal or a faithful $W(E_8)$ action on F_3^4. Equality of coefficients or dimensions requires a separately defined representation before it gains that meaning.

Next questions

Which invariant map is injective on the declared orbit space, rather than merely locally independent?

For a proposed exceptional symmetry, can its generator matrices and exact intertwiner ranks be supplied over a stated field?

Source anchors: 2023:M033, 2023:M036, 2023:M038, 2024:Y24-12-019, 2025:Y25-06-023, 2025:Y25-07-035, 2025:Y25-07-019, 2026:Y26-09-009

Etingof et al., Introduction to Representation Theory — https://math.mit.edu/~etingof/reprbook.pdf
MIT Lie groups and Lie algebras notes — https://ocw.mit.edu/courses/18-745-lie-groups-and-lie-algebras-i-fall-2020/lists/lecture-notes/

D03-F04 · Lie brackets, octonion associators and exceptional weight structure

Which identities survive a nonassociative or graded product, and what additional data turn exceptional dimensions into a Lie algebra?

This family joins two recurring themes: bracket consistency and exceptional symmetry. The starting point is a declared multiplication convention. Associative multiplication produces a commutator satisfying Jacobi; graded associative multiplication produces the super-Jacobi identity with parity signs. Octonionic multiplication is alternative rather than associative, so its commutator carries a nonzero Jacobiator. These are different algebraic structures, each with identities that can be derived and tested exactly.

The octonion model is especially productive because it turns regrouping into a measurable defect. Under a fixed Cayley–Dickson convention, the triple e_1,e_2,e_4 has associator $2e_7$. Norm multiplicativity then yields a sharp universal bound for every triple, with a complete equality criterion. This gives a rigorous invariant of the multiplication, while preventing an illicit reassociation of three unrelated factors. The same discipline applies to superalgebras: in gl(1|1), the bracket of the two odd off-diagonal matrix units is an anticommutator and equals the identity, not an ordinary commutator.

Exceptional Lie algebras enter at the representation level. The decomposition $so_{16}\oplus S^+$ has dimension $120+128=248$, and its D_8 roots together with the half-spin weights form the 240-element E_8 root set. That is a substantial and checkable weight calculation. It is still distinct from constructing the bracket: the so_{16} action on S^+ and an equivariant map $\Lambda^2S^+\to so_{16}$ must be specified with a normalization satisfying spinor Jacobi. The magic-square dimension arithmetic likewise organises candidate structures without replacing their multiplication. This synthesis keeps the genuine exceptional mathematics while making its missing construction obligations visible.

Definitions

Use quaternion units ij=k, jk=i, ki=j and octonion product (a,b)(c,d)=(ac−conj(d)b, da+b conj(c)).

The associator is [a,b,c]=(ab)c−a(bc); the octonion norm is multiplicative.

For homogeneous parity |X|∈{0,1}, [X,Y]_s=XY−(−1)^{|X||Y|}YX.

The E_8 root set is {±e_i±e_j:i<j} together with {(±1/2,...,±1/2):an even number of minus signs}.

Theorem and hypotheses

For real octonions, ||(ab)c−a(bc)||≤2||a||||b||||c||. For nonzero norm product, equality holds exactly when (ab)c=−a(bc), and the constant two is sharp. Independently, the displayed E_8 root set has 240 vectors of squared length two.

Proof

Write u=(ab)c and v=a(bc). Multiplicativity gives ||u||=||v||=n=||a||||b||||c||. Hence ||u−v||≤||u||+||v||=2n. If n>0, equality in the triangle inequality requires u and −v to have the same direction; equal norms make this exactly u=−v. The declared multiplication gives $e_1e_2=e_3$, $e_3e_4=e_7$, $e_2e_4=e_6$ and $e_1e_6=-e_7$, proving sharpness. For the root count, choosing two positions among eight and two independent signs gives $4\cdot\binom{8}{2}=112$ integer roots. Of the 2^8 half-integral sign vectors exactly half have even parity: flipping one chosen sign pairs the two parity classes. Thus there are 128 such roots and 240 total. Each integer root has norm squared 1+1=2; each half-integral vector has 8/4=2. No Jacobi identity for an exceptional Lie bracket is implied by this count.

Worked example

Compare a maximally nonassociative triple, an associative quaternion triple, and a small superalgebra.

For (e_1,e_2,e_4), the two groupings are e_7 and $-e_7$, so the defect is two. For (e_1,e_2,e_3), both groupings equal -1, so the defect is zero. In $gl(1|1)$, $Q_+=E_{12}$ and $Q_-=E_{21}$ are odd. Their superbracket is $E_{11}+E_{22}=I$, while $Q_+^2=Q_-^2=0$. Taking $N=E_{11}$ gives $[N,Q_+]_s=Q_+$ and $[N,Q_-]_s=-Q_-$. Expanding associative triple products with parity signs proves super-Jacobi. A finite exact verifier can check the octonion basis associators and all E_8 root lengths without asserting that either computation constructs a physical model.

Convention-sensitive multiplication and root enumeration are complete finite tasks. A dimension match remains a consistency check until the actual bracket maps are present.

Continuity across the source years

2023 — The octonion multiplication convention and sharp defect bound make nonassociativity quantitative. Sources: 2023:M007, 2023:M008

2024 — Superbrackets, Virasoro central terms and magic-square dimensions introduce graded and exceptional structure with explicit hypotheses. Sources: 2024:Y24-05-004, 2024:Y24-09-026, 2024:Y24-06-002, 2024:Y24-06-013, 2024:Y24-08-007, 2024:Y24-08-012, 2024:Y24-11-005, 2024:Y24-12-015

2025 — The Jacobiator and infinitesimal SO(3) commutator distinguish bracket identities from informal rotation analogies. Sources: 2025:Y25-01-020, 2025:Y25-08-002

2026 — The 120+128 decomposition provides exact E_8 weight accounting while leaving bracket construction separate. Sources: 2026:Y26-07-014

Connections

D06 — Infinitesimal geometry. Matrix rotation generators satisfy $[J_x,J_y]=J_z$ and encode second-order commutation defects. A bracket does not by itself specify a connection or PDE.

D16 — Deformation and quantization. Associators and Jacobi identities are algebraic consistency conditions for proposed deformed products. Ordinary associative matrices cannot faithfully realise a nonassociative product while preserving all multiplications.

D04 — Graded differentials. A square-zero differential needs a graded bracket and its compatibility law. An odd symbol alone does not give a differential graded algebra.

Capacity and interactive scope

Lab ids: []

Inputs: Declared octonion table or graded multiplication, norm and parity assignments; explicit root lattice.

Outputs: Associators, defect bounds, bracket residuals, root counts and weight norms.

Validation: Exact basis multiplication; polarization when extending finite identity checks; root parity and norm checks.

Status: Worked finite multiplication and root-enumeration recipes; no exceptional Lie-algebra constructor is implemented.

Limits

The octonion defect theorem does not classify every zero-associator orbit. Root counts do not prove anomaly cancellation, physical field content or a full E_8 bracket. Real forms and normalization must accompany any further comparison.

Next questions

Can an explicit normalized map $\Lambda^2 S^+ \to so_{16}$ be supplied and its spinor Jacobi identity verified in a chosen basis?

Which automorphism orbits of unit octonion triples attain each fixed associator norm?

Source anchors: 2023:M007, 2023:M008, 2024:Y24-05-004, 2024:Y24-09-026, 2025:Y25-01-020, 2026:Y26-07-014

Baez, The Octonions — https://arxiv.org/abs/math/0105155
MIT Lie groups and Lie algebras — https://ocw.mit.edu/courses/18-745-lie-groups-and-lie-algebras-i-fall-2020/

D03-F05 · Local algebras, jets, path bases and structural flatness

How do relations determine a finite algebra, and which properties remain constant when those relations vary?

The local and quiver examples supply a concrete algebraic laboratory for the atlas. A finite Hilbert function is not determined by a few initial coefficients; it comes from the whole ideal. A constant vector-space dimension is not determined by a reported cochain profile; it comes from a basis or a locally free presentation. A symmetric Hilbert function is not enough to prove Gorenstein duality; the socle and multiplication pairing must be inspected. These distinctions turn several ambitious geometric discussions into tractable structural calculations.

The first model begins with two quadrics. An invertible change of coordinates converts them into u^2 and uv, revealing an infinite tail of powers of v. Adding cubic closure kills v^3 and produces the intended four-dimensional algebra. Its basis gives the Hilbert function, radical powers and socle directly. The second model starts with a finite quiver and a square-zero loop. Path enumeration gives fourteen independent paths, and one projective relation removes exactly one dimension because its source and target prevent further left or right multiplication. The resulting rank thirteen is therefore uniform for a structural reason, not a sampling pattern.

These models also connect finite algebras to representation categories. Dual numbers and their matrix rings have different raw dimensions but equivalent module categories; radical layers and Ext groups retain structure that a state count misses. Conversely, a flat family can change Hochschild cohomology even while its basis size stays fixed. The family is organised so that basis certificates come first, module invariants second and deformation claims third. This ordering makes it possible to build interactive normal-form explorers without advertising an unimplemented general Gröbner, quiver-deformation or moduli solver.

Definitions

For a graded algebra A, H_A(d)=dim_k A_d is its Hilbert function.

For a local Artinian algebra with maximal ideal m, soc(A)={a:ma=0}; a one-dimensional socle is required for the usual commutative Artinian Gorenstein condition.

A quiver path algebra has vertices as orthogonal idempotents and directed paths as a basis before imposing relations.

A finite algebra family is flat when its underlying module is locally free in the finite presentations used here; constant fibre dimension alone is not a general flatness proof.

Theorem and hypotheses

Over any field, $k[u,v]/(u^2,uv,v^3)$ has basis $1,u,v,v^2$, Hilbert function $(1,2,1)$, maximal ideal cube zero and socle $\operatorname{span}\{u,v^2\}$. The two quadratic relations alone give Hilbert function $(1,2,1,1,\ldots)$.

Proof

The monomial ideal (u^2,uv) kills exactly monomials divisible by u^2 or uv. Its surviving monomials are $1,u,v,v^2,v^3,\ldots$, so it has one surviving monomial in every degree at least two. Adding v^3 kills all powers of v of degree at least three and leaves precisely $1,u,v,v^2$. Their independence follows because a monomial ideal contains a polynomial exactly when each of its monomials belongs to the ideal. The maximal ideal is $\operatorname{span}\{u,v,v^2\}$; its square is $\operatorname{span}\{v^2\}$ and its cube is zero. For $a=a_0+a_1u+a_2v+a_3v^2$, multiplication by u gives a_0u and multiplication by v gives $a_0v+a_2v^2$. Both vanish precisely when $a_0=a_2=0$. Hence the socle is $\operatorname{span}\{u,v^2\}$, of dimension two, and the algebra is not Gorenstein despite its symmetric Hilbert function. The coordinate change $u=a-\lambda b, v=b$ is invertible for every λ and turns $(a^2-\lambda^2b^2,ab-\lambda b^2)$ into (u^2,uv), including in characteristic two.

Worked example

Compare the jet algebra with the projective quiver pencil having arrows $\alpha:0\to1$, $\gamma:1\to1$, $\beta,\beta':1\to2$ and relations $\gamma^2=0$, $u\beta\gamma\alpha-v\beta\alpha=0$.

Before the projective relation, count three vertex paths, four arrows, five length-two paths and two length-three paths, for fourteen total. The two paths $\beta\gamma\alpha$ and $\beta\alpha$ both run from source 0 to sink 2. Thus the nonzero relation line generates exactly a one-dimensional two-sided ideal for every $[u:v]\in\mathbb{P}^1$, and every fibre has dimension thirteen. Globally this is a nowhere-zero $\mathcal{O}(-1)$ subbundle in a trivial rank-two path summand; its quotient is $\mathcal{O}(1)$. The full path bundle is therefore $\mathcal{O}^{12}\oplus\mathcal{O}(1)$, a locally free rank-thirteen module. The local chart bases change, but flatness has an explicit bundle certificate.

Basis size, socle type and flatness are separately proved invariants. None determines the full Hochschild profile or a claimed high-order Kuranishi calculation.

Continuity across the source years

2023 — The explicit syzygy resolution makes relations among generators distinguishable from geometric blowups. Sources: 2023:M095

2025 — Dual-number Hochschild and Tor models add derived invariants to finite multiplication data. Sources: 2025:Y25-03-034, 2025:Y25-10-015

2026 — Cubic closure, socle calculations and a projective path-basis proof supply complete local algebra certificates; flatness is kept separate from cohomology constancy. Sources: 2026:Y26-01-034, 2026:Y26-07-002, 2026:Y26-07-004, 2026:Y26-08-011, 2026:Y26-08-012, 2026:Y26-09-014

Connections

D04 — Resolutions and derived invariants. Free resolutions turn relations into Tor, Ext and Hochschild calculations. A module resolution is not a resolution of a geometric singularity.

D02 — Finite jets and flat families. Coordinate changes and locally free presentations identify actual fibres of algebraic families. Higher global moduli claims require actual maps and compatibility data.

D16 — Obstructions. HH^2 may describe infinitesimal changes while higher equations govern extension. Constant dimension does not imply cohomology constancy or unobstructedness.

Capacity and interactive scope

Lab ids: []

Inputs: Polynomial or path generators, coefficient field, relations and optional parameter chart.

Outputs: Standard basis, multiplication table, radical layers, socle and fibre dimension certificate.

Validation: Reduce every product; prove basis spanning and independence; retain transition functions for a global flatness claim.

Status: Worked normal-form and path-basis recipe; no general quiver-cohomology interface is implemented.

Limits

The historical quiver cochain profile and order-twelve deformation packet are not reconstructed by the path count. The tensor extension with Betti numbers (1,4,5,2) is an authored comparison, not an identification with an unseen original packet.

Next questions

For the actual rank-thirteen pencil, what are the exact ranks of the Hochschild differentials on each parameter stratum?

Which additional relations, if any, make the intended jet algebra Gorenstein without changing the required Hilbert function?

Source anchors: 2023:M095, 2026:Y26-01-034, 2026:Y26-07-002, 2026:Y26-07-004, 2026:Y26-08-011, 2026:Y26-08-012

Stacks Project, Locally free sheaves — https://stacks.math.columbia.edu/tag/01C5
Etingof et al., Introduction to Representation Theory — https://math.mit.edu/~etingof/reprbook.pdf

D03-F06 · Finite operator algebras, modular flow and equilibrium

What do a faithful state and its dynamics determine exactly in a finite algebra?

The operator-algebra discussions become most transparent when the algebra, state and representation are separated. In M_d(C), a faithful state is given by a positive density matrix ρ. It defines expectations on observables, an inner-product geometry on matrix space and a modular flow. A thermal Hamiltonian H defines another flow through conjugation. A Gibbs state links the two with a temperature-dependent rescaling and sign, but the state Hamiltonian −logρ is not the same object as the standard-form modular generator acting on matrices.

This distinction is one of the strongest connections across the later years. On the Hilbert space of Hilbert–Schmidt matrices, left multiplication represents the observable algebra and right multiplication represents its commutant. The modular operator is Δ=L_ρR_{ρ^{-1}}. Its eigenvalues on matrix units are ratios p_i/p_j, so its generator has spectral gaps rather than the original energy levels. A second faithful state produces a unitary relative cocycle with a twisted multiplication law. These are exact finite models of powerful general constructions, with all domains automatic and every formula inspectable.

Equilibrium also requires a declared dynamics convention. With α_t(A)=e^{itH}Ae^{−itH}, the Gibbs state satisfies φ(Aα_{iβ}(B))=φ(BA). The boundary identity follows from cyclicity of trace and extends to all matrices because finite-dimensional imaginary-time conjugation is entire. This offers a meaningful interactive calculation: change populations, inspect modular frequencies and test the KMS relation. It does not turn a fitted ratio q=e^{−βE} into a deformed quantum group, or establish infinite type-III modular inclusions, holographic identities or geometric field equations.

Definitions

A faithful density matrix is $\rho>0$ with $\mathrm{tr}\rho=1$, defining $\varphi(A)=\mathrm{tr}(\rho A)$.

On Hilbert–Schmidt matrix space, $L_A(X)=AX$ and $R_A(X)=XA$.

The finite modular operator is $\Delta\rho(X)=\rho X\rho^{-1}$; $J(X)=X^*$ is antiunitary.

For $\alpha_t(A)=e^{itH}Ae^{-itH}$, the KMS convention here is $\varphi(A\alpha_{i\beta}(B))=\varphi(BA)$.

Theorem and hypotheses

For $\Omega=\rho^{1/2}$, the Tomita map $S(A\Omega)=A^*\Omega$ satisfies $S=J\Delta\rho^{1/2}$. If $\rho=Z^{-1}e^{-\beta H}$, φ is α-invariant and obeys the stated KMS identity. The modular generator $-\log\Delta\rho$ has eigenvalues $\log(p_j/p_i)$ on E_{ij}.

Proof

Every matrix X can be written $A\rho^{1/2}$ because ρ is invertible. Therefore $S(X)=(X\rho^{-1/2})^*\rho^{1/2}=\rho^{-1/2}X^*\rho^{1/2}$. Meanwhile $\Delta\rho^{1/2}X=\rho^{1/2}X\rho^{-1/2}$; applying J gives exactly the same expression. In a ρ-eigenbasis, $\Delta\rho E_{ij}=(p_i/p_j)E_{ij}$, yielding the generator formula by finite functional calculus. For KMS, $\alpha_{i\beta}(B)=e^{-\beta H}Be^{\beta H}$. Multiply $\varphi(A\alpha_{i\beta}B)$ by Z and cyclically move the final $e^{\beta H}$ to the front: $\mathrm{tr}(e^{-\beta H}Ae^{-\beta H}Be^{\beta H})=\mathrm{tr}(Ae^{-\beta H}B)=\mathrm{tr}(e^{-\beta H}BA)$. Division by Z proves the identity. Since ρ commutes with e^{itH}, cyclicity also proves $\varphi(\alpha_t A)=\varphi(A)$. Positivity and normalization follow directly from $\rho\geq0$ and $\mathrm{tr}\rho=1$.

Worked example

Take $\rho=\mathrm{diag}(3/4,1/4)$, with $\beta E=\log3$ for $H=\mathrm{diag}(0,E)$, and use $A=E_{01}, B=E_{10}$.

The modular operator multiplies E_{01} by three and E_{10} by one third, fixing both diagonal matrix units. Its generator therefore has eigenvalues $0,0,-\log3,+\log3$ on matrix space. The state Hamiltonian $-\log\rho$ instead has just the two values $\log(4/3),\log4$ on the original two-dimensional space. At imaginary thermal time, $\alpha_{i\beta}(B)=B/3$. Hence $\varphi(A\alpha_{i\beta}B)=(1/3)(3/4)=1/4=\varphi(BA)$. Matrix-unit multiplication checks the same identity for a basis of every A,B. The zero modular frequencies form the diagonal commutant of ρ; degeneracy of populations enlarges that fixed algebra.

An operator state supplies a complete finite equilibrium model, including its faithful-state hypothesis and normalization. Reparametrizing a population ratio changes no multiplication law.

Continuity across the source years

2023 — The Heisenberg local algebra supplies an early operator-and-state theme, although its full Fock space is infinite. Sources: 2023:M103

2025 — Finite modular flow, relative cocycles and Tomita factorization provide exact faithful-state models. Sources: 2025:Y25-03-012, 2025:Y25-03-014, 2025:Y25-11-018, 2025:Y25-07-018

2026 — The Gibbs/KMS calculation and trace-range examples distinguish equilibrium data from algebra deformation and infinite inductive limits. Sources: 2026:Y26-01-015, 2026:Y26-04-018, 2026:Y26-01-026

Connections

D16 — Quantum dynamics. Gibbs states, completely positive channels and modular flow use the same finite observable algebra. A channel need not preserve a given Gibbs state or its modular flow.

D10 — Classical equilibrium. For a commutative diagonal algebra the state is an ordinary finite probability distribution. Noncommutative ordering remains visible outside that subalgebra.

D07 — Spectral functional calculus. Logarithms and imaginary powers of positive matrices are defined by the spectral theorem. Infinite-dimensional domains and trace-class hypotheses need separate treatment.

Capacity and interactive scope

Lab ids: []

Inputs: Positive definite density matrix, Hamiltonian, inverse temperature and observable matrices.

Outputs: Modular ratios, fixed algebra, KMS boundary residual and faithful-state diagnostics.

Validation: Check trace one, strict positivity and the chosen dynamics sign; verify basis identities exactly for rational population ratios.

Status: Worked two-level recipe; no general infinite operator-algebra or thermal simulator is implemented.

Limits

The standard form is finite and faithful. Singular states need support reduction; infinite von Neumann algebras can have no density matrix or trace. The finite derivation does not prove type-III, DHR, JLMS or entanglement-to-geometry claims.

Next questions

For a declared channel, when does it preserve both the Gibbs state and its modular eigenspace decomposition?

Which trace-range and ordered K_0 data are available for a specified inductive-limit algebra, beyond finite matrix approximations?

Source anchors: 2023:M103, 2025:Y25-03-012, 2025:Y25-03-014, 2025:Y25-11-018, 2026:Y26-01-015, 2026:Y26-04-018

Bratteli and Robinson, Operator Algebras and Quantum Statistical Mechanics 2 — https://doi.org/10.1007/978-3-662-09044-1

D04 · Categories and homological methods

Organizing question

Which constructions preserve the object, and what explicit maps certify preservation?

This domain supplies the architecture used to compose the rest of the atlas. Its organizing unit is a construction together with the equations that make it valid. A category needs a declared class of morphisms and a composition law; a dynamical intertwiner must commute with evolution. A quotient needs an equivalence relation; its universal factorization identifies exactly which maps survive the loss of distinctions. A pushout needs two arrows from shared data and produces one object encoding all compatible pairs of outgoing maps. These examples give a precise language for composing mathematics without treating a common label or equal dimension as a structural identification.

The next layer is homological. A complex is more than a sequence of vector spaces because its differential must square to zero. Signed totalization, shifts and cones make this requirement explicit across degrees. A chain contraction gives a finite certificate that a reduction preserves cohomology. Its homotopy exhibits the difference between the identity and a reduced reconstruction as a boundary correction. This is the appropriate bridge to smaller models: one may eliminate a contractible pair while preserving homology, even though total coordinate dimension changes. The chapter therefore separates presentation size, rank information and invariant content.

Derived constructions retain information that ordinary products and kernels erase. Tor and Ext arise by resolving an object before applying a nonexact functor. A nonzero extension class can distinguish modules with identical dimension; the size of the whole Ext group cannot decide whether one selected extension splits. Similarly a tangent complex supplies potential obstruction spaces, while the actual higher-order lifting equation determines an obstruction class. The double-point and finite Maurer–Cartan examples show both outcomes: a nonzero first-order direction can fail to lift, while a nonzero quadratic cocycle can be removable because it is a boundary. These are constructive alternatives with explicit witnesses.

Finally, filtered and infinite constructions state what remains unknown after a partial computation. The associated graded forgets how layers attach. Page differentials can kill apparent classes, and integral extensions can distinguish groups with identical limiting graded pieces. Inverse limits introduce genuinely infinite compatibility constraints that finite truncations do not settle. Perverse shifts, coefficient base change and t-structures belong in this same discipline of declared data: they change the interpretation of degrees and comparisons. The dated source continuity is strongest from the 2024 categorical and spectral questions through the 2025 derived examples to the 2026 exact signs, limits and obstruction calculations. Those dates locate topics in the available corpus; they do not date the creation of the new proofs here.

The interactive capacity is consequently a certificate interface. Finite matrices can test $d^2=0$, construct a cone, verify a contraction and compare homology. They cannot manufacture the missing differential, functor or higher coherence map of an unspecified object. The chapter’s open frontier is to replace those absent operators with explicit constructions and then verify comparison maps at the correct coefficient and categorical level.

Core objects

categories with specified morphisms

quotients and universal mapping properties

complexes and chain homotopies

resolutions, Tor, Ext and cotangent complexes

filtrations, page differentials and inverse systems

Open frontier

Build actual comparison functors, controlling complexes and higher-coherence maps for the proposed structures; compute unknown differentials and extension classes before claiming equivalence or reconstruction.

D04-F01 · Universal constructions, quotients and equivalence

Which maps descend through a quotient, and which reversible constructions preserve the whole mathematical problem?

The categorical thread becomes strongest when a named operation is replaced by a construction with a testable mapping property. A quotient first declares which distinctions are erased. A pushout identifies two copies of shared data. An adjunction converts one kind of mapping problem into another, while an equivalence supplies inverse functors and natural comparison maps. These statements organize mathematics across the archive because they separate an object from a chosen presentation and make compatibility part of the theorem. They also distinguish several levels of preservation: agreement of dimensions is weaker than an isomorphism, an isomorphism of individual objects is weaker than a natural equivalence, and an equivalence of categories must respect the specified morphisms.

The useful starting point is the category of linear dynamical systems. An object is a vector space with an endomorphism A; a morphism L satisfies LA=BL. This single equation guarantees L exp(tA)=exp(tB)L by the power series, so composition preserves dynamics rather than merely relabeling coordinates. The tensor–Hom correspondence then makes composition of structures explicit. Currying is available for arbitrary algebraic vector spaces, whereas a duality datum with coevaluation requires finite dimension. This distinction prevents an infinite formal basis sum from being treated as an algebraic tensor. Reconstruction has a similar demand: a finite group can be recovered from the natural automorphisms of the forgetful functor on all finite G-sets, using the regular G-set and its equivariant maps; one representation or one dimension table cannot carry the same information.

Definitions

For linear maps $i:A\to B$ and $j:A\to C$, their pushout is $P=(B\oplus C)/R$ with $R=\{(i(a),-j(a)):a\in A\}$. The structure maps send b and c to their corresponding quotient classes.

A categorical equivalence consists of functors F and G with natural isomorphisms $GF\cong Id$ and $FG\cong Id$. An adjunction instead supplies natural bijections of Hom sets, with unit and counit satisfying the triangle identities.

In algebraic vector spaces, a duality datum $ev:W\otimes V\to k$ and $coev:k\to V\otimes W$ must reconstruct every vector through both triangle identities. Every algebraic tensor is a finite sum.

Theorem and hypotheses

The stated P is a pushout of i and j. Its dimension is dimB+dimC−rank(a↦(i(a),−j(a))). In particular the subtraction is dimA when at least one of i or j is injective. A vector space is dualizable under algebraic tensor product exactly when it is finite dimensional.

Proof

Given compatible maps u:B→V and v:C→V with ui=vj, the map (b,c)↦u(b)+v(c) kills every (i(a),−j(a)). It therefore descends uniquely to P because the quotient map is surjective. Conversely every map from P gives such a compatible pair by composition. Rank-nullity gives the dimension formula, including the correction needed when both legs have a common kernel. For finite-dimensional V choose a basis e_r and its dual basis; coev(1)=Σe_r⊗e^r and evaluation give v=Σe^r(v)e_r. Conversely, if coev(1)=Σ_{r=1}^n v_r⊗w_r, one triangle identity writes every v as Σv_r ev(w_r⊗v). Thus finitely many v_r span V. This proves the finite-dimensional necessity directly, rather than inferring duality from an informal adjunction symbol.

Worked example

Take A=Q, B=C=Q^2, i(a)=(a,0), j(a)=(0,a). Let u(b_1,b_2)=b_1+2b_2 and v(c_1,c_2)=3c_1+c_2.

The relation is (a,0,0,−a), so P has dimension three. Coordinates (b_1+c_2,b_2,c_1) are invariant under this relation and identify P with Q^3. The compatible functional descends to w(x,y,z)=x+2y+3z. For b=(2,1), c=(3,4), w(6,1,3)=17=u(b)+v(c). The equality is a certificate for this mapping problem, not just a matching dimension count.

Universal properties are practical interfaces: later calculations can use the descended map without choosing representatives. Quotienting is information loss with a declared kernel, while equivalence preserves the specified category. Neither operation automatically supplies a metric, topology, probability model or computationally efficient algorithm.

Continuity across the source years

2024 — Dated topic continuity is supported by: Intertwining maps define a category of dynamical systems. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-05-010

2025 — Dated topic continuity is supported by: A categorical equivalence has explicit units and counits; A fibre functor reconstructs a finite group only with its whole category; A pushout glues compatible maps by a universal property. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-02-013, 2025:Y25-06-007, 2025:Y25-10-006

2026 — Dated topic continuity is supported by: A quotient has a universal factorization only after its equivalence relation is specified; Tensor–Hom adjunction separates currying from dualizability. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-04-033, 2026:Y26-07-024

Connections

D03 — Representations and reconstruction. Intertwiners, induction–restriction and natural automorphisms of a fibre functor. Requires the full declared category and its morphisms.

D14 — Semantic and computational quotients. A map descends exactly when it is constant on every specified equivalence class. Existence does not imply an efficient decision procedure.

D06 — Quotients and geometry. A symplectic quotient adds a moment-map level and a regularity condition to a categorical quotient. Metric and smooth structure require additional data.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: finite i,j,u,v matrices with ui=vj; objects and morphisms for any equivalence claim.

Outputs: Written construction of the pushout, descended map and finite duality certificate. No pushout or categorical-equivalence editor is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

The finite vector-space pushout is not a general homotopy pushout. A quotient metric may be only a pseudometric unless zero-distance classes are further identified. Higher adjunctions need their own coherence data; an equality of dimensions or a formal inverse symbol cannot replace units, counits and naturality.

Next questions

For the actual proposed quotient, list all generating relations and determine whether their closure commutes with passage to each local carrier.

For a proposed equivalence, construct the two comparison natural transformations and verify their components and triangle identities on generators.

Source anchors: 2024:Y24-05-010, 2025:Y25-02-013, 2025:Y25-06-007, 2025:Y25-10-006, 2026:Y26-04-033, 2026:Y26-07-024

Stacks Project Galois categories and fundamental groups — https://stacks.math.columbia.edu/tag/0BQ6
Stacks Project schemes étale over a point — https://stacks.math.columbia.edu/tag/04JI
Stacks Project pushouts of modules — https://stacks.math.columbia.edu/tag/0D12
Milne, Group Theory — https://www.jmilne.org/math/CourseNotes/GT.pdf

D04-F02 · Chain complexes, exactness and certificates of equivalence

How can a reduction preserve homology, and what finite identities certify that nothing essential was lost?

A chain complex records permitted boundaries and the relations among them. Its defining equation $d^2=0$ is the first useful certificate: every boundary is a cycle, so a quotient of cycles by boundaries exists. The archive repeatedly approaches more elaborate constructions through this same requirement. Total complexes need Koszul signs; mapping cones need a genuine chain map; a blocked representation needs maps and a chain homotopy before it can inherit the original cohomology. These are constructive conditions. They provide explicit witnesses that a smaller calculation preserves the invariant instead of only producing similar output on an example.

The key distinction is between changing a presentation and changing an object. Adding a contractible pair increases the number of coordinates while preserving homology. Consequently total chain dimension cannot define an invariant of the derived object. Conversely a cochain map can annihilate a nonzero class, so the fact that an operation is linear or functorial does not prove that it preserves obstruction spaces. The finite Hodge decomposition gives a third view: with declared positive inner products, harmonic vectors represent homology, but that representative depends on the metric while the quotient space is invariant

under chain equivalence. The supertrace index then becomes an exact cancellation statement, because all nonzero differential modes occur in paired even and odd degrees. No asymptotic heat approximation is required in the finite model.

Definitions

A cochain complex has $d:C^n\to C^{n+1}$ with $d^2=0$ and $H^n=\ker d^n/\operatorname{im} d^{n-1}$. A cochain map f satisfies d_D f=f d_C.

A contraction onto H consists of maps i:H→C, p:C→H and degree −1 h with pi=Id and Id−ip=dh+hd. This convention fixes the sign of h.

For commuting raw horizontal and vertical differentials on C^{p,q}, the total differential is δ+(−1)^p∂. $\operatorname{Cone}(f)^n=D^n\oplus C^{n+1}$ has differential (u,v)↦(d_Du+fv,−d_Cv).

Theorem and hypotheses

A contraction makes i and p inverse on cohomology. Signed totalization squares to zero, and the stated cone differential squares to zero exactly when f is a cochain map. In particular Cone(Id) is contractible.

Proof

For a cocycle c, the contraction identity gives c−ip(c)=d h(c), hence the classes of c and ip(c) agree. Since pi=Id, the induced maps are inverse. In a bicomplex the two mixed compositions have coefficients (−1)^p and (−1)^{p+1}; they cancel because the raw differentials commute. The remaining squares are zero. Applying the cone differential twice yields ((d_D f−f d_C)v,0), displaying the precise defect. When f=Id, define h(u,v)=(0,u); then dh(u,v)+hd(u,v)=(u,v). This is an explicit contracting homotopy. In finite inner-product complexes, orthogonality gives $C^n=\operatorname{im} d^{n-1}\oplus\ker \Delta_n\oplus\operatorname{im}(d^n)^*$, because the orthogonal complement of the two images is exactly the simultaneous kernel of d and d*. Every cohomology class has one harmonic representative, although changing the metric changes that vector.

Worked example

Tensor two complexes Q --Id→ Q concentrated in degrees zero and one, using degree-one coordinates ordered horizontal, vertical.

The total dimensions are (1,2,1). The exact matrices are $d^0=(1,1)^{\mathsf T}$ and $d^1=(-1,1)$. Their product is zero, both maps have rank one, and the Betti vector is (0,0,0). Replacing the second matrix with the unsigned (1,1) produces $d^1d^0=2$, so there is no complex over Q. Separately $C^{-1}=Qe$, $C^0=Qu\oplus Qv$ with de=u contracts onto Qv: set p(u)=0, p(v)=v, i(v)=v, h(u)=e, h(v)=0. Both sides of Id−ip=dh+hd send u to u and v to zero. If de=(1+ε)u, the same certificate survives with h(u)=e/(1+ε) exactly when ε≠−1.

The failure at ε=−1 is a real change of homology: the formerly contractible pair becomes two surviving classes. This makes perturbation a testable algebraic question and identifies the parameter where the reduction stops being valid.

Continuity across the source years

2023 — Dated topic continuity is supported by: An exact free resolution and a separate geometric blowup. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2023:M095

2024 — Dated topic continuity is supported by: A categorical decomposition needs a chain complex and an actual invariant. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-10-022

2025 — Dated topic continuity is supported by: Cohomology-preserving blocking requires a chain homotopy certificate; The mapping cone turns an echo equivalence into an acyclicity test. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-03-021, 2025:Y25-10-004

2026 — Dated topic continuity is supported by: A chain contraction proves homology equivalence and survives a controlled elementary perturbation; Total-complex signs and mapping cones are structural data. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-05-002, 2026:Y26-07-023

Connections

D05 — Topological computation. Simplicial and cellular boundary matrices are complexes to which the same contraction and Hodge certificates apply. Integral torsion additionally needs integer normal forms.

D15 — Reduction and observability. A declared reduction can be tested by commuting maps and a homotopy certificate. No equivalence follows merely from similar trajectories.

D03 — Resolutions and syzygies. A free resolution is an exact complex whose differentials encode relations among generators. Exactness and minimality are separate properties.

Capacity and interactive scope

Lab ids: ['L09']

Inputs: Implemented: select triangle, square or K3,3 and optionally fill the triangle. Recipe only: arbitrary complexes, chain maps and homotopies.

Outputs: Implemented: rational boundary ranks, Betti numbers and boundary-square check. Cones, signed bicomplex totalization and contraction parameters are exact written/verifier models, not UI operations.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Partial interactive support through L09; general complex operations remain recipes.

Limits

The finite Hodge statement assumes positive definite inner products. Rational ranks do not retain integer torsion. A general perturbation lemma needs convergence or nilpotence conditions; the one-pair rational rescaling above proves only its own elementary model. Source conventions using ip−Id change the homotopy sign.

Next questions

Construct explicit comparison maps for the proposed blocked complex and solve the chain-homotopy equations over its intended coefficient ring.

Determine the exact parameter set on which a proposed perturbation retains a bounded contraction, including the condition number of the chosen homotopy.

Source anchors: 2023:M095, 2024:Y24-10-022, 2025:Y25-03-021, 2025:Y25-10-004, 2026:Y26-05-002, 2026:Y26-07-023

Singular official tutorial, free resolutions and Schreyer methods — https://www.singular.uni-kl.de/ftp/pub/Math/Singular/doc/tutor.pdf
Stacks Project cones of complexes — https://stacks.math.columbia.edu/tag/0149
The Stacks Project, complexes and homology — https://stacks.math.columbia.edu/tag/010V
Marius Crainic, On the perturbation lemma and deformations — https://arxiv.org/abs/math/0403266

D04-F03 · Derived products, extension classes and deformation obstructions

What information survives in Tor, Ext and a tangent complex that an ordinary product or dimension count discards?

Derived constructions retain the effect of relations that ordinary algebra has suppressed. Tensoring a resolution and applying Hom to it produce Tor and Ext; these groups measure failures of exactness rather than furnishing decorative refinements of a product. A particular extension belongs to an Ext group, so the dimension of the ambient group cannot determine whether that extension splits. Likewise a nonzero obstruction space is a location where an obstruction may live, not a proof that every first-order motion fails. The actual class or nonlinear obstruction map must be computed.

This principle unifies three distinct lines of the source corpus. Dual numbers provide an exact module model with nonzero derived tensor groups in all positive degrees. A singular hypersurface supplies an explicit cotangent complex and a failed second-order lift. A differential graded Lie algebra organizes nonlinear compatibility through the Maurer–Cartan equation. These constructions share a mechanism: solve at first order, identify the next defect as a cocycle, and ask whether a correcting term makes it a boundary. Their coefficient rings and degree conventions remain different. A deformation of a scheme, an extension of modules and a flatness problem for an algebra must therefore be connected by an actual functor or comparison theorem before their obstruction groups are identified. The archive’s elliptic self-extension example is especially useful as a boundary: stable objects can have nontrivial self-extensions, whereas an honestly semisimple abelian category has none in positive Ext degree.

Definitions

For $R=k[\varepsilon]/(\varepsilon^2)$, let $k=R/(\varepsilon)$. A free resolution of k has all positive differentials equal to multiplication by ε.

An element of $\mathrm{Ext}^1_R(C,A)$ classifies an extension $0\to A\to B\to C\to 0$ up to the stated equivalence; the zero class is the split extension.

For $A=k[x]/(x^2)$ in characteristic zero, the cotangent complex is $[A \xrightarrow{2x} A]$ in degrees -1 and zero. Its derived dual at $x=0$ is the tangent complex.

In a differential graded Lie algebra, $d\alpha+[\alpha,\alpha]/2=0$. Writing $\alpha=ta+t^2b$ makes the first two equations $da=0$ and $db=-[a,a]/2$.

Theorem and hypotheses

For dual numbers, $\mathrm{Tor}_n^R(k,k)\cong k$ and $\mathrm{Ext}^n_R(k,k)\cong k$ for every $n\geq 0$. The extension $0\to k\to R\to k\to 0$ is nonsplit. At the double point $x^2=0$, a first-order displacement $x=at$ lifts modulo t^3 only if $a^2=0$.

Proof

The resolution is exact because the kernel and image of multiplication by ε are both εR. Tensoring it with k makes every differential zero, as does applying $\mathrm{Hom}_R(-,k)$, proving the Tor and Ext assertions. An R-linear section of $R\to k$ would send 1 to a vector v with $\varepsilon v=0$, hence $v\in\varepsilon R$, whose quotient is zero; it cannot map back to 1. The direct-sum extension with middle term $k\oplus k$ does split although its endpoint Ext group is identical. For the singularity, differentiating x^2 gives $2x\,dx$. At $x=0$ this map becomes zero, so the tangent and first obstruction spaces are each k. Direct substitution gives $(at+bt^2)^2=a^2t^2$ modulo t^3. This computes the actual obstruction map $a\mapsto a^2$; over a field it kills only $a=0$. In a differential graded Lie algebra, the derivation and Jacobi identities make $[a,a]/2$ closed when $da=0$. The next equation is solvable precisely when this cocycle class vanishes, which is a class-level criterion.

Worked example

Use a finite deformation model $L^1=Qa\oplus Qb$, $L^2=Qc$, with $da=0$, $db=c$, $[a,a]=2c$ and all other relevant brackets zero.

The first-order term ta is closed. Its quadratic defect is t^2c, a nonzero vector that is nevertheless the boundary $d(t^2b)$. The correction $\alpha=ta-t^2b$ satisfies $d\alpha+[\alpha,\alpha]/2=-t^2c+t^2c=0$ exactly. Removing b from the complex makes c a nonzero H^2 class and prevents the same correction. Thus the raw size of the residual does not decide whether the deformation is obstructed.

The constructive output is a lift or a nonzero quotient class. It specifies exactly which information the derived layer adds, and distinguishes algebraic failure from numerical error in a candidate representative.

Continuity across the source years

2024 — Dated topic continuity is supported by: Hochschild homology and K-theory detect different data. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-09-010

2025 — Dated topic continuity is supported by: The derived tangent complex comes from the cotangent complex; Derived tensor products record information an ordinary product erases; The dimension of Ext is not the splitting class of one extension. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-01-015, 2025:Y25-05-011, 2025:Y25-07-013, 2025:Y25-10-017

2026 — Dated topic continuity is supported by: An elliptic curve has nontrivial self-extensions even for a stable line bundle; A second-order Maurer–Cartan obstruction is a cohomology class, not a numerical residual; Tensoring by a dual-number factor gives Betti numbers(1,4,5,2). These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-02-007, 2026:Y26-05-029, 2026:Y26-07-004

Connections

D02 — Singularities and moduli. The cotangent complex and actual lifting equations separate tangent directions from obstructions. The hypersurface calculation assumes a regular-sequence presentation.

D03 — Modules and resolutions. Tor and Ext are computed from exact free resolutions; Betti and socle data constrain Gorenstein claims. A symmetric Hilbert function alone is insufficient.

D16 — Deformation and quantization. Maurer–Cartan-type equations package higher compatibility and obstruction classes. A specific controlling complex and bracket are required.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: dual-number modules, a declared hypersurface or a specified finite differential graded Lie algebra.

Outputs: Written Tor/Ext calculations, obstruction classes and explicit lift or failed-lift witnesses. No derived-functor or deformation solver is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

The finite Maurer–Cartan model is a complete example, not the missing controlling algebra of a historical proposal. A potential obstruction space can be nonzero while the actual class vanishes. Derived equivalence, formal smoothness and physical stability are distinct claims.

Next questions

For a specified singularity, compute the nonlinear obstruction map through third order and determine which first-order directions lift.

For a proposed obstruction energy, show that its zero test factors through the relevant cohomology class rather than depending on a chosen cocycle representative.

Source anchors: 2025:Y25-01-015, 2025:Y25-05-011, 2025:Y25-07-013, 2025:Y25-10-017, 2026:Y26-02-007, 2026:Y26-05-029, 2026:Y26-07-004

Stacks Project The cotangent complex — https://stacks.math.columbia.edu/tag/08P5
Stacks Project homological algebra — https://stacks.math.columbia.edu/tag/010V
de Jong commutative algebra course and Tor constructions — https://www.math.columbia.edu/~dejong/courses/commutative-algebra-fall-2021.html
Etingof Gelaki Nikshych Ostrik Tensor Categories — https://doi.org/10.1090/surv/205
Marius Crainic, On the perturbation lemma and deformations — https://arxiv.org/abs/math/0403266

D04-F04 · Filtrations, spectral sequences and information at infinity

How much of a total object can be recovered from graded pages, truncations and inverse systems?

A filtration is an organized sequence of partial views. Passing to the associated graded records the layers but suppresses how they interact. A spectral sequence progressively restores some of that interaction through differentials. Even after every differential is determined, reconstructing an integral group or ring can require extension and multiplicative data. The source sequence develops this distinction through fibrations, filtered complexes, cohomological shifts and inverse limits. Their common question is not whether the displayed table looks plausible, but which connecting maps remain unknown.

The Hopf fibration and the product $S^2\times S^1$ have the same initial Serre page but different transgressions. The finite model below makes the same phenomenon an exact two-coordinate computation. It also exposes a second loss: Z/4 and Z/2⊕Z/2 have the same two graded layers, although only the former has an element of order four. A filtration therefore supports a partial reconstruction interface, with each unknown differential and extension listed as an unresolved input. Perverse shifts and t-structures add a related bookkeeping lesson. The declaration of a heart or a support/cosupport condition changes which cohomological degrees count as an object; it is not recoverable from an unshifted label. Finally an infinite inverse system has genuinely limiting information. The doubling tower of integers has vanishing ordinary inverse limit but a nonzero derived limit; no finite truncation alone certifies that infinite result. These examples justify a hierarchy of outputs: graded dimensions, actual graded groups, filtered groups, rings, and geometric objects.

Definitions

Use an increasing filtration F_pC preserved by the differential. A component lowering filtration by r may first appear as the corresponding page differential after earlier components vanish.

An associated graded group is gr_F A=⊕F_pA/F_{p−1}A. It does not include a chosen splitting of the filtration.

For a countable inverse system $A_0 \leftarrow A_1 \leftarrow \cdots$, define $\Delta: \prod A_n \to \prod A_n$ by $\Delta(a)_n = a_n - f_n(a_{n+1})$; then $\lim A = \ker \Delta$ and $\lim^1 A = \operatorname{coker} \Delta$ in this setting.

Theorem and hypotheses

There exist filtered complexes with identical associated graded differential but different total cohomology. There exist nonisomorphic integral groups with the same associated graded. For the doubling tower $Z \xleftarrow{\times 2} Z \xleftarrow{\times 2} \cdots$, $\lim = 0$ and $\lim^1 \cong Z_2/Z$.

Proof

Let $C^0 = Qv$ and $C^1 = Qw$ with v in filtration one and w in filtration zero. Both $d = 0$ and $dv = w$ induce zero on each same-filtration quotient, so their initial graded complexes agree. For $dv = w$ the next filtration-dropping differential is an isomorphism and the total is acyclic; for $d = 0$ both classes survive. For the extension example filter $Z/4$ by $0 \subset 2Z/4 \subset Z/4$ and filter $(Z/2)^2$ by one coordinate; each graded layer is $Z/2$, but their exponents differ. For the inverse limit, a compatible sequence satisfies $a_0 = 2^n a_n$ for all n, hence $a_0 = 0$ and all $a_n = 0$. Define $S(b) = \Sigma 2^n b_n$ in Z_2, convergent in the 2-adic topology. If $b = \Delta(a)$, partial sums telescope to $a_0 - 2^{N+1} a_{N+1}$, so $S(b) = a_0 \in Z$. Conversely if $S(b) = m \in Z$, set $a_n = (m - \Sigma_{j<n} 2^j b_j)/2^n$. The numerator is divisible by 2^n, so a_n is an integer and $a_n - 2a_{n+1} = b_n$. Binary digits make S onto Z_2. Thus S induces exactly $\operatorname{coker} \Delta \cong Z_2/Z$.

Worked example

Compare the two filtered two-term complexes and the two order-four groups just described.

For $dv = w$ the ranks are one and the Betti vector is $(0,0)$; for $d = 0$ they are zero and the Betti vector is $(1,1)$. Yet each associated graded has one generator in its listed bidegree and zero initial differential. In the group comparison, enumerate element orders: $Z/4$ has orders $(1,2,4,4)$, whereas $(Z/2)^2$ has $(1,2,2,2)$. The order-four test detects an extension invisible to the common graded ranks.

A page is useful evidence with a declared reconstruction stage. It becomes a complete answer only after the required differentials, convergence, extensions and products have been supplied.

Continuity across the source years

2024 — Dated topic continuity is supported by: An E^2 page is not already the homology of the total space; A cohomology comparison must survive a coefficient check. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-04-013, 2024:Y24-10-025

2025 — Dated topic continuity is supported by: Perverse degree is fixed by support and cosupport; Coherence is an identity of operations, and a spectral page still needs a differential. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-01-016, 2025:Y25-03-011

2026 — Dated topic continuity is supported by: A derived inverse-limit obstruction can be computed without a geometric analogy; A spectral-sequence page does not determine an integral ring or its extension data. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-02-015, 2026:Y26-05-032

Connections

D01 — Integral and p-adic information. Derived inverse limits and integer extension classes retain arithmetic lost under rank computations. A countable tower and its actual bonding maps are essential.

D05 — Fibrations and persistent layers. Filtrations organize homological information, while differential and extension data determine the total. A persistence module is not itself the same invariant as a spectral sequence.

D02 — Derived and perverse structures. Support, cosupport, shifts and coefficient base change determine the valid comparison. No unspecified motivic or perverse object is reconstructed from notation alone.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: filtered complexes, page differentials, integral extensions and actual inverse-system bonding maps.

Outputs: Written comparison of graded information, total cohomology, extension classes and the infinite doubling-tower proof. No spectral-sequence or inverse-limit UI is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

The proof of $\lim^1$ is an infinite algebraic argument, while the executable tests check only finite supporting identities. A finite experiment cannot verify convergence of an arbitrary tower. Multiplicative structures require their own signs, coefficient ring and extension analysis.

Next questions

For a specified filtered complex, compute the first nonzero differential and prove the relevant convergence statement.

Given a proposed integral cohomology ring, determine the additive extensions and cup products rather than reporting only graded ranks.

Source anchors: 2024:Y24-04-013, 2024:Y24-10-025, 2025:Y25-01-016, 2025:Y25-03-011, 2026:Y26-02-015, 2026:Y26-05-032

Allen Hatcher Spectral Sequences in Algebraic Topology — https://pi.math.cornell.edu/~hatcher/SSAT/SSATpage.html
Mark Goresky Lecture notes on sheaves and perverse sheaves — https://arxiv.org/abs/2105.12045
The Stacks Project, Inverse Systems — https://stacks.math.columbia.edu/tag/02MY
Allen Hatcher, Algebraic Topology, Chapter 3 and loopspace material — https://pi.math.cornell.edu/~hatcher/AT/ATpage.html

D05 · Topology, cohomology and gluing

Organizing question

Which global features survive local changes, and which exact obstruction prevents a global construction?

The topological domain is organized around two constructive tasks: calculate an invariant of a specified space, and identify an obstruction to assembling local information globally. The first begins with cells and boundaries. A graph, the surface on which it is drawn and a filling of its cycles are different spaces. Their boundary matrices make the distinction exact. A triangle boundary has a one-dimensional cycle; attaching its face kills that class. The same principle scales to flag complexes, polyhedral boundaries and singular links, where the problem is to state the actual attachment data before assigning a homology group.

Coefficients determine what the calculation can see. Rational Betti numbers retain free rank but erase torsion. Integer normal forms supply the finite abelian groups hidden by that compression, and reduction modulo a prime can reveal a different pattern again. Reduced homology and intersection homology refine different aspects of a space: one adjusts the degree-zero basepoint contribution, while the other controls allowable chains near specified singular strata. The contractible cone on a torus makes the latter concrete, because ordinary bounding disks may be disallowed at the apex. These distinctions improve classification questions by matching the invariant to the equivalence relation and phenomenon under study.

Persistence adds maps and time to this framework. It retains finite-lived classes that disappear in the eventual space. A legal filtration orders cells after their faces, and induced maps determine births and deaths. Small exact examples support a clear interactive model, while shifted sublevel inclusions give a direct perturbation certificate. Random entry times connect topological lifetimes to order statistics in the simple square example. The scope stays explicit: a stable barcode is a controlled invariant of the declared filtration, not a complete reconstruction of the geometry or a validation of the mechanism generating the data.

The local-to-global side starts with the sheaf axiom’s actual content: compatible data glue uniquely. Arbitrary discrepancies need not be compatible, and their failure can occupy a nonzero cohomology group. Incidence equations, cycle periods and rank-one transport give complete exact examples. Discrete Hodge theory chooses harmonic representatives after an inner product is declared; it does not make every harmonic vector a zero class. Consequently a small residual or a successful test vector cannot certify disappearance of an entire obstruction space. A comparison between local models needs a cochain equivalence if it is to preserve those classes.

Periods, group multipliers and characteristic classes connect this algebraic computation to geometric invariance. Rephasing changes a multiplier by a coboundary, nonvanishing loop homotopies preserve winding, and normalized bundle curvature represents characteristic classes. The family treatments keep their degrees and coefficients separate while exposing their shared local-choice versus global-defect mechanism. This makes the domain’s interactive capacity concrete: supply a finite complex or transport system, and return a cycle, a potential, a barcode or a nonzero obstruction certificate. The frontier is to recover missing attachment, coefficient and comparison data in larger historical proposals rather than infer topology from terminology.

Core objects

spaces with declared cells and attachments

integral and field-valued homology

persistence modules and diagrams

Čech and cellular sheaf complexes

periods, cocycles and characteristic classes

Open frontier

Specify the spaces, coefficient systems, legal filtrations and attachment maps in the larger proposals; produce integer certificates and valid cochain comparisons that preserve the intended global obstruction.

D05-F01 · Spaces, chains, torsion and refined homology

Which features belong to the space, which to its cell structure, and which survive a change of coefficients?

The topological material repeatedly becomes more informative after the underlying space is specified. A triangle can mean three vertices, its three-edge boundary, or a filled two-simplex. These objects have different chain groups and therefore different homology. A graph drawn on a torus is still a one-dimensional complex until its faces are attached; its cycle rank need not equal the surface's first Betti number. Polyhedral spheres have the same homology while retaining different symmetry groups. These examples teach how to ask a sharper classification question: first choose the equivalence relation, then ask which invariant can distinguish its classes.

Integer coefficients add a second layer. Rational homology records free rank and erases torsion, so it cannot distinguish every space or quotient. A Smith normal form supplies an exact arithmetic certificate for the torsion, provided row and column operations are unimodular and the relevant quotient is actually the homology group. Reduced homology removes the connected-component contribution in degree zero; intersection homology instead restricts which chains may meet singular strata. The two operations address different questions. The cone on a torus is contractible as an ordinary space, yet lower-middle allowable chains retain two one-dimensional classes because their potential cone fillings meet the forbidden apex. A refinement is useful precisely because its allowed chains and intended invariance are explicit, rather than because the word cohomology was attached to a construction.

Definitions

For an oriented finite simplicial complex, $\partial[v_0,\dots,v_k]=\Sigma(-1)^j[v_0,\dots,\hat{v}_j,\dots,v_k]$. Its cancellation identity is $\partial^2=0$.

A finite connected graph with V vertices and E edges has free H_1 of rank $E-V+1$. Attaching faces adds ∂_2 and can kill cycles.

For an integer matrix M with Smith form $\mathrm{diag}(d_1,\dots,d_r,0)$, the cokernel is $\oplus Z/d_i\oplus Z^{m-r}$; units contribute trivial factors.

Intersection homology fixes a stratification and a perversity; allowable chain dimensions at a singular stratum determine which fillings count.

Theorem and hypotheses

The triangle boundary has Betti vector (1,1), while the filled triangle has (1,0,0). A cellular model of RP^2 with one cell in dimensions zero, one and two has $\partial_1=0$ and $\partial_2=2$, giving $H_1(RP^2;Z)=Z/2$ and $H_2(RP^2;Z)=0$.

Proof

For the triangle use edges 01,12,02 and boundary columns $(-1,1,0)$, $(0,-1,1)$, $(-1,0,1)$. The matrix has rank two and its kernel is generated by $(1,1,-1)$. With no face this is a nonzero H_1 class. Adding the oriented face 012 gives precisely that vector as ∂_2, so the quotient is zero. In the projective plane, the two-cell attaches to the one-skeleton circle by a degree-two map, hence its cellular boundary is multiplication by two. Its kernel on Z is zero and its cokernel is Z/2. Tensoring the cellular complex with Q makes multiplication by two invertible and kills H_1. Tensoring with F_2 makes that differential zero and produces classes in degrees one and two. This change illustrates why coefficients belong in the question. For the cone on T^2, allowable one-cycles and their bounding two-chains cannot reach a codimension-three apex at lower-middle perversity. They therefore see the punctured cone, which retracts onto T^2; the two degree-one classes survive. A three-chain may reach the apex and fills the link’s two-cycle.

Worked example

Compare a triangle boundary, a filled triangle and the integral cellular projective-plane model.

Exact rational matrix ranks yield $(\beta_0,\beta_1,\beta_2)=(1,1,0)$ and $(1,0,0)$ for the two triangle models. In the projective-plane model the Smith entry is 2, so integral first homology has a generator a with $2a=0$. The rational Betti vector is (1,0,0), but the mod-two Betti vector is (1,1,1). Euler characteristics remain consistent: $1-1+1=1$ for the projective-plane cell counts and for its mod-two Betti numbers.

A full answer records space, boundary convention, coefficient ring and the invariant’s scope. Equal Betti numbers are useful compression, not a complete geometric classification.

Continuity across the source years

2023 — Dated topic continuity is supported by: A Moore spectrum as a precise chromatic example. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2023:M043

2024 — Dated topic continuity is supported by: Fundamental groups distinguish boundaries, fillings and quotients; A toroidal toy Tonnetz separates graph cycles from surface homology; Reduced homology and intersection cohomology remove different phenomena. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-06-014, 2024:Y24-06-018, 2024:Y24-07-013, 2024:Y24-12-023

2025 — Dated topic continuity is supported by: Intersection homology retains cycles that an ordinary cone contracts. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-01-014

2026 — Dated topic continuity is supported by: Integral homology retains torsion that rational ranks erase. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-07-021

Connections

D04 — Algebraic certificates. Boundary matrices, exactness and chain homotopies give finite computations of invariants. Homotopy groups require additional information beyond homology.

D13 — Graphs and cell complexes. Cycle rank $E-V+C$ changes when two-cells are attached. A drawing or decision reduction need not induce homology injections.

D01 — Torsion and arithmetic. Smith normal forms retain the finite abelian groups erased by rational rank. The matrix must represent the actual integer quotient.

Capacity and interactive scope

Lab ids: ['L09']

Inputs: Implemented: triangle, square or K3,3; optional triangle face.

Outputs: Implemented: exact rational boundary ranks, Betti numbers and boundary-square check. Integral Smith forms, RP2 torsion and intersection homology are written/verifier models, not UI operations.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Implemented finite rational-homology model; broader invariants remain recipes.

Limits

The RP^2 calculation is cellular; a general integer homology computation must restrict ∂_2 to an integral basis of $\ker\partial_1$, rather than take $\operatorname{coker}\partial_2$ in the wrong ambient group. The intersection-homology cone formula depends on perversity and stratification. No complete homotopy classification is asserted.

Next questions

For the proposed geometric construction, specify every face attachment and compute the integer homology with unimodular certificates.

Determine which spaces with identical Betti numbers in the source examples are distinguished by fundamental group, cup product, torsion or group action.

Source anchors: 2024:Y24-06-014, 2024:Y24-06-018, 2024:Y24-07-013, 2024:Y24-12-023, 2025:Y25-01-014, 2026:Y26-07-021

Allen Hatcher Algebraic Topology, fundamental groups — https://pi.math.cornell.edu/~hatcher/AT/ATchapters.html
Goresky and MacPherson Morse theory and intersection homology theory — https://www.math.ias.edu/~goresky/pdf/MorseIH.pdf
Hatcher, Algebraic Topology — https://pi.math.cornell.edu/~hatcher/AT/ATpage.html
Goresky and MacPherson Intersection homology theory — https://doi.org/10.1016/0040-9383(80)90003-8

D05-F02 · Persistent topology, interleavings and random birth laws

How does a topological feature persist through a filtration, and what perturbation or randomness can be certified?

Persistence studies a sequence of spaces together with the maps between them. A barcode records the life of a class, including classes that disappear before the terminal space is reached. This is why the colimit cannot replace a persistence module: a finite bar contributes information during its lifetime even though its eventual image is zero. The archive moves from explicit small filtrations to diagram distances, stability and random birth times. These developments fit one constructive interface: input a legal filtration, compute boundary maps and induced maps on homology, and then declare which comparison theorem applies.

The legal-filtration requirement is essential. A simplex cannot enter before one of its faces. Assigning a dramatic label such as singularity or cavity to a parameter value does not change homology; the chain complex changes when cells or maps change. A metric between persistence diagrams also needs the diagonal. An unmatched point can die by matching to its closest diagonal point, with cost half its lifetime in the usual $\ell\infty$ convention. Stability then has an exact elementary core: uniformly close functions give mutually shifted sublevel inclusions. Passing to barcode stability adds finiteness or tameness

assumptions. Random filtrations supply another useful bridge, because a feature born when all required edges are present has a birth time given by a maximum of random variables. This connects a topological lifetime to an exact order-statistic law without claiming that all random topology reduces to one scalar distribution.

Definitions

A filtration $K_s \subseteq K_t$ for $s \le t$ induces a persistence module $H_k(K_s) \to H_k(K_t)$ over a fixed field. A bar $[b,d)$ records a feature born at b and killed at d.

For diagrams with the standard bottleneck distance, match points using ℓ^∞ cost and allow arbitrary diagonal points. A point (b,d) has diagonal cost $(d-b)/2$.

For functions f,g on one space, their sublevel filtrations are $F_t=\{f \le t\}$, $G_t=\{g \le t\}$. A uniform bound $\|f-g\|_\infty \le \varepsilon$ gives an ε-interleaving by inclusions.

Theorem and hypotheses

A triangle whose vertices enter at zero, all edges at one and its face at two has one essential H_0 bar, two H_0 bars $[0,1)$, and one H_1 bar $[1,2)$. For a square whose four edge times are independent uniform variables on $[0,1]$ and whose face enters at one, its H_1 birth B has $P(B \le t)=t^4$ and expected lifetime $1/5$.

Proof

Before time one the triangle has three components; at one its edge boundary matrix has rank two, merging them and leaving the one-dimensional cycle kernel. At two the face boundary spans that kernel. This proves the bars, including the two simultaneous deaths. For the random square, the unique graph cycle exists exactly when every edge has arrived, so $B=\max(U_1,U_2,U_3,U_4)$. Independence gives $P(B \le t)=\prod P(U_j \le t)=t^4$. The face kills it at one, hence expected lifetime is $E(1-B)=\int_0^1 P(B \le t)\,dt=1/5$. At a fixed $t<1$ the first Betti number is its birth indicator, with expectation t^4. For stability, if $f(x) \le t$ then $g(x) \le f(x)+\varepsilon \le t+\varepsilon$, giving $F_t \subseteq G_{t+\varepsilon}$. The reverse inclusion is identical, and the composites are exactly the ordinary inclusion maps shifted by 2ε. This is the direct interleaving certificate before any diagram theorem is invoked.

Worked example

Use $D=\{(0,3),(1,2)\}$ and $E=\{(1/10,16/5),(4/5,23/10)\}$ with rational coordinates and the standard diagonal convention.

Pairing long bar to long bar costs $\max(1/10,1/5)=1/5$. Pairing short bar to short bar costs $\max(1/5,3/10)=3/10$, so the bottleneck is at most $3/10$. The point $(1,2)$ has no match of cost below $3/10$: its closest listed point has that cost and its diagonal cost is $1/2$. Therefore the bottleneck is exactly $3/10$. Exhaustive finite matchings including diagonal removals certify this small case.

The stable statement concerns the declared filtration and distance. It controls the persistence invariant; it does not prove a geometric deformation is physically valid or recover all fine geometry from bars.

Continuity across the source years

2023 — Dated topic continuity is supported by: A complete small persistence calculation. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2023:M040

2024 — Dated topic continuity is supported by: Diagram distances require diagonal matching and a stated filtration; What persistence stability can certify for a deformation. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-08-001, 2024:Y24-09-008

2025 — Dated topic continuity is supported by: A persistence module retains finite bars that its colimit forgets; Persistent topology starts with a chain complex whose boundary squares to zero. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-01-017, 2025:Y25-03-003

2026 — Dated topic continuity is supported by: Random birth times connect order statistics to an exact barcode law. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-02-002

Connections

D10 — Random filtrations. Births defined by maxima of independent entry times yield exact order-statistic distributions. Dependent entries or multiple interacting cycles require richer laws.

D04 — Maps and graded information. A persistence module retains maps and finite bars lost by a terminal colimit. Persistence and spectral sequences organize different data.

D11 — Robust inference. Interleaving and bottleneck stability quantify how input perturbations affect a chosen invariant. They do not identify the data-generating model.

Capacity and interactive scope

Lab ids: ['L09']

Inputs: Implemented supporting example: triangle face toggle. Recipe only: time-indexed filtrations, diagram matchings and random birth parameters.

Outputs: The UI shows homology before and after a face attachment. Barcode intervals, bottleneck 3/10 and random lifetime 1/5 are proved and exactly checked separately; there is no arbitrary persistence or diagram-distance UI.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Partial support from two homology snapshots; persistent-homology operations remain recipes.

Limits

Barcode decompositions require an appropriate finite or tame setting. The linked lab uses one finite filtration, not arbitrary point-cloud persistence. Expected Euler characteristic is an alternating aggregate and does not determine expected Betti numbers separately.

Next questions

For a specified noisy filtration, derive a uniform perturbation bound and state the exact tameness assumptions needed to pass from interleaving to diagram distance.

For a multi-cycle random complex, calculate joint births and deaths rather than inferring a barcode law from expected Euler characteristic.

Source anchors: 2023:M040, 2024:Y24-08-001, 2024:Y24-09-008, 2025:Y25-01-017, 2025:Y25-03-003, 2026:Y26-02-002

Allen Hatcher Algebraic Topology homology chapters — https://pi.math.cornell.edu/~hatcher/AT/ATchapters.html
Cohen-Steiner Edelsbrunner and Harer Stability of Persistence Diagrams — https://doi.org/10.1007/s00454-006-1276-5
Edelsbrunner Letscher and Zomorodian Topological persistence and simplification — https://doi.org/10.1007/s00454-002-2885-2
Afra Zomorodian and Gunnar Carlsson, Computing Persistent Homology — https://doi.org/10.1007/s00454-004-1146-y

D05-F03 · Local-to-global gluing, cellular sheaves and Hodge obstructions

When can compatible local data be replaced by one global object, and how can the obstruction be calculated exactly?

Gluing is a mapping problem with two distinct outcomes. A sheaf axiom says a compatible family glues uniquely; it does not say every arbitrary family is compatible or that higher cohomology vanishes. The circle examples make this distinction concrete. Two arcs overlap in two components, allowing two discrepancies; local constants can change both discrepancies only by the same amount. Their difference is therefore a one-dimensional global obstruction. This is an exact finite complex with a transparent quotient, not a violation of sheaf gluing.

The same mechanism extends to graph potentials, cellular sheaves and discrete Hodge theory. Edge values are globally integrable exactly when their cycle sums vanish. A spanning tree constructs the potential, while each remaining edge checks one cycle constraint. With matrix or scalar transport, the coefficient system changes the equation itself; the ordinary cycle of the underlying graph need not yield a nonzero sheaf class. A rank-one local system on a circle has cohomology controlled by its total monodromy. Its harmonic representative is then a metric-dependent choice inside a metric-independent quotient. This is especially relevant to source claims based on residuals: a vector with zero Hodge Laplacian residual may be a nonzero harmonic obstruction. Certifying $H^1=0$ requires a rank or coercivity statement on the entire relevant space, not one successful vector. Transfers between local models similarly need a chain equivalence before they can transport obstruction classes bijectively.

Definitions

For a connected oriented graph, $D:C^0\to C^1$ sends a potential f to edge differences f(head)−f(tail). Gluable edge data are imD; $H^1=C^1/\mathrm{im}D$ when no two-cells are present.

For the constant real sheaf on a two-arc circle cover with disconnected overlap, the Čech map is $R^2\to R^2$, $(a,b)\mapsto(b-a,b-a)$.

A rank-one local system on an m-cycle has nonzero transports r_j and differential $(\delta s)_j=s_{j+1}-r_j s_j$. Its loop multiplier is $h=\prod r_j$.

Theorem and hypotheses

On a connected graph, edge data are a potential difference exactly when every oriented cycle period vanishes; the potential is unique up to a constant. For the rank-one circle system, H^0 and H^1 each have dimension one if $h=1$ and both vanish if $h\neq 1$.

Proof

Necessity of zero periods follows by telescoping potential differences around a closed walk. For sufficiency choose a root and define f(v) by summing edge data along a tree path to v. A zero period on the closed walk formed with any other path proves path independence. Every edge then has the required difference; two solutions differ by a function with zero difference on every edge, hence a constant. For the local system, a global section satisfies $s_{j+1}=r_j s_j$. Once s_0 is chosen all others follow, and closure requires $(h-1)s_0=0$. Thus the kernel is one-dimensional when $h=1$ and zero otherwise. The differential is square, so rank-nullity gives the same cokernel dimension. Rescaling the basis at vertices changes individual transports but telescoping shows their product is unchanged. In a higher-dimensional cellular sheaf, face-composition identities must additionally make $\delta^2=0$; arbitrary transport matrices alone do not guarantee a complex.

Worked example

Use a three-cycle over Q. Compare transports r=(2,3,1/6) and r=(2,3,1), with δ rows expressing the three stated equations.

The first product is one and (1,2,6) spans the kernel. The differential has rank two and a one-dimensional cokernel. The second product is six, forcing $s_0=6s_0$ and hence all section coordinates zero; the differential is invertible over Q. For additive triangle data (1,2,−3), total period is zero and f=(0,1,3) glues them. Replacing the last value by −2 gives period one and no potential. These exact checks distinguish coefficient holonomy from the graph’s ordinary cycle rank.

The correct interactive output is a potential or an obstruction certificate, together with the coefficient system and a gauge convention. A residual can supplement this certificate but cannot replace its quotient meaning.

Continuity across the source years

2024 — Dated topic continuity is supported by: A nonzero H^1 from two arcs without violating sheaf gluing; A complete small sheaf calculation before spectral-sequence claims. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-06-009, 2024:Y24-09-009

2025 — Dated topic continuity is supported by: Phase consistency is a cycle condition, not a missing matrix certificate; Discrete Hodge splitting and the correct cohomology transport group. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-08-017, 2025:Y25-11-022

2026 — Dated topic continuity is supported by: An exact Čech calculation distinguishes local primitives from global gluing; Compatibility and gluing can be tested by a linear obstruction; A rank-one cellular sheaf on a circle is controlled by holonomy. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-01-030, 2026:Y26-02-004, 2026:Y26-07-022

Connections

D04 — Homotopy-invariant transfer. A cochain homotopy equivalence induces an isomorphism on obstruction groups. A single cochain map can kill a nonzero class.

D06 — Connections and local systems. Transport around a loop controls twisted global sections. Curvature requires two-dimensional geometric data.

D15 — Reconstruction. An incidence equation reconstructs potentials modulo constants after cycle compatibility is verified. A coercivity margin and a gluing obstruction are independent unless explicitly coupled.

Capacity and interactive scope

Lab ids: ['L05']

Inputs: Implemented related example: integer shears a,b in [-3,3]. Recipe only: incidence data, local-system transports and Čech restriction maps.

Outputs: The related lab returns the shear commutator and determinant. It does not compute potentials, Čech cohomology, rank-one local-system groups or harmonic representatives; those are written/verifier models.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Related transport example implemented; gluing and sheaf solvers remain recipes.

Limits

The examples concern finite graph complexes and a declared rank-one coefficient system. General sheaf cohomology needs a valid acyclic cover or an appropriate resolution. A vanishing residual for one vector cannot certify vanishing of an entire cohomology group.

Next questions

For a proposed two-dimensional cellular sheaf, verify every face-composition identity and compute the exact cohomology over the intended ring.

Construct an explicit comparison chain equivalence before transferring a nonzero gluing class between two local models.

Source anchors: 2024:Y24-06-009, 2024:Y24-09-009, 2025:Y25-08-017, 2025:Y25-11-022, 2026:Y26-01-030, 2026:Y26-02-004, 2026:Y26-07-022

Milne Lectures on Etale Cohomology sheaves and Cech constructions — https://www.jmilne.org/math/CourseNotes/LEC210.pdf
Hatcher, Algebraic Topology — https://pi.math.cornell.edu/~hatcher/AT/ATpage.html
The Stacks Project, Čech Cohomology — https://stacks.math.columbia.edu/tag/01EW

D05-F04 · Periods, cocycles, winding and characteristic classes

Which global defects survive changes of local choices, and what makes their values topological?

A cocycle expresses consistency of repeated composition; a coboundary expresses a defect removable by changing local choices. Periods and characteristic classes provide ways to detect what remains. The archive contains several versions: winding of a nonvanishing loop, projective phases of group actions, transition functions of bundles and characteristic-form integrals. They share an invariance mechanism, but their coefficients, degrees and products differ. A useful synthesis keeps the actual equation visible rather than treating every nonzero loop quantity as the same obstruction.

Winding is protected by avoidance of the origin: a continuous nonvanishing loop has an integer degree, and a homotopy through nonvanishing loops preserves it. A curl flux is instead a boundary integral depending continuously on the field; equality for two surfaces with the same boundary is a Stokes statement, not automatic quantization. Group projective phases satisfy a two-cocycle equation because multiplication is associative. Rephasing the operators changes the cocycle by a coboundary, while the commutator phase for a commuting pair is unchanged. Bundle transition functions obey their own triple-overlap identity, and characteristic forms connect curvature to integral topology only for a declared bundle and normalization. The source corpus's four-torus example makes the latter quantitative: if $c_1=n\,dx_1\wedge dx_2+m\,dx_3\wedge dx_4$ on the unit torus, then $\int c_1^2=2nm$, with the factor two arising from degree-two commutativity. A topological density can have a variation that is entirely a boundary term, so it need not furnish a bulk field equation.

Definitions

For a nonvanishing piecewise smooth complex loop γ, winding is $(2\pi i)^{-1}\oint\gamma^{-1}d\gamma$. The integral depends on its homotopy class in the punctured complex plane.

For an abelian group written additively, a normalized multiplier σ satisfies $\sigma(g,h)\sigma(g+h,k)=\sigma(h,k)\sigma(g,h+k)$. Rephasing by $b(g)$ multiplies σ by $b(g)b(h)/b(g+h)$.

A closed differential form is exact only if it has a global primitive; nonzero integral over a closed cycle certifies nonexactness.

Theorem and hypotheses

For $G=F_2^2$ and $\sigma((a,b),(c,d))=(-1)^{bc}$, σ is a two-cocycle which is not a coboundary. The angular form $(-y\,dx+x\,dy)/(x^2+y^2)$ on the punctured plane is closed but not exact.

Proof

For the cocycle, write $k=(e,f)$. The left exponent is $bc+(b+d)e$, while the right exponent is $de+b(c+e)$; they agree modulo two. Hence the multiplication is associative. If σ were an abelian coboundary, it would be symmetric in g,h because $b(g)b(h)/b(g+h)$ is symmetric. But for $g=(1,0)$, $h=(0,1)$, $\sigma(g,h)=1$ and $\sigma(h,g)=-1$, a contradiction. The alternating commutator ratio therefore detects the nontrivial projective class and survives rephasing. For the angular form, differentiation gives zero on the punctured plane. Restrict to the unit circle $x=\cos t, y=\sin t$: the form becomes dt and its integral is 2π. Every globally exact form has zero integral on a closed loop by the fundamental theorem, proving nonexactness. This is the same local-to-global pattern as gluing, with a de Rham coefficient model rather than finite incidence coordinates.

Worked example

Enumerate the four elements of F_2^2 and the sixty-four triples, then evaluate the multiplier and its commutator.

Every triple passes the cocycle identity. The chosen coordinate generators anticommute projectively: their multiplier ratio is -1. In the geometric model $\gamma(t)=e^{ikt}$ with integer k, $\gamma^{-1}\gamma'=ik$, so the winding is k. Under a homotopy that crosses zero, γ^{-1} is undefined at the crossing and the certificate ceases to apply. For the unit four-torus with $n=2,m=3$, the normalized characteristic integral is $2\cdot2\cdot3=12$, independently of the chosen connection representing that fixed line-bundle class.

The invariant is the class with its hypotheses and normalization. Local changes of gauge alter representatives, while crossing a singularity, changing coefficients or changing the underlying bundle can alter the class itself.

Continuity across the source years

2023 — Dated topic continuity is supported by: Winding number and Gauss–Bonnet with boundaries explicit. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2023:M089

2024 — Dated topic continuity is supported by: Wedge products, Hodge duality and a genuine closed-form obstruction; Characteristic forms and why a topological density is not a bulk field equation. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-06-015, 2024:Y24-11-022

2025 — Dated topic continuity is supported by: Projective phases require a genuine two-cocycle; A projective multiplier gives a concrete H^2 cocycle; Transition functions satisfy a cocycle equation, and gauge changes preserve it. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-01-009, 2025:Y25-03-020, 2025:Y25-10-005

2026 — Dated topic continuity is supported by: A torus mirror cocycle can carry exactly one binary obstruction. These records support the presence and refinement of the topic in this year; the present

synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-06-023

Connections

D03 — Projective representation theory. Associativity is the multiplier cocycle equation, with scalar rephasing acting by coboundaries. The coefficient action matters for group cohomology.

D06 — Curvature and characteristic forms. Normalized curvature forms represent fixed bundle classes whose cycle integrals can be quantized. Arbitrary matrix loop data do not define a Chern class.

D07 — Complex analysis. The argument principle can certify winding by counting zeros and poles. One must exclude zeros on the integration contour.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: group elements and multipliers, nonvanishing loops, or normalized curvature on a declared bundle.

Outputs: Written cocycle and period certificates, including all 64 multiplier identities in the verifier. No arbitrary cocycle, winding or characteristic-class UI is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

The finite multiplier is an explicit example, not an identification of every historical phase model. Characteristic forms require declared degree, orientation, trace and 2π conventions. A finite cover multiplies a nonzero integer circle class by its degree; it does not erase it.

Next questions

For the proposed transition data, compute the actual cocycle and determine whether a gauge change can trivialize it.

For any claimed integer characteristic value, exhibit the bundle, orientation, integral cohomology class and normalized representative.

Source anchors: 2023:M089, 2024:Y24-06-015, 2024:Y24-11-022, 2025:Y25-01-009, 2025:Y25-03-020, 2025:Y25-10-005, 2026:Y26-06-023

Bott and Tu Differential Forms in Algebraic Topology, publisher record — https://link.springer.com/book/10.1007/978-1-4757-3951-0
Chern, characteristic classes of Hermitian manifolds — https://doi.org/10.2307/1969037
J S Milne Group Theory — https://www.jmilne.org/math/CourseNotes/GT.pdf
Etingof–Gelaki–Nikshych–Ostrik, Tensor Categories — https://math.mit.edu/~etingof/egnobookfinal.pdf
Stacks Project descent data — https://stacks.math.columbia.edu/tag/023U
J. S. Milne, Class Field Theory, group cohomology background — https://www.jmilne.org/math/CourseNotes/CFT.pdf

D06 · Differential and symplectic geometry

Organizing question

What becomes computable once the metric, form, connection and symmetry are explicitly declared?

This domain connects local geometric structure to calculable global behavior. The first input is a declared metric, differential form or connection. Coordinates alone supply none of these. Once a metric is given, one can derive length, geodesics, curvature, volume growth and spectral scaling; once a symplectic form and Hamiltonian are given, one can derive a vector field and test conservation. This order of construction turns the archive’s geometric vocabulary into a set of mathematical interfaces with explicit assumptions and outputs.

Metric geometry separates representation changes from genuine geometric deformation. The hyperbolic half-plane has computable Christoffel symbols, vertical unit-speed geodesics and curvature minus one. Its exponential volume growth prevents a global Euclidean doubling argument even though every smooth point is locally Euclidean in the appropriate sense. Constant scaling links length, curvature and spectral thresholds through exact powers. Discrete transport curvature and prefix ultrametrics then show how different metric inputs answer different questions: probability measures matter for Ollivier curvature, while digit carries obstruct the inverse inference from real distance to prefix agreement. Neither a positive weight table nor an unexplained singular value is a metric-independent invariant.

Symplectic and contact geometry organize mechanics and reduction. Conservation of one Hamiltonian does not make it a Casimir. Removing a cyclic coordinate is a legitimate reduction only at a fixed momentum level, where the restricted two-form’s null directions are precisely the symmetry directions being quotiented. A regular moment-map quotient can be identified with a cotangent bundle in explicit charts. The oscillator demonstrates another kind of geometric change: a generating function blows up at a projection caustic while the underlying phase-space trajectory remains smooth. A Legendrian curve and a thermodynamic potential similarly require the contact form and tangent condition before their consequences follow.

Connections supply transport, curvature measures local two-dimensional noncommutation, and holonomy records a global loop. A flat circle system can have nontrivial monodromy; a curved connection on the plane can live on a trivial bundle; a Berry eigenline over the sphere can obstruct a global smooth potential. The exact shear loop makes noncommutativity and gauge conjugacy visible in finite matrices. It is the appropriate worked model for the interactive atlas, with a clear gap between this calculation and an empirical connection whose actual matrices or two-dimensional refinement are unavailable.

Curves and variational geometry complete the domain by integrating local data into shapes. Frenet frames are rotation equations tied to regularity, moving affine frames introduce explicit derivative terms, and linear curvature uniquely minimizes one declared curvature-rate functional. Endpoint positions or tangents add constraints and create a different problem. The same care with assumptions governs Willmore normalization and the distinction between immersions and embeddings. The source dates support continuity from 2023 Berry and affine questions through the 2024 geometric forms to the 2026 exact transport and reduction examples. The present proofs unify those dated topics without attributing new historical arguments. The open frontier is the construction of declared higher-dimensional metrics, transports and quotient data with quantitative verification.

Core objects

metrics, path lengths and curvature

symplectic and contact forms

moment maps and reduced spaces

connections, parallel transport and holonomy

moving frames, curvature functionals and geometric curves

Open frontier

Construct genuine two-dimensional transport refinements, verify regular quotient hypotheses and solve geometric variational problems with the full intended boundary conditions; distinguish representation singularities from failure of the underlying structure.

D06-F01 · Metrics, curvature, scale and transport geometry

Which conclusions follow from a declared metric, and which survive a change of scale or representation?

Geometry begins with enough data to measure. A coordinate chart does not select a Riemannian metric, a positive symmetric weight table need not satisfy the triangle inequality, and a Jacobian alone does not determine singular values without input and output inner products. These distinctions let the metric material in the archive be organized by constructive outputs: path length, geodesics, curvature, volume growth and transport distance. The archive’s hyperbolic examples are particularly coherent because the metric is explicit and each claimed phenomenon can be computed from it.

On the upper half-plane with $ds^2=(dx^2+dy^2)/y^2$, the connection, vertical unit-speed geodesics and curvature -1 agree with the disk model under the Cayley map. Constant metric rescaling changes distances, volumes, curvature and Laplace spectra in related but distinct ways. This blocks a common confusion between a geometric normalization and a dimensionless threshold. Hyperbolic volume growth gives another exact boundary: local Euclidean behavior does not imply a global doubling bound. Discrete Ollivier curvature introduces still another metric construction, comparing probability distributions by optimal transport. Its value depends on the measure convention as well as the graph, and it cannot certify unrelated computational hardness. Prefix ultrametrics supply a complementary geometry: distance is controlled by the first differing digit, while a many-to-one real coding map can erase prefix information through carries. The best synthesis therefore treats metrics as inputs whose consequences can be tested, rather than as labels attached to algebraic expressions.

Definitions

A Riemannian metric g assigns positive inner products to tangent spaces; its path distance is the infimum of lengths of curves joining two points.

For a constant $c>0$, scaling g to c^2g multiplies lengths by c, n-dimensional volume by c^n, sectional curvature by c^{-2} and the scalar Laplacian by c^{-2}.

For graph measures m_x and m_y, Ollivier curvature on an edge is $\kappa(x,y)=1-W_1(m_x,m_y)/d(x,y)$, with the measure convention declared.

On b-ary sequences, d(x,y)=a^{N(x,y)} for 0<a<1 and first-difference index N gives an ultrametric, since common prefixes concatenate by a minimum rule.

Theorem and hypotheses

The metric $g=(dx^2+dy^2)/y^2$ has Gaussian curvature -1 and vertical unit-speed geodesics $y=y_0e^t$. Hyperbolic three-space has ball volume $V_3(R)=\pi(\sinh 2R-2R)$, and consequently $V_3(2R)/V_3(R)$ is unbounded. For nonlazy uniform-neighbor measures, K_2 has edge curvature zero and K_3 has curvature 1/2.

Proof

For the conformal plane metric e^{2u}(dx²+dy²), $K=-e^{-2u}\Delta u$. Here $u=-\log y$, so $\Delta u=1/y^2$ and $K=-1$. A vertical path $y=y_0e^t$ has speed $|y'|/y=1$ and satisfies the geodesic equation from $\Gamma^y_{yy}=-1/y$. In H^3 the sphere area is $4\pi\sinh^2 r$. Integrating from zero to R gives $\pi(\sinh 2R-2R)$. Its exponential leading term is (π/2)e^{2R}, so doubling the radius produces an unbounded ratio. For K_2, each nonlazy neighbor distribution is a point mass at the opposite endpoint, so $W_1=1$ and $\kappa=0$. For K_3, the neighbor measures share mass 1/2 at the third vertex; only the other mass 1/2 must move distance one, giving $W_1=1/2$. A transport plan attains that cost, and the mass discrepancy at one endpoint gives the matching lower bound.

Worked example

Fix the nonlazy neighbor convention on K_2 and K_3, and compare a constant rescaling $g\mapsto 9g$ in the hyperbolic metric.

The exact transport costs are 1 and 1/2, yielding curvatures 0 and 1/2. Under 9g, the old unit vertical geodesic has speed three in the same parameter, and reparametrization by t/3 restores unit speed. Gaussian curvature becomes $-1/9$. A Laplace eigenvalue or quadratic-form spectral threshold scales by 1/9 as well, whereas a dimensionless ratio formed with the squared length scale can remain unchanged.

A metric-dependent value is meaningful when its convention is part of the result. Topological equivalence, geometric equivalence and equality of a few scalar observables remain different comparison problems.

Continuity across the source years

2024 — Dated topic continuity is supported by: Hyperbolic curvature and geodesics in a declared metric; SU(2) rotations, area asymptotics, and a metric counterexample. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-06-012, 2024:Y24-12-017

2025 — Dated topic continuity is supported by: Spectral thresholds and curvature must survive changes of scale; Hyperbolic volume growth prevents a global Euclidean doubling argument; Ollivier curvature is a transport calculation, not a hardness oracle. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-03-008, 2025:Y25-04-014, 2025:Y25-07-032

2026 — Dated topic continuity is supported by: Prefix geometry has an exact Hausdorff dimension and a carry obstruction. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-02-005

Connections

D07 — Spectral geometry. Quadratic forms and Laplace spectra depend on the declared metric and rescale by length^{-2}. Curvature bounds alone do not determine the full spectrum.

D10 — Optimal transport. Wasserstein distance compares the chosen probability distributions. Changing laziness changes discrete curvature.

D14 — Representation and coding. A prefix metric measures symbolic agreement that a real coding map may collapse. A small image distance does not imply a long common prefix.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: a declared metric, scaling factor or graph-neighbor measure convention.

Outputs: Written metric-scaling, hyperbolic and finite transport calculations. No arbitrary geodesic, curvature or Wasserstein solver is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

The exact graph checks cover two finite graphs, not a general optimal-transport solver. The hyperbolic formula assumes curvature −1 and the stated dimension. A positive weight matrix can fail metric axioms, and no unprovided metric is inferred from a historical coordinate description.

Next questions

For a proposed metric deformation, prove positivity and the triangle property, then identify which quantities are invariant under the allowed coordinate changes.

For a claimed curvature-based certificate, state the measure convention and prove an explicit theorem linking that curvature quantity to the target property.

Source anchors: 2024:Y24-06-012, 2024:Y24-12-017, 2025:Y25-03-008, 2025:Y25-04-014, 2025:Y25-07-032, 2026:Y26-02-005

John Lee Introduction to Riemannian Manifolds, author page — https://sites.math.washington.edu/~lee/Books/RM/
Carbonaro–Mauceri–Meda, H1 and BMO for locally doubling spaces — https://www.numdam.org/item/ASNSP_2009_5_8_3_543_0/
Kenneth Falconer, Fractal Geometry — https://doi.org/10.1002/9781118762868
Yann Ollivier, Ricci curvature of Markov chains on metric spaces — https://arxiv.org/abs/math/0701886

D06-F02 · Hamiltonian, symplectic and contact structures

How do conservation, reduction and projection follow from a specified geometric form?

Symplectic geometry makes dynamical rules structural: a nondegenerate closed two-form converts the derivative of a Hamiltonian into a vector field. Contact geometry replaces this even-dimensional structure with a maximally nonintegrable hyperplane distribution in odd dimension. The archive’s examples become coherent when these forms are written before their consequences. Hamiltonian conservation follows from antisymmetry, a Casimir is a much stronger object, and a cyclic coordinate can be removed only after its conjugate momentum is fixed. A dimension count for a quotient needs a regular constraint and a free action, not just a formal subtraction.

The oscillator links Hamilton’s equations, Hamilton–Jacobi graphs and Riccati evolution without identifying their coordinate descriptions. Its full phase-space flow is smooth for all times. The generating function for the image of one initial graph fails at a caustic because projection to configuration space ceases to be invertible. The divergence is in the graph representation rather than in the trajectory itself. The quiver-style cotangent quotient supplies an equally concrete reduction: solving a moment-map equation and quotienting a free scaling action produces cotangent coordinates, including their canonical one-form. Contact examples then add a compatible but distinct structure. A Legendrian curve satisfies a tangent condition such as dz−y dx=0; on a thermodynamic equilibrium surface the same contact logic yields a

potential identity and a Maxwell relation, provided the stated differentiability and change of variables hold.

Definitions

Use $\omega=\Sigma dq_j\wedge dp_j$ and define $i_{X_H}\omega=dH$, giving $\dot{q}_j=\partial H/\partial p_j$ and $\dot{p}_j=-\partial H/\partial q_j$.

A canonical Casimir has zero Poisson bracket with every differentiable function; nondegeneracy forces it to be constant on a connected domain.

For a symmetry generated by translation of q_2, its momentum is p_2. The reduced space at $p_2=\mu$ is the momentum level modulo that translation.

A contact form α in dimension $2n+1$ satisfies $\alpha\wedge(d\alpha)^n\neq0$. A Legendrian submanifold has dimension n and tangent bundle inside $\ker\alpha$.

Theorem and hypotheses

For $H=(p_1^2+p_2^2)/2+V(q_1)$, reduction at $p_2=\mu$ yields $\omega_{red}=dq_1\wedge dp_1$ and $H_\mu=p_1^2/2+V(q_1)+\mu^2/2$. For the scaling action on $(z,w)\in\mathbb{C}^m\oplus(\mathbb{C}^m)^*$, the regular quotient $\{w(z)=0,z\neq0\}/\mathbb{C}^*$ is T^*P^{m-1}.

Proof

On the level $p_2=\mu$ every tangent variation has $dp_2=0$, so the pullback of ω is $dq_1\wedge dp_1$. Its kernel is precisely the q_2 translation direction. Quotienting removes that kernel and leaves a nondegenerate form. The restricted H is independent of q_2, so it descends; the projected full equations are exactly Hamilton’s equations of H_μ. In the complex quotient, z selects a line ℓ. The constraint makes w vanish on ℓ, and $z\otimes w$ defines an invariant map $\mathbb{C}^m/\ell\to\ell$, canonically a cotangent vector at ℓ. Conversely a cotangent vector has such a representation after choosing a nonzero z, with different choices exactly the scaling action. The moment-map derivative is surjective on $z\neq0$ by varying w, and the action is free there, giving dimension $2m-2$. In a chart $z_1=1$ with $z=(1,u)$, the constraint gives $w=(-up,p)$ and the canonical one-form $w\cdot dz$ is $p\,du$.

Worked example

For the unit oscillator $H=(q^2+p^2)/2$ start with $p_0=0$, and compare the full trajectory with its Hamilton–Jacobi graph.

Hamilton’s equations give $q=q_0\cos t$, $p=-q_0\sin t$. When $\cos t\neq0$ this graph is $p=-q\tan t$, with generating function $S=-q^2\tan t/2$. Direct substitution gives $S_t+(S_q^2+q^2)/2=0$. At $t=\pi/2$ all initial q_0 project to $q=0$ while their momenta differ, so no single-valued $p(q)$ describes the line. The phase-space map is still the exact symplectic rotation. For $\alpha=dz-y\,dx$, $\gamma(t)=(t,t,t^2/2)$ satisfies $\gamma^*\alpha=t\,dt-t\,dt=0$ and is Legendrian.

Reduction removes a declared symmetry after fixing conserved momentum. A caustic changes a coordinate description. Neither phenomenon means that the original smooth dynamics has ceased to exist.

Continuity across the source years

2024 — Dated topic continuity is supported by: Hamilton, Hamilton–Jacobi and Riccati equations meet at a caustic; Hamiltonian conservation is not the Casimir property; Legendrian structure begins with a contact form and a tangent condition. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-03-031, 2024:Y24-06-003, 2024:Y24-07-014, 2024:Y24-09-027

2026 — Dated topic continuity is supported by: Thermodynamic contact geometry derives a Maxwell relation; Symplectic reduction removes a cyclic coordinate only after fixing momentum. These records

support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-01-018, 2026:Y26-05-017

Connections

D09 — Mechanics and dynamics. Hamiltonian vector fields and conserved momenta define the reduced dynamical model. Energy conservation is weaker than the Casimir property.

D12 — Optimization and Hamilton–Jacobi. Generating functions and Riccati equations describe projections of Lagrangian evolution. A graph chart can fail at a caustic while the geometric solution persists.

D02 — Geometric quotients. A moment-map equation and a free stable locus identify an actual cotangent bundle. Singular strata require a separate quotient analysis.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: symplectic/contact form, Hamiltonian, symmetry action and fixed momentum value.

Outputs: Written reduction, cotangent quotient, oscillator-caustic and Legendrian certificates. No symplectic-reduction or Hamilton-Jacobi UI is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

The cotangent quotient is a Higgs-type complex symplectic model; identifying a hyperkähler metric or a Coulomb branch needs additional constructions. Singular momentum levels and nonfree actions require stratified reduction. The oscillator graph is valid only away from its caustic.

Next questions

For a specified symmetry action, prove regularity and freeness on the chosen momentum level, and calculate the reduced form in compatible charts.

For a proposed Hamilton–Jacobi model, locate every projection singularity and replace failed graph coordinates with another local generating family.

Source anchors: 2024:Y24-03-031, 2024:Y24-06-003, 2024:Y24-07-014, 2024:Y24-09-027, 2026:Y26-01-018, 2026:Y26-05-017

MIT Advanced Classical Mechanics notes — https://ocw.mit.edu/courses/8-09-classical-mechanics-iii-fall-2014/d9bac33f6c60b304dc0398e99b327102_MIT8_09F14_full.pdf

MIT optimal-control notes for comparison of HJ and Riccati formulations — https://ocw.mit.edu/courses/16-323-principles-of-optimal-control-spring-2008/pages/lecture-notes/

Meinrenken Introduction to Poisson Geometry — https://www.math.toronto.edu/mein/teaching/MAT1341_PoissonGeometry/Poisson7.pdf

John Etnyre Legendrian and Transversal Knots — https://arxiv.org/abs/math/0306256

Hausel–Sturmfels, Toric Hyperkähler Varieties — https://arxiv.org/abs/math/0203096

Vladimir Arnold, Mathematical Methods of Classical Mechanics — https://doi.org/10.1007/978-1-4757-2063-1

Ana Cannas da Silva, Lectures on Symplectic Geometry — https://www.math.ist.utl.pt/~acannas/Books/lsg.pdf

D06-F03 · Connections, gauge covariance, curvature and holonomy

How can local transport and global loop effects be distinguished and checked without confusing flatness with triviality?

A connection supplies a rule for differentiating and transporting sections. Curvature measures its local failure to commute in two independent directions; holonomy measures transport around a loop. They interact, but neither is interchangeable with the other or with the topology of the underlying space. A one-dimensional circle supports no curvature two-form and can still have nontrivial monodromy. On a simply connected surface, a nonflat connection can have loop holonomy arising from curvature even when the bundle is topologically trivial. A nontrivial bundle can additionally obstruct a global gauge potential. These distinctions give the archive's connection and Berry material a common, precise architecture.

Gauge transformations change coordinates in the fibers. They conjugate loop holonomy at its base point, leaving conjugacy-invariant information such as trace, determinant and eigenvalues unchanged. The ordered product matters for matrices; scalar circulation cannot stand in for a nonabelian calculation. The source's nilpotent shear example is especially effective because its exponentials terminate and its loop product is an exact polynomial. Berry transport adds a geometric realization: eigenlines form a bundle over a parameter space, and local eigenvector choices are gauge choices. A Chern number requires that declared bundle and a normalized curvature integral; it cannot be recovered from a phase function on an unidentified state space. For the spin-half sphere example the integral is one, proving that no globally smooth potential can represent that curvature on the whole sphere.

Definitions

For $\nabla=d+A$ on a vector bundle, curvature is $F=dA+A\wedge A$ and ∇^2 acts by F. Parallel sections satisfy $Fv=0$, a necessary condition rather than a general converse.

For edge transport $U_j:V_j\to V_{j+1}$, basis changes h_j act by $U_j\mapsto h_{j+1}U_j h_j^{-1}$. Ordered loop holonomy is the composite returning to its base fiber.

Flat means $F=0$. Trivial holonomy means every relevant loop acts as identity. Topological triviality means the underlying bundle has a global trivialization; these are different properties.

Theorem and hypotheses

A loop holonomy transforms by conjugation at the base vertex. For $X=[[0,1],[0,0]]$ and $Y=[[0,0],[1,0]]$, the loop $(I+xX)(I+yY)(I-xX)(I-yY)$ equals $[[1+xy+x^2y^2,-x^2y],[xy^2,1-xy]]$. Its mixed quadratic coefficient is $[X,Y]=\mathrm{diag}(1,-1)$.

Proof

Compose transformed edge maps in path order. Each interior h_j^{-1} cancels its neighboring h_j, leaving only $h_0 H h_0^{-1}$; this proves gauge covariance and explains why entries themselves are not invariant. Since $X^2=Y^2=0$, $\exp(xX)=I+xX$ and $\exp(yY)=I+yY$ exactly. Multiplying the four factors gives the displayed polynomial, whose determinant is one and trace is $2+x^2y^2$. Setting $x=y=\varepsilon$ yields $I+\varepsilon^2[X,Y]$ plus explicit cubic and quartic terms. This is the exact algebraic mechanism behind a small plaquette limit; a smooth curvature interpretation additionally needs specified base directions, derivative terms and a transport sign convention. On a circle $A=i\alpha\,d\theta$ is flat by dimension, but with parallel transport defined by $s'+i\alpha s=0$ its holonomy is $\exp(-2\pi i\alpha)$, nontrivial for noninteger α. Thus flatness alone does not force global trivial monodromy.

Worked example

Take x=y=1 in the shear loop and a base gauge h=diag(2,1). Compare it with a rank-one flat circle connection.

The shear holonomy is H=[[3,−1],[1,0]], with determinant one and trace three. Conjugation gives hHh^{-1}=[[3,−2],[1/2,0]], retaining both invariants. For α=1/2 the flat circle holonomy is −1. As a separate curved example on R^2, A=ix dy has F=i dx∧dy. Parallel transport around a positively oriented unit square is exp(−i) in the stated convention by the scalar Stokes theorem, despite the bundle and base being topologically trivial.

These examples give three independent checks: fiber gauge covariance, local curvature and global monodromy. A numerical loop becomes geometric evidence only after these inputs and conventions are declared.

Continuity across the source years

2023 — Dated topic continuity is supported by: A two-level Berry holonomy with a gauge check. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2023:M041

2024 — Dated topic continuity is supported by: Bundle curvature, parallel sections, and a determinant identity that needs evidence. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-12-006

2025 — Dated topic continuity is supported by: Berry curvature and a Chern number require a state bundle. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-02-016

2026 — Dated topic continuity is supported by: Noncommuting shears have an exact loop commutator; Holonomy on a loop and curvature on a surface are different invariants. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2026:Y26-04-005, 2026:Y26-08-018

Connections

D05 — Monodromy and gluing. Flat local systems can have nontrivial global holonomy and twisted cohomology. The coefficient representation must be supplied.

D16 — Quantum state geometry. Berry connections live on declared eigenstate bundles, and their normalized curvature can yield Chern classes. An adiabatic physical prediction needs its spectral-gap hypotheses.

D15 — Transport diagnostics. Loop matrices and area refinement provide candidate tests for a specified learned transport. Unrecovered matrices or missing two-dimensional operators remain gaps.

Capacity and interactive scope

Lab ids: ['L05']

Inputs: Implemented: shear parameters a,b in [-3,3] and shear-loop mode. The separate projection mode selects a line angle from 5 to 85 degrees.

Outputs: Implemented: exact integer shear commutator and determinant; projection mode gives a floating contraction factor, commutator norm and iteration bound. Gauge conjugation, circle holonomy

and curvature refinements are written/verifier models, not UI operations.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Finite shear and projection models implemented; general connections and curvature reconstruction remain recipes.

Limits

The lab’s shear loop and rank-one circle are exact worked models. They do not certify an empirical connection whose matrices, base space or refinement rule are absent. A full curvature limit needs derivative contributions and quantitative area-normalized remainder control.

Next questions

Construct actual plaquette transports on a two-dimensional parameter mesh and bound the difference between area-normalized holonomy and the declared curvature.

For a flat connection on a specified non-simply-connected base, compute its monodromy representation and determine whether a global parallel frame exists.

Source anchors: 2023:M041, 2024:Y24-12-006, 2025:Y25-02-016, 2026:Y26-04-005, 2026:Y26-08-018

Michael Berry Quantal phase factors accompanying adiabatic changes — https://doi.org/10.1098/rspa.1984.0023
Chern, characteristic classes — https://doi.org/10.2307/1969037

D06-F04 · Curves, moving frames and geometric variational models

Which curve geometry is fixed by curvature, and when does a variational principle actually select a shape?

Curves provide a manageable laboratory for separating geometry, coordinates and dynamics. An affine transformation composes by the semidirect rule, while a time-dependent coordinate frame adds velocity and acceleration terms. A Frenet frame follows a curve’s tangent, normal and binormal; its rotation can be written using a Darboux angular-velocity vector. Neither operation by itself is a new physical law. The archive’s most useful curve examples become explicit once the parameter, regularity and objective are fixed: winding requires a nonvanishing loop, spirograph closure requires commensurate frequencies, and a clothoid minimizes one declared curvature-rate energy with endpoint curvatures.

The variational distinction is especially important. Minimizing the squared derivative of curvature under fixed endpoint curvatures selects linear curvature, but prescribing endpoint positions and tangent directions adds integral constraints and changes the admissible problem. A clothoid is therefore an exact solution of a clearly stated functional, not a universal optimal connector. Extrinsic surface geometry makes a parallel point about normalization: if mean curvature is the average of the principal curvatures, a round sphere has Willmore energy 4π at every radius; using their sum changes the scale by four. Surface immersion, embedding and eversion are also different questions. The curve and surface material consequently contributes a practical discipline: identify the geometric object, pick invariant quantities, state the boundary data and only then optimize or classify.

Definitions

For a unit-speed regular space curve with nonzero curvature, the Frenet equations are $T'=\kappa N$, $N'=-\kappa T+\tau B$ and $B'=-\tau N$. Their Darboux vector is $\Omega=\tau T+\kappa B$.

For a planar unit-speed curve, $\theta'=\kappa$ and $\gamma'=(\cos\theta,\sin\theta)$. A curvature function plus an initial position and tangent determines the curve by integration.

Fix $L>0$ and endpoint curvatures $\kappa(0)=\kappa_0$, $\kappa(L)=\kappa_1$. The curvature-rate objective is $J[\kappa]=(1/2)\int_0^L(\kappa')^2ds$ on absolutely continuous κ with square-integrable derivative.

With averaged mean curvature $H=(\kappa_1+\kappa_2)/2$, the Willmore functional is $\int H^2dA$. Its normalization must be stated before numerical comparison.

Theorem and hypotheses

The unique minimizer of the stated curvature-rate problem is $\kappa(s)=\kappa_0+(\kappa_1-\kappa_0)s/L$, with $J_{min}=(\kappa_1-\kappa_0)^2/(2L)$. The associated tangent angle is $\theta_0+\kappa_0 s+(\kappa_1-\kappa_0)s^2/(2L)$. A round sphere has averaged-mean-curvature Willmore energy 4π independently of radius.

Proof

Write $\kappa=\kappa_{lin}+\eta$ with $\eta(0)=\eta(L)=0$. Expanding J gives $J[\kappa]=J[\kappa_{lin}]+(\kappa_1-\kappa_0)\int\eta'/L+(1/2)\int(\eta')^2$. The cross term vanishes because the endpoint difference of η is zero. The last term is nonnegative, with equality only when η is constant, hence zero. This proves existence and uniqueness without assuming an unproved Euler–Lagrange conclusion. Integrating $\theta'=\kappa$ gives the quadratic phase, and integrating the unit tangent yields Fresnel-type coordinates. For a sphere of radius R, both principal curvatures have magnitude $1/R$, so $H^2=1/R^2$ and area $=4\pi R^2$, giving 4π. The Frenet–Darboux identity follows directly from the oriented orthonormal cross-product relations: $\Omega\times T=\kappa N$, $\Omega\times N=-\kappa T+\tau B$ and $\Omega\times B=-\tau N$. It describes the same frame rotation rather than an additional dynamical postulate.

Worked example

Set $L=2$, $\kappa_0=0$, $\kappa_1=2$ and $\theta_0=0$. Compare linear curvature with $\kappa_a(s)=s+a\,s(2-s)$.

The minimizer is $\kappa=s$ with $\theta=s^2/2$ and $J=1$. Since $\kappa_a'=1+2a-2as$, direct integration gives $J[\kappa_a]=1+4a^2/3$, strictly larger for $a\neq0$. For a moving affine frame $y=M(t)x+b(t)$, differentiation gives $\dot y=M\dot x+Mx+\dot b$ and $\ddot y=M\ddot x+2M\dot x+Mx+\ddot b$; the extra terms are coordinate effects determined by the frame. For a two-frequency spirograph with both coefficients nonzero, rational frequency ratio gives a common period, while irrational ratio prevents one by uniqueness of finite exponential sums.

The constructive geometric object comes from integrating declared local data. Optimality, closure and coordinate covariance each have their own explicit conditions, allowing the interactive atlas to vary one assumption at a time.

Continuity across the source years

2023 — Dated topic continuity is supported by: Affine coordinate changes and time evolution are different operations. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2023:M046

2024 — Dated topic continuity is supported by: Frenet frames, Darboux rotation, and covariant differentiation; Sphere eversion and a correctly normalized Willmore bound. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2024:Y24-11-021, 2024:Y24-12-001

2025 — Dated topic continuity is supported by: Spirograph closure is a commensurability theorem. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument. Sources: 2025:Y25-08-001

2026 — Dated topic continuity is supported by: A clothoid is the minimizer of a specified curvature-rate functional. These records support the presence and refinement of the topic in this year; the present synthesis and proof do not establish historical priority, novelty or a recovered verbatim argument.

Sources: 2026:Y26-01-013

Connections

D08 — Differential equations. Curvature and torsion determine frame ODEs; a chosen functional yields a boundary-value problem. Additional endpoint-position constraints alter the minimizer.

D12 — Variational optimization. A square-completion proof gives a unique minimizer and an exact objective gap. Optimality applies only to the declared admissible class.

D05 — Global closure and winding. Periodicity and origin avoidance supply global conditions beyond local curvature. Equal local invariants need not determine identical global placement.

Capacity and interactive scope

Lab ids: []

Inputs: Proof recipe: curvature endpoints, length and admissible curvature function, or declared moving-frame data.

Outputs: Written unique minimizer, objective-gap and frame identities. No curve drawing, constrained connector optimization or surface eversion UI is implemented.

Validation: Standard-library exact checks in domains/geometry/verify.py. L05 and L09 are finite worked models; an unsupported arbitrary-input solver is not claimed.

Status: Proof-backed recipe; no dedicated interactive lab.

Limits

Frenet coordinates fail where curvature vanishes and need another frame. The clothoid theorem does not impose endpoint positions or fixed total turning. Sphere eversion uses immersions and can pass through self-intersections; it is not an isotopy of embedded spheres.

Next questions

For a connector with fixed endpoint positions and tangents, formulate the additional integral constraints and derive the constrained Euler–Lagrange system.

For a proposed curve classification, determine whether the chosen invariants separate curves up to the exact rigid-motion, reparametrization or homotopy relation intended.

Source anchors: 2023:M046, 2024:Y24-11-021, 2024:Y24-12-001, 2025:Y25-08-001, 2026:Y26-01-013

MIT differential geometry course — https://ocw.mit.edu/courses/18-950-differential-geometry-fall-2008/
Smale, A classification of immersions of the two-sphere — https://doi.org/10.1090/S0002-9947-1959-0104227-9
NIST DLMF, Fresnel Integrals — https://dlmf.nist.gov/7.2

D07 · Analysis, transforms and special functions

Organizing question

What analytic representation makes the desired quantity computable without losing its domain, norm or normalization?

This domain organizes the part of the archive concerned with representing an object so that a question about it becomes a controlled calculation. The representation may be a coefficient sequence, a contour integral, a special-function identity or a spectral measure. In every case, the first task is to state what is being represented and in which space. Bessel’s inequality sees an orthogonal projection; Parseval requires completeness. An atomic pairing uses cancellation in a particular measure. A Fourier transform needs a normalization, and a sampled FFT describes an interpolant before it describes any unknown continuous function. These distinctions provide constructive working rules, not merely objections to formulas. They show how to calculate the information retained by a representation and the error introduced by discarding the rest.

The first family develops that rule through transforms, frames and cancellation. Its four-coordinate Haar calculation accounts exactly for all energy and for the residual after discarding fine detail. A lower frame bound then converts a measurement discrepancy into a state discrepancy with an explicit constant. This is the analytical bridge to reconstruction, while the possibility of a nullspace explains why coefficient summaries can fail to identify an object. Weighted atoms and local Hardy–BMO estimates keep the same measure in the cancellation and pairing; directional image differences explain why a transform may be useful for an edge geometry without being a complete representation.

The second family puts local analytic certificates next to the global information they need. Cauchy extraction requires a holomorphic neighborhood; logarithmic derivatives record zeros or poles with multiplicity; contour counts do not distinguish two simple zeros from one double zero. Positive finite atomic measures give a particularly productive exact class. Their Herglotz functions have simple real poles, a fixed half-plane sign and one simple zero between each pair of poles. The proof uses only explicit signs and derivatives. Branch periods then demonstrate why locally closed differentials can still carry global continuation data.

The third family collects exact integration and special functions by method. Mellin moments, theta and modular reindexing, logarithmic integrals, dilogarithms, elliptic limits and finite trigonometric sums each retain their convergence and branch conditions. The elementary logarithmic moment supplies factorial weights and a rational tail bound for a zeta integral. Its role is to teach a reproducible reduction with a stopping rule, while the original member treatments retain the additional hypotheses needed for their richer constants.

The fourth family treats spectra as data of an operator with a domain and multiplicity. A diagonal model connects a counting function, a heat-trace exponent and a zeta half-plane. It separates fixed-time trace class from a singular short-time limit and shows why changing multiplicity changes dimension. Together, the four families suggest a practical order: fix the representation and its information loss, establish the analytic certificate, calculate with controlled tails, and only then interpret a spectral or geometric invariant. The progression entries identify dated source topics across 2023–2026; they do not date the reconstructed proofs or assert historical discovery priority.

Core objects

Hilbert and atomic function spaces

holomorphic functions and positive spectral functions

Mellin/Fourier transforms and exact periods

self-adjoint operators and spectral measures

Open frontier

The main frontier is a portable certificate that carries domains, measures, branches, multiplicities and uniform error bounds through changes of representation. The recovered material does not yet provide one general automated solver.

D07-F01 · Transforms, frames and cancellation in a declared norm

When do coefficients preserve information, and how does the chosen measure control a reconstruction error?

The recurring analytical object is a map from functions or vectors to coefficients. Fourier, Haar, wavelet, shearlet and character systems make different geometries visible, but every useful reconstruction statement has an ambient space, a normalization and a completeness hypothesis. The early Bessel–Parseval and Hardy–BMO questions already distinguish coefficient energy from the energy of the original object. Later sampling, weighted atoms and coherence questions add perturbations and observation loss. These are constructive ingredients for an analysis workflow: first identify which subspace the coefficients see, then calculate its norm identity or frame bounds, and finally account for noise and discarded directions. A finite example is valuable because every coefficient and residual can be inspected exactly. It does not automatically establish a continuous transform theorem or a global sampling theorem.

Definitions

In a real or complex Hilbert space H, an orthonormal family (e_j) satisfies ⟨e_i,e_j⟩=δ_ij. Its finite projection is P_m x=Σ_{j≤m}⟨x,e_j⟩e_j.

A frame satisfies A||x||²≤Σ_j|⟨x,f_j⟩|²≤B||x||² with 0<A≤B<∞. A vanishing lower bound exposes invisible directions.

An atom supported in a set B cancels in the pairing measure μ when ∫_B a dμ=0; using cancellation in a different measure changes the argument.

Theorem and hypotheses

For a finite orthonormal family, ||x||²=Σ|⟨x,e_j⟩|²+||x−P_mx||². For a finite analysis matrix T with lower frame bound A, any two candidates obey ||x−y||≤A^(−1/2)||Tx−Ty||. If each is consistent with observations to error ε, their distance is at most 2ε/√A.

Proof

The projection residual r=x−P_mx is orthogonal to every e_j, because taking its inner product with e_k subtracts exactly the kth coefficient. Thus P_mx and r are orthogonal. Expanding ||P_mx+r||² removes the cross term, and orthonormality evaluates the projection norm as the sum of coefficient squares. This proves the identity and Bessel inequality without a completeness assumption. For a countable family, monotone convergence of the nonnegative partial sums gives the inequality; equality for every x additionally requires dense closed span. For the frame assertion apply its lower inequality to x−y and take square roots. If ||Tx−b||≤ε and ||Ty−b||≤ε, the triangle inequality bounds ||T(x−y)|| by 2ε. The conclusion is about measurement residual in the declared norm: a good residual has no such implication if A=0. Cancellation follows a parallel mechanism. Writing b_B for the μ-average, ∫ab dμ=∫a(b−b_B)dμ, so an atom bounded by 1/μ(B) pairs at most with the mean oscillation of b. This elementary estimate is the finite or single-atom part of Hardy–BMO duality; the complete representation theorem has additional content.

Worked example

Use the orthonormal Haar basis of R^4: $h_0=(1,1,1,1)/2$, $h_1=(1,1,-1,-1)/2$, $h_2=(1,-1,0,0)/\sqrt{2}$ and $h_3=(0,0,1,-1)/\sqrt{2}$, applied to x=(1,2,3,4).

The coefficients are 5, −2, $-1/\sqrt{2}$ and $-1/\sqrt{2}$. Their squared sum is $25+4+1/2+1/2=30=1^2+2^2+3^2+4^2$. Keeping only h_0,h_1 gives the block-average reconstruction (3/2,3/2,7/2,7/2); its residual has squared norm one, exactly the energy of the omitted detail coefficients.

This calculation makes approximation loss measurable before any interpretation of the four entries. A prime-gap vector, image row or sensor trace uses the same finite identity, while the statistical meaning of its coefficients requires separate assumptions.

Continuity across the source years

2023 — The source topics establish cancellation, equality conditions and the distinction between directional and complete information. Sources: 2023:M019, 2023:M066, 2023:M067, 2023:M084

2024 — Atomic estimates and a finite Haar decomposition turn those distinctions into inspectable calculations. Sources: 2024:Y24-09-023, 2024:Y24-10-014

2025 — Weighted measures, coordinate changes and interpolation make the observation model explicit. Sources: 2025:Y25-02-009, 2025:Y25-02-011, 2025:Y25-09-008

2026 — Power-spectrum ambiguity and Fourier modal rank show exactly which coefficient summaries lose information. Sources: 2026:Y26-02-012, 2026:Y26-08-023

Connections

D11 — Observation loss. An analysis map induces a kernel or equivalence class of indistinguishable signals. A norm identity alone supplies neither statistical identifiability nor a noise distribution.

D15 — Stable reconstruction. A lower frame bound is a finite observability inequality. L04 uses its own singular-value cutoff model, not a universal inverse for arbitrary frames.

D13 — Discrete harmonic analysis. Finite character orthogonality and circulant diagonalization implement exact coefficient maps. The finite formulas do not prove a continuous wavelet or shearlet theorem.

Capacity and interactive scope

Lab ids: ['L04']

Inputs: L04 selects diagonal singular values k^(−p) or exp(−ck), smoothness r and decay p/c in [0.5,3], noise exponent L in [4,14], and integer cutoff N in [1,60]. The severe lab fixes β=1.

Outputs: Floating evaluations of the proved bias bound (N+1)^(−r), noise bound δ/σ_N, their sum, a 60-cutoff comparison curve and a suggested cutoff. This is an upper envelope, not measured reconstruction error.

Validation: The lab engine is replayed on declared diagonal examples; the family verifier separately checks its Haar or diffusion-stencil model. Bounds follow from the proofs rather than sampled plots.

Status: Implemented diagonal cutoff worked model linked by an analysis/approximation bridge. Haar, general frames and PDE discretization remain documented recipes; the lab does not implement them.

Limits

Completeness, a positive lower frame bound and a matched measure cannot be inferred from a visually plausible transform. FFT differentiation differentiates the trigonometric interpolant; aliasing and unknown high frequencies remain outside that certificate.

Next questions

Which observation subspaces retain the task-relevant quantity with the largest provable lower bound?

Can local weighted atoms be assembled with uniform constants on the specified noncompact geometry?

Source anchors: 2023:M019, 2023:M066, 2023:M067, 2023:M084, 2024:Y24-10-014, 2025:Y25-02-009, 2025:Y25-02-011, 2025:Y25-09-008

University of Bonn Hardy and BMO notes — https://www.math.uni-bonn.de/ag/ana/WiSe1415/V4B5_hardy.pdf
MIT OpenCourseWare Functional Analysis Hilbert spaces — https://ocw.mit.edu/courses/18-102-introduction-to-functional-analysis-spring-2021/
Kutyniok, Lemvig and Lim, Shearlets and Optimally Sparse Approximations — https://arxiv.org/abs/1106.1325
I. Daubechies, Orthonormal bases of compactly supported wavelets — https://doi.org/10.1002/cpa.3160410705
Coifman and Weiss Extensions of Hardy spaces and their use in analysis — https://www.degruyterbrill.com/document/doi/10.1515/9781400827268.295/html

D07-F02 · Analytic domains, contour certificates and positive spectral functions

What information is certified by local analytic data, and which global conclusions require a domain or positivity theorem?

Analytic reasoning becomes especially effective when its domain is part of the input. Cauchy integrals extract coefficients only when the disk and singularity assumptions hold. Logarithmic derivatives locate zeros and poles locally but also carry branch and continuation data. A contour argument counts zeros with multiplicity; it need not show that two counted zeros coincide. The 2026 phase and positive-atom discussions add a useful constructive class: a rational function built from positive real atoms has a fixed imaginary sign and strictly ordered real zeros. That model connects coefficient extraction, monotonicity and spectral measures without making any claim about an unspecified zeta operator. It also provides a small exact laboratory in which poles, residues, roots and signs can all be checked.

Definitions

A holomorphic function is complex differentiable on an open set. Cauchy extraction on |z−a|=r requires holomorphy on a neighborhood of the closed disk.

A Herglotz function maps the upper half-plane into itself. For distinct real t_j and positive μ_j, M(z)=Σ_j μ_j/(t_j−z) is a rational example.

The argument principle counts zeros minus poles inside a contour that avoids both; the count includes multiplicity.

Theorem and hypotheses

For $t_1<\cdots<t_m$ and $\mu_j>0$, $M(z)=\Sigma\ \mu_j/(t_j-z)$ is Herglotz. On every interval (t_j,t_{j+1}) it is strictly increasing from $-\infty$ to $+\infty$ and therefore has exactly one simple zero. Its poles are simple with residues $-\mu_j$.

Proof

Write $z=x+iy$ with $y>0$. The imaginary part of $1/(t-z)$ is $y/((t-x)^2+y^2)>0$; positive weighted summation proves the half-plane property. For real x away from the poles, $M'(x)=\Sigma\ \mu_j/(t_j-x)^2>0$. Immediately to the right of t_j its own term tends to $-\infty$, and immediately to the left of t_{j+1} that pole contributes $+\infty$. Continuity and the intermediate value theorem give a zero, strict monotonicity gives uniqueness, and a positive derivative at that zero makes it simple. Multiplying the jth term by $z-t_j$ and taking the limit gives residue $-\mu_j$. Outside the extreme poles all denominators have the same sign, so no further real zero occurs. The argument proves the entire finite interlacing structure without plotting a dense scatter of points. A difference of two such functions does not preserve positive weights and has to be reanalyzed. Separately, the contour integral $f^{(n)}(a)=n!/(2\pi i)\oint f(z)/(z-a)^{n+1}dz$ follows by the convergent Taylor expansion on a smaller disk and termwise integration; its size bound is $n!M/r^n$. These two exact mechanisms explain why sign, locality and the contour hypothesis should travel with any calculated residue or root count.

Worked example

Take $M(z)=1/(-1-z)+2/(2-z)$, with atoms of masses one and two at -1 and 2.

Combining terms gives $M(z)=-3z/[(-1-z)(2-z)]$. The unique real zero between the poles is zero, $M'(0)=1+2/4=3/2$, and the residues are -1 and -2. The reciprocal slope at the zero is $2/3$. By comparison, $p(z)=z^2-1$ has contour count two on $|z|=2$ but two distinct simple roots; its count does not certify a double root.

The atom model supplies an exact positive spectral function and a root certificate. A phase formula may reproduce its local masses only after the chosen A/B convention and derivative sign are fixed.

Continuity across the source years

2023 — The chain-rule and level-set topics supply local derivative control, including the failure of threshold coordinates at a vanishing derivative. Sources: 2023:M079, 2023:M114

2024 — Coefficient extraction and the distinction between finite checks and a global zero theorem sharpen analytic hypotheses. Sources: 2024:Y24-03-026, 2024:Y24-08-008

2025 — Logarithmic poles and contour multiplicity become explicit local certificates. Sources: 2025:Y25-02-018, 2025:Y25-04-012, 2025:Y25-09-009

2026 — Branch periods, positive atoms, monotone phases and critical-point limits connect local calculations to global continuation data. Sources: 2026:Y26-06-012, 2026:Y26-07-016, 2026:Y26-07-017, 2026:Y26-07-019

Connections

D05 — Local-to-global obstruction. A closed logarithmic differential can have a nonzero loop period. Local curl zero does not remove monodromy or establish a single global primitive.

D15 — Root continuation. A simple-root slope bounds local sensitivity through implicit differentiation. A continuation certificate needs a covering or remainder bound along the whole parameter path.

D01 — Special L-functions. A logarithmic derivative records local order of vanishing and leading coefficients. The rational Herglotz proof is not a proof of a Riemann hypothesis.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

The finite positive-atom theorem requires distinct real poles and strictly positive masses. Limits of such functions need convergence hypotheses. Formal positivity of a kernel and pointwise positivity do not imply positive definiteness.

Next questions

Which proposed spectral quotients have a proved Herglotz normalization?

Can every claimed positivity region be covered by interval or derivative bounds rather than sampled points?

Source anchors: 2024:Y24-03-026, 2024:Y24-08-008, 2025:Y25-04-012, 2025:Y25-09-009, 2026:Y26-06-012, 2026:Y26-07-016, 2026:Y26-07-017, 2026:Y26-07-019

MIT Jeremy Orloff, Cauchy integral formula notes — https://ocw.mit.edu/courses/18-04-complex-variables-with-applications-spring-2018/b1a90df19f643b974555bbbb93138a48_MIT18_04S18_topic4.pdf
NIST DLMF Riemann Zeta Zeros — https://dlmf.nist.gov/25.10
NIST DLMF, Polylogarithms — https://dlmf.nist.gov/25.12
Baranov et al., Spectral synthesis in de Branges spaces — https://arxiv.org/abs/1309.6915

D07-F03 · Exact integrals, special functions and normalized periods

How can an integral be reduced to an exact constant while keeping convergence, branch and normalization information visible?

The archive contains a sustained exact-integration strand: elementary logarithmic moments, sine quotients, Catalan and dilogarithm values, theta and modular transformations, Mellin central values, elliptic logarithms and a final finite-phase integral. These belong together as calculational methods rather than as one claimed transcendence theory. A useful reduction identifies an integration domain, a convergent expansion, the interchange theorem and a final finite or controlled infinite expression. The resulting number can then be checked independently by rational recurrences, symmetry or tail enclosures. The elementary Mellin moment below is a central reusable example. It supplies factorial weights exactly and explains why positive geometric expansions turn certain logarithmic integrals into zeta sums. More elaborate modular and elliptic examples retain additional transformation and branch information that this elementary calculation does not replace.

Definitions

For a>0 and integer m≥0, the Mellin logarithmic moment is J_m(a)=∫₀¹x^(a−1)(−log x)^m dx.

An exact special-function identity includes its parameter region and branch convention. A numerical agreement at one parameter is a check, not the identity itself.

A period representation here means a specified convergent integral or analytic continuation; no algebraic independence claim follows from that name.

Theorem and hypotheses

For real a>0 and integer m≥0, J_m(a)=m!/a^(m+1). For m≥1, ∫₀¹(−log x)^m/(1−x) dx=m!ζ(m+1). Truncating the latter positive series after n=N leaves a tail between zero and (m!/m)N^(−m).

Proof

Set x=e^(−t). Then dx=−e^(−t)dt and x^(a−1)dx=−e^(−at)dt, so the first integral becomes ∫₀∞e^(−at)t^m dt. The substitution u=at factors out a^(−m−1). Integration by parts gives Γ(m+1)=mΓ(m), starting from Γ(1)=1; the boundary terms vanish because an exponential dominates any fixed power. This proves the first formula and convergence at both endpoints. For 0<x<1, 1/(1−x)=Σ_{k≥0}x^k. All summands multiplied by (−log x)^m are nonnegative, so monotone convergence justifies exchanging integral and sum without an unproved uniform bound near x=1. Each resulting moment is m!/(k+1)^(m+1). The series converges exactly for m≥1 in the nonnegative-integer range. The tail is m!Σ_{n>N}n^(−m−1), at most m!∫_N∞t^(−m−1)dt=(m!/m)N^(−m). The m=0 integral diverges logarithmically, showing why the endpoint restriction is mathematical content. Complex a or a noninteger logarithmic exponent can be handled only after choosing powers and checking a new convergence region. This proof is deliberately transparent enough that its finite partial sums and error bounds can be reproduced with rational arithmetic.

Worked example

Compute J₃(2) and the integral K=∫₀¹(−log x)/(1−x) dx.

J₃(2)=3!/2⁴=3/8 exactly. For K, the first four rational summands give 1+1/4+1/9+1/16=205/144. The remaining positive tail is at most 1/4, hence 205/144<K<241/144. Euler’s separately established ζ(2)=π²/6 identifies the exact constant, while the rational enclosure already checks the reduction without using that evaluation.

This is a reproducible reduction with an explicit stopping rule. The same workflow is appropriate for the archive’s accelerated arctangent, product-variable and finite-phase identities, with their own expansions and convergence arguments.

Continuity across the source years

2023 — The q-derivative, affine denominator identity and modular lattice reindexing make normalization and convergence part of the calculation. Sources: 2023:M014, 2023:M034, 2023:M050

2024 — Cauchy extraction and Euler-product tails establish complementary local and infinite-product controls. Sources: 2024:Y24-03-026, 2024:Y24-09-018

2025 — Dilogarithms, Mellin splitting and product-variable integration yield independently checkable exact constants. Sources: 2025:Y25-04-013, 2025:Y25-08-906, 2025:Y25-10-036, 2025:Y25-10-024

2026 — Acceleration, a logarithmic-cosine convergence strip, an elliptic log constant and finite Bernoulli sums extend the same disciplined workflow. Sources: 2026:Y26-01-908, 2026:Y26-03-008, 2026:Y26-10-038, 2026:Y26-10-034

Connections

D01 — Arithmetic transforms. Mellin transforms, theta series and Euler products relate analytic formulas to arithmetic coefficients. A convergent formula does not establish transcendence or algebraic independence.

D02 — Period equations. Parameter-dependent integrals can satisfy Picard–Fuchs equations. The operator and boundary cycle must be supplied; an isolated exact value does not recover them.

D14 — Exact certification. Rational partial sums and monotone tail bounds create independently reproducible enclosures. Quadrature or symbolic simplification without endpoint control is insufficient.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

Different definitions of K, Fourier transforms, q-parameters and logarithm branches alter constants. Each member treatment retains its own conventions. The elementary theorem is not a common proof of every special-function identity in this family.

Next questions

Which exact identities admit rational enclosures whose cost can be predicted before calculation?

How can branch and cycle metadata be carried automatically through parameter differentiation?

Source anchors: 2023:M014, 2023:M034, 2023:M050, 2025:Y25-04-013, 2025:Y25-08-906, 2025:Y25-10-036, 2026:Y26-10-038, 2026:Y26-10-034

NIST DLMF q derivatives — https://dlmf.nist.gov/17.2#iv
NIST DLMF Gudermannian — https://dlmf.nist.gov/4.23#viii
NIST DLMF Jacobi triple product — https://dlmf.nist.gov/17.8
J S Milne Modular Functions and Modular Forms — https://www.jmilne.org/math/CourseNotes/MF.pdf
NIST DLMF, Polylogarithms — https://dlmf.nist.gov/25.12

D07-F04 · Self-adjoint spectra, heat traces and spectral dimension

What does a spectrum actually determine, and which operator-domain and multiplicity data are needed before using it?

The spectral strand moves from concrete Jacobi and Toeplitz matrices to unbounded operators, residual estimates, heat traces, zeta transforms and spectral dimensions. Its organizing object is not a free-standing list of eigenvalues but an operator with a domain and a functional calculus. Self-adjointness makes a residual meaningful as distance to the spectrum; nonnormal matrices behave differently. Multiplicities change counting functions and commutants even when a list of distinct eigenvalues looks

unchanged. Fixed-time trace class is also distinct from short-time heat asymptotics. A diagonal Hilbert-space model makes all these distinctions constructive and supplies a precise bridge between counting growth, heat decay and a zeta convergence half-plane. It is an exact model, not a proposed identification with zeros of a number-theoretic function.

Definitions

On $\ell^2(\mathbb{N})$, define $Ae_n=n^2e_n$ with domain $D(A)=\{x:\Sigma n^4|x_n|^2<\infty\}$. This domain makes A self-adjoint and nonnegative.

The counting function with multiplicity is $N_A(\Lambda)=\#\{n:n^2\le\Lambda\}$. The heat operator $H_t=e^{-tA}$ has diagonal entries e^{-tn^2}.

The spectral zeta series is $\zeta_A(s)=\Sigma n^{-2s}$, initially defined where it converges absolutely.

Theorem and hypotheses

For the specified A, $N_A(\Lambda)=\lfloor\sqrt{\Lambda}\rfloor$ for $\Lambda\ge1$, H_t is trace class for every $t>0$, and $\sqrt{t}\,\mathrm{Tr}(H_t)\to\sqrt{\pi}/2$ as $t\downarrow0$. The defining zeta series converges absolutely exactly when $\mathrm{Re}\,s>1/2$. Its positive heat trace obeys $\int_1^\infty e^{-tx^2}dx\le\Sigma_{n\ge1}e^{-tn^2}\le\int_0^\infty e^{-tx^2}dx$.

Proof

Coordinatewise multiplication by a real sequence is self-adjoint on its maximal square-summability domain; equivalently the adjoint condition forces the same weighted square sum. Its resolvent at -1 has diagonal entries $1/(n^2+1)$, and finite coordinate truncations converge in operator norm because the remaining supremum tends to zero. Thus the resolvent is compact. The counting identity is immediate from $n^2\le\Lambda$. For $t>0$, the diagonal entries of H_t are positive and summable by the integral test, so its trace equals their sum. The function $x\mapsto e^{-tx^2}$ decreases on $[0,\infty)$; rectangle comparison gives the two integral bounds. Their difference is at most $\int_0^1 1\,dx=1$. Multiplying by $\sqrt{t}$ and substituting $u=\sqrt{t}\,x$ makes both bounds tend to $\int_0^\infty e^{-u^2}du=\sqrt{\pi}/2$. This proves the leading heat asymptotic without asserting an unproved full expansion. Finally the absolute zeta series is $\Sigma n^{-2\mathrm{Re}\,s}$, and the p-series test gives precisely $\mathrm{Re}\,s>1/2$. Repeating every eigenvalue n times would instead give $N(\Lambda)$ of order Λ and heat growth of order t^{-1}, so multiplicity changes the inferred dimension. The proof exposes where positivity, compactness and the actual counting law enter.

Worked example

At $t=1$, retain $n=1,2$ in the heat trace.

The partial trace is $e^{-1}+e^{-4}$. Its omitted tail is $\Sigma_{n\ge3}e^{-n^2}$, bounded above by $\int_2^\infty e^{-x^2}dx\le e^{-4}/4$, using $x/2\ge1$ for $x\ge2$. The true leading exponent at small t is $t^{-1/2}$, while for every fixed t the omitted tail tends to zero as the coordinate cutoff grows.

A certified finite approximation and a dimension limit answer different questions. Their order of quantifiers must be specified when both a time parameter and a spectral cutoff vary.

Continuity across the source years

2023 — Jacobi orthogonality and Parseval supply concrete spectral measures and completeness questions. Sources: 2023:M024, 2023:M066

2024 — Toeplitz laws, residual bounds, operator domains and Weyl growth constrain proposed spectral models. Sources: 2024:Y24-09-021, 2024:Y24-09-016, 2024:Y24-10-001, 2024:Y24-10-002

2025 — Heat dimension, Mellin residues and multiplicity separate several formerly conflated spectral invariants. Sources: 2025:Y25-02-017, 2025:Y25-03-035, 2025:Y25-10-010, 2025:Y25-11-019

2026 — Spectral measure near zero, fixed-time trace tails and a confining Bessel operator make limiting and boundary data explicit. Sources: 2026:Y26-03-005, 2026:Y26-05-015, 2026:Y26-09-034

Connections

D08 — Evolution. The semigroup e^(−tA) solves the linear heat equation in the spectral representation. A spectral gap controls only the modes to which the chosen observable couples.

D01 — Spectral proposals. Counting asymptotics can disprove an equality of spectral multisets. Matching a few eigenvalues or a functional symmetry cannot establish a global spectral identity.

D16 — Operator observables. Multiplicity controls the commutant and the available invariant subspaces. A scalar spectrum alone does not specify a quantum observable algebra.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

The diagonal model omits continuous spectrum, nonself-adjoint pseudospectra and singular perturbations of the domain. Mellin residues need a justified continuation and are not automatically integer-valued indices.

Next questions

Which source spectral proposals survive a direct counting-function comparison?

Can joint time–cutoff bounds be made uniform for the particular observable and perturbation class?

Source anchors: 2023:M024, 2024:Y24-09-021, 2024:Y24-09-016, 2024:Y24-10-001, 2025:Y25-02-017, 2025:Y25-03-035, 2025:Y25-11-019, 2026:Y26-05-015

NIST DLMF continuous q-Hermite polynomials — https://dlmf.nist.gov/18.28#vi
NIST Digital Library of Mathematical Functions — https://dlmf.nist.gov/
NIST DLMF, Zeros of the Riemann zeta function — https://dlmf.nist.gov/25.10
Alain Connes Noncommutative Geometry — https://alainconnes.org/wp-content/uploads/book94bigpdf.pdf
Connes, Noncommutative geometry and reality — https://alainconnes.org/wp-content/uploads/reality.pdf

D08 · Differential equations and asymptotics

Organizing question

How does a stated equation produce a controlled solution, approximation or limiting law?

The differential-equation material becomes coherent when the equation, its state space and its boundary or initial data are treated as one object. A familiar profile is not yet a solution, and a formal operator is not yet a closed evolution law. The early hyperbolic-identity discussion is an instructive starting point: substituting a sech-squared wave into the stated KdV normalization forces its amplitude and speed. That same direct method extends to energy laws, diffusion stencils, local Frobenius recurrences and asymptotic remainders. The aim of this chapter is to make each conclusion executable as a sequence of explicit mathematical steps while preserving the assumptions that make those steps valid.

The first family begins with balances. On a periodic circle, the heat equation preserves the mean and decreases the squared fluctuation norm at a rate fixed by the first nonzero Fourier mode. The worked example has only two modes, so its solution and energy derivative can be calculated independently. Nonlinear members then alter the energy or the mobility: a Cahn–Hilliard flow preserves mass while dissipating an energy through a gradient of the chemical potential, whereas an L^2 gradient flow has a different dissipation law. Coupled diffusion and rotating waves add exchange or flux terms. Boundary conditions and regularity remain part of every calculation.

The second family separates the finite numerical map from its continuum interpretation. The periodic heat stencil has an exact mass identity, a sharp positivity range and an independent Fourier-symbol calculation. Taylor expansion supplies consistency only for sufficiently smooth solutions; stability and defect accumulation are then needed for a finite-time error bound. Distributed communication tests, anisotropy estimates, interpolation and inverse cutoffs are related by this error-budget viewpoint. They are not interchangeable certificates. In particular, the linked cutoff lab balances bias and amplified noise for its own observation model rather than providing a universal PDE solver.

The third family treats local singular equations through normalized bases. A coalescing Euler pair becomes regular after taking a divided difference, and its limiting logarithm has nontrivial unipotent monodromy. The Wronskian distinguishes an actual loss of independence from a bad basis. The recovered Picard–Fuchs examples supply more specific recurrences, while value–derivative matching and gauge conventions determine basis comparison. Extra harmonics in a Hill reduction remain visible instead of being silently renamed Mathieu.

The fourth family makes limiting statements quantitative. Its factorial Borel model has a divergent power series, a valid finite remainder estimate and an explicitly calculable exponential ambiguity. A cubic fold supplies a separate exact three-halves action difference and two-thirds transition scale. The elliptic logarithmic constant and perturbative discriminant questions belong to the same discipline of specified limits, although each uses a different proof. Reading these families in order moves from exact identities to stable approximation, then to local continuation and singular limits. The open task is to make estimates uniform when several of these operations are combined, rather than infer a global theorem from a local germ or a formal expansion.

Core objects

Evolution equations and energy identities

positive finite update operators

Fuchsian equations and fundamental bases

finite asymptotic expansions and sectorial integrals

Open frontier

Uniform multi-parameter error bounds, resonant basis normalization, nonlinear invariant-preserving schemes and global selection of asymptotic branches remain the central extensions.

D08-F01 · Evolution equations through conservation and energy

Which quantities are transported, conserved or dissipated by the stated evolution law and boundary conditions?

Heat, nonlinear waves, reaction–diffusion and gradient flows recur across the four years. The useful common language is an evolution equation accompanied by a balance law. A conserved mass and a decreasing energy are different statements: a reaction term can destroy the first, and changing the metric of a gradient flow changes the second. Spatial boundary conditions determine whether integration by parts closes the calculation. The simplest periodic heat equation supplies a complete constructive model with a solution formula, a conserved mean and a quantitative decay estimate for every nonconstant Fourier mode. The nonlinear wave and Cahn–Hilliard members then show how the same workflow changes with the equation. An explicit profile must be substituted into that equation, rather than recognized by its shape.

Definitions

On the circle $x\in[0,2\pi]$, let $u_t=\kappa u_{xx}$ with $\kappa>0$ and periodic boundary conditions. Its spatial mean is $\bar{u}=(2\pi)^{-1}\int u\,dx$.

The fluctuation energy is $E(t)=\tfrac{1}{2}\int|u-\bar{u}|^2dx$. For sufficiently regular u, integration by parts has no boundary contribution.

A gradient flow is defined by both an energy functional and a metric or mobility operator; L^2 and H^{-1} gradient flows need not conserve the same quantities.

Theorem and hypotheses

Periodic heat flow conserves the mean and satisfies $E'=-\kappa\int|u_x|^2dx\le-2\kappa E$. Consequently $\|u(t)-\bar{u}\|_2\le e^{-\kappa t}\|u(0)-\bar{u}\|_2$. For a Fourier mode $\cos(nx)$, the decay rate is exactly κn^2.

Proof

Integrating the PDE over one period yields $d\int u/dt=\kappa[u_x]_0^{(2\pi)}=0$, so the mean is constant. Differentiating E and integrating by parts gives $E'=\kappa\int(u-\bar{u})u_{xx}\,dx=-\kappa\int u_x^2dx$ for a real solution; the complex case uses the real part. Expand the mean-zero fluctuation in Fourier modes. Parseval gives $\int|u_x|^2=2\pi\Sigma_{n\neq0}n^2|\hat{u}_n|^2\geq2\pi\Sigma_{n\neq0}|\hat{u}_n|^2=2E$. This Poincaré inequality yields $E'\leq-2\kappa E$; multiplying by $e^{(2\kappa t)}$ and integrating proves the stated norm estimate. Each Fourier coefficient solves $\hat{u}_n'=-\kappa n^2\hat{u}_n$, giving the exact mode formula and a spectral construction of the solution. For rough L^2 data, approximate by finite Fourier sums, apply the same inequality, and pass to the L^2 limit. This argument also exhibits its boundary dependence. For Cahn–Hilliard, $u_t=\Delta\mu$ with $\mu=\delta E/\delta u$, the analogous calculation gives $E'=-\int|\nabla\mu|^2$ and a conserved mean. For the L^2 flow $u_t=-\mu$ it instead gives $E'=-\int\mu^2$, usually with no mass conservation. Naming an energy without specifying the mobility would miss this distinction.

Worked example

Let $\kappa=1/2$ and $u(x,0)=3+2\cos x-\cos2x$.

The exact solution is $u(x,t)=3+2e^{(-t/2)}\cos x-e^{(-2t)}\cos2x$. Its mean is three and $E(t)=\pi[2e^{(-t)}+(1/2)e^{(-4t)}]$. Direct differentiation gives $E'=-\pi[2e^{(-t)}+2e^{(-4t)}]$, matching $-\kappa\int|u_x|^2$. The $n=1$ term makes the asymptotic norm decay rate exactly 1/2.

The balance law is an independent diagnostic for simulation. A purported time step or additional coupling that violates the claimed mean or energy behavior requires a new model and a new proof.

Continuity across the source years

2023 — An exact travelling wave and a positive heat stencil begin the equation-by-equation verification strand. Sources: 2023:M009, 2023:M044

2024 — Rotating waves, diffusion-driven instability and coupled diffusion distinguish balance laws from spectral stability. Sources: 2024:Y24-03-030, 2024:Y24-04-020, 2024:Y24-04-021, 2024:Y24-07-010

2025 — Fixed geometry, exact solitons and conservative smoothing make boundary and convolution assumptions explicit. Sources: 2025:Y25-07-012, 2025:Y25-09-021, 2025:Y25-10-028, 2025:Y25-11-031

2026 — Vorticity, circle diffusion, parametrices and ignition front questions expose where local identities need further global analysis. Sources: 2026:Y26-01-010, 2026:Y26-01-028, 2026:Y26-07-013, 2026:Y26-08-906

Connections

D07 — Spectral evolution. Fourier diagonalization turns the heat equation into scalar decay laws. A nonlinear reaction or variable geometry couples those modes and changes the certificate.

D09 — Stability versus invariants. Energy dissipation and mode stability provide separate dynamical conclusions. Volume contraction or a decreasing energy alone is not a chaos certificate.

D15 — Model validation. Conservation defects provide computable residual diagnostics. Small residual needs a stability estimate before it gives a trajectory-error bound.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

The proved norm estimate uses a periodic fixed geometry and constant positive diffusivity. Moving domains, weak solutions, nonlinear advection and boundaries require their own regularity or flux arguments. The ignition integral is proved separately from its conditional front-speed law.

Next questions

Which conservative discretizations retain the stated nonlinear energy inequality?

What uniform estimates justify a front law or vorticity criterion beyond its local formal identity?

Source anchors: 2023:M009, 2024:Y24-04-020, 2024:Y24-04-021, 2024:Y24-07-010, 2025:Y25-07-012, 2025:Y25-10-028, 2025:Y25-09-021, 2026:Y26-01-028

Alan Turing The Chemical Basis of Morphogenesis — https://doi.org/10.1098/rstb.1952.0012
John Cahn and John Hilliard Free Energy of a Nonuniform System — https://doi.org/10.1063/1.1744102
Swift and Hohenberg Hydrodynamic fluctuations at the convective instability — https://doi.org/10.1103/PhysRevA.15.319
Hirota Exact solution of the Korteweg–de Vries equation for multiple collisions of solitons — https://doi.org/10.1103/PhysRevLett.27.1192
Leonard Gross, Logarithmic Sobolev Inequalities — https://doi.org/10.2307/1970986

D08-F02 · Discretization, truncation and controlled approximation

Which finite approximation preserves the model’s invariants, and what bound connects it to the intended continuous object?

Approximation becomes usable when its error sources are separated and bounded. The archive includes positive diffusion stencils, CFL conditions, distributed halo exchange, lattice anisotropy, FFT differentiation, Newton error and inverse cutoffs. They share a need to distinguish consistency from stability and implementation correctness. A correct communication pattern cannot repair an unstable time step; a stable matrix cannot recover frequencies never observed. The periodic heat stencil below has sharp positivity conditions, exact mass conservation and an explicit Fourier symbol. It is therefore a fully inspectable worked model. Its continuum error additionally needs smoothness and a consistency estimate. The inverse-cutoff lab L04 uses a different, explicitly declared observation model: bias decreases with cutoff while noise amplification increases. The common capability is selecting a finite scale under a proved error budget, not one universal algorithm.

Definitions

For $N \geq 3$ with indices modulo N, define $(T_r\, u)_j = r u_{j-1} + (1-2r) u_j + r u_{j+1}$, $r = \kappa \Delta t / (\Delta x)^2$.

Positivity means T_r maps every coordinatewise nonnegative vector to a nonnegative vector. Fixed-grid identities use the sum-of-squares norm; continuum error rates use the normalized norm $(N^{-1} \Sigma |u_j|^2)^{1/2}$.

Consistency compares the discrete equation with a sufficiently smooth exact solution; it is distinct from finite-dimensional conservation and positivity.

Theorem and hypotheses

T_r preserves Σu_j for every r. It preserves nonnegativity exactly for $0 \le r \le 1/2$. In that range it contracts the discrete ℓ^2 norm, with Fourier amplification $1-4r\sin^2(\pi k/N)$. For a smooth heat solution, the forward-Euler centered-space residual is $O(\Delta t+\Delta x^2)$ on a fixed bounded time interval with bounded derivatives.

Proof

Cyclic summation makes each old value appear with coefficient $r+(1-2r)+r=1$, proving mass conservation. If all three coefficients are nonnegative, every new coordinate is a convex combination. Conversely a coordinate basis vector detects a negative off-diagonal coefficient when $r<0$ or a negative central coefficient when $r>1/2$, proving the sharp positivity condition. Convexity of $z \mapsto |z|^2$ then gives $|(T_r u)_j|^2 \le r|u_{j-1}|^2+(1-2r)|u_j|^2+r|u_{j+1}|^2$. Sum over j to get contraction. Substituting the discrete character $e^{2\pi i k j/N}$ gives the amplification symbol $r e^{-2\pi i k/N}+(1-2r)+r e^{2\pi i k/N}=1-4r\sin^2(\pi k/N)$, an independent spectral check. Finally Taylor expansion yields $[u(t+\Delta t)-u(t)]/\Delta t=u_t+O(\Delta t)$, and $[u(x+\Delta x)-2u(x)+u(x-\Delta x)]/\Delta x^2=u_{xx}+O(\Delta x^2)$. The heat equation cancels the leading terms. In the stable range, summing the resulting one-step defects gives a global fixed-time error of the same order under the stated smoothness in the normalized grid norm, with a uniform bound on the sampled consistency defect. A reaction term or boundary update changes these steps and must be analyzed separately.

Worked example

Use $N=5$, $r=1/4$ and $u=(1,0,0,0,0)$.

The next vector is $(1/2,1/4,0,0,1/4)$. Its mass remains one, and its squared norm falls from one to $1/4+1/16+1/16=3/8$. At $r=3/4$ the same central value becomes $-1/2$, disproving positivity immediately. For a separate cutoff recipe with bound $B(N)+\delta S(N)$, a valid choice must drive both terms to zero; specifying only $N\to\infty$ is insufficient.

The finite stencil certificate is exact and independent of a continuum limit. The cutoff lab exposes the parallel bias–amplification tradeoff with its own supplied singular values and source-class assumptions.

Continuity across the source years

2023 — A finite positive stencil supplies exact conservation and a sharp local step restriction. Sources: 2023:M044

2024 — Optimization residuals and spectral residuals show why an approximation needs the correct certification theorem. Sources: 2024:Y24-03-005, 2024:Y24-09-016

2025 — CFL, communication, lattice dispersion and interpolation separate the numerical and implementation layers. Sources: 2025:Y25-02-007, 2025:Y25-05-006, 2025:Y25-08-026, 2025:Y25-09-008

2026 — Parametrices, covering bounds and regularization quantifiers supply explicit remainder and noise controls. Sources: 2026:Y26-07-013, 2026:Y26-07-018, 2026:Y26-08-901, 2026:Y26-08-903

Connections

D15 — Regularization. A cutoff balances source-class bias against inverse noise amplification. L04 is a worked cutoff model and does not instantiate every PDE discretization in this family.

D14 — Execution correctness. An exact stencil or halo ownership test checks the implemented finite map. Communication correctness is independent of the CFL condition.

D07 — Spectral symbols. Discrete Fourier symbols quantify dispersion and directional anisotropy. A low-wave-number expansion must include its valid wave-number range and remainder.

Capacity and interactive scope

Lab ids: ['L04']

Inputs: L04 selects diagonal singular values k^(−p) or exp(−ck), smoothness r and decay p/c in [0.5,3], noise exponent L in [4,14], and integer cutoff N in [1,60]. The severe lab fixes β=1.

Outputs: Floating evaluations of the proved bias bound (N+1)^(−r), noise bound δ/σ_N, their sum, a 60-cutoff comparison curve and a suggested cutoff. This is an upper envelope, not measured reconstruction error.

Validation: The lab engine is replayed on declared diagonal examples; the family verifier separately checks its Haar or diffusion-stencil model. Bounds follow from the proofs rather than sampled plots.

Status: Implemented diagonal cutoff worked model linked by an analysis/approximation bridge. Haar, general frames and PDE discretization remain documented recipes; the lab does not implement them.

Limits

For odd N the sharp ℓ^2 spectral stability threshold can exceed the positivity threshold; the two notions are not equivalent. The O estimate requires bounded derivatives, a fixed time interval and consistent initial data, not arbitrary nonsmooth inputs.

Next questions

Can positivity, accuracy and a nonlinear energy law be retained simultaneously at a useful step size?

Which logarithmic safety margin is sufficient for a specific severely ill-posed observation operator?

Source anchors: 2023:M044, 2025:Y25-02-007, 2025:Y25-05-006, 2025:Y25-08-026, 2025:Y25-09-008, 2026:Y26-08-901, 2026:Y26-08-903

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NIST DLMF rational approximation — https://dlmf.nist.gov/3.11
MIT numerical analysis Newton method — https://ocw.mit.edu/courses/18-335j-introduction-to-numerical-methods-spring-2019/
Lloyd Trefethen Spectral Methods in MATLAB, author resources — https://people.maths.ox.ac.uk/trefethen/spectral.html
Christian Clason, Regularization of Inverse Problems (author lecture notes) — https://arxiv.org/abs/2001.00617

D08-F03 · Regular-singular equations, normalized bases and monodromy

How should a parameter-dependent local basis be chosen so logarithms, resonances and continuation data remain meaningful?

Frobenius and Picard–Fuchs questions recur in scalar connections, elliptic periods, gauge changes and Hill equations. A differential equation is not determined by an evocative period name: its coefficients, singularities, normalization and initial or cycle data must be supplied. Once the operator is known, coefficient recurrence and a Wronskian provide concrete local information. When exponents collide, a raw basis can appear singular even if the solution space varies regularly. The divided-difference model below makes the logarithm appear explicitly and proves which pole is removable. It also exhibits nontrivial unipotent monodromy with eigenvalues equal to one, explaining why those eigenvalues do not imply trivial continuation. General Fuchsian systems require additional resonance compatibility conditions; the model is a constructive baseline rather than a replacement for the missing operator.

Definitions

For t>0 let θ=t d/dt and L_ε=θ(θ+ε). Its indicial exponents are 0 and −ε.

For ε≠0 define g_ε(t)=(t^(−ε)−1)/ε. A branch of log t must be fixed for complex continuation.

A fundamental pair of scalar solutions has nonzero Wronskian. Monodromy records how a chosen basis changes after analytic continuation around a singular point.

Theorem and hypotheses

g_ε extends analytically in ε to $g_0=-\log t$. The pair $(1,g_\varepsilon)$ remains independent with Wronskian $-t^{-\varepsilon-1}$. Writing $g_\varepsilon=\sum_{r\ge0}\varepsilon^r g_r$ yields $g_r=(-1)^{r+1}(\log t)^{r+1}/(r+1)!$, $\theta^2 g_0=0$ and $\theta^2 g_r=-\theta g_{r-1}$ for $r\ge1$.

Proof

Choose a simply connected punctured domain and one analytic logarithm. Then $t^{-\varepsilon}=\exp(-\varepsilon \log t)$ is entire in ε for each fixed t. Its exponential Taylor expansion shows that the numerator vanishes to first order at $\varepsilon=0$, so division removes the apparent pole and gives the stated coefficients. Both 1 and $t^{-\varepsilon}$ solve $L_\varepsilon f=0$, hence so does their divided difference. Direct differentiation gives $g_\varepsilon'=-t^{-\varepsilon-1}$, never zero on the chosen domain; the Wronskian therefore has a finite nonzero limit $-1/t$. This proves independence even when the original pair $(1,t^{-\varepsilon})$ coalesces. Substituting the ε-series into $(\theta^2+\varepsilon\theta)g=0$ and equating coefficients gives the triangular hierarchy. It can also be checked directly using $\theta(\log t)^k=k(\log t)^{k-1}$. After a positive loop around zero, $\log t$ gains $2\pi i$, so the limiting second solution $-\log t$ gains $-2\pi i$ times the first. In the basis $(1,-\log t)$, continuation is represented by a nonidentity unipotent matrix, although both of its eigenvalues are one. The construction distinguishes a removable basis pole from genuine singular boundary data. For an actual Picard–Fuchs operator, its Frobenius recurrence and resonance equations must still be derived from that operator, rather than imported from this Euler example.

Worked example

Take $t=e^2$ and $\varepsilon=1/10$, and compare the exact divided difference with its first three ε terms.

$g_{1/10}(e^2)=10(e^{-1/5}-1)\approx-1.812692469$. The first three terms give $-2+1/5-1/75=-136/75\approx-1.813333333$. Taylor’s theorem for $e^{-1/5}$ bounds the absolute error after those three terms by $10(1/5)^4/4!=1/1500$. The actual error is approximately 0.000640864 and lies within that bound.

The regularized basis, its differential equation, its Wronskian and its continuation law can all be inspected. A symbolic basis with an explicit $1/\varepsilon$ should not be labeled a genuinely singular solution before this test.

Continuity across the source years

2024 — Laplace and Bessel equations establish the need for a stated operator and its boundary or initial conditions. Sources: 2024:Y24-03-020, 2024:Y24-03-022

2025 — Monodromy identity, scalar gauge changes and value–derivative matching distinguish bases from solutions. Sources: 2025:Y25-02-014, 2025:Y25-07-014, 2025:Y25-08-016, 2025:Y25-10-012

2026 — Colliding exponents, an explicit recovered recurrence and triangular period systems make logarithmic and extra-harmonic terms computable. Sources: 2026:Y26-01-024, 2026:Y26-01-907, 2026:Y26-01-910, 2026:Y26-02-020

Connections

D02 — Gauss–Manin and periods. Period bases satisfy parameter-dependent differential systems with connection matrices. A basis comparison needs values and derivatives or a full fundamental matrix at a regular point.

D05 — Continuation. Monodromy is transport around a loop in a punctured domain. Eigenvalues one do not establish identity monodromy; a nilpotent part may remain.

D16 — Integrable spectral equations. Hill and Lamé equations relate mode spectra to monodromy. A Hill coefficient with extra harmonics cannot simply be called Mathieu.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

General resonant families may have compatibility obstructions or genuine parameter singularities. This model does not identify a historical elliptic integral whose differential operator or cycle data were absent.

Next questions

For the supplied operator, which resonance equations force logarithms and which allow an analytic basis?

Can basis normalization and numerical continuation be certified uniformly near coalescing exponents?

Source anchors: 2025:Y25-02-014, 2025:Y25-07-014, 2025:Y25-08-016, 2025:Y25-10-012, 2026:Y26-01-024, 2026:Y26-01-907, 2026:Y26-01-910, 2026:Y26-02-020

Edward Frenkel Langlands Correspondence for Loop Groups — https://doi.org/10.1017/CBO9780511543371
Zhou, Hodge-theoretic mirror symmetry for the quintic — https://sites.math.northwestern.edu/~byzhou01/MS_quintic_E.pdf
Isidro et al Picard–Fuchs equations for Seiberg–Witten models — https://arxiv.org/abs/hep-th/9703186
NIST DLMF, Fuchs–Frobenius Theory — https://dlmf.nist.gov/2.7
NIST DLMF, Hill’s Equation — https://dlmf.nist.gov/28.29

D08-F04 · Asymptotic expansions, singular scales and exponentially small data

What does a finite asymptotic expansion determine, and how can its remainder and missing exponential contribution be made explicit?

The asymptotic strand includes renormalization flows, spectral gaps, elliptic logarithms, cubic folds, Mathieu perturbation, and a factorial Borel model. Its common requirement is to state the small parameter, the approach sector and the uniformity of a remainder. A formal series can be useful while diverging everywhere; convergence and asymptotic accuracy are distinct. Conversely, a few perturbative coefficients cannot determine a global continuation, a Stokes constant or a nonlinear front law. The one-pole Borel example makes the distinction fully constructive. Its coefficients, exact integral, finite error bound and exponentially small solution ambiguity are all explicit. The cubic fold supplies a complementary algebraic scale: coalescing critical points produce a three-halves action difference and a two-thirds transition parameter. Each model has its own hypotheses and should not be promoted to a universal description by analogy.

Definitions

The formal series F(z)=Σ_{n≥0}n!z^(n+1) has zero radius of convergence. Its Borel transform is B(t)=Σt^n=1/(1−t) near t=0.

For a>0 define the nonsingular negative-ray sum F(−a)=−∫₀∞e^(−u/a)/(1+u)du.

An asymptotic expansion controls the error after each fixed finite truncation as a parameter tends to a limit; it need not converge when infinitely many terms are added.

Theorem and hypotheses

For every a>0 and N≥1, |F(−a)−Σ_{n=0}^{N−1}n!(−a)^(n+1)|≤N!a^(N+1). The corresponding Laplace sums solve z²F′−F=−z. Any two solutions differ by Ce^(−1/z); with above-minus-below bypasses of the positive Borel pole, the lateral difference is 2πi e^(−1/z).

Proof

The exact finite geometric identity gives 1/(1+u)=Σ_{n=0}^{N−1}(−u)^n+(−u)^N/(1+u). Integrate against −e^(−u/a). The nth term evaluates to (−1)^(n+1)n!a^(n+1) by the Mellin moment. The absolute remainder is bounded by ∫₀∞e^(−u/a)u^Ndu=N!a^(N+1), since 1/(1+u)≤1. This proves a genuine finite-order error estimate on the negative ray even though the coefficient ratio makes the infinite power series divergent for every nonzero argument. Differentiation under the convergent integral, or integration by parts in its Laplace form, gives the inhomogeneous equation. Subtracting two solutions gives H′=H/z², hence H=Ce^(−1/z), which is smaller than every power as z↓0 on the positive ray. To compare the two lateral integrals, their difference closes clockwise around t=1. The residue of e^(−t/z)/(1−t) is −e^(−1/z), giving +2πi e^(−1/z) in that convention. The exponential ambiguity is therefore visible without guessing it from several factorial coefficients. A complete sectorial asymptotic proof must keep the bypass and decay contour fixed in a compatible sector; the explicit negative-ray bound is the directly verified remainder used here.

Worked example

Use a=1/10 and N=5. Separately consider Φ(x;t)=x³/3−tx for t>0.

The Borel-model truncation is −1/10+1/100−2/1000+6/10000−24/100000=−11455/125000=−0.09164, with error at most 120/10⁶=0.00012. The fold saddles are x_±=±√t and Φ(x_−;t)−Φ(x_+;t)=4t^(3/2)/3. Scaling x=h^(1/3)y in Φ/h yields y³/3−(t/h^(2/3))y, so the nontrivial transition occurs at t of order h^(2/3).

These examples attach a bound or exact scaling identity to an asymptotic statement. They do not identify the Stokes data or thimbles of an unspecified amplitude.

Continuity across the source years

2023 — Exact flow coordinates and singular threshold coordinates expose where a limiting parameter becomes nonuniform. Sources: 2023:M113, 2023:M114

2024 — Fractional memory and logarithmic running require explicit derivative and scale conventions. Sources: 2024:Y24-09-011, 2024:Y24-12-036

2025 — Conditional series convergence, Riccati wall times and two-loop logarithms provide concrete control of singular behavior. Sources: 2025:Y25-03-006, 2025:Y25-08-027, 2025:Y25-10-001

2026 — Factorial remainders, discriminant perturbation, fold scaling and the elliptic log constant distinguish several asymptotic mechanisms. Sources: 2026:Y26-01-909, 2026:Y26-02-021, 2026:Y26-06-013, 2026:Y26-10-038

Connections

D07 — Mellin and Laplace transforms. Factorial coefficients are Laplace moments, and logarithmic constants can be isolated by an exact integral split. An integral identity must specify its sector and branch.

D09 — Critical scaling. A cubic fold gives an exact action difference and degeneration of the Hessian. A local fold normal form does not determine global attractors or transition rates.

D14 — Certified truncation. The explicit remainder supplies a stopping rule for a divergent expansion. More terms can increase the error once the least term has been passed.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

The factorial theorem is a stipulated one-pole model. The printed fold scale concerns that cubic phase. Uniform multi-parameter asymptotics, resonance denominators and global contour topology need further estimates.

Next questions

Can the expansion be truncated near its least term with a uniform computable sectorial bound?

Which global contour and boundary data select the exponentially small constant in the intended application?

Source anchors: 2023:M113, 2025:Y25-03-006, 2025:Y25-08-027, 2025:Y25-10-001, 2026:Y26-01-909, 2026:Y26-02-021, 2026:Y26-06-013, 2026:Y26-10-038

Beisert Quantum Field Theory II notes renormalization sections — https://people.phys.ethz.ch/~nbeisert/lectures/QFT2-17FS-Notes.pdf
Milne, Modular Functions and Modular Forms — https://www.jmilne.org/math/CourseNotes/MF.pdf
NIST DLMF logarithms — https://dlmf.nist.gov/4.2
NIST DLMF, Small-q Expansions for Noninteger Mathieu Order — https://dlmf.nist.gov/28.15
NIST DLMF, Mathieu Characteristic Equation — https://dlmf.nist.gov/28.2

D09 · Dynamics, mechanics and mathematical modeling

Organizing question

What state, metric and invariant make a dynamical claim precise and testable?

The dynamics material is organized around four choices that a model must make explicit: its state, its clock, its metric and its observations. The same eigenvalues can coexist with very different transient gains; the same geometric path can have different time parametrizations; the same observation can come from different hidden states. The archive’s strongest recurrent ideas are constructive ways to resolve these ambiguities. They include exact Lyapunov matrices, explicit period-two branches, conserved Hamiltonians, dimensionally complete mass balances and derived memory kernels. Each answers a particular question. Their connections become useful once the question and the retained state are stated before the calculation.

The stability family leads with an exact triangular matrix. Its eigenvalues are stable, its Euclidean norm can initially grow, and a positive matrix certifies decay in another norm. This gives a complete example of nonnormal amplification rather than a slogan about it. Continuous and discrete Gramians, common storage functions and hybrid dwell-time bounds extend the same reasoning. Switching incurs a logarithmic metric cost, so stability of the individual modes alone does not establish stability of every switching sequence. A two-line projection calculation supplies a further geometric model: the cycle contraction and the commutator norm are different functions of the same angle. The linked projection lab keeps that distinction visible.

The bifurcation family follows invariant branches and their multipliers. Exact logistic algebra proves the first fixed-point threshold, the born two-cycle and the next flip boundary. That substantial result is deliberately separated from the deeper infinite-cascade and universality questions. Return-map jets identify the cubic term left after a multiplier-minus-one map is composed with itself. Square-root cycle branches and the tanh pitchfork provide further local mechanisms with explicit conditions. None of these local facts alone identifies a global chaotic attractor or establishes a physical renormalization equivalence.

The mechanics and modeling family starts from conservation and units. Hamilton’s equations preserve a stated energy, and a simple turn gives different exponents for elapsed time and action because those quantities integrate reciprocal momentum and momentum respectively. Scalar RG coordinates preserve a fixed-point derivative under an invertible state-coordinate change with the same time variable; a time rescaling changes the rate. Population recurrences, radiation fluxes, drag, chemical yield and volume-to-mass calculations are organized by similarly explicit balance laws. Dimensional agreement narrows a model but does not supply its missing measurements or dimensionless coefficient.

The memory family finally asks whether a reduced state determines its future. Eliminating a hidden linear coordinate produces both a convolution kernel and a source from the hidden initial condition. Omitting them changes the equation. Mechanical words, substitutions and relative periodicity pose discrete versions of the retained-information question. A balanced aperiodic rotation coding can be completely deterministic and have an exact frequency, so irregular appearance is not a chaos certificate. Across these four families, the dated progression traces changing source topics from 2023 onward; the present proofs and worked models are a new organization of that material, with their scope stated independently of historical attribution.

Core objects

Flows, maps and return maps

Lyapunov functions and switching budgets

Hamiltonians and dimensioned balance laws

memory kernels and symbolic codings

Open frontier

The most useful next step is to combine certified basins, physically meaningful transient norms and minimal sufficient reduced states for specific models, with quantitative memory-approximation bounds.

D09-F01 · Stability in the right metric, including transient and hybrid behavior

Which norm or storage function certifies decay when eigenvalues, Euclidean growth and switching tell different stories?

The stability strand is unusually consistent across the years: feedback can be asymptotically stable and still amplify an initial perturbation, a spectral gap depends on the modes actually observed, and changing metrics can reveal coercivity absent in the Euclidean symmetric part. A positive quadratic Lyapunov function supplies a constructive certificate. Its matrix also quantifies the cost of a coordinate change, forcing, switching and resets. These are capabilities with distinct inputs: an eigenvalue calculation, a transient-gain calculation and a hybrid dwell-time bound are related but answer different questions. The two-dimensional triangular model below is deliberately small enough to solve exactly. It proves decay in one metric while displaying temporary growth in another, giving a positive explanation for the archive's repeated cautions about nonnormality.

Definitions

For $P=P^{T}>0$, define $V(x)=x^{T}Px$ and $||x||_P=\sqrt{V(x)}$. A Hurwitz matrix has all eigenvalues with negative real part.

A quadratic Lyapunov certificate is $A^{T}P+PA\leq-2\alpha P$ for $\alpha>0$. It implies $||e^(tA)x||_P\leq e^(-\alpha t)||x||_P$.

For hybrid switching, a metric jump ratio $\rho\geq1$ bounds $V_{after}\leq\rho V_{before}$; an average dwell-time bound controls how often this cost can be incurred.

Theorem and hypotheses

For $A=[[-1,3],[0,-1]]$ and $P=[[1/2,3/4],[3/4,11/4]]$, $P>0$ and $A^{T}P+PA=-I$. Since $P\leq3I$, $||e^(tA)x||_P\leq e^(-t/6)||x||_P$. Nevertheless the Euclidean squared norm has positive derivative at $x=(1,1)$.

Proof

The leading principal entry of P is 1/2 and det P=11/8−9/16=13/16>0, so the two-by-two Sylvester criterion proves positive definiteness. Direct multiplication gives $A^TP+PA=-I$. Along $x'=Ax$, $V'=x^T(A^TP+PA)x=-||x||_2^2$. The matrix 3I−P has positive first entry 5/2 and determinant (5/2)(1/4)−9/16=1/16>0, so P≤3I. Therefore $||x||_2^2 \geq V/3$ and $V' \leq -V/3$. Integration and square roots give the rate 1/6 in the P norm. In contrast, at x=(1,1), Ax=(2,−1), so $d||x||_2^2/dt=2x^TAx=2>0$. The eigenvalues of A are both −1; their location does not prohibit this initial Euclidean amplification. Because A=−I+3N with $N^2=0$, its exact exponential is e^(−t)(I+3tN), explicitly displaying the polynomial transient. If continuous segments obey $V' \leq -2\alpha V$ and jumps cost at most ρ, then $V(t) \leq \rho^{N(t)} e^{(-2\alpha t)} V(0)$. Under $N(t) \leq N_0 + t/\tau$, this becomes $\rho^{N_0} \exp[-2(\alpha - \log\rho/(2\tau))t]V(0)$. Thus switching consumes a precise logarithmic decay budget rather than preserving stability merely because each mode is stable.

Worked example

Use $x_0=(0,1)$ for the continuous model, and compare two rank-one orthogonal projections whose lines meet at angle θ.

The exact trajectory is x(t)=e^(−t)(3t,1). At t=1 its Euclidean norm is √10/e>1 although it eventually tends to zero. For projections P_0 onto (1,0) and Q onto (cosθ,sinθ), $P_0QP_0=\cos^2\theta\, P_0$. Thus a cycle restricted to im P_0 contracts by $\cos^2\theta$, while $||[P_0,Q]||=|\sin\theta\cos\theta|$. These are different quantities, even though both depend on the same angle.

The Lyapunov example has a globally valid certificate for its linear flow. The L05 projection component provides another exact model of convergence and noncommutation, without identifying either number as a universal holonomy invariant.

Continuity across the source years

2023 — Locking branches, triangular feedback and a dissipative nonlinear example separate local attraction from amplification and identifiability. Sources: 2023:M053, 2023:M058, 2023:M107

2024 — Polynomial stability boundaries and frequency filtering supply concrete norm-dependent tests. Sources: 2024:Y24-03-032, 2024:Y24-05-011, 2024:Y24-11-002

2025 — Multiplicative updates, parametric switching, damping conventions and volume contraction refine what is actually controlled. Sources: 2025:Y25-04-020, 2025:Y25-04-027, 2025:Y25-10-032, 2025:Y25-12-012

2026 — Continuous and discrete Gramians, adapted metrics and hybrid budgets form a coherent certificate toolkit. Sources: 2026:Y26-03-002, 2026:Y26-03-004, 2026:Y26-05-011, 2026:Y26-05-026

Connections

D15 — Control and observability. Lyapunov and observability Gramians solve different matrix equations with related energy meanings. A decay certificate does not imply that every state coordinate is measured.

D06 — Projection geometry. Principal angles determine a two-line projection contraction. L05 separately labels transport or holonomy quantities and the projection model.

D12 — Robust margins. A metric perturbation bound δ<α retains a decay rate at least α−δ. The norm and uncertainty set must be stated before calling a margin robust.

Capacity and interactive scope

Lab ids: ['L05']

Inputs: L05 has a shear-loop mode with integer a,b in [−3,3] and a projection mode with line angle 5–85 degrees. The displayed projection iteration target is ε=0.01.

Outputs: For shears: UVU^(−1)V^(−1), its entries and determinant. For projections: $\cos^2\theta$, $|\sin(2\theta)|/2$, the number of cycles reaching ε and the residual bound.

Validation: Integer shears use exactly represented small integer operations; projection trigonometry is floating arithmetic. The companion verifier checks the distinct contraction 9/25 and commutator norm 12/25 for cosθ=3/5; engine replay checks the implemented formulas.

Status: Two implemented explicit 2×2 worked models. General Lyapunov synthesis, switching simulation and geometric holonomy remain mathematical recipes, not lab solvers.

Limits

The exact matrix certificate is global for its stated linear model. A Jacobian certificate for a nonlinear system is usually local unless its inequality holds throughout a forward-invariant region. Common storage, reset inequalities and switching bounds are additional hybrid hypotheses.

Next questions

Which common or multiple storage functions certify the complete switching architecture?

How large is the physically relevant transient gain in the observation norm rather than only the state norm?

Source anchors: 2023:M058, 2024:Y24-03-032, 2025:Y25-04-027, 2026:Y26-03-002, 2026:Y26-03-004, 2026:Y26-05-011, 2026:Y26-05-026, 2026:Y26-06-026

Boyd and Vandenberghe Introduction to Applied Linear Algebra — https://stanford.edu/~boyd/vmls/
Boyd and coauthors Linear Matrix Inequalities in System and Control Theory — https://web.stanford.edu/~boyd/lmibook/index.html
MIT Dynamics and Control II, Routh–Hurwitz lecture — https://ocw.mit.edu/courses/2-004-dynamics-and-control-ii-spring-2008/resources/lecture_25/
MIT Routh–Hurwitz course notes — https://sites.mit.edu/meche2004/news/routh-hurwitz-stability-criterion/
Moore, Quantum theory of the electromagnetic field in a variable-length one-dimensional cavity — https://doi.org/10.1063/1.1665432

D09-F02 · Local bifurcation, exact cycles and the boundary of chaos claims

Which changes of stability can be derived from a return map, and what additional work separates a local bifurcation from global chaos?

The dynamics archive repeatedly returns to logistic period doubling, oscillator locking, pitchfork regulation, scalar flip equations and return-map jets. These are best organized around a local map or flow, its invariant branches and its multipliers. A nonlinearity is not itself chaos, a symbolic analogy is not a conjugacy, and a local germ is not a global attractor. The logistic family offers an unusually complete positive example: the first fixed-point instability, the born two-cycle and its entire linear-stability interval follow from exact algebra. Later 2026 questions enrich this with explicit transcendental flip candidates, square-root branching and cubic coefficients of the second iterate. Those mechanisms can be compared once the local coordinates and nondegeneracy conditions are stated.

Definitions

Let $f_r(x)=rx(1-x)$, $r>0$. A point has exact period two when $f_r^2(x)=x$ and $f_r(x)\neq x$.

The multiplier of a k-cycle is the product of derivatives along it. Absolute multiplier less than one proves local attraction; equality is a boundary case requiring higher-order analysis.

A local flip occurs when a fixed-point multiplier crosses -1 transversely with an appropriate nonzero nonlinear coefficient.

Theorem and hypotheses

The nonzero logistic fixed point is stable for $1<r<3$. For $r>3$ its genuine two-cycle has points $x_{\pm}=[r+1\pm\sqrt{((r-3)(r+1))}]/(2r)$ and multiplier $m_2=-r^2+2r+4$. It is linearly attracting exactly for $3<r<1+\sqrt{6}$.

Proof

Solving $f_r(x)=x$ gives zero and $x^*=1-1/r$. At x^*, $f_r'(x^*)=r(1-2+2/r)=2-r$, whose absolute value is less than one exactly when $1<r<3$. To locate period-two points, expand $f_r^2(x)-x$ and divide by the known fixed-point factors. The remaining equation is $r^2x^2-r(r+1)x+(r+1)=0$. Its discriminant is $r^2(r-3)(r+1)$, giving the displayed distinct real roots for $r>3$. Their sum is $(r+1)/r$ and product is $(r+1)/r^2$. Consequently $f_r'(x_+)f_r'(x_-)=r^2[1-2(x_++x_-)+4x_+x_-]=-r^2+2r+4$. For $r>3$ this polynomial decreases strictly, starts at $+1$ at $r=3$ and reaches -1 at $r=1+\sqrt{6}$. Hence $|m_2|<1$ precisely on the stated interval. At the first endpoint the two-cycle merges with the fixed branch; at the second, its multiplier reaches the next flip boundary. This calculation establishes neither every later doubling nor Feigenbaum universality. A complementary jet calculation explains why higher terms matter: if $R(s)=-s+as^2+bs^3+O(s^4)$, direct composition gives $R^2(s)=s-2(a^2+b)s^3+O(s^4)$; the quadratic term cancels and the sign or vanishing of a^2+b is relevant. A multiplier of -1 alone omits that information.

Worked example

Take $r=16/5$. Also compare the regulator $m\mapsto\tanh(am)$ near $a=1$.

The logistic two-cycle is $(21\pm\sqrt{21})/32$, and its multiplier is $4/25$. It is therefore locally attracting. For the regulator, $\tanh(am)=am-a^3m^3/3+O(m^5)$; a nonzero fixed point satisfies $m^2=3(a-1)/a^3+O((a-1)^2)$ as $a\downarrow1$. This square-root branch is a pitchfork mechanism, distinct from the logistic two-cycle despite a shared critical-scaling vocabulary.

Exact branch formulas make hypotheses testable. They provide a foundation for a numerical bifurcation diagram, while the diagram itself cannot certify an infinite cascade or a global chaotic invariant set.

Continuity across the source years

2023 — Locking, logistic algebra and volume contraction distinguish several dynamical mechanisms. Sources: 2023:M053, 2023:M091, 2023:M092, 2023:M107

2024 — Fixed-point criteria and a homoclinic horseshoe question separate local tests from global symbolic conclusions. Sources: 2024:Y24-04-018, 2024:Y24-04-019, 2024:Y24-11-013

2025 — An interval map, repeated exact logistic algebra and a noisy pitchfork provide explicitly stated models. Sources: 2025:Y25-09-015, 2025:Y25-12-006, 2025:Y25-12-017

2026 — Scalar flip equations, branched cycles and second-iterate jets sharpen the local computation. Sources: 2026:Y26-03-010, 2026:Y26-03-024, 2026:Y26-03-025, 2026:Y26-04-006

Connections

D02 — Branched families. A born two-cycle can be exchanged by continuation around a square-root branch point. This local permutation does not force the fixed branch itself to be multivalued.

D12 — Iterative optimization. Gradient descent on a quadratic reaches multiplier -1 at step $2/\lambda_{max}$. At the endpoint the top mode alternates instead of converging.

D16 — Renormalization proposals. A return-map rescaling can be defined on an explicit function space. Writing a conjugation formula does not establish Feigenbaum-class universality.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

All stated nonlinear stability conclusions are local unless a global invariant region and basin argument are supplied. Chaotic horseshoes, positive entropy and universality require different theorems from the first two logistic thresholds.

Next questions

What basin of attraction can be certified for the explicit nonpolynomial flip model?

Which nondegeneracy and compactness hypotheses hold for a proposed renormalization operator?

Source anchors: 2023:M091, 2024:Y24-04-018, 2024:Y24-04-019, 2025:Y25-12-006, 2026:Y26-03-025, 2026:Y26-03-023, 2026:Y26-04-006, 2026:Y26-08-013

Yuri Kuznetsov, Elements of Applied Bifurcation Theory, flip theorem — https://www.ma.imperial.ac.uk/~dturaev/kuznetsov.pdf
MIT Nonlinear Dynamics and Chaos — https://ocw.mit.edu/courses/18-353j-nonlinear-dynamics-and-chaos-fall-2012/
Edward Lorenz Deterministic Nonperiodic Flow — https://doi.org/10.1175/1520-0469(1963)020%3C0130:DNF%3E2.0.CO;2
Otto Rössler An equation for continuous chaos — https://doi.org/10.1016/0375-9601(76)90101-8
Lanford, computer-assisted Feigenbaum proof — https://doi.org/10.1090/S0273-0979-1982-15008-9

D09-F03 · Mechanics, dimensional modeling and invariant flow coordinates

Which quantities and scaling laws survive a change of coordinates, units or time parameter?

The modeling strand includes mechanics, population and asset recurrences, conductance gates, mass balances and renormalization flows. Its common discipline is to identify state variables, units, conserved quantities and the meaning of the time parameter before interpreting a formula. Dimensional consistency narrows possible equations but does not determine numerical coefficients, boundary conditions or a physical mechanism. Hamiltonian mechanics supplies a precise constructive invariant: energy conservation. A simple turning point then exhibits two different exponents for physical time and abbreviated action. Their difference is caused by the measure being integrated, not a contradiction. The RG examples use the same chain-rule discipline to distinguish an invariant linearization exponent from coordinate-dependent higher coefficients. These are typed bridges between calculations, rather than a claim that every model in the family describes the same physics.

Definitions

For a unit-mass one-dimensional Hamiltonian $H(q,p)=p^2/2+V(q)$, Hamilton's equations are $q'=p$ and $p'=-V'(q)$.

At energy E, a simple turning point q^* has $E-V(q)\approx F(q^*-q)$ with $F>0$ on the allowed side.

A scalar autonomous flow $g' = \beta(g)$ transforms under $h = \varphi(g)$, $\varphi' \neq 0$, to $\beta_h(h) = \varphi'(g)\beta(g)$. The meaning of the time variable remains part of this definition.

Theorem and hypotheses

For the stated Hamiltonian, H is conserved. If $V(q) = E - F(q^* - q)$ exactly on $q \leq q^*$, then with $\delta = q^* - q$ the remaining travel time is $T(\delta) = \sqrt{(2/F)}\delta^{(1/2)}$, while the action length is $S(\delta) = (2/3)\sqrt{(2F)}\delta^{(3/2)}$. At any scalar-flow fixed point, $d\beta_h/dh = \beta'(g^*)$ under a smooth invertible state-coordinate change.

Proof

Energy conservation is a direct chain-rule calculation: $H' = V'(q)q' + pp' = V'p - pV' = 0$. On the increasing branch $p = \sqrt{[2(E - V(q))]} = \sqrt{(2F\delta)}$. Physical travel time integrates dq/p, so $T(\delta) = \int_0^{\delta} (2Fs)^{(-1/2)} ds = \sqrt{(2/F)}\sqrt{\delta}$. The abbreviated action integrates $p\,dq$, giving $S(\delta) = \int_0^{\delta} \sqrt{(2Fs)} ds = (2/3)\sqrt{(2F)}\delta^{(3/2)}$. The two powers therefore concern different observables. For a smooth potential with a nonzero turning derivative, Taylor's theorem produces these leading terms with controlled local remainders; a degenerate turning point has different exponents. For the scalar coordinate calculation, differentiate $\beta_h = \varphi'\beta$ with respect to h using $d/dh = (\varphi')^{(-1)} d/dg$. The result is $\beta' + (\varphi''/\varphi')\beta$. At a fixed point $\beta(g^*) = 0$, the second term vanishes and the linearized exponent is unchanged. This conclusion uses a state-coordinate change with φ' nonzero and the same time. A rescaling of time multiplies the rate and must not be silently included. Exact flow integration makes the same point: for $g' = bg^2$, $1/g(t) = 1/g_0 - bt$ on every interval avoiding its pole. The finite-time wall is a property of that stated flow, not a physically meaningful singularity without an identified scale convention.

Worked example

Let $F = 2$ in the turning-point model, and separately let $g' = g^2$ with $g_0 = 1/4$.

The travel time from distance δ to the turn is $\sqrt{\delta}$, whereas the action is $(4/3)\delta^{(3/2)}$. The exact flow is $g(t) = 1/(4 - t)$ for $t < 4$. Replacing g by $h = 2g$ changes the beta function to $h' = h^2/2$, but the same solution and wall time remain. At a nonzero fixed point of a flow with $\beta(g) = -\varepsilon g + bg^2$, the derivative is ε, and the state-coordinate argument preserves that number.

Each formula carries its own dimensions and clock. A mass balance similarly needs an actual volume, molar conversion and yield; an area alone cannot recover an amount.

Continuity across the source years

2023 — Logarithmic crossing time, dimensional checks and exact scalar RG establish quantitative modeling conventions. Sources: 2023:M064, 2023:M070, 2023:M077, 2023:M113

2024 — Missing geometric depth, drag coefficients, population first integrals and radiation flux distinguish units from a complete law. Sources: 2024:Y24-02-001, 2024:Y24-03-034, 2024:Y24-04-017, 2024:Y24-08-004

2025 — Energy bounds, dimensionful predictions and two-loop flow coordinates make mechanism and convention explicit. Sources: 2025:Y25-03-026, 2025:Y25-04-024, 2025:Y25-06-015, 2025:Y25-10-001

2026 — Turning-point exponents, an exact pendulum separatrix and stoichiometric balances consolidate invariant-based modeling. Sources: 2026:Y26-01-017, 2026:Y26-05-016, 2026:Y26-05-018, 2026:Y26-06-021

Connections

D06 — Hamiltonian geometry. The symplectic convention generates conservation through the Poisson bracket. The mechanical state space and symplectic form must be specified before transporting the argument.

D08 — Singular flow integration. Separating dg/β(g) supplies exact first integrals and identifies logarithmic terms. A local expansion or coordinate pole need not be a global physical singularity.

D11 — Model identifiability. A dimensionally valid equation can still leave a volume, yield or parameter combination unidentified. Units do not supply missing measurements or select dimensionless numerical constants.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

The turning-point theorem assumes a simple nondegenerate turn and the stated mass normalization. Time-dependent forcing breaks this energy conservation unless its work is included. Phenomenological recurrences retain their stipulated interpretation and are not validated by algebra alone.

Next questions

Which measured invariant distinguishes alternative physically dimensionally consistent laws?

How do degenerate turning points and state-dependent time changes alter the observable exponents?

Source anchors: 2023:M064, 2023:M070, 2023:M113, 2024:Y24-02-001, 2024:Y24-04-017, 2025:Y25-10-001, 2026:Y26-01-017, 2026:Y26-05-016

Sean Carroll General Relativity lecture notes — https://www.preposterousuniverse.com/wp-content/uploads/2015/08/grtinypdf.pdf
Beisert Quantum Field Theory II notes renormalization sections — https://people.phys.ethz.ch/~nbeisert/lectures/QFT2-17FS-Notes.pdf
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D09-F04 · Memory, recurrence and the information needed for a closed evolution

When does a reduced state determine its own future, and what exact memory or symbolic data repair a failure of closure?

A final dynamics strand connects hidden-variable elimination, damped recurrences, rotations, substitutions, cellular automata and path-dependent accumulators. The shared question is what information must be retained for a well-defined next state. Eliminating a hidden coordinate usually creates a memory kernel and an initial-state source. Forgetting either can turn an exact reduction into a different equation. Symbolic models pose the discrete analogue: an observed label may not encode the phase or history needed for the future. At the same time, deterministic aperiodicity can be highly structured. A rotation coding has exact balance, and a dense quasiperiodic orbit can have zero entropy. The family therefore organizes recurrence by a declared state, observation and equivalence relation rather than grouping every irregular-looking sequence under chaos.

Definitions

Consider $x'=-ax+by$ and $y'=cx-dy$ with constants a,b,c,d and given (x_0,y_0). A closed scalar evolution in x alone must determine x' from retained data.

The memory convolution $(K*x)(t)=\int_0^t K(t-s)x(s)ds$ retains past x values. The source $be^{(-dt)}y_0$ carries the hidden initial condition.

A mechanical word is $w_n=\lfloor(n+1)\alpha+\rho\rfloor-\lfloor n\alpha+\rho\rfloor$ for $0<\alpha<1$. It codes a rotation with an explicitly retained phase.

Theorem and hypotheses

Eliminating y gives exactly $x'(t)=-ax(t)+be^{(-dt)}y_0+bc\int_0^t e^{(-d(t-s))}x(s)ds$. For $a=d=b=c=1$ and $(x_0,y_0)=(1,0)$, $x(t)=(1+e^{(-2t)})/2$. Moreover every length-m block of the mechanical word has either $\lfloor m\alpha\rfloor$ or $\lceil m\alpha\rceil$ ones.

Proof

Multiply the y-equation by $e^{(dt)}$. Its derivative is $c\,e^{(dt)}x(t)$; integrate from zero to t and divide by $e^{(dt)}$ to obtain $y(t)=e^{(-dt)}y_0+c\int_0^t e^{(-d(t-s))}x(s)ds$. Substitution into the x-equation proves the exact Volterra equation and shows why the initial source cannot be omitted. In the stated symmetric example, $s=x+y$ satisfies $s'=0$ and $r=x-y$ satisfies $r'=-2r$. With $s_0=r_0=1$, $x=(s+r)/2=(1+e^{(-2t)})/2$ and $y=(1-e^{(-2t)})/2$. Dropping the memory would give $e^{(-t)}$, which has the wrong long-time limit. This is an exact closure failure, not an approximation estimate. For the mechanical word, sum from $n=j$ to $j+m-1$: the floors telescope to $\lfloor(j+m)\alpha+\rho\rfloor-\lfloor j\alpha+\rho\rfloor$. Writing $j\alpha+\rho$ as integer plus $\theta\in[0,1)$ gives $\lfloor m\alpha+\theta\rfloor$, which is either floor or ceiling of $m\alpha$. Any two length-m blocks therefore differ in their number of ones by at most one. Dividing the prefix count by m yields limiting frequency α. If α is irrational, a periodic binary word would have rational frequency, so the word is aperiodic. These two examples demonstrate distinct constructive repairs: keep a memory term for the continuous reduction, or keep the rotation phase for the symbolic coding. Neither irregular recurrence nor aperiodicity alone proves positive entropy or unpredictability.

Worked example

Compare the exact continuous reduction at $t=\log 2$ with a mechanical word of slope $\alpha=(\sqrt{5}-1)/2$ and $\rho=0$.

At $t=\log 2$, $x=(1+1/4)/2=5/8$, while the memory-free approximation is $1/2$. The mechanical word begins $0,1,0,1,1,0,1,0$; its first eight symbols contain $\lfloor 8\alpha\rfloor=4$ ones. Every block of eight contains four or five ones. The compact floor formula supplies all finite blocks without asserting an independent random-symbol model.

The archive’s symbolic Perron models, substitutions and relative periodic cellular automata can be compared through their declared coding maps and retained state. They are not automatically interchangeable realizations of the same process.

Continuity across the source years

2023 — Affine p-adic iteration and an unidentifiable nonlinear observation already distinguish a state trajectory from its measured image. Sources: 2023:M003, 2023:M107

2024 — Transported invariant measures and Caputo memory make the evolution structure explicit. Sources: 2024:Y24-07-001, 2024:Y24-09-011

2025 — Hidden kernels, Markov measures, relative periodicity and quasiperiodic recurrence provide several exact model classes. Sources: 2025:Y25-03-023, 2025:Y25-08-004, 2025:Y25-08-010, 2025:Y25-11-016

2026 — Path-dependent accumulators, damped memory, substitution frequencies and balanced words develop constructive finite descriptions. Sources: 2026:Y26-02-031, 2026:Y26-02-019, 2026:Y26-04-029, 2026:Y26-05-006

Connections

D15 — Sufficient state. An observation determines a closed target evolution only when indistinguishable states have indistinguishable futures. State reconstruction and target reconstruction can have different minimal dimensions.

D13 — Symbolic counting. Transfer matrices and substitution incidence matrices count admissible words and frequencies. A symbolic coding must be proved faithful before its entropy is attributed to a physical flow.

D10 — Stochastic descriptions. A specified Perron transition matrix defines a Markov measure. A deterministic balanced word is not an independent Bernoulli sample merely because it has a frequency.

Capacity and interactive scope

Lab ids: []

Inputs: A fully specified finite model, its domain, norm, parameter range and requested tolerance; exact arithmetic where available.

Outputs: Declared invariants, computed model quantities, residual or truncation bound, and a statement of the model assumptions.

Validation: The companion stdlib verifier checks the stated worked model. Annual entry verifiers remain the checks for their respective treatments.

Status: Documented recipe and reproducible worked example; no general-purpose solver is claimed.

Limits

The memory formula is exact for the stated linear system. A Markovian approximation needs an additional scale-separation error bound. The balance proof establishes frequency and aperiodicity for irrational slope, not a blanket entropy theorem for all symbolic systems.

Next questions

What is the smallest retained statistic that closes the specific target evolution?

Can the hidden-state memory be approximated by finitely many modes with a uniform trajectory-error bound?

Source anchors: 2024:Y24-07-001, 2024:Y24-09-011, 2025:Y25-03-023, 2025:Y25-08-004, 2025:Y25-08-010, 2025:Y25-11-016, 2026:Y26-02-019, 2026:Y26-05-006

Mainardi–Gorenflo, On Mittag-Leffler-type functions in fractional evolution processes — https://arxiv.org/abs/0706.3866

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NIST DLMF, Floquet Theory — https://dlmf.nist.gov/28.29

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D10 · Probability and stochastic processes

Organizing question

What is the complete law, and which operations preserve or transform the uncertainty needed by the question?

The probability material is organized by mathematical operation rather than by a single chronological narrative. A model first specifies its sample space and law. Expectations then integrate against that law, dependence connects its coordinates, transitions evolve it in time, and limit theorems approximate it under explicit scalings. This order makes the archive readable without making every probabilistic topic an instance of one favored physical metaphor. The four families progress from compatible outcomes to combinatorial randomness, from processes to asymptotic simplification.

The first family supplies the indispensable foundation. Marginal event probabilities determine an expected count but usually leave the probability of complete success unresolved. Population weights enter an expected square through square roots in its Euclidean representation; amplitude attenuation is a different operation. Posterior predictive dependence gives a second mechanism: future outcomes independent conditional on a latent parameter become dependent after it is integrated out. These exact finite distinctions continue through the three-item surplus model at the end of the source period. Their significance is constructive: the revised question can ask for the range over all admissible couplings instead of assuming an unprovided independence law.

Random sampling and filtrations move the same discipline into combinatorics and topology. Weighted fixed-cardinality samples are normalized by elementary symmetric polynomials, and inclusion probabilities yield expected Euler characteristic by linearity. A random barcode needs more: the actual birth rule and boundary ranks. The four-cycle fixture makes one complete persistence distribution accessible, while the general weighted formula remains an explicit mathematical recipe. This is an important boundary between a finite worked lab and a universal persistent-homology tool.

Stochastic dynamics specify how the law changes. A generator determines expectation evolution, while Itô's second derivative produces an extra term invisible to deterministic differentiation. The OU process provides exact solutions, stationary variances and a transparent comparison with Euler–Maruyama. Linear SDE propagation separates present posterior uncertainty from fresh process noise. Sensor choices are meaningful only when compared against one prior, one target and one terminal horizon. A driver sensor can beat a direct present-target sensor for predicting a future target without violating any monotonicity theorem.

The final family studies simplification under scaling. Finite variance yields the Gaussian averaging fixed point; Cauchy stability shows why that conclusion cannot be universal. Logarithms turn random products into sums with their own transform threshold. Trace-summable Gaussian covariances permit genuine Hilbert-space samples, whereas finite-dimensional consistency alone permits only cylindrical data. Direct couplings give a different approximation method, bounding Lipschitz observables without a central limit theorem. Across all four families, exact finite identities, limiting laws, numerical approximations and empirical calibration remain distinct claims. The atlas executes representative model checks and labels the broader extensions as recipes.

Core objects

Finite laws and couplings; moments and characteristic functions; weighted subset measures; random filtrations; Markov generators; Wiener integrals; Gaussian covariance operators.

Open frontier

Combine exact finite combinatorial expectations with quantitative approximation of stochastic dynamics, retaining explicit dependence, topology and error metrics rather than replacing them by one scalar uncertainty score.

D10-F01 · Probability laws, moments and joint dependence

Which parts of a conclusion follow from marginal probabilities, and which require a full joint law?

A probability model is an assignment of mass to compatible outcomes before it is a formula for an average. The earliest finite probability questions, the distinction between population and amplitude weights, reliability calculations, Bayesian prediction and the final three-item surplus model all converge on this point. Linearity of expectation survives dependence, whereas the probability of a conjunction generally does not. The most productive revision is to preserve the entire compatibility structure until the requested functional has been calculated. A mean, a variance, a tail probability and an optimal action use different parts of that structure. This family therefore treats uncertainty as a mathematical input with explicitly stated moments, rather than as a single decorative error bar.

Definitions

A finite law is $p(\omega)\geq 0$ with $\Sigma_\omega p(\omega)=1$; $E\, f=\Sigma_\omega p(\omega)f(\omega)$. A covariance requires finite second moments.

For Bernoulli indicators B_i, marginal success probabilities $p_i=E\, B_i$ specify only n linear constraints on their joint table.

Population-weighted energy is $\Sigma p_i|a_i|^2=\|\mathrm{diag}(\sqrt{p})a\|^2$; replacing $\sqrt{p}$ by p changes the quantity.

A posterior predictive probability integrates a conditional law against the posterior; its repeated outcomes can be dependent after the latent parameter is integrated out.

Theorem and hypotheses

For events $A_1,\dots,A_n$ with probabilities p_i, $\max(0,\Sigma p_i-n+1)\leq P(\cap A_i)\leq \min_i p_i$. For any integrable B_i, $E\Sigma B_i=\Sigma E\, B_i$ without independence. If p is random, conditionally independent future Bernoulli outcomes have unconditional covariance $\mathrm{Var}(p)$, so their joint success probability is $E\, p^2$ rather than $(E\, p)^2$.

Proof

The upper intersection bound follows from containment. For the lower bound apply the union bound to the complementary events: $P(\cap A_i)=1-P(\cup A_i^c)\geq 1-\Sigma(1-p_i)$, then combine with nonnegativity. Linearity is a finite rearrangement of sums, or the linearity of the Lebesgue integral. For two future indicators, conditional independence gives $E[B_1B_2|p]=p^2$, and the tower property yields $E\, B_1B_2=E\, p^2$. Each mean is $E\, p$; subtraction gives $\mathrm{Var}(p)$. Thus independence conditional on one variable cannot be silently transported through marginalization. The proof also identifies the missing datum: a coupling or a latent-parameter distribution.

Worked example

Three indicators each have success probability $4/5$, and an illustrative surplus is $\Pi=10(B_1+B_2+B_3)-95/4$.

The mean is $30\cdot 4/5-95/4=1/4$ for every compatible coupling. Profit requires three successes, whose probability lies between $2/5$ and $4/5$. The lower endpoint is attained by assigning mass $2/5$ to 111 and $1/5$ each to 110,101,011. The upper endpoint is attained by mass $4/5$ at 111 and $1/5$ at 000. Independence

instead yields 64/125. In a Beta(4,2) posterior, the mean success probability is 2/3 but the probability of two future successes is 4·5/(6·7)=10/21, exceeding 4/9 by 2/63.

A positive expectation and a frequent positive outcome are separate objectives. These exact fixtures give a practical diagnostic: ask whether the claimed conclusion remains constant over all joint laws with the supplied marginals. If it does not, the dependence model belongs in the question itself. This is also the common starting point for distributionally robust decisions.

Continuity across the source years

2023 — Dated topic continuity is evidenced by Population weights versus amplitude weights; Joint reliability is a dependence problem. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M060, 2023:M106, 2023:M057

2024 — Dated topic continuity is evidenced by Ratios, percentages and lottery events require a denominator. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-04-008

2025 — Dated topic continuity is evidenced by Bayesian prediction differs from plug-in independence; Second-order correlations are factorial moments with distinguishable models. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-07-009, 2025:Y25-03-025

2026 — Dated topic continuity is evidenced by Two-sector failure has a correlation parameter and a rigorous confidence envelope; Expected surplus and probability of profit require different dependence assumptions. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-06-007, 2026:Y26-10-040

Connections

D12 — Robust decisions over couplings. Optimize an expected loss or event probability over the polytope of joint tables with fixed marginals. Finite marginals define a feasible polytope, not calibrated empirical uncertainty.

D03 — Covariance is a positive quadratic form. $\mathrm{Var}(w^{T}X)=w^{T}\Sigma w$ and $\Sigma \geq 0$ when second moments exist. The Cauchy example lies outside the second-moment hypothesis.

Capacity and interactive scope

Lab ids: []

Inputs: A declared finite probability table, covariance model, convex objective, or common uncertainty set, as specified in the worked example. Parameters without a sampling or error model are treated as assumptions.

Outputs: Exact finite values and rational certificates when available; otherwise numerical values explicitly restricted to the stated worked model.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Mathematical recipe with executable scoped checks; no general interactive estimator or solver is implemented.

Limits

Finite moment calculations do not establish that observations follow the proposed law. A Cauchy sample mean remains Cauchy because its characteristic function is stable under averaging, so even increasing sample size does not repair a nonexistent mean. Signed weights need not define probability or positive energy. Marginal confidence statements also require simultaneous coverage before they define a valid joint uncertainty set.

Next questions

Characterize extremizing couplings when both marginals and pairwise intersections are fixed.

For a specified posterior family, bound the gap E p^k−(E p)^k and state how it changes a chosen finite loss.

Source anchors: 2023:M060, 2023:M106, 2024:Y24-04-008, 2025:Y25-07-009, 2026:Y26-06-007, 2026:Y26-10-040, 2023:M057, 2025:Y25-03-025

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D10-F02 · Random filtrations, weighted sampling and exact inclusion laws

How can random selection be transported into an exact expectation of a combinatorial or topological observable?

Random filtrations connect probability with topology by making the event that a face is present explicit. The source trajectory runs from elementary denominators and collision probabilities to fractal measures, random birth times and weighted fixed-size samples. A random complex cannot be understood solely by calling it random: its vertex law, face rule, dependence and filtration parameter must be declared. Once those inputs are fixed, indicator expansion often turns a seemingly global expected invariant into a short exact sum. This family organizes three scales of the same idea: finite subset enumeration, order statistics of random birth times, and weak approximation of measures by coupled partial constructions.

Definitions

For positive weights w_1,…,w_N and fixed size m, assign a subset S probability proportional to ∏_{i∈S}w_i. The normalizer e_m(w) is the degree-m elementary symmetric polynomial.

For a prescribed face F with b vertices, the inclusion indicator is 1_{F⊆S}. A simplicial complex is induced by the selected vertices only after the face rule is specified.

The Euler characteristic of a finite complex is χ=Σ_F(−1)^{dim F}; expectation commutes with this finite sum.

A filtration K_t is increasing in t. Independent uniform vertex birth times create correlated face births via the maximum over each face.

Theorem and hypotheses

Under weighted fixed-size sampling, P(F⊆S)=w_F e_{m−b}(w_{F^c})/e_m(w), taking e_r=0 outside valid degrees. Consequently Eχ(K[S])=Σ_F(−1)^{dim F}P(F⊆S). In the equal-weight case this becomes binom(N−b,m−b)/binom(N,m).

Proof

Every selected subset containing F is uniquely F∪T with T⊆F^c and |T|=m−b. Its unnormalized mass is w_F∏_{i∈T}w_i. Summing over T gives the numerator; summing over all size-m subsets gives the denominator. The Euler formula follows by inserting one inclusion indicator per face and using linearity, with no independence assumption between faces. The normalizers satisfy E_{j,r}=E_{j−1,r}+w_jE_{j−1,r−1}, because each size-r subset either excludes j or includes it. This dynamic program is an exact coefficient computation, not a Monte Carlo estimate. Equal weights reduce both symmetric polynomials to binomial coefficients and recover the hypergeometric law.

Worked example

Select two vertices from four with weights (1,2,3,4). Separately, let the four vertices of a cycle appear at independent uniform times, with edges appearing when both endpoints are present and the filling two-cell arriving at time one.

Here e_2=2+3+4+6+8+12=35. Vertex one is selected with probability (2+3+4)/35=9/35; the pair {1,2} is selected with probability 2/35. If K is the four-cycle, Eχ=Σ_iP(i∈S)−Σ_edgesP(edge⊆S)=2−(2+6+12+4)/35=46/35. In the birth-time model, the loop appears at B=max(U_1,…,U_4), hence P(B≤t)=t^4. Its barcode is [B,1), and E(1−B)=∫_0^1t^4dt=1/5.

The expected Euler characteristic needs only face inclusion probabilities, whereas expected Betti numbers usually require joint rank information. The barcode example is exactly solvable because the topology has one possible loop. The distinction prevents a facewise expectation formula from being mistaken for a general persistent-homology estimator.

Continuity across the source years

2024 — Dated topic continuity is evidenced by Landscape division is a collision probability with exact merge laws. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-04-011

2025 — Dated topic continuity is evidenced by A fractal probability measure has a quantitative weak-convergence proof. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-03-029

2026 — Dated topic continuity is evidenced by Weighted sampling replaces hypergeometric ratios by symmetric polynomials; Random birth times connect order statistics to an exact barcode law. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-02-003, 2026:Y26-02-002, 2026:Y26-10-016

Connections

D05 — Random topology. Face indicators give Eχ, while boundary-matrix ranks determine Betti numbers. The implemented homology lab is a finite worked complex; these weighted formulas are a recipe beyond it.

D13 — Exact combinatorial normalizers. Elementary symmetric polynomials enumerate weighted subsets by coefficient extraction. Strictly positive weights ensure the stated probability law.

D14 — Count before sampling. Completion counts give exact uniform addresses and rejection sampling. A deterministic seed alone supplies reproducibility rather than uniformity.

Capacity and interactive scope

Lab ids: ['L02', 'L09']

Inputs: Related current UI only: L02 varies the integer budget 0–27 and requested feasible rank 0–63 for six fixed items; L09 selects triangle, square or K3,3 and whether to fill the triangle.

Outputs: L02 enumerates 64 masks, counts feasible subsets, returns optimum masks and one ranked subset. L09 returns exact rational boundary ranks and Betti numbers of the selected fixed complex. Weighted sampling, random birth times and expected barcode laws remain recipes.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Related deterministic finite labs are implemented; the random-filtration and weighted-sampling models in this family are not interactive.

Limits

The fixed-cardinality law is different from independent Bernoulli selection or sequential probability-proportional-to-size sampling. Faces are strongly dependent. Expected χ can conceal simultaneous changes in several Betti numbers. Random birth-time claims need their continuity and independence assumptions; ties, delayed edges or an early filling cell change the barcode. Weighted expected persistence is not implemented by the atlas.

Next questions

Derive exact expected Betti curves for a supplied small complex by grouping subsets according to boundary rank.

Compute weighted barcode distributions with rational weights and compare coefficient dynamic programming with exhaustive enumeration.

Source anchors: 2026:Y26-02-003, 2026:Y26-02-002, 2025:Y25-03-029, 2024:Y24-04-011, 2026:Y26-10-016

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Afra Zomorodian and Gunnar Carlsson, Computing Persistent Homology — https://doi.org/10.1007/s00454-004-1146-y
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Bryant: graph-based Boolean-function algorithms — https://doi.org/10.1184/r1/6605990

D10-F03 · Stochastic dynamics, Itô correction and finite-horizon prediction

Which drift, noise and observation model determines the uncertainty at a future time?

The stochastic strand becomes coherent when a process is specified before its spectrum, stationary law or prediction error is computed. Finite reversible chains, a two-spin Gibbs model, Ornstein–Uhlenbeck dynamics, bridge functionals and exact linear propagation supply complementary levels of resolution. The key unification is the generator: transition rates in a finite state space and second derivatives in a diffusion both encode how expectations evolve. They do not make deterministic gradient descent, white noise and colored fluctuations interchangeable. The prediction problem adds a second layer: an observation changes the present conditional law, then a common transition propagates that law and introduces fresh process noise. Reversibility supplies a complementary finite certificate: detailed balance makes a Markov generator symmetric after weighting by its stationary law, so its relaxation modes can be checked spectrally. That self-adjoint reduction is specific to the declared chain and is not available for arbitrary driven dynamics.

Definitions

An Itô diffusion satisfies $dX=b(X,t)dt+\sigma(X,t)dW$; its generator is $Lf=b\cdot\nabla f+(1/2)\mathrm{tr}(\sigma\sigma^{T}D^{2}f)$.

For $dX=AXdt+LdW$ with deterministic coefficients, $F=e^{A\Delta}$ and $Q=\int_0^\Delta e^{As}LL^Te^{A^Ts}ds$ describe the exact transition.

An OU process has drift $-\kappa X$, $\kappa>0$; a stationary variance solves the scalar Lyapunov equation $-2\kappa v+\sigma^2=0$.

A Brownian bridge $B_t=W_t-(t/T)W_T$ conditions the endpoint, changing covariances and all integrals sensitive to constants.

Theorem and hypotheses

For $dX=-\kappa Xdt+\sigma dW$ and deterministic $X_t=x$, $E[X_T^2|X_t=x]=e^{-2\kappa\tau}x^2+\sigma^2(1-e^{-2\kappa\tau})/(2\kappa)$, $\tau=T-t$. More generally a linear Gaussian prior covariance P propagates to FPF^T+Q. A common propagation preserves Loewner order between two posterior covariances.

Proof

The integrating factor gives $X_T=e^{-\kappa\tau}x+\sigma\int_t^Te^{-\kappa(T-s)}dW_s$. The stochastic integral has mean zero and, by Itô isometry, variance $\sigma^2\int_0^\tau e^{-2\kappa s}ds$. Squaring and taking expectations gives the displayed expression. The generator gives the same conclusion: $L(x^2)=-2\kappa x^2+\sigma^2$, where the constant σ^2 is the Itô correction. For the matrix process use the variation-of-constants formula, independence of future Wiener increments and the matrix isometry. If $P_1\le P_2$, then $F(P_2-P_1)F^T\ge0$. Adding the same Q preserves the order. Different terminal times require different F and Q and therefore are outside that comparison.

Worked example

Let $X=(\text{position},\text{driver})$, $A=[[0,1],[0,0]]$, prior $P=I$, horizon $\Delta=10$, and a scalar sensor with noise variance $r=1/100$. Compare measuring position with measuring driver, both at the present time.

$F=[[1,10],[0,1]]$. A measurement of position gives posterior $\mathrm{diag}(1/101,1)$, so terminal position variance is $100+1/101$. A driver measurement gives $\mathrm{diag}(1,1/101)$, hence variance $1+100/101=201/101$. The indirect sensor is better for this future target. If $L=(0,1)^T$, process covariance is $[[\Delta^3/3,\Delta^2/2],[\Delta^2/2,\Delta]]$; at $\Delta=2$ it has determinant $4/3$ and must be added for either sensor. For OU Euler–Maruyama, stationary variance is $\sigma^2/(2\kappa-\kappa^2h)$, existing only for $0<\kappa h<2$.

A good present measurement and a good future-target measurement are different design questions. The exact transition separates structural ignorance, current sensor noise, future noise and numerical discretization bias. Each has a different remedy and a different mathematical certificate.

Continuity across the source years

2023 — Dated topic continuity is evidenced by A reversible q-dependent chain with an exact gap. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M023

2024 — Dated topic continuity is evidenced by A specified noisy dynamics and its stationary covariance; A complete two-spin Gibbs and Metropolis model. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-05-012, 2024:Y24-06-021

2025 — Dated topic continuity is evidenced by Euler–Maruyama and an exact Ornstein–Uhlenbeck comparison; Cumulant evolution is triangular for an actual OU process. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-12-007, 2025:Y25-12-027, 2025:Y25-11-004, 2025:Y25-03-005, 2025:Y25-03-022, 2025:Y25-12-011

2026 — Dated topic continuity is evidenced by Itô correction and a Feynman–Kac model with an exact solution; Exact SDE propagation can make a driver measurement more informative than a direct present-target measurement. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-01-901, 2026:Y26-09-021, 2026:Y26-04-015

Connections

D08 — Backward equations. The conditional second moment solves $u_t+Lu=0$ with terminal data x^2. Feynman–Kac needs the stated integrability or martingale conditions.

D15 — Sensor design. Compare $l^T(FP_postF^T+Q)l$ at one fixed terminal time. The observability lab is deterministic and does not implement arbitrary Kalman filtering.

D09 — Numerical dynamics. The Euler multiplier $1-\kappa h$ controls stability and stationary bias. Continuous dissipation does not prove discrete stability.

Capacity and interactive scope

Lab ids: ['L01']

Inputs: Related current UI only: L01 selects the target x1+x2 or (x1,x2) and a mask of six fixed linear probes on four real coordinates.

Outputs: Exact rational row-space observability and all 64 fixed probe-subset comparisons. No stochastic transition, covariance update, OU simulation or filter is exposed in the UI.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: L01 is a related deterministic observation model; all stochastic dynamics in this family remain mathematical recipes with scoped Python checks.

Limits

Linearity and Gaussianity make these covariance calculations closed. Nonlinear filtering needs a different theorem. Spectral color additionally needs stationarity or a declared integrated process and finite cutoffs when variance would diverge. A bridge cannot be treated as independent Brownian noise: its constant test functional vanishes. None of the illustrative matrices validates a historical weather or sensor-performance number.

Next questions

Bound finite-horizon prediction error when A and L range over a declared compact uncertainty set.

For an OU discretization, choose a step to control stationary variance bias and computational cost simultaneously.

Source anchors: 2024:Y24-05-012, 2025:Y25-12-007, 2025:Y25-12-027, 2026:Y26-01-901, 2026:Y26-09-021, 2025:Y25-11-004, 2023:M023, 2024:Y24-06-021

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D10-F04 · Scaling limits, stable laws and the geometry of approximation

When does repeated averaging simplify a law, and which quantitative information survives the limit?

Averaging, renormalization, heavy tails, multiplicative products and infinite-dimensional Gaussian models share a demand for explicit scaling. A Gaussian fixed point is a theorem about normalized sums with finite variance, not a universal fate of every noise source. Cauchy stability provides an exact countermodel, and multiplicative products become additive only after taking logarithms. The fractal measure construction adds a different kind of approximation: convergence follows from a direct coupling bound rather than from a central limit theorem. This family distinguishes exact finite identities, limiting distributions and numerical approximations, preserving the assumptions that make each useful.

Definitions

For independent copies X_1,X_2 of a centered variable, the averaging renormalization is $R(X)=(X_1+X_2)/\sqrt{2}$.

The characteristic function is $\varphi(t)=E\, e^{itX}$; independent summation multiplies characteristic functions.

An α-stable fixed law for $R_\alpha=(X_1+X_2)/2^{1/\alpha}$ has characteristic function $\exp(-c|t|^\alpha)$ in the symmetric model.

A Gaussian series $X=\Sigma\sqrt{q_n}Z_ne_n$ is H-valued in mean square when $\Sigma q_n<\infty$; cylindrical coordinate consistency alone does not guarantee H-valued samples.

Theorem and hypotheses

If a centered nondegenerate finite-variance law is fixed under R_α, then $\alpha=2$; under R_2 its characteristic function is Gaussian. More generally $R^k(X)$ converges in distribution to the centered Gaussian of variance v when $EX=0$ and $EX^2=v<\infty$.

Proof

A fixed law has $v=2\cdot 2^{-2/\alpha}v$, so nonzero finite v forces $\alpha=2$. In that case the characteristic-function equation implies $\varphi(t)=[\varphi(t/\sqrt{n})]^n$ for dyadic n. Finite variance and centering give $\varphi(s)=1-vs^2/2+o(s^2)$. Inserting $s=t/\sqrt{n}$ and letting n grow yields $\exp(-vt^2/2)$, proving the fixed-law assertion. The same expansion applied to the normalized sum of n independent copies proves convergence of characteristic functions, and hence convergence in distribution, for dyadic n and for the usual all-n central limit theorem. If higher cumulants exist, the rth cumulant scales as $n^{1-r/2}$; this records the disappearance of finite higher-order cumulants without asserting a uniform density approximation or a rate from only two moments.

Worked example

Let U_j be independent uniform variables on $(0,1)$, $\kappa_k=\prod_{j=1}^kU_j$ and $Y_j=-\log U_j$. Also compare a standard Cauchy sample mean.

Each Y_j has exponential density e^{-y} for $y>0$, mean and variance one. Therefore $k^{-1}\log\kappa_k\to-1$ almost surely and $(\log\kappa_k+k)/\sqrt{k}$ converges to $N(0,1)$. The transform $E\kappa_k^{-\beta\gamma}=(1-\beta\gamma)^{-k}$ is finite precisely for $\beta\gamma<1$. For a Cauchy variable, $\varphi(t)=e^{-|t|}$; averaging n copies yields $[\varphi(t/n)]^n=e^{-|t|}$, so the mean never concentrates and $P(|\bar{X}_n|\le 1)=1/2$ for every n. For a Cantor coupling whose omitted tail is at most 3^{-n}, every L-Lipschitz test has expectation error at most $L3^{-n}$.

Different mechanisms supply different error statements: a weak Gaussian limit, exact Cauchy stability, a transform convergence threshold, or a deterministic transport bound. Choosing among them begins by

stating which observable must be approximated and in which metric. A visual similarity between histograms does not substitute for that choice.

Continuity across the source years

2023 — Dated topic continuity is evidenced by A Cauchy model has no ordinary mean or variance. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M057

2024 — Dated topic continuity is evidenced by A renormalization map with a proved Gaussian fixed point; Cylindrical Gaussian data versus a Radon Gaussian on Hilbert space. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-06-019, 2024:Y24-11-014, 2024:Y24-09-015, 2024:Y24-03-027

2025 — Dated topic continuity is evidenced by A fractal probability measure has a quantitative weak-convergence proof. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-03-029

2026 — Dated topic continuity is evidenced by The RG characteristic-function map has explicit stable fixed points and a finite-variance Gaussian obstruction; Multiplicative random products give a testable CLT and convergence threshold. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-05-003, 2026:Y26-01-027

Connections

D07 — Transforms and convergence. Characteristic functions turn independent convolution into products; logarithms turn positive products into sums. Transform convergence and moment convergence are distinct.

D06 — Measure transport. A coupling with $|X-X_n|\le\varepsilon$ controls Lipschitz observables by $L\varepsilon$. This does not imply total-variation convergence for singular and discrete measures.

D01 — Leading digits. Benford frequencies follow from equidistribution of fractional logarithms. No decimal-normality conclusion follows from a finite entropy calculation.

Capacity and interactive scope

Lab ids: []

Inputs: A declared finite probability table, covariance model, convex objective, or common uncertainty set, as specified in the worked example. Parameters without a sampling or error model are treated as assumptions.

Outputs: Exact finite values and rational certificates when available; otherwise numerical values explicitly restricted to the stated worked model.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Mathematical recipe with executable scoped checks; no general interactive estimator or solver is implemented.

Limits

Finite variance supplies no universal finite-sample normal-approximation rate. A third absolute moment enables further theorems, but is an additional hypothesis. Infinite covariance trace obstructs an ℓ^2 Gaussian even though each finite coordinate vector is valid. The product fixture is an iid uniform construction; dependence, atoms at zero or heavy logarithmic tails change its law. Its exact transform threshold is unrelated to an empirical phase transition unless a model is supplied.

Next questions

For a specified non-Gaussian input with finite third moment, derive an explicit approximation error for the chosen normalized sum.

For a covariance sequence q_n, bound the mean-square tail and the error of a supplied Lipschitz functional.

Source anchors: 2024:Y24-06-019, 2026:Y26-05-003, 2026:Y26-01-027, 2023:M057, 2024:Y24-11-014, 2025:Y25-03-029, 2024:Y24-09-015

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D11 · Information, statistics and inference

Organizing question

What does the available observation distinguish, and what claim does its evidence actually justify?

Information and inference begin with an observation map and a target. The map decides which states can be distinguished; the target and loss decide whether those distinctions matter. A sampling law then determines which statistical guarantees are available. This domain therefore organizes covariance, entropy, diagnostics and validation around a sequence of questions: what remains hidden, what transformation loses information, what a finite sample can establish, and what a summary preserves. These questions connect closely but their answers are not interchangeable.

Covariance geometry is the first layer. A covariance is a positive semidefinite quadratic form when second moments exist. Null directions record exact linear dependence, and positive averaging preserves the covariance cone. Nonlinear maps add a decisive qualification: a vanishing Jacobian can conceal a nonzero leading uncertainty at the next order. The squared-Gaussian example is fully calculable. It then becomes a hidden-target model, where observing one coordinate perfectly still leaves a squared hidden coordinate random. Conditional expectation gives the optimal estimator and an exact orthogonal decomposition of risk. This identifies separately the part due to structural invisibility and the part due to measurement noise.

Entropy and information geometry describe how laws compare. Relative entropy contracts under a specified stochastic channel; Shannon entropy increases under doubly stochastic mixing but can decrease on approach to a concentrated equilibrium. Mutual information is statistical dependence, not a causal arrow. In parameter space, local KL expansion produces the Fisher quadratic form, with a null direction for common softmax logits. These results explain what may be monitored during optimization while preserving the need to evaluate actual prediction loss independently. Matrix entropy has related exact identities, but the quantum algebra and positivity hypotheses remain visible.

Statistical inference adds sampling and selection. Identifiability is a statement about whether parameters produce distinct observation laws at the stated design. Calibration is a statement about a test statistic under a null. Generalization is a statement about unseen draws. A nested fit automatically improves training residuals, and a prescribed spectrum can always be embedded in a matrix; neither fact proves prediction. Finite uniform bounds and PAC–Bayes methods supply genuine guarantees under declared bounded-loss sampling models. Exact paired randomization supplies another route when the sign-exchangeability null is justified. Its finite p-value remains separate from effect magnitude and practical benefit.

The final family treats the capacity and limitations of summaries. Lossless reconstruction requires an injective encoding and therefore a counting lower bound. Prefix entropy grows by conditional information, independently of a chosen symbolic metric. Spectral flatness, circular concentration, AUROC and pooled correlation each retain particular features and discard others. Their precise invariances are useful; promoting one to a universal certificate is not. The resulting interactive direction is a panel of observation fibres, conditional-risk calculations and finite diagnostic populations. The current atlas supplies small verified fixtures and cross-links to observability and exact-counting labs, while general statistical estimation and historical empirical validation remain unclaimed.

Core objects

Observation sigma-algebras; conditional expectation and Bayes risk; PSD covariance; KL divergence and channels; Fisher matrices; iid validation laws; randomization populations; finite encodings and diagnostic invariants.

Open frontier

Design observations sufficient for a specified target or action, then combine exact structural guarantees with independent finite-sample calibration without conflating identifiability, uncertainty and predictive performance.

D11-F01 · Covariance geometry, nonlinear uncertainty and hidden-target risk

Which uncertainty is reducible by precision, and which requires a different observation?

The covariance strand connects finite quadratic forms, data fusion, singular maps and statistical observability. Its central revision is to separate an exact law from a linearized uncertainty approximation, and to separate noise in an observed coordinate from a target component that the observation never determines. Positive semidefiniteness is a structural constraint on covariances, not merely a numerical preference. The observation sigma-algebra determines what can be estimated; the loss determines what error means. Together these ideas give a precise route from a hidden coordinate to an irreducible risk, while preventing a zero Jacobian from being read as zero nonlinear uncertainty.

Definitions

For $X \in L^2$, $\Sigma = E[(X-EX)(X-EX)^T]$ is positive semidefinite; singularity records an almost-sure linear relation.

The exact output law under f is the pushforward $f_{\#}\mu$. The delta approximation $\varepsilon^2 J \Sigma J^T$ applies to $f(x_0+\varepsilon Z)$ only as an asymptotic statement with appropriate moments and remainders.

For target $T \in L^2$ and observation Z, the squared-loss Bayes estimator is $E[T|Z]$ and its risk is $E\,\mathrm{Var}(T|Z)$.

The Markov condition $T \to Z \to Y$ means Y supplies no target information beyond Z; this includes adding independent measurement noise to Z.

Theorem and hypotheses

Under $T \to Z \to Y$, $E\,\mathrm{Var}(T|Y) = E\,\mathrm{Var}(T|Z) + E\,\mathrm{Var}(E[T|Z]|Y)$. Thus the best noisy-observation risk is at least the noise-free hidden-target floor. Separately, for $X \sim N(0,\sigma^2)$, $\mathrm{Var}(X^2) = 2\sigma^4$ even though the derivative of x^2 at zero vanishes.

Proof

Write $m = E[T|Z]$, $e = T - m$. Conditional independence gives $E[e|Z,Y] = 0$, hence $E[T|Y] = E[m|Y]$. Expand $T - E[T|Y] = e + m - E[m|Y]$. Conditioning on Z,Y kills the cross term, so the mean squared error is $E\,e^2 + E(m - E[m|Y])^2$. Conditional-mean optimality itself follows by decomposing $T - a(Y)$ into $T - E[T|Y]$ and $E[T|Y] - a(Y)$. The nonlinear example uses $E\,X^2 = \sigma^2$ and $E\,X^4 = 3\sigma^4$. Its full density follows by the two inverse branches $\pm\sqrt{y}$ and their Jacobians. The nonzero variance is of order σ^4; the failed linear term would have been order σ^2, so the issue is an altered leading order rather than a contradiction.

Worked example

Take independent X_1 and $Z_3 \sim N(0,\sigma^2)$, target $T = X_1 + Z_3^2$, and observe X_1 exactly. Compare independent repeated sensors with identical copies of one sensor.

The best estimate is $X_1+\sigma^2$ and risk is $2\sigma^4$. Repeating a noiseless observation of X_1 leaves the observation sigma-algebra unchanged. A sensor informative about Z_3 can improve the risk because it enlarges that sigma-algebra. For n independent observations of a scalar mean with noise variance s^2, averaging has variance s^2/n. For n identical copies of one noise realization, $\Sigma=s^2 1 1^T$ and every unbiased weighted average has variance s^2. If local covariances C_j and means μ_j are mixed with $p_j \geq 0$, total covariance is $\Sigma p_j C_j+\Sigma p_j(\mu_j-\mu)(\mu_j-\mu)^T$, preserving positivity.

This family connects experimental design directly to target information. More repetitions can remove measurement noise; a new kind of observation can remove hidden-target ambiguity. A coordinate covariance, a worst-case fibre width and a Bayes risk are distinct quantities with distinct units and quantifiers. The atlas makes those distinctions operational in its worked examples.

Continuity across the source years

2023 — Dated topic continuity is evidenced by Covariance inequalities with finite moments. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M061

2024 — Dated topic continuity is evidenced by Bayesian uncertainty and data fusion depend on covariance. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-04-006, 2024:Y24-08-003

2025 — Dated topic continuity is evidenced by Uncertainty propagation needs covariance and does not create a field. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-05-015

2026 — Dated topic continuity is evidenced by A hidden-target conditional variance is a genuine noise-free error floor; Linear covariance transport can vanish while nonlinear uncertainty remains. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-09-022, 2026:Y26-02-025, 2026:Y26-09-027, 2026:Y26-08-009, 2026:Y26-09-023, 2026:Y26-04-013

Connections

D15 — Observability and target fibres. A target constant on each observation fibre has zero noise-free ambiguity; probabilistically, T measurable with respect to Z has zero conditional variance. Zero Bayes risk is distribution-dependent; deterministic identifiability quantifies over all admissible states.

D06 — Singular nonlinear maps. A fold can annihilate the Jacobian while transporting nonzero fourth-order uncertainty. The metric alone does not select a coordinate noise law.

D12 — Decision-focused experiments. A sensor need only resolve the sign of the decision difference, which may require less than reconstructing T. A robust fibre certificate and an expected risk reduction are different design objectives.

Capacity and interactive scope

Lab ids: ['L01', 'L08']

Inputs: Related UI: L01 target and six-probe mask; L08 common radius C from 0 to 30 and differential endpoints from −5 to 5.

Outputs: Deterministic observable-target certificates in L01 and coupled score intervals with robust sign flags in L08. The Gaussian-square pushforward, conditional variance and posterior covariance calculations are text-and-check recipes.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Related deterministic observation and additive-cancellation labs are implemented; no interactive covariance estimator, nonlinear filter or conditional-risk engine is implemented.

Limits

Second moments and the Markov hypothesis are essential. An extra sensor correlated with the hidden residual is not a noisier version of the old observation. A signed correction can leave the PSD cone even if it improves one scalar statistic. The Gaussian example gives an exact model, not a calibration for unprovided observations. General nonlinear filtering and empirical conditional-variance estimation remain recipes.

Next questions

For a specified polynomial target and Gaussian hidden variables, compute the first nonzero uncertainty order and bound its Taylor remainder.

Design the cheapest finite sensor panel whose observation fibres reduce a stated risk or decision margin below a specified threshold.

Source anchors: 2026:Y26-09-022, 2026:Y26-02-025, 2024:Y24-04-006, 2025:Y25-05-015, 2023:M061, 2026:Y26-09-027, 2026:Y26-08-009, 2026:Y26-09-023

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D11-F02 · Entropy contraction, channel information and Fisher geometry

Which information quantity is monotone under a declared transformation, and what geometry does it induce?

Entropy appears across finite chains, communication channels, softmax models, quantum states and gradient diffusion. The synthesis needs a reference measure and a transformation before it needs a slogan about disorder. Shannon entropy can decrease under a nonuniform Markov equilibrium while relative entropy to that equilibrium decreases as well. Mutual information describes a joint distribution and contracts along a Markov channel; it does not assign a causal arrow. At small scale, relative entropy generates Fisher geometry, whose null directions express parameter redundancy. These facts connect probabilistic coarse-graining, differential geometry and optimization through explicit maps rather than through a shared word.

Definitions

For positive finite q, $D(p||q)=\Sigma p_i \log(p_i/q_i)$, with $0 \log 0=0$; Shannon entropy is $H(p)=-\Sigma p_i \log p_i$.

A stochastic channel T maps input distributions to output distributions. Mutual information is $D(p_{XY}||p_X p_Y)$.

For logits u, $A(u)=\log\Sigma e^{u_i}$, $p_i=e^{u_i-A(u)}$, and $F=\mathrm{diag}(p)-pp^{T}$. A common additive logit is unidentifiable.

For a faithful density matrix σ, its modular Hamiltonian is −logσ. Matrix entropy and classical entropy agree on diagonal states, but their full differential geometry requires operator calculus.

Theorem and hypotheses

Every finite stochastic channel satisfies $D(Tp||Tq) \le D(p||q)$. For softmax probabilities, $F \ge 0$, $F1 = 0$, and $D(p(u)||p(u+h)) = (1/2)h^{T}Fh + O(\|h\|^{3})$. At uniform p on m outcomes, $F = (I - 11^{T}/m)/m$ and $H = \log m$.

Proof

For each output j, apply the log-sum inequality to $a_i = T_{ji} p_i$ and $b_i = T_{ji} q_i$. Summing the inequalities over j and using $\Sigma_j T_{ji} = 1$ gives contraction. Equality requires the likelihood ratio p_i/q_i to be constant across every collection of inputs mixed into a positive output. For the geometric statement, differentiating A gives $\nabla A = p$ and $\nabla^2 A = F$. The quadratic form $v^{T}Fv = \Sigma p_i v_i^2 - (\Sigma p_i v_i)^2$ is a variance, hence nonnegative and zero on constants. The KL expression is exactly $A(u+h) - A(u) - p(u)\cdot h$; Taylor expansion supplies the local quadratic term. This proves the geometry on the quotient by common logits rather than pretending the full Fisher matrix is invertible.

Worked example

Take a Markov matrix whose every row is $\pi = (9/10, 1/10)$, and initial $p = (1/2, 1/2)$. Then compare a fair bit passed through two binary symmetric channels with flip probability 1/10.

One Markov step sends p to π, reducing H from log2 to $-.9\log.9 - .1\log.1$, while $D(p||\pi) = \log(5/3)$ falls to zero. In the channel cascade the effective flip probability is $2(.1)(.9) = .18$. Information therefore falls from $1 - h_2(.1) \approx .531004$ bits to $1 - h_2(.18) \approx .319923$ bits. At $m = 97$ uniform classes, entropy is $\log 97 \approx 4.574711$ nats and the 96 nonzero Fisher eigenvalues are 1/97. Neither uniform entropy nor a small parameter step reports classification accuracy.

The correct monotone depends on the experiment: relative entropy for a fixed reference law, mutual information for a channel, or KL locally for a parameterized model. This provides a useful common language while preserving their distinct arguments, units and equality conditions. An information quantity can be mathematically exact even when its proposed application lacks data.

Continuity across the source years

2023 — Dated topic continuity is evidenced by Relative entropy along a finite Markov chain; Mutual information detects dependence rather than a causal channel. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M026, 2023:M047

2024 — Dated topic continuity is evidenced by Ψ mutual information and capacity need a channel; Entropy is a concave function with an exact uniform maximum. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-07-002, 2024:Y24-10-031, 2024:Y24-03-004, 2024:Y24-10-004

2025 — Dated topic continuity is evidenced by Softmax Fisher geometry and a uniform fixed-point diagnostic; The entropy first law is an exact first derivative at a faithful state. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-12-014, 2025:Y25-03-013, 2025:Y25-01-023

2026 — Dated topic continuity is evidenced by A computational monotone needs a specified coarse-graining map; Fokker–Planck entropy decay follows from an explicit gradient structure. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-01-023, 2026:Y26-01-020, 2026:Y26-01-026

Connections

D10 — Markov evolution. A stationary π makes D(p_t||π) decrease under the Markov semigroup. A quantitative rate requires an additional functional inequality or finite spectral estimate.

D06 — Information geometry. The Hessian of log-partition defines a positive metric after quotienting redundant parameters. The local quadratic approximation is not a global distance identity.

D16 — State entropy. For a faithful reference, $D(\rho||\sigma)=\mathrm{Tr}((\rho-\sigma)(-\log\sigma))-(S(\rho)-S(\sigma))$. The quantum matrix identity needs positivity and the declared algebra.

Capacity and interactive scope

Lab ids: []

Inputs: A declared finite probability table, covariance model, convex objective, or common uncertainty set, as specified in the worked example. Parameters without a sampling or error model are treated as assumptions.

Outputs: Exact finite values and rational certificates when available; otherwise numerical values explicitly restricted to the stated worked model.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Mathematical recipe with executable scoped checks; no general interactive estimator or solver is implemented.

Limits

Finite alphabets avoid differential-entropy coordinate issues. Continuous densities must first be normalized and have adequate integrability for differentiation. Mutual information alone does not identify causal direction, and a proposed logarithmic transform of mutual information need not be a metric. Fisher, empirical Fisher, Gauss–Newton and the objective Hessian are different matrices outside special model assumptions.

Next questions

Find the exact equality fibres for a supplied stochastic coarse-graining and determine its sufficient statistic.

Compare an actual KL-constrained step with a Fisher quadratic step using an explicit third-order remainder bound.

Source anchors: 2023:M026, 2023:M047, 2024:Y24-07-002, 2024:Y24-10-031, 2025:Y25-12-014, 2026:Y26-01-023, 2026:Y26-01-020, 2025:Y25-03-013

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Claude Shannon A Mathematical Theory of Communication — https://www.princeton.edu/~wbialek/rome/refs/shannon_48.pdf

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C. E. Shannon, A mathematical theory of communication — https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

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Imre Csiszár, Information-Type Measures of Difference of Probability Distributions — https://doi.org/10.1007/BF02159459

Richard Jordan, David Kinderlehrer and Felix Otto, The Variational Formulation of the Fokker–Planck Equation — https://doi.org/10.1137/S0036141096303359

Araki, Relative entropy of states of von Neumann algebras — https://doi.org/10.2977/prims/1195191148

D11-F03 · Identifiability, calibration and honest statistical validation

What mathematical contract turns a fitted pattern into a justified inference about unseen outcomes?

The inference strand develops from an identifiable logistic design and calibrated residual energy to finite strategy search, PAC–Bayes bounds, rank statistics and exact randomization. Its organizing distinction is between fitting a chosen sample, identifying a model parameter, calibrating a test under a null, and predicting a new draw. Each asks a different question. A low residual proves a relation within a specified sample geometry; it does not automatically validate a causal mechanism or a future performance claim. This family supplies positive mathematical tools for stronger questions: declare the sampling law, expose the selection step, reserve independent data, and report an uncertainty statement whose quantifiers survive that selection.

Definitions

A parameter is identifiable if distinct admissible values produce distinct observation laws at the stated design.

For a fixed predictor f and bounded loss in [0,1], true risk R(f) is the expected loss under a declared sampling distribution, while empirical risk is the sample mean.

A valid p-value obeys $P_0(p \le \alpha) \le \alpha$ for every $\alpha \in [0,1]$ under the stated null. Pairing alone does not supply the null.

AUROC is $P(s_+ > s_-) + (1/2)P(s_+ = s_-)$ for independent uniform empirical positive and negative choices; its tie rule belongs to the definition.

Theorem and hypotheses

For K fixed candidates evaluated on n iid validation observations with loss in [0,1], with probability at least $1-\delta$ all risk errors are at most $\varepsilon = \sqrt{(\log(2K/\delta)/(2n))}$. The empirically best candidate then has true risk at most 2ε above the best candidate. A paired fair-sign randomization test has a finite superuniform p-value under its exchangeable sign null.

Proof

Hoeffding's bound gives $P(|\hat{R}_k - R_k| > \varepsilon) \le 2e^{-2n\varepsilon^2}$ for one fixed candidate. A union bound over K choices yields δ at the stated ε. On the simultaneous event, $R(\hat{f}) \le \hat{R}(\hat{f}) + \varepsilon \le \hat{R}(f^*) + \varepsilon \le R(f^*) + 2\varepsilon$. The candidates must be fixed independently of this validation sample, or their entire adaptive selection must be included in a stronger valid argument. For the randomization result, condition on the observed magnitudes and enumerate every equally likely sign assignment. Ranking the test statistic within that finite null population makes the upper-tail rank p superuniform; ties only make it conservative. This proof is finite and requires no normal approximation.

Worked example

Eight paired differences have illustrative magnitudes all one, and the observed signs are all positive. A separate two-class score example has positive scores (2,1) and negative scores (1,0).

There are $2^8 = 256$ sign vectors. Only one gives signed sum eight, so the one-sided exact p-value is 1/256. The absolute statistic has two extremes, giving 1/128. For AUROC, the four positive-negative comparisons yield wins 1,1,0,1 with the tied comparison replaced by 1/2, so the total is 7/2 and AUROC=7/8. Every strictly increasing score transformation preserves these comparisons. Multiplying all paired magnitudes by a positive constant leaves randomization p unchanged, demonstrating that statistical evidence and practical effect size are different outputs.

The sources on regressions, periodic contrasts, spectral fits and normalized residuals become useful when attached to this contract. A period-matched difference can identify an exact trend in a declared model; a nested regression always improves training fit; neither alone supplies a future-risk bound. The best revised question names both the estimand and the independent evidence that could challenge it.

Continuity across the source years

2023 — Dated topic continuity is evidenced by A valid probability model and identifiable parameters; Residual energy and exact Gaussian calibration. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M004, 2023:M078

2024 — Dated topic continuity is evidenced by Spectral fitting is not prediction: six-point statistics and prescribed masses. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-12-016, 2024:Y24-04-007, 2024:Y24-09-017, 2024:Y24-10-006

2025 — Dated topic continuity is evidenced by Searching many strategies needs an independent validation contract; A PAC–Bayes guarantee needs a sampling law, prior and bounded loss. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-02-012, 2025:Y25-05-021, 2025:Y25-07-028, 2025:Y25-06-020, 2025:Y25-09-018

2026 — Dated topic continuity is evidenced by A paired sign-randomization test has a finite exact null certificate; AUROC is an exact pairwise statistic with a specified tie convention. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-09-004, 2026:Y26-04-021, 2026:Y26-04-024, 2026:Y26-09-003, 2026:Y26-03-022, 2026:Y26-01-025, 2026:Y26-01-008

Connections

D12 — Selection versus evaluation. Uniform validation bounds survive choosing the best of a fixed finite candidate family. Searching on the validation labels violates that fixed-family contract.

D15 — Experiment design. A full-rank design separates structural parameter identifiability from estimator variance. Rank conditions do not imply predictive adequacy.

D10 — Null probabilities. Exact enumeration gives finite reference distributions and confidence inversion. Exchangeability is an assumption to justify, not a consequence of pairing.

Capacity and interactive scope

Lab ids: []

Inputs: A declared finite probability table, covariance model, convex objective, or common uncertainty set, as specified in the worked example. Parameters without a sampling or error model are treated as assumptions.

Outputs: Exact finite values and rational certificates when available; otherwise numerical values explicitly restricted to the stated worked model.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Mathematical recipe with executable scoped checks; no general interactive estimator or solver is implemented.

Limits

No historical raw paired samples, strategy logs or model checkpoints were provided for replay. These are mathematical fixtures and guarantees under stated laws. Covariate transfer additionally needs support coverage and conditional invariance. A large correlation can arise from a common cause; an intervention requires causal assumptions beyond its joint table. Zero observed failures yields an upper confidence bound, never a deterministic zero-risk certificate.

Next questions

For a specified adaptive search, construct a fresh holdout or a selection-aware guarantee and compare its sample cost.

Combine exact randomization inference with a prespecified practical-effect threshold and an independent correctness gate.

Source anchors: 2025:Y25-02-012, 2025:Y25-05-021, 2026:Y26-09-004, 2026:Y26-04-021, 2023:M004, 2023:M078, 2026:Y26-04-024, 2026:Y26-09-003

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Montgomery, pair correlation of zeros — https://doi.org/10.1090/pspum/024

D11-F04 · Finite information, observation equivalence and diagnostic statistics

How much can an observation distinguish, and what information does a summary discard?

Several source questions use entropy, spectral flatness, phase concentration, compression or correlation as a proxy for a richer structure. Their strongest mathematical development begins by specifying the observation map and its fibres. A summary can be exactly computed and still fail to separate the objects relevant to the question. Conversely, an injective finite encoding supports exact recovery regardless of whether its values resemble a preferred statistical distribution. This family organizes capacity and diagnostics as distinct operations: counting distinguishable messages, preserving information under a reversible map, and measuring one chosen feature of a data object.

Definitions

An observation $h:X\to Y$ induces equivalence $x\sim x'$ when $h(x)=h(x')$. Exact reconstruction from h exists precisely when the required target is constant on each fibre.

A lossless finite encoder E has a decoder D with $D(E(x))=x$, hence must be injective.

A b-ary prefix $X_1,\dots,X_k$ has entropy H_k; its increment is conditional entropy, lying between zero and $\log b$.

For positive powers P_i, spectral flatness is $(\prod P_i)^{1/n}/((\sum P_i)/n)$. For phases θ_j, circular resultant is $|\sum e^{i\theta_j}|/n$.

Theorem and hypotheses

A lossless fixed-length binary code for N objects needs b≥ceil(log_2N). No injective binary encoder can strictly shorten every binary string of lengths at most b. Prefix entropy obeys H_{k+1}−H_k=H(X_{k+1}|X_1,...,X_k), independently of a metric contraction placed on the symbolic space.

Proof

Injectivity follows by applying D to a collision E(x)=E(x′). A b-bit codomain contains only 2^b addresses, so N≤2^b. For universal shortening, the number of strings of lengths at most b is 2^{b+1}−1, whereas all shorter outputs through length b−1 number only 2^b−1; the pigeonhole principle rules out injectivity. The entropy chain rule follows by factoring p(x_1,...,x_{k+1}) into p(x_1,...,x_k)p(x_{k+1}|x_1,...,x_k) and expanding the logarithm in the expectation. Conditional entropy is nonnegative and at most log b. No property of an externally imposed ultrametric enters this algebra, so a metric contraction alone cannot prove an entropy decrease.

Worked example

Suppose five feasible objects are indexed by three bits, and examine two phase populations: all phases zero, or half zero and half π. Compare flat and nonflat power arrays.

Three bits suffice because ceil(log_25)=3. Drawing a uniform three-bit address and rejecting 5,6,7 yields each valid object with probability 1/5. Taking the address modulo five instead gives probabilities (2,2,2,1,1)/8. The all-zero phase population has resultant one; the two-opposite-cluster population has resultant zero despite being sharply concentrated in two directions. Powers (1,1,1,1) have flatness one, while (1,1,1,9) have flatness √3/3<1. Neither flatness nor resultant defines a false-acceptance exponent without an observation and null model.

The useful output is a hierarchy of observables, each with a declared invariance and a known failure mode. AUROC preserves ranking but discards calibration. Spectral power discards phase. A finite entropy calculation records empirical symbol balance but cannot establish decimal normality. A reversible code preserves object identity but does not imply shorter descriptions for every object. These distinctions turn diagnostics into well-typed tools.

Continuity across the source years

2024 — Dated topic continuity is evidenced by A linear chirp has quadratic phase, with a precise sampling ambiguity. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-03-033

2025 — Dated topic continuity is evidenced by Circular concentration measures evidence, not a motivic sector; CRT and finite entropy do not prove decimal normality. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-11-015, 2025:Y25-08-025, 2025:Y25-01-025, 2025:Y25-02-005

2026 — Dated topic continuity is evidenced by Lossless reconstruction forces injectivity and gives a counting lower bound on compression; Prefix entropy grows by conditional information, not by metric contraction alone. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-06-028, 2026:Y26-02-034, 2026:Y26-02-023, 2026:Y26-02-024, 2026:Y26-10-016, 2026:Y26-04-022

Connections

D14 — Exact encoding and quotienting. An address bijection recovers every source object; a quotient only recovers its equivalence class. Completion counts support finite exact indexing, not a universal compression claim.

D07 — Phase and spectra. Fourier powers retain magnitudes while discarding relative phases. A spectral statistic needs its window, normalization and reference ensemble.

D15 — Observation sufficiency. A target factors through h exactly when it is constant on observation fibres. Finite message capacity and continuous-state error need a declared resolution model.

Capacity and interactive scope

Lab ids: ['L02']

Inputs: L02 integer budget 0–27 and requested feasible rank 0–63 for six fixed weighted items.

Outputs: Feasible count, optimum masks and the item subset at the clamped rank. The current UI does not draw random addresses, encode arbitrary objects, calculate entropy, or analyze an uploaded spectrum.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: A related finite count-and-rank model is implemented; information capacity, random sampling and diagnostic calculations remain recipes.

Limits

A real number contains arbitrarily many formal digits but a finite-resolution device distinguishes only finitely many separated values. Strict spacing conventions matter at endpoints. Differential entropy changes under coordinates, while message count does not. The examples supply no empirical hardware capacity, compression benchmark or false-acceptance rate. Those require raw data and operational noise models.

Next questions

For a supplied finite observation map, enumerate fibres and compute which target functions factor through it.

Design a sufficient diagnostic panel for a specified finite family, minimizing measurement cost while preserving target distinctions.

Source anchors: 2026:Y26-06-028, 2026:Y26-02-034, 2026:Y26-02-023, 2026:Y26-02-024, 2025:Y25-11-015, 2025:Y25-08-025, 2025:Y25-01-025, 2024:Y24-03-033

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NIST DLMF modified Bessel functions — https://dlmf.nist.gov/10.25
NIST Guide for the Use of the International System of Units — https://www.nist.gov/pml/special-publication-811
Georgia Tech DSP First, AM/FM and chirp signals — https://dspfirst.gatech.edu/chapters/DSP1st2eLabs/AMFMSignalsLab.pdf
UCLA IPAM wavelet tutorial on chirps — https://helper.ipam.ucla.edu/publications/mpstut/mpstut_13975.pdf
Bryant: graph-based Boolean-function algorithms — https://doi.org/10.1184/r1/6605990

D12 · Optimization and decision theory

Organizing question

Which objective, feasible set and certificate turn a proposed choice into a justified global decision?

Optimization unifies much of the archive only after its inputs are typed. An objective assigns preference, a feasible set defines admissible alternatives, and a certificate supplies a bound valid over all of them. Algorithms search for candidates but do not replace these definitions. This domain is arranged from global convex certificates through constructive updates and finite search to robust decisions under shared uncertainty. It connects the practical question of choosing an action with the mathematical question of which uncertainties can change that action.

Convex duality supplies the cleanest starting point. A feasible primal point and a feasible dual bound can prove global optimality when their values agree. The constrained covariance quadratic makes nullspaces central: a feasible zero-energy null vector changes the answer completely, while a budget orthogonal to the nullspace leads to a pseudoinverse formula. Fenchel conjugates expose the effect of domains and nonsmooth penalties. An SDP zero-gap pair provides a finite exact certificate even when an iterative solver's convergence is irrelevant. In contrast, a stationary cubic equation or an indefinite quadratic requires a separate global analysis.

Proximal and projected methods turn an objective into a reproducible update. Soft thresholding, radial shrinkage and ridge scaling differ because their regularizers differ. The projection variational inequality and smooth descent lemma give a direct finite-step bound. A resolvent step damps curvature according to its eigenvalues and needs a positive quadratic model. Graph heat flow makes the distinction between continuous dissipation and discrete stability explicit. Metric projections, subspace restrictions and Wasserstein pushforwards add useful geometry while preserving their own feasibility and contraction hypotheses. These ideas explain why clipping a gradient, projecting a state and constraining a trajectory are different operations.

Objective design and finite search then expose the combinatorial side. The total-scatter identity can collapse seemingly independent clustering weights into the same objective. Capacity constraints and weighted preferences require different formulations. Pareto nondominance contains legitimate alternatives that no positive linear selector can choose. Exact QUBO enumeration and min-plus transfer provide finite certificates of a chosen problem. Frontier and equivalence-class compression are valid when they preserve complete continuation behavior, not merely a current best answer. Search error, physical-model error and measurement error consequently remain separate parts of an implementation claim.

The final family asks for the cheapest information that settles a decision. When two scores share nuisance uncertainty, subtract them at the same state before optimizing. Their joint difference can have a strict positive lower bound even when separately computed intervals overlap widely. Measurements restrict the compatible-state set, and a finite experiment tree can stop when each remaining fibre certifies one action. The implemented L08 model makes this cancellation visible without claiming a general optimizer or empirical forecast. This endpoint joins the other domains: exact counting can enumerate alternatives, observability can expose decisive fibres, covariance can describe noise, and duality can certify a robust margin. The result is an organized program of mathematical choices whose guarantees state their actual scope.

Core objects

Convex objectives and conjugates; affine constraints and PSD cones; primal-dual gaps; proximal maps and variational inequalities; finite Pareto sets; frontier signatures; common uncertainty sets and robust margins.

Open frontier

Construct exact positivity and dual certificates for nonlinear decision differences, then optimize finite measurement trees whose outcome fibres preserve those certificates at minimum guaranteed cost.

D12-F01 · Convex duality, singular quadratic programs and global certificates

What finite certificate proves global optimality, including null directions and boundary cases?

The optimization strand is strongest when an answer arrives with a lower bound and a feasible point attaining it. The source sequence links constrained covariance minimization, Fenchel conjugates, polyhedral actions and semidefinite primal-dual gaps. These are several forms of one principle: weak duality gives an inequality valid over the entire feasible set, and a matching construction turns it into a global theorem. The singular cases deserve equal status with the invertible ones. A pseudoinverse is meaningful only after the nullspace has been checked against the constraint; stationarity alone cannot provide the missing boundedness or convexity.

Definitions

For convex f, its conjugate is f*(p)=sup_x(p·x−f(x)); the domain of x is part of f and changes the conjugate.

A symmetric covariance matrix $\Sigma \geq 0$ defines the objective $w^{T}\Sigma w$. The affine budget is $c^{T}w=1$ with $c \neq 0$.

The Moore–Penrose inverse Σ^{+} inverts positive eigenspaces and vanishes on the nullspace.

An SDP primal feasible $X \geq 0$ and dual slack $S=C-\Sigma_i y_i A_i \geq 0$ satisfy constraints $\mathrm{Tr}(A_i X)=b_i$.

Theorem and hypotheses

If some $z \in \ker\Sigma$ has $c^{T}z \neq 0$, the constrained quadratic minimum is zero. Otherwise $c \in \mathrm{range}\Sigma$ and, with $q=c^{T}\Sigma^{+}c>0$, all minimizers are $\Sigma^{+}c/q+z$ for $z \in \ker\Sigma$, with minimum $1/q$. For every SDP feasible pair, the objective gap is $\mathrm{Tr}(SX) \geq 0$; a zero gap certifies global optimality without requiring a separate strong-duality theorem.

Proof

In the first quadratic case scale z by $1/(c^{T}z)$; feasibility and zero energy follow, while nonnegativity proves minimality. Otherwise decompose w into positive spectral and null components. The constraint sees only the positive part. Cauchy–Schwarz applied to $\Sigma^{1/2}w$ and $\Sigma^{+1/2}c$ gives $1=(c^{T}w)^{2} \leq (w^{T}\Sigma w)(c^{T}\Sigma^{+}c)$, so the objective is at least $1/q$. Equality gives the displayed positive component, with arbitrary invisible null addition. For the SDP, subtract the objectives, insert the primal equalities, and obtain $\mathrm{Tr}(SX)$. This is nonnegative because $\mathrm{Tr}(SX)=\mathrm{Tr}(X^{1/2}SX^{1/2})$ and the enclosed matrix is PSD. A feasible zero-gap pair therefore sandwiches the optimum exactly.

Worked example

Minimize $w_1^{2}+4w_2^{2}$ subject to $w_1+w_2=1$. Compare $\Sigma=\mathrm{diag}(0,4)$, then $\Sigma=11^{T}$. A separate SDP minimizes $\mathrm{Tr}(\mathrm{diag}(1,2)X)$ subject to $\mathrm{Tr}\,X=1$ and $X \geq 0$.

For the positive definite quadratic, q=1+1/4=5/4 and w*=(4/5,1/5), giving value 4/5. For diag(0,4), the null vector e_1 satisfies the budget and achieves zero. For 11^T, every feasible point has value one; the nullspace is budget-invisible. In the SDP, X*=diag(1,0), y=1 and S=diag(0,1) are feasible and give Tr(SX*)=0. Finally, the conjugate of kx^2 on all real x is $p^2/(4k)$, but restricting $x \geq 0$ changes it to $\max(p,0)^2/(4k)$.

The same computational discipline applies across these examples: inspect feasibility, preserve the actual domain, identify nullspaces, and produce a global inequality. A solver residual can guide a candidate, while an exact rational primal-dual pair can turn it into a reviewable certificate. Boundary constraints may alter the optimum and cannot be repaired by postprocessing an unconstrained answer.

Continuity across the source years

2023 — Dated topic continuity is evidenced by A quadratic program with singular cases included; Convex conjugates with their actual domains. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M062, 2023:M093

2024 — Dated topic continuity is evidenced by Finite mixtures and typed kernel pairings. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-05-001

2025 — Dated topic continuity is evidenced by Polyhedral Fenchel duality and the drift-action distinction. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-12-009, 2025:Y25-08-007

2026 — Dated topic continuity is evidenced by An SDP primal-dual pair gives an exact certificate and a traceable gap; A danger-to-information quotient is finite only after its null directions are checked. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-05-012, 2026:Y26-08-007, 2026:Y26-09-035

Connections

D03 — PSD geometry. Nullspaces and ranges determine whether quadratic objectives see feasible directions. Indefinite forms require a different boundedness analysis.

D11 — Information-danger quotients. A generalized quotient is finite only when the information nullspace lies in the danger nullspace. The finite quotient is a largest generalized eigenvalue on the supported subspace.

D07 — Spectral convexity. Ky Fan eigenvalue sums are convex and their smooth edge remainder is a weighted curvature integral. Eigenvalue crossings may require a nonsmooth formulation.

Capacity and interactive scope

Lab ids: []

Inputs: A declared finite probability table, covariance model, convex objective, or common uncertainty set, as specified in the worked example. Parameters without a sampling or error model are treated as assumptions.

Outputs: Exact finite values and rational certificates when available; otherwise numerical values explicitly restricted to the stated worked model.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Mathematical recipe with executable scoped checks; no general interactive estimator or solver is implemented.

Limits

KKT equations are sufficient only under the appropriate convexity and regularity conditions; a cubic stationary system may contain saddles or fail to have a global maximizer. Zero-gap feasible certificates avoid an appeal to strong duality but do require exact feasibility. Floating-point near-feasibility needs explicit error bounds before it becomes a certificate. A convex relaxation need not attain the original discrete optimum.

Next questions

Convert an approximate rational SDP pair into certified feasible points with a bound on the resulting gap.

Classify how an additional nonnegativity or box constraint changes the singular quadratic minimizer set.

Source anchors: 2023:M062, 2023:M093, 2025:Y25-12-009, 2026:Y26-05-012, 2026:Y26-08-007, 2026:Y26-09-035, 2024:Y24-05-001

Boyd and Vandenberghe Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/
Rockafellar, Convex Analysis — https://doi.org/10.1515/9781400873173
Ky Fan On a theorem of Weyl concerning eigenvalues — https://doi.org/10.1073/pnas.35.11.652
NIST DLMF integral operators and numerical quadrature background — https://dlmf.nist.gov/3.5

D12-F02 · Proximal maps, projected descent and stable optimization dynamics

Which update respects the feasible geometry and comes with a descent or contraction theorem?

The source development repeatedly distinguishes an intended variational objective from an informal update rule. Soft thresholding, radial shrinkage, ridge regularization, projected gradient, Mahalanobis steps, resolvents and Wasserstein proximal maps all become coherent when the update is defined as a minimizer. This family uses that minimization property to derive an inequality, then studies what remains invisible in a projected or singular geometry. Continuous dissipation and discrete stability are related but separate claims. A small gradient, a clipped direction or an Armijo line search is not a substitute for the exact hypotheses needed by the chosen method. Exact proximal examples also show what the objective preserves: radial shrinkage keeps direction, soft thresholding can change support, and a subspace projector can permanently preserve invisible components.

Definitions

For a proper convex g, prox_{ηg}(z)=argmin_x[g(x)+‖x−z‖²/(2η)] where the minimizer exists; the quadratic supplies strict convexity.

Projection Π_C is the proximal map of the indicator of a nonempty closed convex set C.

An M-weighted step uses ‖d‖_M²=dᵀMd with M>0; the metric is part of the algorithm.

For differentiable f with L-Lipschitz gradient, the descent lemma is f(y)≤f(x)+∇f(x)·(y−x)+(L/2)‖y−x‖².

Theorem and hypotheses

For x∈C closed convex and x+=Π_C(x−η∇f(x)), f(x+)≤f(x)−(1/η−L/2)‖x+−x‖². Thus 0<η<2/L gives strict descent unless the step vanishes. For a quadratic local model with Hessian H, the resolvent step δ=−η(I+ηH)^{-1}g is the unique minimizer when I+ηH>0.

Proof

The projection variational inequality gives $(x-\eta\nabla f(x)-x_+)\cdot(x-x_+)\le 0$. Rearranging yields $\nabla f(x)\cdot(x_+-x)\le -\|x_+-x\|^2/\eta$. Substitute this into the descent lemma to obtain the bound. Convexity is used for the projection inequality; smoothness controls the objective. The resolvent follows by differentiating $q(\delta)=g\cdot\delta+(1/2)\delta^{T}H\delta+\|\delta\|^2/(2\eta)$. Its Hessian $H+I/\eta$ is positive definite under the stated condition, so the stationary point is the global unique minimizer. In an H eigenbasis each component is $-\eta g_i/(1+\eta\lambda_i)$, making curvature damping and a possible negative-curvature obstruction explicit.

Worked example

For $z=(3,4)$, compare the minimizers of $(1/2)\|x-z\|^2+\lambda\|x\|_1$, with $\lambda\|x\|_2$, and with $(\lambda/2)\|x\|^2$ at $\lambda=1$. Also inspect a four-cycle quadratic energy.

The three updates are respectively $(2,3)$, $(12/5,16/5)$, and $(3/2,2)$. Componentwise sparsity, radial shrinkage and uniform rescaling are different objectives. For a graph Laplacian with spectrum $(0,2,2,4)$, a gradient step has factors $1,1-2\eta,1-2\eta,1-4\eta$. At $\eta=.4$ every nonconstant mode contracts; at $\eta=.6$ the top factor is -1.4 and grows. For $\eta>0$ and convex potential $a|x|^2/2$ with $a\ge 0$, the quadratic Wasserstein proximal map is the pushforward $x\mapsto x/(1+\eta a)$. W_2 distances scale exactly by $q=1/(1+\eta a)$; this is strict contraction when $a>0$ and the identity when $a=0$.

The source sequence now has a common constructive workflow: state the objective and metric, derive its minimizing map, then prove feasibility, descent and any stronger convergence property separately. A projected update can vanish because the gradient points into an excluded subspace, even when the original objective is not stationary. That may be a desired constraint, but its interpretation must be declared.

Continuity across the source years

2023 — Dated topic continuity is evidenced by Graph heat flow and the stability of a discrete gradient step. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M080

2024 — Dated topic continuity is evidenced by Three regularizers produce three different shrinkage rules. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-04-003, 2024:Y24-04-004

2025 — Dated topic continuity is evidenced by Projecting the updated state has a descent proof; clipping the gradient is a different operation; Mahalanobis proximal steps and what a tube projector preserves. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2025:Y25-10-021, 2025:Y25-12-015, 2025:Y25-08-020, 2025:Y25-08-021, 2025:Y25-06-017

2026 — Dated topic continuity is evidenced by Quasi-static elimination is a resolvent step with a stability condition; A quadratic Wasserstein proximal step has an exact pushforward solution. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-02-026, 2026:Y26-02-035, 2026:Y26-04-023, 2026:Y26-02-017, 2026:Y26-01-012, 2026:Y26-05-004

Connections

D09 — Stable discretization. Spectral multipliers $1-\eta\lambda$ determine whether discrete descent follows continuous dissipation. Positivity preservation can impose a stricter step bound.

D06 — Metric projection. $M>0$ gives unique convex projection and its variational inequality. A nonconvex admissible set can have several nearest branches.

D15 — Control geometry. A projected gradient can miss directions that control a target or objective. Descent on the selected subspace is not full-state identifiability.

Capacity and interactive scope

Lab ids: []

Inputs: A declared finite probability table, covariance model, convex objective, or common uncertainty set, as specified in the worked example. Parameters without a sampling or error model are treated as assumptions.

Outputs: Exact finite values and rational certificates when available; otherwise numerical values explicitly restricted to the stated worked model.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: Mathematical recipe with executable scoped proximal and gradient-step checks; no general interactive optimizer, proximal map or Wasserstein solver is implemented.

Limits

The projection theorem does not cover arbitrary nonconvex sets or projecting the gradient instead of the updated state. BFGS positive definiteness needs $y^{T}s>0$; Armijo decrease alone does not ensure it. Fisher, Gauss–Newton and Hessian substitutions need their own approximation analysis. These worked maps are not an implementation of a general optimizer or Wasserstein solver.

Next questions

For a supplied metric M and nonsmooth separable regularizer, derive the proximal map or an exact dual certificate.

Choose the largest stable step compatible with objective descent, coordinate positivity and a specified approximation error.

Source anchors: 2024:Y24-04-003, 2025:Y25-10-021, 2025:Y25-12-015, 2026:Y26-02-026, 2026:Y26-02-035, 2026:Y26-04-023, 2023:M080, 2026:Y26-02-017

Parikh and Boyd Proximal Algorithms — https://web.stanford.edu/~boyd/papers/prox_algs.html
Boyd and Vandenberghe Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/
R. Tyrrell Rockafellar, Monotone Operators and the Proximal Point Algorithm — https://doi.org/10.1137/0314056
Richard Jordan, David Kinderlehrer and Felix Otto, The Variational Formulation of the Fokker–Planck Equation — https://doi.org/10.1137/S0036141096303359
Boyd and Vandenberghe, Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf
James Martens, New Insights and Perspectives on the Natural Gradient Method — https://www.jmlr.org/papers/v21/17-678.html
Nocedal, Updating quasi-Newton matrices with limited storage — https://courses.engr.illinois.edu/ece544na/fa2014/nocedal80.pdf

D12-F03 · Objective design, discrete selection and certified finite search

Which feasible set and objective make a proposed improvement mathematically meaningful?

Many optimization questions initially hide their most consequential choice: what is being optimized over which alternatives. Clustering weights can duplicate one scatter identity; capacity quotas are different from score weights; Pareto points can resist every linear selector; a QUBO is meaningful only after the binary variables and constraints are fixed. The later equivalence-class and separator constructions improve search by preserving the cost of every legal continuation. This family unifies these questions around a typed feasible set and a certified objective, while distinguishing an abstract optimum from a measured implementation cost.

Definitions

A feasible optimization problem is (X,J), with optimum inf_{x∈X}J(x); changing X or J changes the problem.

A vector objective is Pareto nondominated when no feasible competitor improves all coordinates and at least one strictly.

A QUBO minimizes a quadratic polynomial on {0,1}^n; a graph-cut objective is one exact finite instance.

A frontier state in dynamic programming is sufficient only when its signature retains all information connecting processed choices to future feasibility and cost.

Theorem and hypotheses

For fixed data, total scatter T equals within-cluster scatter W plus between-cluster scatter B. Therefore maximizing aB−bW with a,b≥0 and a+b>0 is equivalent to minimizing W over the same partitions. Separately, if a frontier signature determines every future interaction, retaining only the cheapest prefix per signature preserves the global optimum.

Proof

For the scatter identity, insert x_i−μ=(x_i−μ_{c(i)})+(μ_{c(i)}−μ). Cross terms vanish within each cluster because Σ_{i∈c}(x_i−μ_c)=0. Hence T=W+B and aB−bW=aT−(a+b)W, differing from −W only by a positive scale and constant. For frontier compression, take two partial solutions with the same sufficient signature. Every legal completion contributes the same added cost and has the same feasibility for both; replacing the dearer prefix with the cheaper one can only improve the total. Induct over processed layers. The proof requires equality of continuation behavior, not merely equality of the currently best continuation or the present objective value.

Worked example

Use the triangle maximum-cut problem and a two-state repeated-layer transfer. Also consider minimization of two costs at feasible points A=(0,2), B=(1,3/2), C=(2,0).

The triangle has eight binary labelings: the two constant ones cut zero edges and all six nonconstant ones cut two, proving optimum two by enumeration. In min-plus algebra, M=[[∞,2],[3,∞]] has M⊗M=5I_minplus. Thus with initial costs (0,∞), the minimum terminal sequence is 0,2,5,7,10,… and d_{L+2}=d_L+5. In the Pareto fixture all three points are nondominated. For positive weights α,β, choosing B over A requires α≤β/2, while choosing B over C requires α≥3β/2, impossible. The middle Pareto point cannot be reached by a positive linear selector.

The scatter identity shows how a proposed extra criterion can collapse algebraically; the Pareto example shows how a single scalarization can omit legitimate tradeoffs; the transfer identity shows how finite search can be compressed without losing its optimum. Together they support a stronger design practice: simplify the objective symbolically, preserve deployment constraints, and certify both the search reduction and the final feasible candidate.

Continuity across the source years

2024 — Dated topic continuity is evidenced by Specify the clustering objective before adding weights; Spatial coherence gives an exact cut-versus-fit threshold. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2024:Y24-04-001, 2024:Y24-04-002

2025 — Dated topic continuity is evidenced by Weighted scores and capacity quotas need different constraints; A QUBO and a one-edge QAOA instance can be solved exactly. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved

discovery to that exchange. Sources: 2025:Y25-02-003, 2025:Y25-03-004, 2025:Y25-10-019

2026 — Dated topic continuity is evidenced by Semantic fibers enlarge the feasible optimum only when deployment constraints are preserved; A frontier state supports exact min-plus transfer when it separates past from future. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-08-801, 2026:Y26-08-805, 2026:Y26-08-803, 2026:Y26-08-806

Connections

D13 — Finite enumeration. Binary labels, graph cuts and frontier signatures give exact finite search spaces. Enumeration proves only the declared finite instance.

D14 — Continuation semantics. Safe merging is a quotient preserving all relevant future costs. A current optimum is insufficient to define a stable quotient.

D11 — Selection and measurement. Abstract optimum gap, search error, model error and measurement error are separate quantities. No runtime or generalization claim follows from a smaller state count alone.

Capacity and interactive scope

Lab ids: ['L02', 'L03']

Inputs: L02 varies budget and requested rank for six fixed items. L03 varies total nonnegative integer height 0–6 and q in [0,1] for two fixed seven-vertex graphs.

Outputs: L02 returns exhaustive feasible counts, the exact finite optimum and its masks, and a ranked subset. L03 returns exact integer energy censuses and a floating evaluation of their difference. Pareto optimization, QUBO, min-plus transfer and automatic frontier compression remain recipes.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: The fixed knapsack and graph-census labs are implemented; no general search, dynamic-programming or Pareto solver is exposed.

Limits

The exact examples do not replay absent historical placement tables, graph instances or runtime benchmarks. A sufficient separator may need auxiliary connectivity state, and algebraic compression preserves only its stated future predicate. A quantum circuit formula for one edge is a finite calculation, not an asymptotic algorithmic advantage. Nonlinear objectives or changed deployment constraints require a new feasible-set argument.

Next questions

For a supplied search system, prove that its frontier signature separates past and future, or exhibit a minimal counterexample.

Enumerate a finite Pareto set and compare which members can be selected by linear, constrained and nonlinear preferences.

Source anchors: 2024:Y24-04-001, 2024:Y24-04-002, 2025:Y25-02-003, 2025:Y25-03-004, 2025:Y25-10-019, 2026:Y26-08-801, 2026:Y26-08-805, 2026:Y26-08-803

Boyd and Vandenberghe Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/

Farhi–Goldstone–Gutmann, A quantum approximate optimization algorithm — https://arxiv.org/abs/1411.4028

Bodlaender, Cygan, Kratsch and Nederlof, Solving weighted and counting variants of connectivity problems parameterized by treewidth deterministically in single exponential time — https://arxiv.org/abs/1211.1505

D12-F04 · Shared uncertainty, robust margins and decision-sufficient measurement

What must be measured to certify the decision, rather than reconstruct every uncertain score?

The final decision strand ties the archive together especially well. Earlier finite thresholds, singular covariance optimization, normalized residual margins and later target-observation questions lead to a precise endpoint: compare the competing actions under the same compatible state. Independent score intervals can erase cancellation and make an easy decision look unresolved. A decision difference may depend on only a small part of a high-dimensional uncertainty set. The best experiment can therefore be one that resolves the sign of that difference, even if it leaves most of the system unknown. This is a mathematical design principle with deterministic quantifiers, distinct from a posterior preference or an empirical forecast.

Definitions

Let X be a nonempty uncertainty set and J_A,J_B:X→R be scores, with larger preferred; δ(x)=J_A(x)−J_B(x).

A strict robust certificate is a proved m>0 with δ(x)≥m for every x∈X; the infimum need not be attained.

A bounded-error measurement h(x)≈z restricts compatibility to X_z={x∈X:|h(x)−z|≤ε}.

A decision-sufficient finite test makes each nonempty outcome fibre certify one action. Its objective may be worst-case cost rather than expected estimation error.

Theorem and hypotheses

Joint comparison satisfies inf_X(J_A−J_B)≥inf_XJ_A−sup_XJ_B, often strictly. Restricting to a measurement fibre cannot decrease the lower margin for A. In the shared model J_A=c+d, J_B=c, c∈[0,100], d∈[7/4,9/4], the exact robust A-margin is 7/4 although subtraction of separate score intervals yields lower bound −393/4.

Proof

For every x, J_A(x)≥inf J_A and J_B(x)≤sup J_B, so subtraction and then infimization prove the inequality. Separate extrema may occur at incompatible x, explaining strictness. If X_z⊆X, every lower bound valid on X remains valid on X_z. In the shared example δ=d because the common c cancels before minimization. The independent ranges are J_A∈[7/4,409/4] and J_B∈[0,100], whose separate lower difference is 7/4−100=−393/4. The distinction is about preserving the same state in both scores. In a finite test tree, a recurrence V(S)=min_t[c_t+max_oV(S∩t^{-1}(o))], with V=0 on decision-certified states, computes the cheapest guaranteed procedure when the admissible tests and stopping certificates are fixed and a terminating strategy exists.

Worked example

The L08 worked model uses a common nuisance score and an interval for the decision difference. A second fixture measures d with absolute error at most 1/10.

For c∈[0,100] and d∈[7/4,9/4], every compatible state prefers A with margin at least 1.75, regardless of nuisance width. Changing c's range changes the separate-score bound but leaves the joint margin unchanged. A measurement z=1/4 implies d≥z−1/10=3/20>0, certifying A. A result z=1/20 yields d∈[−1/20,3/20], which certifies neither direction. With affine δ=a$^{\top}$x+b on a box, its exact lower bound is b+Σ_j min(a_jl_j,a_ju_j); shared-variable simplification should precede this bound.

This creates a bridge from information to action: the required observation need distinguish states only when they demand different actions. Reconstructing nuisance coordinates can be unnecessary.

Conversely a confident score estimate without a defensible compatibility set cannot become a robust theorem. L08 exposes the common-variable cancellation directly; arbitrary semialgebraic positivity and automatic measurement-tree synthesis remain specified extensions. L08 centers the nuisance interval as $[-C,C]$; $C=50$ gives the same width and independent lower bound as $[0,100]$, with both scores translated by -50. It does not implement the bounded-error measurement or test-tree extensions.

Continuity across the source years

2023 — Dated topic continuity is evidenced by Decision thresholds and genuine strategic equilibria; Infinite utilities through controlled finite limits. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2023:M055, 2023:M056

2026 — Dated topic continuity is evidenced by Shared uncertainty should be optimized jointly in a decision difference; A normalized maximum residual gives a robust acceptance margin. The theorem and worked synthesis here are newly authored; this date does not attribute the present proof or an unobserved discovery to that exchange. Sources: 2026:Y26-10-023, 2026:Y26-01-033, 2026:Y26-09-022, 2026:Y26-08-007, 2026:Y26-08-801

Connections

D11 — Information sufficient for action. Observation fibres need preserve the sign of δ, a weaker requirement than reconstructing the whole state. A Bayes risk uses a distribution; robust certification quantifies over X.

D15 — Target observability. A decision target can be observable even when the physical state is not. The admissible state set and measurement error must be stated.

D14 — Optimal experiment trees. Finite Bellman recursion compares test cost against the resulting compatible-state fibres. This requires declared finite tests and termination; no general test planner is implemented.

Capacity and interactive scope

Lab ids: ['L08']

Inputs: Current L08 UI: $C\in[0,30]$ with integer steps, $lo,hi\in[-5,5]$ with decimal step .1; reversed endpoints are ordered. The nuisance is $c\in[-C,C]$. The source text separately uses $c\in[0,100]$, whose common translation leaves the decision difference unchanged.

Outputs: Decimal score intervals $A=[-C+lo,C+hi]$, $B=[-C,C]$, interval overlap and strict robust preference flags from $lo>0$ or $hi<0$. The algebraic formulas are exact; their displayed evaluation uses floating arithmetic. No measurement-error fibre, positivity solver or test tree is implemented.

Validation: The domain verify.py independently checks representative finite identities and numerical formulas. No historical program, empirical dataset, or out-of-sample claim is replayed.

Status: The specified additive shared-variable comparison is implemented in L08. Measurement and decision-tree extensions remain mathematical recipes.

Limits

An uncertainty set inferred from data needs its own coverage justification. A posterior credible set and a uniform deterministic error bound are different inputs. Infinite utilities require a controlled limiting model; even a common limiting prior may produce different preferences if utility scales simultaneously. A strict sign certificate differs from a numerical optimizer returning a small positive value. These examples concern abstract scores and carry no empirical or financial-performance claim.

Next questions

Produce an exact positivity certificate for a supplied rational or polynomial δ on a compact semialgebraic set.

Given a finite family of bounded-error tests with costs, solve and certify the cheapest terminating decision tree.

Source anchors: 2026:Y26-10-023, 2026:Y26-01-033, 2023:M055, 2023:M056, 2026:Y26-09-022, 2026:Y26-08-007, 2026:Y26-08-801

Boyd and Vandenberghe: convex optimization — https://web.stanford.edu/~boyd/cvxbook/
John Nash Equilibrium points in n-person games — https://pmc.ncbi.nlm.nih.gov/articles/PMC1063129/
Williams Probability with Martingales, conditional expectation — https://doi.org/10.1017/CBO9780511813658

D13 · Combinatorics and graph mathematics

Organizing question

What exact information must be retained to count, distinguish or compose finite structures?

This domain supplies exact finite objects on which the atlas’s more abstract ideas can be tested. Its organizing operation is a disjoint decomposition: remove the largest tableau entry, choose the next binary project coordinate, separate a graph along a boundary or decompose a path flow. Such a decomposition is useful only if its pieces represent the declared original objects exactly once. Generating products and completion recurrences make those multiplicities explicit. The resulting counts then support a stronger interface: reversible addresses, exact sampling, conditional populations and a unique optimum proved by a singleton threshold census. This is why population definitions and skipped-coordinate factors belong in the mathematics rather than being deferred to implementation.

Graphs contribute a complementary lesson about observations. An adjacency matrix contains more than its eigenvalues, and a full heat operator contains more than its trace or diagonal. The 2023 rook–Shrikhande collision demonstrates an exact loss of structure. The 2026 seven-vertex pair refines the issue: the entire binary size-cut census agrees, while one integer-height coefficient separates the graphs. This is a model for constructive research throughout the report. First prove the collision at the weaker observation; then identify a richer observable and calculate a finite separating witness. It gives a defensible improvement without claiming that the stronger invariant solves every graph-isomorphism problem.

Incidence matrices link this graph thread to topology and optimization. Their rank yields cycle dimension, and spanning forests produce explicit integer cycle bases. The finite DAG path-difference theorem explains which conservation directions can be expressed as differences of admissible routes. Directed acyclicity does not remove undirected circulation: the diamond is the smallest useful reminder. At the optimization boundary, a matching witnesses a necessary number of vertex calls and a cover supplies a sufficient number. The two meet in a complete proof only after the permitted action is fixed. Unrecovered historical call rules and extremal point lists therefore remain concrete retrieval obligations.

The final family turns these principles into composition. Sum–product and min–plus elimination preserve different semiring values through distributivity. A compact separator stores exactly the interaction that future factors can observe. Erasing a physical lane or adding an auxiliary gate is safe only under a corresponding fibre theorem. Equal optimum values can coexist with unequal counting measures, as the 43,680 physical channel assignments and 256 bank tuples show. The interactive capacities follow directly: L02 exposes counts and addresses, L03 inspects a complete graph-invariant collision, and L09 computes boundaries and homology on small declared complexes. These are bounded, inspectable mathematical models. The general elimination, large Petersen optimization and unrecovered routing benchmarks remain recipes or source claims until their actual instances and complete certificates are supplied.

Core objects

Labeled finite populations

Generating functions and completion recurrences

Incidence and Laplacian matrices

Cycle and cut spaces

Factor scopes and separator states

Multiplicity-bearing quotient fibres

Open frontier

Recover original large-instance tables and proof dependencies; characterize the distinguishing strength of bounded-height graph observations; build minimal boundary quotients that preserve counts as well as values.

D13-F01 · Exact populations: products, recurrences and reversible addresses

How can a finite combinatorial population be counted, optimized and indexed without confusing objects with their statistics?

The recurring constructive idea is to make an object contribute exactly once to a finite sum. A partition contributes a monomial through its part multiplicities; a tableau contributes through the position of its largest entry; a project selection contributes through one factor for every labeled project. These constructions give independent descriptions of the same population rather than a fit to its reported size. They also explain why equal weights do not merge labeled selections and why a terminal in a reduced decision diagram can represent many assignments. Once completion counts are available, enumeration becomes an address system: the same integers support counting, conditional queries, exact uniform sampling, and an optimum proved by a threshold population of size one. The mathematical advance is the unification of these operations under a single disjoint decomposition. It does not require trusting an optimizer or assuming that a compact diagram has already retained every multiplicity.

Definitions

A completion count C(q) is the number of feasible objects extending a prefix q; it counts original objects, unless a quotient population has explicitly been chosen.

For labeled binary choices with nonnegative vector weights w_i and capacity b, C(i,b) counts feasible assignments of coordinates i,…,n−1.

The generating polynomial ∏_i(1+z^{w_i}t^{p_i}) records vector weight and integer profit with multiplicity; a coefficient counts choices, not distinct numerical sums.

A species a_1≥a_2≥…≥0 corresponds bijectively to differences d_j=a_j−a_{j+1}; its least representative is ∏_j P_j^{d_j}, where P_j is the jth primorial.

Theorem and hypotheses

For nonnegative weights, C(i,b)=C(i+1,b)+C(i+1,b−w_i), with C(i,b)=0 if a capacity coordinate is negative and C(n,b)=1 otherwise. These counts yield inverse bijections rank:Ω→{0,…,|Ω|−1} and unrank. A feasible object of profit v is the unique maximizer if the count of feasible objects with profit at least v equals one.

Proof

Split each population according to its next bit. The two pieces are disjoint and correspond exactly to the smaller residual capacities, proving the recurrence by induction. If the entire suffix fits, every remaining bit is free and contributes 2^{n−i}; an omitted block of g coordinates therefore contributes a factor 2^g. In lexicographic order the zero branch occupies an initial address interval of length C(q0); the one branch occupies the next interval. Unranking compares an address with C(q0), subtracting that count in the one branch; ranking performs the inverse additions. Induction down the tree proves a bijection, including empty branches. A uniform address consequently induces a uniform object. Finally a singleton threshold population includes the supplied feasible witness, excludes every higher-profit object and excludes every competing equal-profit object. This proves optimum and uniqueness together.

Worked example

Use three distinct projects with weights (2,3,4), capacity 5 and profits (3,4,5). Also compare a small partition problem.

The feasible words in zero-before-one order are 000,001,010,100,110. Their profits are 0,5,4,3,7, so addresses 0,…,4 enumerate the complete population and 110 is uniquely optimal. The prefix 1 has two completions. Three unbiased random bits can sample an address by rejecting 5,6,7; direct reduction modulo 5 instead gives probabilities (2,2,2,1,1)/8. Independently, for species with least representative at most 12, the inequality 2^{d_1}6^{d_2}≤12 gives representatives 1,2,4,6,8,12. A partition dynamic program gives p(5)=7, while corner removal gives five standard tableaux of shape (3,2). These different integers count different declared objects.

An exact count is useful infrastructure: it supports reversible inspection, unbiased sampling and optimality witnesses while preserving which population was actually counted.

Continuity across the source years

2023 — Tableaux and set partitions establish the need to declare the population and distinguish a counting statistic from an energy. Sources: 2023:M035, 2023:M075

2024 — Formal products become exact finite coefficient recurrences; the classical pentagonal identity is the stated input, not something established by finite tests. Sources: 2024:Y24-07-008

2026 — Fibonacci decompositions, primorial species and project populations supply several realizations of disjoint recursion. The October treatments add threshold certificates and reversible addresses. This retrospective organization does not date the present proof to earlier conversations. Sources: 2026:Y26-03-027, 2026:Y26-05-021, 2026:Y26-10-015, 2026:Y26-10-016

Connections

D01 — Prime factorizations encode partition species. a_i=Σ_{j≥i}d_j changes ∏p_i^{a_i} into ∏P_j^{d_j}. This is an exact bijection; its sharp asymptotic prefactor requires further analysis.

D12 — Optimization is a threshold census. Count {x feasible:profit(x)≥v} and check a witness at v. A particular large historical coefficient table must still be supplied.

D14 — A completion oracle is an address interpreter. The identities C(q)=C(q0)+C(q1) certify rank/unrank semantics. Implementation correctness includes skipped-variable multiplicity and integer bounds.

Capacity and interactive scope

Lab ids: ['L02']

Inputs: Small labeled weights, capacities, profits and an address or bit word.

Outputs: L02: exact six-item population, optimum multiplicity and object at an integer address. General rank/unrank and completion queries are the checked recipe.

Validation: Exhaustive enumeration independently checks completion counts and both inverse identities on every small feasible word.

Status: Implemented small finite lab; general population and historical large instance remain separate.

Limits

The Petersen total 702,508,860,122,761, optimum 16,537 and uniqueness are historical source claims. The fifty input rows, full 9,190,151-node DAG and complete formal bridge were unavailable in the annual source; this family does not claim their replay. Exact uniformity assumes a uniform integer source. A deterministic seed supplies reproducibility alone.

Next questions

Which separator or variable order minimizes intermediate state while preserving the exact count?

Can the full historical Petersen table, DAG and proof dependencies be recovered and checked independently?

Source anchors: 2023:M035, 2024:Y24-07-008, 2026:Y26-05-021, 2026:Y26-10-015, 2026:Y26-10-016

NIST DLMF, unrestricted partitions — https://dlmf.nist.gov/27.14
Bryant, Graph-Based Algorithms for Boolean Function Manipulation — https://doi.org/10.1184/r1/6605990

D13-F03 · Incidence kernels, path differences and primal–dual graph witnesses

How do local incidence constraints produce global cycles, and when does a short witness prove an optimal graph action count?

Incidence is the common finite language for conservation, cycle spaces, graph gauge classes and many covering arguments. An oriented edge is sent to the difference of its endpoint vertices; cancellation at internal vertices explains why a path has only source and sink boundary. Subtracting two paths therefore produces a circulation. The converse requires hypotheses: in a finite source-to-sink directed acyclic graph with every edge covered by a path, these differences span the entire real circulation space. Acyclic direction does not erase undirected cycles; a diamond already has a one-dimensional incidence kernel. The complementary optimization theme is that an admissible construction establishes an upper bound, while a matching or other dual witness establishes a lower bound. A support size or observed action count has no optimality meaning until the permitted action is stated. This family makes that action model and its certificate visible.

Definitions

For an oriented graph with n vertices, m edges and c connected components, B is the vertex-by-edge matrix having -1 at a tail and $+1$ at a head.

The circulation space is ker B. Over any field its dimension is $m-n+c$; over the integers a spanning forest gives a fundamental cycle basis.

A vertex-call covers every edge incident to the chosen vertex. A matching consists of edges with disjoint endpoints, so any one vertex-call covers at most one matching edge.

For a graph regarded as a one-dimensional complex, $H_1=\ker B$ because there are no two-cells. Adding filled faces changes the quotient by their boundaries.

Theorem and hypotheses

The incidence rank is n−c, hence the circulation dimension is m−n+c. On a finite directed acyclic s-to-t graph with every edge on an s-to-t path, ker B is spanned over R by χ(p)−χ(p_0). In the vertex-call model on K_{a,b}, the exact minimum call count is min(a,b).

Proof

On each connected component, incidence columns sum to zero and hence rank is at most n_component−1. A spanning tree contributes independent columns: remove a leaf and inspect its unique incident coefficient, inductively forcing every coefficient to zero. Summing components proves the rank formula. Every path difference has zero boundary. For the converse sum covering path indicators to obtain a strictly positive feasible flow f. For z∈ker B, both f and f+εz are nonnegative for small enough ε. A finite DAG permits a decomposition of each into nonnegative s-to-t path flows: follow positive flow, use conservation internally, and subtract the least path-edge value. At least one edge is removed at each stage. Their source divergence is the same, so subtracting the decompositions gives a combination of path differences. Finally assume a≤b. An a-edge matching demands at least a vertex-calls, and all a left vertices cover every edge. Matching lower bound and cover upper bound meet.

Worked example

Use the diamond with edges (sa,at,sb,bt), then K_{3,3} and the call model K_{4,7}.

The diamond has four vertices and four edges, so ker B is one-dimensional, generated by (1,1,−1,−1), the difference of its two directed paths. It is a DAG even though this undirected cycle survives. For K_{3,3}, m−n+1=9−6+1=4; a spanning tree and each of its four non-tree edges produce an integral fundamental cycle. For K_{4,7}, 28 edges are covered by four left-vertex calls. A four-edge matching proves that three calls cannot suffice. If a call instead selected one edge, the answer would be 28, demonstrating why action semantics changes the optimization problem.

Global invariants become replayable through integer matrices, explicit cycles and matching witnesses; pictures alone do not establish homology or cost.

Continuity across the source years

2024 — Discrete differential identities are stated as actual incidence compositions and conservation laws. Sources: 2024:Y24-10-032

2025 — Edge-count constraints provide elementary but decisive consistency checks before a fitted graph label is interpreted. Sources: 2025:Y25-08-028

2026 — Explicit cycle bases, DAG path differences and matching witnesses supply constructive certificates for the refined graph questions. Sources: 2026:Y26-04-001, 2026:Y26-04-027, 2026:Y26-05-025, 2026:Y26-09-011

Connections

D05 — Graph cycles are a first homology calculation. H_1=ker B/im ∂_2; an unfilled graph has ∂_2=0. A drawing or a directed orientation does not determine whether faces are filled.

D12 — Optimal calls require matching lower bounds. Disjoint required edges constrain every admissible cover. The exact K_{a,b} theorem does not reconstruct the historical 81-call semantics.

D03 — Integral lattices refine rational kernels. A rational span records dimension but can miss a finite saturation index. A toric lattice certificate needs the actual integer matrix, not its rank alone.

Capacity and interactive scope

Lab ids: ['L09']

Inputs: Small oriented graph or simplicial complex and its declared filled faces.

Outputs: Boundary matrices, ranks, cycle representatives and homology dimensions.

Validation: Check $B\partial_2=0$ and exact ranks; the standalone verifier checks diamond circulation and the four-versus-seven call certificate.

Status: Small homology lab implemented; graph cover and integral lattice tasks are mathematical recipes.

Limits

Hypergraph ordinary matchings differ from induced matchings or tensor restrictions. The corner count 3,840 uses $V\times V$ with $V=F_2^4$ and does not establish an extremal $\beta_4=137$ claim. The historical incidence fibres and call rules were missing. Z_2 gauge count 2^{m-n+c} applies to graph edge labels under vertex gauge action; added constraints can change it.

Next questions

Which integer cycles, rather than only real cycle coordinates, are preserved by a proposed graph quotient?

Can the omitted historical call semantics be recovered to produce a matching or LP dual witness for its claimed optimum?

Source anchors: 2024:Y24-10-032, 2026:Y26-04-001, 2026:Y26-04-027, 2026:Y26-05-008, 2026:Y26-05-025, 2026:Y26-09-011

Hatcher, Algebraic Topology — https://pi.math.cornell.edu/~hatcher/AT/AT.pdf
Boyd and Vandenberghe, Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/

D13-F02 · Graph observations: spectra, cut censuses and stronger energy probes

Which graph features survive an observation, and what extra measurement separates a known collision?

The graph thread develops from a warning about incomplete spectral observations into an exact hierarchy of finite invariants. A graph can be encoded by its adjacency matrix, its Laplacian, a heat operator, a heat trace, diagonal heat signatures, binary cut statistics or integer-height energy coefficients. These are different observation maps. Equality at one level need not imply equality at a richer level, and a numerical agreement at several parameters does not prove equality of a polynomial invariant. The productive method is to construct a full exact collision, identify the discarded correlations, then design a stronger finite observation. Incidence identities provide a parallel constructive route: singular values of a point-line incidence matrix determine the spectrum of its bipartite Levi graph. The centered Green matrix adds a variational capacity, translating a graph operator into an exact energy required to force one coordinate while maintaining zero mean.

Definitions

For a simple graph G, $L=D-A$ and $E_G(h)=h^TLh=\Sigma_{uv\in E}(h_u-h_v)^2$.

The binary census records all pairs $(\Sigma h_i,E_G(h))$ for $h\in\{0,1\}^V$; on binary heights E_G is a cut boundary size.

The formal integer-height series $F_G(x,q)=\Sigma_{h\in Z_{\geq 0}^V}x^{\Sigma h_i}q^{E_G(h)}$ has finitely many terms at every fixed x-degree.

For $\lambda>0$ and $\Pi=I-11^{T}/n$, the centered Green matrix $K=\Pi(\lambda I+L)^{\{-1\}}\Pi$ is the inverse of $\lambda I+L$ on $1^{\perp}$.

Theorem and hypotheses

The explicit seven-vertex graphs G and H below have identical complete binary size-energy censuses but different integer-height series: $[x^{3}](F_G-F_H)=q^{3}(1-q)(1-q^{3})^{2}(1-q^{4})^{2}$. Separately, for any graph with $n\geq 2$, $\min\{v^{T}(\lambda I+L)v : 1^{T}v=0, v_i=a\}=a^{2}/K_{ii}$.

Proof

For the collision, enumerate every one of the 128 binary subsets and store its size and cut size as an integer histogram. For total height R enumerate all weak compositions of R into seven coordinates, recording energy. Exact subtraction at R=0,1,2 is zero; at R=3 its coefficient polynomial factors as stated. This finite computation is a proof certificate for these explicit graphs. For the variational theorem, $QK=\Pi$ on $1^{\perp}$. Thus $v_i=(Ke_i)^{T}Qv$ and Cauchy–Schwarz in the positive Q-inner product gives $a^{2}\leq K_{ii}\, v^{T}Qv$. Equality holds at $v=aKe_i/K_{ii}$; its mean is zero and its ith component is a. The inverse is positive on $1^{\perp}$, so $K_{ii}>0$. This proves both the sharp bound and its unique minimizer.

Worked example

Use $G=\{(0,3),(0,4),(0,6),(1,2),(1,3),(2,3),(2,4),(3,5)\}$ and $H=\{(0,1),(0,4),(0,6),(1,5),(1,6),(2,6),(3,6),(4,5)\}$. For a second model use the path P_3 and $\lambda=2$.

The census collision is exact over all binary assignments. At $q=1/2$, the first height separation equals $(1/8)(1/2)(7/8)^{2}(15/16)^{2}=11025/262144$. Support-three terms already cancel because they are binary; support-one terms cancel through the degree census, so the discrepancy comes from support two. For P_3, exact inversion gives $\operatorname{diag} K=(1/5,2/15,1/5)$. Forcing an endpoint to a under zero mean therefore costs $5a^{2}$, while forcing the center costs $15a^{2}/2$. As a separate incidence example, the Fano matrix satisfies $BB^{T}=2I+J$, giving Heawood adjacency eigenvalues ± 3 and $\pm\sqrt{2}$, with the latter each sixfold.

The same graph can support exact reconstruction obstructions, new distinguishing observations and sharp constrained energy calculations; each conclusion is tied to a particular observation map.

Continuity across the source years

2023 — Rook and Shrikhande graphs supply a complete heat-signature collision, while the full labeled heat operator retains enough information to recover its Laplacian. Sources: 2023:M020

2024 — A weighted Laplacian is specified through its actual edge weights and positive energy. Sources: 2024:Y24-10-020

2025 — Incidence blocks and conjugate algebraic spectra explain exact graph zeta moments without arithmetic-zero analogies. Sources: 2025:Y25-08-023, 2025:Y25-08-024, 2025:Y25-11-021

2026 — Finite reconstruction, an explicit binary-census collision and sharp Green costs turn invariant comparison into exact finite tasks. Sources: 2026:Y26-07-011, 2026:Y26-10-027, 2026:Y26-10-028

Connections

D15 — Graph comparison is an observability problem. An invariant I is a many-to-one observation; a collision is a pair in one fibre. A stronger separating coefficient is proved for this pair, not universal graph reconstruction.

D07 — Spectral transforms compress operator information. tr exp(−tL) retains eigenvalues; the full symmetric heat matrix permits $L=-\log K_t/t$. Exact identifiability does not imply stable numerical inversion.

D12 — Green kernels solve constrained quadratic optimization. Evaluation is represented by Ke_i in the Q-inner product. Real quadratic minimization does not settle integer-height asymptotics.

Capacity and interactive scope

Lab ids: ['L03']

Inputs: The fixed pair of seven-vertex graphs, height cutoff and q.

Outputs: Exact census tables and a distinguishing height coefficient.

Validation: All 128 subsets and all weak compositions at the declared height are enumerated with integer energy; the q=1/2 value is rational.

Status: Implemented finite graph-energy lab; Green and incidence constructions are mathematical recipes.

Limits

Finite spectral moments need not determine a graph. The five-vertex uniqueness theorem uses its finite simple-graph scope; it is not a universal inverse spectral theorem. A distance-regular intersection array yields a conditional spectrum only if the graph and relation matrices satisfy the declared axioms. No unspecified Euler product, singularity or graph-to-knot correspondence is inferred.

Next questions

Which additional low-height coefficients distinguish all graphs in a bounded vertex class?

Can an incidence or resolvent representation give a short structural explanation of the particular seven-vertex collision?

Source anchors: 2023:M020, 2025:Y25-08-023, 2025:Y25-08-024, 2025:Y25-11-021, 2026:Y26-07-011, 2026:Y26-10-027, 2026:Y26-10-028

Hatcher, Algebraic Topology: graph incidence foundations — https://pi.math.cornell.edu/~hatcher/AT/AT.pdf

D13-F04 · Separators, semiring elimination and multiplicity-preserving quotients

What boundary state is sufficient to compose local computations without changing a count or optimum?

A large combinatorial computation is often tractable because a separator carries only the information that the unprocessed part can still observe. The mathematics is distributivity plus a correct boundary state, rather than a promise that every graph is sparse. Sum–product elimination computes counts or partition sums; min–plus elimination computes minimum additive costs. Both depend on the same factor scopes, but they answer different questions and require different identities. An auxiliary gate with a unique extension preserves the number of original objects; an arbitrary satisfiability-preserving encoding may not. A channel quotient is legitimate only if forgotten lane identities cannot affect future feasibility or cost. These are not independent implementation details: they specify the exact relation between a compact state and the population or value it represents. The useful design process proves a local elimination identity, proves the quotient lift condition and retains enough separator data to make every future continuation well defined.

Definitions

A factor model over a commutative semiring has Z=⊕_x ⊗_j f_j(x_scope(j)). Ordinary nonnegative integers give counting; min–plus uses min as addition and numerical + as multiplication.

Eliminating a variable v produces a new factor on its surviving neighboring scope: g(y)=⊕_v ⊗_{j:v∈scope(j)}f_j.

A separator state stores the values or equivalence classes through which processed and unprocessed factors can interact. Its adequacy is a statement about every compatible continuation.

For a relation F(x,a), the auxiliary fibre m(x)=#{a:F(x,a)} measures whether projecting auxiliaries preserves a count.

Theorem and hypotheses

Finite semiring elimination preserves Z by distributivity, in any elimination order. Over counting integers, an encoding preserves #P exactly when its satisfying auxiliary fibre is one over each x∈P and zero otherwise. If a quotient map is surjective onto feasible quotient states and feasibility and objective are constant on its fibres, the original and quotient optimum values are equal.

Proof

Separate the factors involving v from the others. The independent factors can be taken outside ⊕_v, and the remaining expression is g by definition. Repetition proves exact elimination; changing order changes intermediate scopes but not the final algebraic value. Over integers, the number of satisfying pairs is Σ_x m(x), so unique extension gives #F=#P; mere existential equivalence does not control m(x). For optimization, every original object projects to a quotient object with equal cost, so the quotient infimum is no larger. Every feasible quotient object has a feasible lift with that cost, proving the reverse inequality. Future operations descend only when they preserve fibres as well, which is a stronger requirement than preserving the present objective.

Worked example

Introduce $a=x_1$ XOR x_2, then erase lane identities from four labeled objects injected into sixteen channels arranged as four banks of four.

The XOR gate has one auxiliary extension for each of four original inputs; adding a=1 leaves exactly 01 and 10. An unconstrained auxiliary would double the count to eight. In the channel model there are 16·15·14·13=43,680 physical assignments and 4^4=256 bank tuples. A tuple with bank occupancies n_b has ∏_b P(4,n_b) lifts. The all-in-one-bank tuple has 24 lifts; the four-distinct-banks tuple has 256. Summing all fibre sizes returns 43,680. The full quotient prefix tree has 1+4+16+64+256=341 nodes. Thus uniform quotient sampling is not uniform physical sampling. On a binary path, a min–plus message m_{i+1}(t)=min_s[m_i(s)+φ_i(s,t)] retains the boundary bit; collapsing two boundary bits is legal only when all future costs agree after the collapse.

The separator is a mathematical interface: exact elimination, optimization and quotient sampling each require the interface to retain the information relevant to that task.

Continuity across the source years

2024 — Branching diagrams provide exact path counts when their vertices and edge multiplicities are declared. Sources: 2024:Y24-11-015

2026 — Channel erasure, retained continuation costs and unique-extension factor projection give distinct precise sufficient conditions for compact computation. Their retrospective synthesis supplies the separator interpretation. Sources: 2026:Y26-08-804, 2026:Y26-09-006, 2026:Y26-10-015, 2026:Y26-10-017

Connections

D14 — Continuation semantics defines legal state merging. Boundary states may merge only when every permitted continuation has the same observations and residual costs. State count alone proves neither congruence nor speedup.

D12 — Min–plus elimination is exact optimization. Replace sum by min and product by additive cost, respecting infinities and feasible fibres. A broad historical tropic label is not automatically a defined min–plus operator.

D10 — A quotient changes counting measure. Physical probability pushes forward with fibre multiplicities. Uniformity on 256 quotient states does not mean uniformity on 43,680 assignments.

Capacity and interactive scope

Lab ids: ['L02']

Inputs: Finite factor tables or a declared original-to-quotient map.

Outputs: Eliminated factors, original/quotient counts, minimum values and fibre multiplicities.

Validation: Standalone exact gate, channel-fibre and min–plus path checks; L02 exercises the related completion-count interface.

Status: Mathematical recipe with independently verified finite examples; no general factor-graph solver in the atlas.

Limits

Intermediate factor size depends on elimination width; distributivity does not imply polynomial complexity. Semiring choice changes the output. Auxiliary existential projection preserves satisfiability more weakly than counts. The original routing graph, byte-hop objective and claimed 9,765,632 optimum were not recovered, so neither those values nor timing claims are certified here.

Next questions

What is the smallest boundary equivalence that preserves every future constraint and cost in a supplied routing instance?

Can quotient-aware weighted sampling preserve the original physical counting measure without reconstructing every lift?

Source anchors: 2026:Y26-08-804, 2026:Y26-09-006, 2026:Y26-10-015, 2026:Y26-10-017

Bryant, Graph-Based Algorithms for Boolean Function Manipulation — https://doi.org/10.1184/r1/6605990
Gaubert and Plus, Max-plus linear algebra — https://doi.org/10.1007/BFb0023465

D14 · Logic, computation and exact certification

Organizing question

Which semantics and finite witnesses justify a global computational claim?

The computation domain joins three questions that must remain distinct: what behavior a machine is intended to preserve, what smaller representation reproduces it, and what certificate establishes that conclusion. The observable state is relative to a declared action alphabet. Present outputs do not determine it. Future equivalence compares outputs after every allowed continuation, and the finite refinement theorem makes that infinite-looking condition executable on a supplied finite transition table. Additive costs enlarge the semantics and must descend with transitions. This explains why a state quotient can be exact for one observer and fail when a reset, query or pinned physical identity is added.

Linear realization gives the parallel quantitative theory of memory. An intertwining identity proves that a smaller state reproduces every time step, while a nonzero Hankel minor proves that no still smaller linear state suffices. The distinction between the reachable subspace and the invisible subspace matters: their intersection quotient determines zero-initial driven order. A recurrence for one observed trajectory, a universal scalar observation recurrence and a multi-output state dimension are different mathematical objects. The min–plus two-by-two gap illustrates the same upper-bound/lower-bound strategy in another algebra, with its scope explicitly limited to additive register factorization.

The certificate family supplies a common proof interface across the archive. Truth tables require a finite domain; quantifiers specify whether a witness is uniform; a polynomial function and a formal polynomial need not coincide; a feasible point is not an optimum; an exact witness is not a complete census. A small checker can validate useful discoveries made by an untrusted search procedure, but its conclusion depends on its declared certificate type. A Bézout identity, a rank factorization and a primal–dual pair establish different propositions. Historical proof diagrams, public recursion identities and two-prime checks retain their narrower meanings unless the missing completeness or dependency arguments are supplied.

The raw-generator theorem is a substantial instance where those obligations are actually met. The annual 2026 bundle contains all 126 integer lattice bases and exact determinant and inverse-column checks. They exclude every collision for any five known positions among nine raw steps, not merely the fixture states. A puncturing argument then proves exact distance $n-4$ and length-optimal one- and two-word correction. The decoder fixtures separately exercise the implementation. Accepted/rejected sequences have unknown hidden step counts and remain outside that theorem. This distinction should not obscure the genuine strength of the raw certificate.

The atlas’s interactive interface reflects this proof architecture. L01 makes a fixed linear quotient test and cheapest static probe panels inspectable; the finite-machine refinement and adaptive policies are checked recipes. L02 exposes a complete small census and address selection, while its completion-count and inverse-ranking theorems are proved separately. The broader exact certificates remain replayable source computations and mathematical recipes, rather than a claimed general theorem prover or universal SMT replacement. Their unification is a disciplined route from behavior to reduction to proof, with every change of model or quantifier visible.

Core objects

Action alphabets and deterministic transition systems

Future equivalence and right congruences

Reachable–observable realizations

Hankel matrices and semiring response tables

Typed witnesses and dependency records

Error-correcting trajectory codes

Open frontier

Formalize supplied exact certificates without losing their input semantics; recover operational tables and missing completeness proofs; quantify approximate state reduction separately from exact equivalence.

D14-F01 · Future equivalence, bisimulation and executable quotient states

When can two internal states be merged while preserving every permitted future observation?

The correct state of a computation is relative to its observer and action alphabet. Two memories that currently print the same value may behave differently after a tick, reset, allocation or later query. Future equivalence compares every such continuation. In a finite deterministic machine, its apparently infinite definition has a finite certificate: iterative partition refinement stops at a stable partition, and stability implies agreement for every word. Weighted futures require immediate costs as well as visible outputs to descend. Bisimulation supplies a useful transition-level proof principle but must be stated for the actual operational semantics. The resulting quotient is an executable machine, not merely a partition of a log. This family is the central bridge between semantic state reduction, target-relative observations and efficient separator states. It explains both why some reductions are exact and why a dramatic reduction in state count alone establishes no behavioral equivalence.

Definitions

A deterministic observation machine has finite states S, actions A, transitions $\delta_a:S\to S$ and output $o:S\to O$. A word acts in its given order.

Future equivalence is $s\sim t$ iff $o(\delta_w(s))=o(\delta_w(t))$ for every finite word w, including the empty word.

Depth-r equivalence checks words of length at most r. Refinement maps a state to its present output together with the current classes of all its successors.

With additive action costs, $C_s(aw)=c(s,a)+C_{\delta_a(s)}(w)$. A cost-preserving quotient must preserve these immediate costs and all successor classes.

Theorem and hypotheses

Future equivalence is the coarsest equivalence respecting outputs and every action transition. Finite partition refinement computes it and has at most $|S|-1$ strict refinements. Once two successive partitions agree, the stable partition checks all word lengths. Equality of all accumulated continuation costs similarly defines a right congruence for deterministic weighted machines.

Proof

If s and t have equal futures, then their present outputs agree. For every action a and every word w, the output after aw agrees; hence δ_a(s)~δ_a(t), making the quotient transition independent of representatives. Conversely, any output-preserving transition congruence preserves the output after every word by induction, so it refines future equivalence. Starting from output classes, the next refinement checks exactly one more leading action by the same induction. A stable partition therefore preserves transitions and already equals the full relation. Every strict refinement increases the number of nonempty blocks, giving the finite bound. For additive costs, compare one-letter words first to obtain c(s,a)=c(t,a), then subtract this finite common value from the equality for aw to obtain future equality at the successors. This subtraction argument needs finite deterministic costs; nondeterministic min–plus machines require their own semantics.

Worked example

Use a TTL counter with states 0,1,2. Tick sends positive values down by one and leaves zero fixed. Read costs one at zero and zero otherwise, then resets the counter to two.

States 1 and 2 have identical one-step costs for read and tick, but tick-read costs one from 1 and zero from 2. Present-output merging therefore loses a future distinction. Refinement retaining both action-cost vectors and successor classes separates all three states. In a separate four-state cycle δ(s)=s+1 mod 4 with output s mod 2, the exact quotient has two classes {0,2} and {1,3}, swapped by the action. Changing only the output to the indicator of state 3 makes every state distinguishable within three actions. Thus the same transition graph has different minimal observable quotients for different observers.

A quotient certificate combines a declared alphabet, output map and closure proof. A finite logged prefix is evidence only within a proved completeness bound.

Continuity across the source years

2025 — The bisimulation thread is reconstructed using a finite labeled transition relation; absent garbage-collection operational semantics are not inferred. Sources: 2025:Y25-09-014

2026 — The conversation topics sharpen from semantic quotients to additive future costs, action-dependent refinement and explicit finite closure hypotheses. The chapter unifies these as one observer-relative construction. Sources: 2026:Y26-01-003, 2026:Y26-09-006, 2026:Y26-10-021, 2026:Y26-10-029

Connections

D15 — Hidden fibres require dynamic closure. Linear closure HA=TH is the linear counterpart of δ_a preserving quotient classes. Present target recovery alone is weaker than a predictive quotient.

D13 — Separator states summarize all future interactions. Equivalence of continuations licenses message-state merging. The relevant continuation alphabet must include every later legal choice.

D03 — Quotient maps are algebraic actions. A right congruence lets the free action monoid act on the quotient. A log-fold monoid action or projection does not by itself supply this observation equivalence.

Capacity and interactive scope

Lab ids: ['L01']

Inputs: Recipe: finite states, action tables, outputs and costs. Related L01 lab: a fixed four-dimensional state, six rational probes and two targets.

Outputs: Recipe: stable future classes and distinguishing words. L01: exact observation ranks and cheapest static probe panels.

Validation: Exact refinement is compared with all words up to the proved finite stabilization bound; the standalone verifier checks the TTL and four-cycle examples.

Status: Finite transition refinement is a mathematical recipe checked by verify.py. L01 implements the related fixed linear observation/probe model, not an action-table editor.

Limits

Changing the action alphabet can split previously valid classes. Trace equivalence and bisimulation are distinct for nondeterministic systems. Infinite-state refinement may not terminate. A broad tropic name is not sufficient to specify weighted, min–plus or nonlinear transition semantics. No unprovided TTL machine table or benchmark quotient count is reproduced.

Next questions

Which shortest action word distinguishes each inequivalent pair in a supplied implementation?

Can a proposed abstraction provide a transition-respecting lift or simulation relation for all operations, including resets and allocation?

Source anchors: 2025:Y25-09-014, 2026:Y26-01-003, 2026:Y26-09-006, 2026:Y26-10-021, 2026:Y26-10-029

Rutten, Universal Coalgebra: A Theory of Systems — https://doi.org/10.1016/S0304-3975(00)00056-6
Eilenberg, Automata, Languages, and Machines — https://www.sciencedirect.com/book/9780122340017/automata-languages-and-machines

D14-F02 · Hankel witnesses and minimal observable memory

How can an exact realization and an independent lower bound prove the smallest recurrence or register dimension?

State reduction has two logically different tasks. A proposed smaller machine must reproduce every output, and every still smaller machine must be ruled out within a declared encoding class. For linear systems these tasks meet through intertwiners and Hankel rank. A matrix identity transports the entire dynamics, while a nonzero finite minor prevents a lower-dimensional realization. This is stronger than fitting a recurrence to a prefix and more informative than quoting the dimension of the original state. Reachable and observable modes must be considered jointly: the zero-initial input-output order lives on the reachable space modulo its invisible part. A scalar universal output recurrence differs from a recurrence for one initial state, and multiple-output recurrence degree need not equal state dimension. In the min–plus setting a two-by-two response table provides a separate additive-register obstruction; ordinary field rank must not be imported into that semiring without justification.

Definitions

A linear sequence realization is $y_n=cA^n b$ over Q. Its Hankel matrix has entries $H_{ij}=y_{i+j}$.

A reduction R with $RA=GR$, $Rb=\beta$ and $c=dR$ is an intertwining certificate for $y_n=dG^n\beta$ at every n.

The reachable space is $\operatorname{span}\{A^k \operatorname{im} B\}$; the invisible space is $\bigcap_{k\geq 0}\ker(CA^k)$. The minimal zero-initial driven state is $R/(R\cap N)$.

A finite min–plus rank-one table has entries $M_{ij}=u_i+v_j$. Such a table has equal diagonal and off-diagonal sums on every two-by-two submatrix.

Theorem and hypotheses

An r-state linear realization has Hankel rank at most r. An order-r forward recurrence has the same rank bound. Therefore an r-state realization together with a nonzero r-by-r Hankel minor proves exact minimal state dimension and recurrence order r. For scalar universal outputs on an observable quotient, the universal recurrence is the quotient operator’s minimal polynomial; its degree equals that observable quotient dimension.

Proof

The intertwiner identity gives RA^n=G^nR by induction, and the input/output identities yield equality of every sequence term. Any finite Hankel block factors as rows cA^i times columns A^jb through the realization state space, so its rank cannot exceed the state dimension. A recurrence of order s expresses every later Hankel column as a fixed combination of the preceding s, giving the same lower bound. Conversely Cayley–Hamilton supplies a recurrence of order at most r for an r-state realization. For universal observation recurrences, N is A-invariant because CA^k(Av)=CA^{k+1}v. A polynomial p annihilates every future output exactly when p(A)V⊆N, or p(A_bar)=0. In the scalar observable case a lower-degree polynomial would make all observation rows dependent on fewer initial rows, contradicting observability.

Worked example

Take A=diag(1,2,3,4,5,6), b=(1,1,1,0,0,0)^T and c=b^T. Also test the stipulated min–plus table [[0,2],[2,0]].

Projection onto the first three coordinates gives y_n=1+2^n+3^n, beginning 3,6,14,36,98. The leading three-by-three Hankel determinant is 4, the squared Vandermonde determinant on 1,2,3. The recurrence y_{n+3}=6y_{n+2}−11y_{n+1}+6y_n is globally certified and has exact order three. For the min–plus table the diagonal sum is 0 and the other sum is 4, excluding a one-register additive factorization. A two-register factorization through the tropical identity gives a matching upper bound. This last obstruction concerns that specified linear min–plus encoding class, not arbitrary nonlinear memory.

A finite witness proves an infinite-time minimality claim only because a global realization identity supplies the model class and upper bound.

Continuity across the source years

2023 — The annual source separates universal observable recurrences from individual traces and identifies the reachable–observable intersection quotient. Sources: 2023:M016, 2023:M017

2026 — Unfitted coefficients test a proposed recurrence; explicit minor witnesses later certify minimality in ordinary linear and stipulated min–plus models. Sources: 2026:Y26-03-012, 2026:Y26-09-007, 2026:Y26-10-018

Connections

D15 — Minimal realizations discard different kinds of hidden state. Reachability restricts the driven state and observation then takes a quotient. A stable transfer function can hide an unstable internal canceled mode.

D03 — Intertwining makes dynamics descend. RA=GR is a representation morphism. Exact matrix identities differ from approximate singular-value truncation.

D13 — Hankel lower bounds complement compact state counts. A finite response minor can force retained boundary dimension. Semiring-specific lower bounds need a declared rank notion.

Capacity and interactive scope

Lab ids: ['L01']

Inputs: Recipe: rational realization matrices or an explicit response table. L01 accepts selections from its fixed six-probe menu.

Outputs: Recipe: intertwiner, Hankel witness and recurrence; L01 illustrates target-relative observation ranks.

Validation: The standalone verifier checks the determinant four, recurrence over a long exact prefix and the nonzero tropical gap; the global proof is the intertwiner argument.

Status: Mathematical recipe with checked exact examples; L01 illustrates the related quotient principle.

Limits

Finite-prefix rank can underestimate later behavior without a proven finite-dimensional model. Approximate numerical rank does not establish exact recurrence order. Multiple-output universal polynomial degree may be smaller than observable dimension: A=C=I has degree one in any dimension. The historical determinant-four tropical matrix was not recovered; the two-by-two example is newly stipulated.

Next questions

Can an explicit intertwiner and matching exact minor be recovered for each claimed large state reduction?

How does noise alter a useful approximate realization guarantee when exact equivalence is no longer appropriate?

Source anchors: 2023:M016, 2023:M017, 2026:Y26-03-012, 2026:Y26-09-007, 2026:Y26-10-018

Balle et al., Canonical forms for weighted automata — https://arxiv.org/abs/1501.06841
Gaubert and Plus, Max-plus linear algebra — https://doi.org/10.1007/BFb0023465

D14-F03 · Typed proof certificates and the boundary between examples and completeness

What exact claim does each certificate establish, and which dependencies remain outside its proof boundary?

The archive repeatedly reaches the same useful discipline from different directions: separate a search procedure from a small exact checker, and state the checker’s theorem with its quantifiers and input type. A feasible point proves existence; a primal–dual equality proves optimality; a rank minor plus factorization proves rank; a Bézout identity proves inconsistency; a complete Gröbner replay proves an ideal basis. These witnesses are not interchangeable. Nor does checking many examples establish a universal identity unless a degree bound or other completeness theorem reduces the universal statement to those examples. This family turns that discipline into a reusable proof interface. It begins with truth tables and bound variables, continues through arithmetic and floating-point envelopes, and reaches exact algebraic and finite-state certificates. The goal is constructive: make a solver’s useful discoveries portable across a trust boundary while exposing the precise assumptions on which the conclusion rests.

Definitions

A certificate is data c for an explicitly specified predicate Check(input,c). Soundness proves Check⇒Claim; completeness, if needed, proves every true claim has an admissible certificate.

A witness establishes ∃x P(x); a complete census establishes both listed⇒valid and valid⇒listed. These are distinct quantifier directions.

A polynomial over F_q reduced to degree below q in every variable is a different object from an arbitrary formal polynomial or a polynomial function modulo x_i^q−x_i.

A proof dependency record states the encoded instance, arithmetic domain, theorem hypotheses, checker implementation and any external mathematical theorem used.

Theorem and hypotheses

If f∈F_q[x_1,...,x_n] has degree <q in each variable and vanishes on all q^n points, then f is the zero formal polynomial. Without this degree condition exhaustive functional agreement need not prove formal polynomial equality. Separately, exact primal and dual feasible points with equal objectives certify an LP optimum by weak duality.

Proof

For the polynomial theorem, induct on n. A one-variable nonzero polynomial of degree <q cannot have q roots. Write f as a polynomial in the last variable. At every fixed earlier tuple all q evaluations vanish, so every coefficient evaluates to zero. Each coefficient retains the coordinate degree bounds, and induction makes all coefficient polynomials zero. Without the bound, x_1^q−x_1 vanishes everywhere but has nonzero coefficient dictionary. For LP minimization Ax=b,x≥0 and dual maximization $b^{T}y$ with $A^{T}y≤c$, every feasible pair satisfies $b^{T}y=y^{T}Ax≤c^{T}x$. Equality therefore sandwiches the optimum. The proof uses feasibility, not closeness of objective values. A floating candidate can guide search, but exact rational feasibility or a rigorous residual enclosure must justify the final theorem.

Worked example

Compare polynomial functions over F_{3^4} and certify min{x+2y:x+y=3,x,y≥0}.

There are 81 assignments. The identity (x_1+x_2)^3=x_1^3+x_2^3 is true formally in characteristic three by coefficient cancellation. The expression x_1^3−x_1 vanishes at all 81 assignments but is not the zero formal polynomial, so an evaluator alone confuses two types. For the linear program, the primal point (3,0) has value 3, and dual scalar λ=1 satisfies λ≤1 and λ≤2 with dual value 3. This exact pair proves the optimum. As a different certificate, 1=y·x−(xy−1) proves that x=0 and xy−1=0 have no common solution over a field; it does not certify a census of a different polynomial system.

Certificate types make the conclusion and its omitted obligations inspectable, enabling exact replay of useful solver output without trusting the solver’s search strategy.

Continuity across the source years

2023 — Object types and efficient public inverses prevent a shared letter or complicated notation from becoming a hardness theorem. Sources: 2023:M001, 2023:M005, 2023:M032

2024 — Truth tables, quantifier order and encoded complexity inputs make the logical claim explicit. Sources: 2024:Y24-10-026, 2024:Y24-10-027, 2024:Y24-11-016

2025 — Finite tests gain force from degree bounds and exact certificate pairs; unproved dependencies remain conditional. Sources: 2025:Y25-06-024, 2025:Y25-07-007, 2025:Y25-12-023

2026 — Ideal identities, rational membership and complete-census obligations refine the distinction between sound witnesses and universal closure. Sources: 2026:Y26-05-030, 2026:Y26-07-005, 2026:Y26-09-032

Connections

D03 — Algebraic identities are compact certificates. Bézout combinations, ideal reductions and nonzero minors are checked in a declared exact ring. A finite-prime test need not prove rational membership or integral saturation.

D12 — Optimality is a two-sided certificate. Primal feasibility supplies an upper bound and dual feasibility a lower bound for minimization. The historical optimization instance must be encoded and linked to its checker.

D04 — Proofs compose only when endpoints and hypotheses match. A dependency chain is valid after every antecedent is independently proved. A diagram of proposed lemmas is not closure of the final theorem.

Capacity and interactive scope

Lab ids: ['L02']

Inputs: Typed finite statements, exact witnesses and explicit domains or completeness bounds.

Outputs: Checked truth tables, polynomial identities, feasibility and certificate conclusions with dependencies.

Validation: Standalone checker exhausts the 81-point counterexample, verifies exact LP equality and the Bézout identity; L02 checks the rank/unrank certificate interface.

Status: Certificate recipes and finite checks; no general theorem prover or complete SMT replacement is implemented.

Limits

Formal proof claims require an actual proof term or complete trusted replay, not a label such as Lean bridge. Two-prime agreement, fitted residuals and public recursion consistency each establish narrower statements. Hardness requires an input distribution, adversary model and reduction; constructing a lattice encoding alone does not produce an efficient decoder. Sequential work and cryptographic provenance are not inferred from an algebraic identity.

Next questions

Can every historical claim be assigned a certificate type and a complete list of unchecked premises?

Which solver outputs admit small exact certificates whose checking cost is substantially lower than search?

Source anchors: 2024:Y24-10-026, 2024:Y24-10-027, 2025:Y25-06-024, 2025:Y25-07-007, 2025:Y25-12-023, 2026:Y26-05-030, 2026:Y26-09-032

SMT-LIB, official integer arithmetic theory — https://smt-lib.org/theories-Ints.shtml
Boyd and Vandenberghe, Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/
MIT OpenCourseWare, Algebra I — https://ocw.mit.edu/courses/18-701-algebra-i-fall-2010/

D14-F04 · Coding and raw-state recovery by exact lattice certificates

When do finitely many output words determine a hidden state, and how does that injectivity become optimal error correction?

The coding thread begins with concrete binary codes whose distance can be independently enumerated and culminates in a universal certificate for a specified 107-bit raw generator. It is important to separate

three achievements. Alphabet capacity gives a necessary observation count. A short-vector exclusion proves injectivity for every state, rather than only tested fixtures. Finally a distance argument converts injectivity of every five-position projection into optimal correction of corrupted words. The large proof has a compact trusted core: integer lattice membership, an exact determinant and rational inverse-column bounds. A basis reduction algorithm may discover useful matrices, but the checker does not trust that search. This makes the raw-state theorem a strong example of portable exact certification. The accepted-output sequence has hidden rejected steps and remains a different observation model; its earlier threshold claim cannot inherit the raw theorem.

Definitions

Set $M=2^{107}-1$, $A=3^{121} \bmod M$, $b=2^{26}$, $x_{i+1}=Ax_i \bmod M$ and $y_i=x_i \bmod b$, with $0\le x_i<M$.

For five known positions $p_0<\dots<p_4$ in $\{0,\dots,8\}$, let $g_j=p_j-p_0$ and $\Lambda_p=\{d\in Z^5 : d_j\equiv A^{g_j}d_0 \bmod M\}$. Set $D=\operatorname{floor}((M-1)/b)=2^{81}-1$.

The raw trajectory code C_n consists of the first n output words of every state. Hamming distance counts changed words, even when many bits in a word are changed.

A supplied row basis B_p is certified by lattice row membership, $|\det B_p|=M^4$, and $D\Sigma_i|(B_p^{-1})_{ij}|<3/8$ for every column j.

Theorem and hypotheses

For every five-position subset of the first nine raw steps, the selected words uniquely determine the initial state. Four words cannot determine every state. For $5\le n\le 9$ the raw trajectory code has minimum distance exactly $n-4$. Thus seven words correct one arbitrary word error and nine correct two, and these lengths are optimal for any code representing all M states over this alphabet.

Proof

The map $d\mapsto(d_j-A^{g_j}d_0)_{j=1}^4$ onto $(Z/MZ)^4$ has kernel Λ_p, so the lattice index is M^4. Membership and matching determinant establish that the supplied rows span the whole lattice. If two states produce the same selected words, their selected state differences are bd_j with $|d_j|\le D$. Since b is invertible modulo M, $d\in\Lambda_p$, hence $d=cB_p$ for integer c. The dual bound gives $|c_j|\le D\Sigma_i|(B_p^{-1})_{ij}|<1$, forcing $c=0$ and then equality of the initial states because A is invertible. Four words are impossible by $b^4<M-1$. Distinct length-n codewords agree in at most four positions, so $d_{\min}\ge n-4$. Puncturing $d_{\min}-1$ positions is injective, giving $M\le b^{n-d_{\min}+1}$; since $M>b^4$, $d_{\min}\le n-4$. Disjoint radius-t Hamming balls require $d_{\min}>2t$. Puncturing $2t$ positions then gives $n\ge 5+2t$, proving optimal lengths.

Worked example

Use the actual 126 supplied five-position bases. For decoding a received exact five-word vector, express selected states as y_j+bz_j with $0\le z_j\le D$.

The congruences define an affine lattice $z\in v+\Lambda_p$. Applying B_p^{-1} to the bounded z-box places each integer coefficient in an exact rational interval of width below $3/8$. Each contains at most one integer. Exact endpoint rounding therefore yields at most one candidate; re-encoding verifies every state bound and observation. The annual certificate checks all 126 bases, and 630 designated decoder fixtures include states $0,1,123456789012345678{,}9,\operatorname{floor}(M/3),M-1$. For one or two corruptions, enumerate five-position subsets, decode candidates, re-encode and retain a state within the allowed word radius. The distance theorem proves uniqueness. The small analogue $M=31,A=3,b=8$ has 8,24,31 distinct one-, two-, three-word outputs and independently illustrates why capacity and actual injectivity are separate.

The universal theorem rests on exact lattice checks; fixtures separately exercise decoder code. This is observability and error correction for a specified deterministic raw recurrence, not a claim of cryptographic unpredictability.

Continuity across the source years

2023 — A fully specified [15,7,5] BCH code establishes two-error correction through exact generator and weight enumeration. Sources: 2023:M037

2024 — Functorial relabeling is separated from Hamming-distance preservation. Sources: 2024:Y24-07-003

2026 — Parity-map and algebraic-geometric code bounds precede the October raw-generator lattice certificate. The supplemental entries contain the actual all-state theorem, whereas accepted-output claims retain an unresolved observation rule. Sources: 2026:Y26-05-028, 2026:Y26-07-026, 2026:Y26-10-024, 2026:Y26-10-901, 2026:Y26-10-902

Connections

D15 — Raw words are finite-state observations. Injectivity of a multi-time observation map gives exact state recovery. Unknown rejection times change that map and invalidate direct transfer of the theorem.

D01 — Congruences become a lattice exclusion. The modular recurrence and output residue produce a bounded integer-lattice problem. A short lattice vector need not by itself be a realizable collision.

D12 — Code lengths have matching lower bounds. Puncturing compares all-state population with remaining alphabet capacity. Optimal information length does not assert optimal decoder running time.

Capacity and interactive scope

Lab ids: []

Inputs: Known raw positions and words, plus supplied exact lattice bases.

Outputs: Certified state or no compatible state; unique radius-one or radius-two decoder when appropriate.

Validation: The annual source bundle contains the exact all-126-basis checker and independent review. This domain verifier replays all 126 basis checks and 630 decoder fixtures, repeats all determinants with an independent Bareiss calculation, and checks a small exhaustive analogue.

Status: Actual large all-state certificate in the included 2026 source bundle; no live FreeLIP browser lab.

Limits

The raw-output theorem assumes the stated recurrence, output convention and known positions. No recovered acceptance/rejection rule supports the previous six-accepted-output claim. A designed distance lower bound for an algebraic-geometric code is not an exhibited minimum-weight codeword or efficient decoder. Structural transport can preserve composition while changing Hamming distance.

Next questions

Can the historical acceptance rule and its hidden-step semantics be recovered for a separate theorem?

Can the all-state lattice proof be translated into a formal proof assistant while retaining its explicit determinant and rounding bounds?

Source anchors: 2023:M037, 2024:Y24-07-003, 2026:Y26-05-028, 2026:Y26-10-024, 2026:Y26-10-901, 2026:Y26-10-902

SageMath, BCH code reference — https://doc.sagemath.org/html/en/reference/coding/sage/coding/bch_code.html
Brouwer, Riemann–Roch and algebraic geometry codes — https://aeb.win.tue.nl/2WF02/rr.pdf

D15 · Control, observability and inverse problems

Organizing question

What can be recovered or controlled from declared observations, at what cost and with what guarantee?

The central object in this domain is a requested target seen through an observation. Full-state reconstruction is only one possible target and often demands unnecessary measurements. In finite dimensions, exact target recovery is equivalent to a kernel inclusion and a factorization G=LH. The minimum recovery norm quantifies the amplification of bounded measurement error. A violating invisible vector is an equally useful certificate: it proves that two states have the same data and different targets. The late WIND and observation-geometry topics sharpen this perspective, while the earlier recurrence questions already show its algebraic roots.

Present recovery, prediction and nonlinear identifiability are successive obligations. A present factorization does not make the observation a closed dynamical state; HA=TH additionally requires the invisible kernel to be invariant. Moving observation coordinates acquire B′. A vanishing hidden directional derivative at one point does not establish target constancy on an entire nonlinear fibre. Counting the coefficients of a local wind ansatz does not guarantee a full-rank beam design. These distinctions organize practical sensing questions into exact mathematical tasks instead of letting one rank statistic carry all of their meaning.

Control introduces the dual constructive problem: choose inputs that change the state or target with a certified cost. Gramians characterize finite-horizon reachability and observability without assuming asymptotic stability. Lyapunov identities establish decay in a compatible metric, including nonnormal examples whose Euclidean energy grows transiently. Completion of squares gives full scalar Riccati value proofs with terminal conditions. Descriptor pencils show why the actual constrained model matters: singular E can encode a regular problem, while adding εI can create an unstable artificial mode. These examples make matrix identities useful certificates of behavior, not merely computational diagnostics.

Inverse problems replace exact inversion by controlled approximation. The diagonal models expose the bias-noise split and matching indistinguishable-pair lower bounds. Polynomially decaying singular values give a power rate; exponentially decaying values require a logarithmic rate and an explicit safety margin. Tikhonov’s parameter and Landweber’s stopping time are two ways to choose the same compromise. The guarantee always names a source class and noise model. Neither a tiny residual nor a leading-order cutoff equivalence supplies those assumptions.

Finally, the experiment-design family asks what information should be purchased next. A finite Bellman theorem gives a policy and its matching worst-case lower bound when tests are deterministic and targets explicit. Correlated ellipsoids and hidden conditional variation explain why some extra observations are useful while repeated deficient directions remain irrelevant. L01 combines a fixed linear quotient test with cheapest static probe panels, L04 explores cutoff envelopes, and L08 retains shared uncertainty in robust decisions. These labs demonstrate exact bounded capacities. General nonlinear reconstruction, continuous noisy adaptive design and historical sensor-performance claims require their own supplied models, data and validation arguments.

Core objects

Observation fibres and target maps

Kernel inclusions and recovery operators

Invariant invisible subspaces

Lyapunov metrics and Gramians

Descriptor pencils and Riccati values

Spectral filters and source classes

Adaptive experiment trees

Open frontier

Extend finite fibre certificates to supplied nonlinear or noisy models; derive robust experiment policies with computable lower bounds; select regularization from data without unproved source assumptions.

D15-F01 · Target-relative observation: fibres, factorization and dynamic closure

Which hidden directions matter for the requested target, and when can the observation itself evolve as a closed state?

The strongest common thread in the later archive is that full-state recovery is often unnecessary. An observation can leave a large invisible fibre while still determining a particular target exactly. The right question asks whether the target is constant on those fibres. In finite-dimensional linear algebra this becomes a kernel inclusion, a factorization and a sharp stability inequality. This also separates three levels frequently merged in informal discussions: present target identifiability, stable noisy recovery and closed predictive dynamics. Present factorization $G=LH$ does not imply that Hx can predict its own future; closure requires the invisible space to be invariant under the state dynamics. In nonlinear models a derivative test is only local and can miss second-order target variation along a hidden fibre. These distinctions guide sensor design constructively: retain or measure directions that change the target, then verify the closure or robustness property actually required by the task.

Definitions

For state $x\in V$, observation $H:V\to W$ and target $G:V\to Z$, a fibre is $\{x:Hx=y\}$. The target is identifiable if $Hx=Hx'$ implies $Gx=Gx'$.

The invisible space of a linear observation is $\ker H$. Target-relative observability is $\ker H\subseteq\ker G$, not necessarily $\ker H=\{0\}$.

For dynamics $x'=Ax$, observation closure means there is T with $HA=TH$. A moving observation $B(t)$ closes when $B'+BA=TB$.

For Euclidean finite-dimensional spaces, the minimum-norm recovery on $\operatorname{im} H$ is $L_0=GH^{\dagger}$; its extension can vanish on the orthogonal complement of $\operatorname{im} H$.

Theorem and hypotheses

The following are equivalent: G is identifiable from H; $\ker H\subseteq\ker G$; $G=LH$ for some linear L; and $G^{T}G\preccurlyeq\lambda H^{T}H$ for some finite $\lambda\geq 0$. The smallest such λ is the squared minimum recovery norm. Separately, $HA=TH$ exists exactly when $\ker H$ is A-invariant.

Proof

Identifiability applied to $x \in \ker H$ and $x' = 0$ gives the kernel inclusion. Conversely, equal observations differ by a vector in $\ker H$, so their targets agree. Define $L(Hx) = Gx$ on $\operatorname{im} H$; the inclusion makes this independent of the chosen representative and gives linearity. Extend L to W. Factorization gives $||Gx|| \leq ||L|| \, ||Hx||$ and hence the quadratic inequality. Conversely the inequality forces $Gx = 0$ whenever $Hx = 0$. The least possible norm is the supremum of $||Gx|| / ||Hx||$ over nonzero observed vectors, exactly the least square-root constant in the inequality; the zero extension attains it in finite dimensions. For closure, HA must vanish on $\ker H$, which is equivalent to $A(\ker H) \subseteq \ker H$. Apply the same factorization construction with target HA. In moving coordinates differentiate Bx and obtain $(B' + BA)x$; this is a new target to which the criterion applies.

Worked example

Observe $Hx = (x_1 + x_2, x_2 + x_3)$ and request $Gx = x_1 + 2x_2 + x_3$. Then compare a nonlinear target at a tangent point.

The observation kernel is $\operatorname{span}((1, -1, 1))$; G annihilates it, so $Gx = y_1 + y_2$ is exactly recoverable even though x is not. The minimum recovery norm is $\sqrt{2}$ because H is onto $\mathbb{R}^2$ and $L = (1, 1)$. Requesting x_1 instead fails: x and $x + (1, -1, 1)$ have the same observation and different first coordinates. For $H(x, z) = x$ and nonlinear target $g(x, z) = z^2$, the hidden derivative $\partial_z g$ vanishes at $z = 0$, yet g changes along that same observation fibre. Thus a zero directional derivative at one point is not finite-amplitude identifiability. In periodic velocity reconstruction, divergence and curl similarly leave a constant harmonic mean invisible unless it is separately observed.

A target can be recovered with fewer channels than a whole state, but stable recovery, future prediction and nonlinear global identifiability each need their own theorem.

Continuity across the source years

2023 — Observable recurrence and parameter-response questions already require distinguishing hidden state, hidden dynamics and fitted parameters. Sources: 2023:M016, 2023:M017, 2023:M082

2024 — Spatial averaging can erase a diffusion parameter exactly; the observation, rather than merely the equation, controls identifiability. Sources: 2024:Y24-07-015

2025 — Sampling aliasing and coordinate-dependent weighted rank expose two different observation-model obligations. Sources: 2025:Y25-06-018, 2025:Y25-12-024

2026 — The later topics make fibre factorization explicit, add noise constants and distinguish tangent tests from finite-amplitude target variation. Sources: 2026:Y26-02-014, 2026:Y26-05-010, 2026:Y26-09-018, 2026:Y26-09-019, 2026:Y26-10-019

Connections

D14 — Linear factorization is the analogue of semantic quotienting. A quotient permits exact prediction only if transitions respect its fibres. A current-output merge can fail under a later action.

D11 — Noise amplification is target-specific. The recovery norm bounds $||L\eta||$, while hidden conditional variation can create an additional irreducible floor. A deterministic bound and a probabilistic confidence statement use different assumptions.

D06 — Moving coordinate observations acquire a derivative term. $d(Bx)/dt = (B' + BA)x$ yields the moving closure equation. A mere relabeling cannot discard B' or change a rank invariant without its transformation law.

Capacity and interactive scope

Lab ids: ['L01']

Inputs: Recipe: rational H,G and optionally A. L01 uses four-dimensional state, six fixed probes and a choice of sum or pair target.

Outputs: Recipe: L or an invisible counterexample and closure checks. L01: exact rank test and cheapest static probe panels.

Validation: Exact kernel and factorization identities are checked; the standalone verifier supplies both the recoverable target and failing target and checks a closure counterexample.

Status: Small quotient/probe lab implemented; nonlinear fibres and field reconstruction are mathematical recipes.

Limits

In infinite-dimensional Hilbert spaces kernel inclusion alone need not give bounded recovery without a range/majorization condition. Local Jacobian rank does not prove a nonlinear target constant on entire fibres. Fifteen coefficients in an affine wind model require a full-rank observation design, not just fifteen arbitrarily repeated readings. Exact optimization on development data does not prove sensor reduction or out-of-sample accuracy.

Next questions

For a supplied target, what is the smallest additional observation subspace that removes all target-relevant hidden directions?

When can local differential factorization be integrated into a global fibre-constant target on a restricted nonlinear state manifold?

Source anchors: 2023:M016, 2024:Y24-07-015, 2026:Y26-05-010, 2026:Y26-09-024, 2026:Y26-09-019, 2026:Y26-09-018, 2026:Y26-10-019

Douglas, On majorization, factorization, and range inclusion — https://doi.org/10.1090/S0002-9939-1966-0203464-1

D15-F02 · Control energies, compatible metrics and constrained descriptor dynamics

How can a declared system carry a complete stability, reachability or optimal-control certificate?

Control claims become concrete only after the plant, allowed inputs, output and state domain are specified. A positive Lyapunov matrix proves decay in one chosen metric; stable eigenvalues alone need not make Euclidean energy decrease at every instant. Finite-horizon Gramians quantify reachability and observation without requiring asymptotic stability. Riccati equations certify quadratic control values through completion of squares, including the terminal condition. Descriptor systems add algebraic constraints: a singular mass matrix can encode a regular constrained evolution, and replacing it by an invertible approximation can introduce a wholly new unstable mode. The common constructive capacity is to certify the property of the actual model with exact matrix identities. It is not enough to import a gain, a condition number or a contraction claim from a nearby surrogate. Norm bounds and dynamics must refer to the same metric and the same invariant domain.

Definitions

For $x'=Ax$, $V(x)=x^{T}Px$ with $P\succ 0$ is a quadratic energy. $A^{T}P+PA\preccurlyeq -Q$, $Q\succ 0$, gives a quantitative decay estimate.

For $x'=Ax+Bu$, $W_c(T)=\int_0^T e^{At}BB^Te^{A^Tt}dt$; the observation Gramian is $W_o(T)=\int_0^T e^{A^Tt}C^TCe^{At}dt$.

A descriptor system $E\,x'=Ax$ has regular pencil when $\det(sE-A)$ is not identically zero. Consistent initial states must satisfy its algebraic constraints.

For scalar $x'=u$ and cost $\int(x^2+u^2)dt$, a value coefficient P solves a Riccati equation; the admissible horizon and terminal value are part of its definition.

Theorem and hypotheses

If $P>0$ and $A^TP+PA=-Q$ with $Q>0$, then $V(t)\le e^{-\lambda_{min}(Q)t/\lambda_{max}(P)}V(0)$. For any $T>0$, $W_c(T)>0$ iff $[B,AB,\dots,A^{n-1}B]$ has rank n; the analogous observation statement holds. For $x'=u$ on $[0,T]$ with zero terminal cost, $P(t)=\tanh(T-t)$ gives the optimal feedback $u=-P(t)x$ and value $P(0)x(0)^2$.

Proof

Differentiate V to get $V'=-x^TQx\le-[\lambda_{min}(Q)/\lambda_{max}(P)]V$ and integrate. For the Gramian, $v^TW_cv=\int||B^Te^{A^Tt}v||^2dt$. It vanishes exactly when this analytic function is zero on the interval, equivalently $B^T(A^T)^kv=0$ for every k. Cayley–Hamilton reduces this to $k<n$, proving the rank equivalence and the dual observation result. For the Riccati model, $P'=P^2-1$ and $P(T)=0$. Along any admissible control, $x^2+u^2+d(Px^2)/dt=(u+Px)^2$. Integration gives $\text{cost}=P(0)x(0)^2+\int(u+Px)^2dt$ because the terminal term vanishes. The proposed feedback makes the square zero and therefore attains the exact lower bound.

Worked example

Use the nonnormal stable matrix $A=[[-1,4],[0,-1]]$, then the constrained pencil $E=\mathrm{diag}(1,0)$, $A_d=\mathrm{diag}(-1,1)$.

At $x=(1,1)$, $d||x||^2/dt=4>0$ although both eigenvalues are -1. The exact positive matrix $P=[[1/2,1],[1,9/2]]$ has determinant $5/4$ and satisfies $A^TP+PA=-I$. It supplies a valid metric certificate without denying transient Euclidean growth. For the descriptor pencil $\det(sE-A_d)=-(s+1)$, so its only finite eigenvalue is -1 and consistent solutions obey $x_2=0$. Replacing E by $E+\varepsilon I$ produces a new second eigenvalue $1/\varepsilon$. This unstable mode belongs to the altered model. For $A=\mathrm{diag}(-1,-2)$,$B=(1,1)^T$, the controllability determinant is -1, and the infinite-horizon Gramian $[[1/2,1/3],[1/3,1/4]]$ has determinant $1/72$.

Exact energy and rank identities reveal which property is proved and which apparent instability or stability belongs only to a changed model.

Continuity across the source years

2023 — The source distinguishes sufficient decay in a correct metric from the false claim that every stable system decreases Euclidean energy. Sources: 2023:M069

2024 — PID questions acquire an actual plant and characteristic-polynomial stability calculation. Sources: 2024:Y24-07-005

2025 — Scalar LQR and Riccati examples provide complete value proofs; linear action becomes a Gramian control-energy problem. Sources: 2025:Y25-07-010, 2025:Y25-11-030, 2025:Y25-12-008

2026 — Metric-compatible semigroup statements and descriptor pencil examples distinguish dynamics, constraints, minimal indices and observations. Sources: 2026:Y26-03-001, 2026:Y26-08-001, 2026:Y26-08-002, 2026:Y26-08-006

Connections

D09 — Stability depends on dynamics and metric. A Hurwitz matrix admits a suitable quadratic certificate but can exhibit nonnormal transient amplification. A decay theorem in one norm does not transfer unchanged to every norm.

D12 — Riccati feedback attains a global lower bound. Completion of squares certifies the value and the minimizer. Terminal and admissibility conditions cannot be omitted.

D03 — Matrix pencils encode constraints and hidden indices. Polynomial nullspaces and finite eigenvalue sensitivities have different algebraic certificates. A singular E need not make the pencil singular; simple-eigenvalue formulas need nonzero normalization.

Capacity and interactive scope

Lab ids: []

Inputs: Declared A,B,C or descriptor E,A; positive metric and horizon.

Outputs: Exact Lyapunov identities, rank conditions, Gramian energy or Riccati value; descriptor consistency checks.

Validation: Standalone rational matrix verification checks P, transient growth, Gramian determinant and descriptor roots; the Riccati square identity is derived analytically.

Status: Mathematical recipes with exact finite checks; no general control simulator is implemented.

Limits

Stability of a transfer function need not imply internal stability in canceled modes. A singular-vector projection is an iteration limit only under appropriate spectral hypotheses. A descriptor intervention gradient assumes a simple eigenvalue and nonzero left-right normalization. Historical benchmark margins, gains and speedups require the unrecovered system and measurements.

Next questions

Can a claimed contraction product be checked in one compatible norm on every component’s actual invariant domain?

Which descriptor regularizations preserve the consistent constrained dynamics instead of creating spurious modes?

Source anchors: 2023:M069, 2025:Y25-07-010, 2025:Y25-11-030, 2025:Y25-12-008, 2026:Y26-08-001, 2026:Y26-08-006

Boyd, Linear quadratic Lyapunov theory — https://web.stanford.edu/class/ee363/lectures/lq-lyap.pdf
Berger and Trenn, The quasi-Kronecker form for matrix pencils — https://stephantrenn.net/wp-content/uploads/2017/09/BergTren12.pdf

D15-F03 · Regularized inverse problems: bias, noise and certified stopping

How should finite resolution be chosen when inversion amplifies measurement error?

Inverse problems make a precise tradeoff visible: a richer inverse reduces approximation bias but amplifies noise. A source class says which signals are admissible; a deterministic noise ball says which data can be trusted. The archive’s strongest late analytic development derives matching upper and lower recovery rates for concrete diagonal models rather than asserting convergence from a suggestive cutoff formula. Mild singular-value decay gives a power-law rate, severe decay gives a logarithmic rate, and the latter requires a lower-order safety margin. Tikhonov and Landweber express the same controlled

approximation through different spectral filters. The constructive payoff is a parameter or stopping rule with a declared worst-case bound. A rate is not just a fitted slope: an indistinguishable pair shows that no recovery map can uniformly do better on the stated signal class.

Definitions

Observe $y^\delta=Ax+\eta$ in Hilbert norm with $||\eta||\le\delta$ and x in a source class M. A linear reconstruction R has bias $B_M(R)=\sup_{x\in M}||(RA-I)x||$.

For diagonal $Ae_k=\sigma_k e_k$, spectral cutoff R_N retains $k\le N$ and divides those coordinates by σ_k.

Use the ellipsoid $M_r=\{x:\Sigma k^{2r}|x_k|^2\le 1\}$. Mild decay is $\sigma_k=k^{-p}$; severe decay is $\sigma_k=e^{-ck^\beta}$, with all parameters positive.

Tikhonov $R_\alpha=(A^* A+\alpha I)^{-1}A^*$ and Landweber early stopping are bounded spectral filters approximating the unbounded inverse.

Theorem and hypotheses

For any bounded R, the worst-case error is at most $B_M(R)+\delta||R||$. Mild cutoff $N=\operatorname{floor}(\delta^{-1/(p+r)})$ has minimax order $\delta^{r/(p+r)}$. Severe cutoff with $L=\log(1/\delta)$ and $N=\operatorname{floor}(((L-(r/\beta)\log L)/c)^{1/\beta})$ has minimax order $L^{-r/\beta}$ for sufficiently large L. Merely requiring $N\sim(L/c)^{1/\beta}$ does not guarantee convergence.

Proof

Decompose $Ry^\delta-x=(RA-I)x+R\eta$ and use the triangle inequality. On M_r the cutoff bias is $(N+1)^{-r}$; its noise norm is δ/σ_N. For polynomial singular values, balance N^{-r} with δN^p to obtain the power-law upper bound. Choose K of order $\delta^{-1/(p+r)}$ and $x_\pm=\pm K^{-r}e_K$. Both signals lie in the ellipsoid and have $||Ax_\pm||\le\delta$ after a harmless integer adjustment, so the same datum zero is admissible for both. Any estimator must err by at least K^{-r} on one, proving the matching order. For severe decay, the stated safety margin gives $\delta e^{cN^\beta}\le L^{-r/\beta}$ while N^{-r} has the same order. A K of order $(L/c)^{1/\beta}$ gives the analogous two-point lower bound. But with $c=\beta=1,\delta=e^{-j},N=j$, the leading asymptotic cutoff holds and noise amplification equals one; convergence fails.

Worked example

Compare mild $p=r=1$ at $\delta=10^{-4}$ with severe $c=\beta=r=1$ at $\delta=e^{-100}$. Then inspect two alternative filters.

Mild cutoff $N=100$ gives error $\le 1/101+1/100<0.02$, while the indistinguishable signals $\pm e_{100}/100$ force worst-case error ≥ 0.01. Severe safe cutoff gives $N=95$ and bound $1/96+e^{-5}<0.01716$; $N=100$ instead has noise bound one. Under $x=(A^*A)^\nu w$, $||w||\le\rho$ and $0<\nu\le 1$, Tikhonov gives bias $\le\rho c_\nu\alpha^\nu$ and noise $\le\delta/(2\sqrt{\alpha})$, where $c_\nu=\nu^\nu(1-\nu)^{1-\nu}$ and $c_1=1$. Balancing gives rate $\delta^{2\nu/(2\nu+1)}$ with the corresponding ρ scaling. For Landweber with $0<\omega\le||A||^{-2}$, m steps give bias $\le\rho(\nu/(e\omega m))^\nu$ and noise $\le\delta\sqrt{(\omega m)}$. More iterations can therefore worsen the total guarantee after the bias-noise balance.

The admissible class, operator spectrum and noise model jointly determine useful resolution; a cutoff or iteration count has no universal meaning on its own.

Continuity across the source years

2024 — A filter is tied to an actual frequency response and stability certificate. Sources: 2024:Y24-11-002

2025 — ISTA and conjugate gradients are assigned different objective, stopping and convergence contracts. Sources: 2025:Y25-08-021

2026 — Moment inversion emphasizes model order and conditioning; the supplemental August treatments make cutoff quantifiers, minimax lower bounds, qualification and early stopping explicit. Their dates refer to source topics rather than proof discovery. Sources: 2026:Y26-07-020, 2026:Y26-08-901, 2026:Y26-08-902, 2026:Y26-08-903, 2026:Y26-08-904, 2026:Y26-08-905

Connections

D07 — Spectral calculus supplies reconstruction filters. The norms sup σ/(σ²+α) and residual sup λ^ν α/(λ+α) give explicit Tikhonov constants. The stated simple qualification proof uses 0<ν≤1.

D11 — Indistinguishable data impose error floors. Opposite source signals and admissible noise produce the same datum. This deterministic minimax argument does not assume a Bayesian prior.

D12 — Parameter choice balances competing errors. Minimize a proved bias-plus-noise envelope over cutoff, α or iteration count. Oracle source smoothness or noise bounds must be known or estimated under new assumptions.

Capacity and interactive scope

Lab ids: ['L04']

Inputs: Positive decay and smoothness parameters, noise level and cutoff.

Outputs: Bias, noise amplification, total envelope and admissible cutoff comparisons.

Validation: Closed-form bounds are checked against exact finite diagonal models; the severe counterexample is exposed explicitly.

Status: Implemented cutoff lab; Tikhonov and Landweber are documented mathematical recipes.

Limits

Theorems assume the declared source class and norm-bounded noise. Arbitrary smoothness does not transfer through Tikhonov beyond its qualification without additional treatment. Tuning formulas using ρ assume ρ>0; if ρ=0 the source is known zero. A local Newton enclosure requires its own domain, inverse-Jacobian and contraction checks. Small numerical residuals alone neither establish a root count nor certify inversion.

Next questions

Can a data-dependent parameter choice achieve the same guarantees without knowing source smoothness?

Which non-diagonal observation models admit computable upper bounds and matching indistinguishable-pair lower bounds?

Source anchors: 2026:Y26-08-901, 2026:Y26-08-902, 2026:Y26-08-903, 2026:Y26-08-904, 2026:Y26-08-905

Clason, Regularization of Inverse Problems, author lecture notes — https://arxiv.org/abs/2001.00617

D15-F04 · Cheapest distinguishing experiments and uncertainty-aware decisions

Which next observation resolves the target at minimum guaranteed cost, and when does additional data remain irrelevant?

Observation and decision become a single constructive problem when the remaining states and available tests are finite and explicit. A test splits the current feasible set into outcome fibres. A target is resolved when it is constant on the surviving set, even if the full hidden state remains unknown. Positive test costs

lead to an exact worst-case Bellman recurrence with matching lower and upper proofs. This is a rigorous version of choosing the cheapest distinguishing experiment, and it complements linear target-factorization: one asks which extra measurement destroys the harmful ambiguity. Correlated uncertainty adds a geometric version. An ellipsoid has exact target extrema, while a hidden-target conditional variance can remain positive even with noiseless recorded channels. More readings do not automatically remove uncertainty if they repeat the same direction or measure variables unrelated to the target’s hidden component. The archive’s decision examples become strongest when their uncertainty sets and terminal target criterion are stated together.

Definitions

Let S be a finite feasible state set, τ:S→T a target, and e a deterministic experiment of positive cost c_e with outcome e(s). Its outcome fibre is S_{e,y}={s∈S:e(s)=y}.

A terminal set is one on which τ is constant. A useful test has every nonempty outcome fibre a proper subset of the current nonterminal set.

For correlated uncertainty E={μ+Σ^{1/2}z:||z||≤r}, Σ is positive definite and a linear target is $a^{T}x$.

Worst-case decision cost, expected cost under a prior and statistical information gain are different objectives.

Theorem and hypotheses

If the available tests separate every pair of states having different target values, the least worst-case cost is V(S)=0 on terminal sets and otherwise V(S)=min_e[c_e+max_{y:S_{e,y}≠∅}V(S_{e,y})], over useful tests. For the ellipsoid E, the exact linear-target range is $a^{T}μ±r√(a^{T}Σa)$.

Proof

Any decision tree chooses a first test e. An adversary can return the outcome whose remaining subtree has largest optimal cost, so every tree costs at least c_e+max_yV(S_{e,y}). Minimizing gives the lower bound. Conversely attach optimal subtrees to a minimizing test. Every nonempty fibre is smaller, so induction on |S| constructs a tree attaining that same value. Pairwise target separation guarantees that a nonterminal set has a splitting test; otherwise two different target values could never be distinguished. Positive cost lets us discard useless first tests. This proves existence and exact optimality. For the ellipsoid write $a^{T}x=a^{T}μ+(Σ^{1/2}a)^{T}z$. Cauchy–Schwarz gives the stated extrema, attained at $z=±rΣ^{1/2}a/√(a^{T}Σa)$ when a≠0. Correlation is retained through the quadratic form rather than discarded in independent coordinate bars.

Worked example

Identify one of states a,b,c. Test e_1 costs one and isolates a; test e_2 costs two and isolates b; a direct three-way test costs four. Separately use μ=0,Σ=[[1,ρ],[ρ,1]], target x_1+x_2.

Testing e_1 first costs one if the answer is a and otherwise requires e_2, for worst-case cost three. Starting with e_2 also has worst-case cost three, while the direct test costs four. Neither cheaper one-test partition resolves every state, so the Bellman lower bound matches three. If the target merely distinguishes a from {b,c}, e_1 alone costs one; state identification was an unnecessarily demanding task. For the ellipsoid target, the exact range is ±r√(2+2ρ). Thus positive and negative correlations have opposite effects on the sum, and treating the axes independently can conceal this difference. In the linear observation model Hx=(x_1,x_2), target x_3 remains unobservable regardless of how often the first two coordinates are read.

An experiment is valuable relative to the target and the current uncertainty set. The optimal policy comes with a lower-bound argument, not just an appealing test order.

Continuity across the source years

2023 — Competing response laws and identifiable parameters establish that an experiment needs a model whose alternatives it can distinguish. Sources: 2023:M082

2025 — Finite sensor aliasing gives exact examples of measurements that do not separate candidate states. Sources: 2025:Y25-06-018

2026 — Joint enclosures, correlated target ranges and conditional error floors enrich the uncertainty model; October adds an exact finite worst-case adaptive policy theorem. Sources: 2026:Y26-08-016, 2026:Y26-09-020, 2026:Y26-09-021, 2026:Y26-09-022, 2026:Y26-10-022

Connections

D12 — Observation choice is an optimization problem. Bellman recursion couples test costs with worst-case remaining fibres. An expected-cost objective under a prior requires a different recurrence.

D11 — Recorded variables can leave a target error floor. Conditional target variance quantifies uncertainty surviving an observation σ-algebra. The finite deterministic test theorem does not assume a statistical distribution.

D14 — A distinguishing word is an adaptive experiment. Future-equivalence counterexamples identify experiments that split otherwise merged states. Allowed actions and their costs must match the actual implementation.

Capacity and interactive scope

Lab ids: ['L01', 'L08']

Inputs: Recipe: finite states, targets and costly tests. L01 uses a fixed linear probe menu; L08 uses one shared-uncertainty model.

Outputs: Recipe: exact adaptive decision policy. L01 outputs cheapest static panels; L08 outputs exact interval decision margins.

Validation: Standalone Bellman enumeration checks the three-state optimum and target-only variant; ellipsoid support values use exact quadratic forms when rational.

Status: L01 implements static cheapest linear probe panels and L08 shared scalar uncertainty. Adaptive Bellman design is a checked recipe, not an implemented decision-tree editor.

Limits

The finite theorem assumes deterministic tests and a complete state/test table. Noisy outcomes require posterior feasible sets and a robust terminal condition. Repeated samples of the same deficient geometry do not create new linear rank. A development-set optimum does not establish future prediction performance, causal benefit or hardware savings. Historical panel timings need their original datasets and evaluation protocol.

Next questions

Which targets can be resolved without identifying the complete state in a supplied experiment menu?

How does one preserve a computable lower bound when observations are noisy, costs depend on history or uncertainty sets are continuous?

Source anchors: 2026:Y26-08-016, 2026:Y26-09-020, 2026:Y26-09-022, 2026:Y26-09-024, 2026:Y26-10-019, 2026:Y26-10-022

Boyd and Vandenberghe, Convex Optimization — https://web.stanford.edu/~boyd/cvxbook/
Douglas, On majorization, factorization, and range inclusion — https://doi.org/10.1090/S0002-9939-1966-0203464-1

D16 · Quantum, deformation and integrable structures: finite models with explicit boundaries

Organizing question

Which state, product, coherence equation or spectral model is actually implemented, and what finite certificate proves its claimed behaviour?

This domain brings together quantum states and channels, finite noncommutative systems, deformation quantization, braid and fusion data, integrability and formal obstruction theory. Their common feature is that an appealing symbolic relation becomes useful only after its model class and normalization are fixed. The six families supply explicit finite examples and derive their identities, while distinguishing mathematical structure from physical realization, asymptotic convergence and general classification.

Quantum states make the atlas-wide observation problem especially concrete. The Bell state and its dephased mixture agree on all diagonal observations and both one-qubit marginals, but an off-diagonal correlation separates them. This gives a complete finite collision and a constructive additional measurement. Kraus formulas then supply the correct transformation standard: complete positivity is preserved with arbitrary ancillas, and the completeness equation preserves total probability. Instruments retain every measurement branch, so a conditional normalized success state cannot conceal its success probability. Error correction similarly depends on the specified error list, as the repetition code's exact bit-flip correction coexists with quantum distance one.

Finite Weyl systems identify what can hold exactly in finite dimension. Trace forbids the canonical additive commutator, and determinants force a multiplicative commutation phase to be a root of unity. Clock and shift matrices then generate a full finite matrix algebra. Binary symplectic kernels compute Pauli symmetries, while finite quadratic forms and Gauss sums retain the central phase in a projective modular relation. These calculations distinguish finite rational models from irrational operator algebras and distinguish alternating commutation forms from symmetric quadratic polarizations.

Polynomial Moyal calculus supplies the most direct implemented interaction. Its formal star product has exact finite falling-factorial coefficients for monomials. Degree bounds determine which odd corrections can survive, so the exact degree-four family and a genuine fifth-order obstruction are compatible. The new bounded sextic monomial atlas exhausts a declared 784-pair universe and exposes four cases missed by the cubic test. Its complete bounded scope is valuable precisely because general polynomial cancellations and unbounded higher degree remain separate questions.

Braid and fusion examples then make coherence visible. An algebraic Hecke matrix needs no unitarity, a fusion ring needs no associator, and an associator needs pentagon and hexagon equations. The four-object pointed model provides complete finite data and proofs, while Temperley–Lieb Gram forms test positivity independently. Integrable models provide another bridge between descriptions: a finite XXX matrix agrees with regular Bethe roots when the infinite-root state is included, and a two-site Gaudin eigenvector matches a scalar differential potential. KdV tau-functions illustrate nonlinear interaction through a verified exponential coefficient identity.

Finally, obstruction theory explains why first-order success is insufficient. A characteristic-two character cannot lift, a free rank-two family has jumping Hochschild cohomology and an order-twelve equation can fail at order thirteen. These are exact worked models, not negative verdicts about unrecovered larger objects. The domain's interactive capacities are therefore bounded and auditable: compute, classify within

a stated finite universe, inspect a witness and identify the next unresolved equation.

Core objects

Density matrices, effects, Kraus maps and instruments

Clock–shift operators and binary symplectic supports

Polynomial Moyal coefficients

Fusion labels and F/R/S coherence data

Finite integrable Hamiltonians and tau-functions

Formal parameter rings and obstruction equations

Open frontier

Extend the bounded monomial Moyal classification to arbitrary polynomial coefficient varieties; recover full anyon and monodromy matrices; prove all-order or completeness statements only with their actual obstruction or spectral data.

D16-F01 · Quantum states, coherence, channels and complete instruments

Which observations distinguish states, and which transformations preserve their physical probability structure?

The quantum information strand is best organised around a state, an observation map and an allowed transformation. A density matrix carries both populations and coherences; a measurement sees only selected linear functionals of that matrix. A channel must preserve positivity even when the system is coupled to an arbitrary ancilla, which is stronger than preserving trace or ordinary positivity. An instrument additionally records the classical measurement branch. These distinctions connect the early coherence example to later dephasing, error correction, leakage and sequential postselection discussions.

The Bell state and its fully dephased mixture are an especially clear organising pair. Computational-basis probabilities, diagonal parity and both one-qubit marginals agree exactly, while an off-diagonal correlation separates them. This is a complete finite instance of observation-relative identifiability, linking quantum tomography to the atlas-wide question of what a target can be reconstructed from. It does not imply that all measurements are blind; it identifies one observation algebra and an explicit additional observable that resolves its collision.

The transformation side is equally constructive. Kraus operators certify complete positivity by turning the channel tensored with any ancilla into a sum of positive conjugations. Their completeness equation certifies total probability. Dephasing changes only selected coherences, amplitude damping transfers populations and depolarisation contracts every traceless component. Error correction requires an actual error family: the three-qubit repetition code corrects single bit flips but has a weight-one logical phase error, hence quantum distance one. Sequential instruments force the success probability to remain visible through every branch. Normalizing a successful state is legitimate conditional probability, but cannot convert a rare success into an unconditional implementation guarantee.

Definitions

A density matrix ρ is positive semidefinite with $\mathrm{tr}\rho=1$; an effect satisfies $0\le E\le I$ and has probability $\mathrm{tr}\rho E$.

A Kraus channel is $E(\rho)=\Sigma_r K_r \rho K_r^*$, with $\Sigma_r K_r^* K_r=I$.

An instrument has branches $I_a(\rho)=\Sigma_r K_{ar}\rho K_{ar}^*$; only their sum is required to preserve trace.

Knill–Laflamme correction for code projector P requires $PE_a^*E_bP=c_{ab}P$ for the specified error family.

Theorem and hypotheses

For Bell state $\rho_B=|\Phi^+\rangle\langle\Phi^+|$ and mixture $\rho_M=(|00\rangle\langle 00|+|11\rangle\langle 11|)/2$, all diagonal observations and both one-qubit marginals agree, but $\mathrm{tr}(\rho_B XX)=1$ and $\mathrm{tr}(\rho_M XX)=0$. Their trace distance is $1/2$. Sequential complete instruments preserve total probability after summing all branches.

Proof

The difference $D=\rho_B-\rho_M=(|00\rangle\langle 11|+|11\rangle\langle 00|)/2$ has zero diagonal, so $\mathrm{tr}(DE)=0$ for every diagonal E. Partial trace removes the off-diagonal tensor terms, proving marginal equality. XX exchanges 00 and 11, so $\mathrm{tr}(DXX)=1$. On their two-dimensional span D has eigenvalues $\pm 1/2$ and vanishes elsewhere; $\|D\|_1/2=1/2$. For sequential instruments, let first-stage Kraus operators be K_{ar} and second-stage ones $L_{bs|a}$ with $\Sigma_r K_{ar}^*K_{ar}=I$ and $\Sigma_s L_{bs|a}^*L_{bs|a}=I$ for each a. The composed operators are $L_{bs|a}K_{ar}$. Summing their adjoint products yields $\Sigma_r K_{ar}^*(\Sigma_s L_{bs|a}^*L_{bs|a})K_{ar}=I$. The same Kraus form proves complete positivity. This algebra establishes the unconditional map; a selected branch has trace equal to its actual success probability.

Worked example

Apply dephasing to the first qubit of a Bell pair, with rational coherence parameter $q=1/3$.

The channel is $E_q(\rho)=((1+q)/2)\rho+((1-q)/2)Z\rho Z$. It produces Bell populations $1/2,1/2$ and cross terms $q/2$. Hence the reduced states remain $I/2$, ZZ stays one and XX becomes $1/3$. Its Kraus squared weights are $2/3$ and $1/3$, nonnegative and summing to one. The dephased state has eigenvalues $(1\pm q)/2=2/3,1/3$ on the Bell subspace, so it is positive. At $q=0$ it equals ρ_M; at $q=1$ it equals ρ_B. An independent repetition-code check gives $PE_a^*E_bP=\delta_{ab}P$ for $\{I,X_1,X_2,X_3\}$, while $PZ_1P=\mathrm{diag}(1,-1)$ on the code, demonstrating the exact boundary of the protection claim.

Coherence, entanglement, observation and protection each have explicit witnesses. Their connection is mathematical rather than an inference from a gate or code name.

Continuity across the source years

2023 — Measurement-dependent coherence and physical open-system dynamics establish state and channel definitions. Sources: 2023:M048, 2023:M063, 2023:M049, 2023:M085

2024 — Two-qubit dephasing, GHZ coherence, Stinespring dilation and stabilizer models develop observation and protection examples. Sources: 2024:Y24-06-023, 2024:Y24-12-037, 2024:Y24-12-029, 2024:Y24-12-035, 2024:Y24-11-024

2025 — Exact correction conditions, mixing laws and parity collisions expose the error model and missing observations. Sources: 2025:Y25-02-019, 2025:Y25-02-023, 2025:Y25-01-024, 2025:Y25-02-021, 2025:Y25-01-003

2026 — Sequential instruments and orthogonal Schmidt blocks keep branch probabilities and sector hypotheses explicit. Sources: 2026:Y26-06-006, 2026:Y26-07-008, 2026:Y26-05-024, 2026:Y26-06-005

Connections

D15 — Observation-relative reconstruction. A state difference is invisible exactly when it lies in the kernel of all measured effects. A classical observability lab supplies an analogy; its matrices are not quantum tomography data.

D11 — Tomographic uncertainty. Finite samples estimate probabilities; exact state separation does not provide sample complexity. Noise and measurement allocation need a declared statistical model.

D05 — Stabilizer homology. Toric-code logical loops are cycles modulo boundaries. The code distance theorem assumes the specified periodic lattice and local Pauli errors.

Capacity and interactive scope

Lab ids: ['L01', 'L09']

Inputs: State matrix, effects, Kraus branches, code projector and explicit error list.

Outputs: Probabilities, remaining coherence, branch success weights and exact correction residuals.

Validation: Check positivity or Kraus construction, completeness, normalization and every declared error pair.

Status: Linked labs illustrate observability and homology; quantum states and instruments are worked mathematical models.

Limits

This is finite-dimensional mathematics, not a hardware threshold or a leakage-free anyon implementation. Negative dephasing time need not define a channel. Postselection costs and unobserved branches cannot be discarded.

Next questions

What minimal added effects make the actual proposed observation map injective on its admissible state family?

For an actual leakage instrument, what is the total success probability after all conditional stages and recovery branches?

Source anchors: 2023:M048, 2023:M063, 2024:Y24-06-023, 2024:Y24-12-037, 2025:Y25-02-019, 2025:Y25-02-023, 2026:Y26-06-006, 2026:Y26-07-008

Knill and Laflamme, Theory of quantum error-correcting codes — https://arxiv.org/abs/quant-ph/9604034
Lindblad, On the generators of quantum dynamical semigroups — https://doi.org/10.1007/BF01608499

D16-F02 · Finite Weyl systems, symplectic Pauli geometry and projective phases

What finite operator relations can hold exactly, and how are their commutation and central phases certified?

Finite quantum algebra has a precise boundary. The canonical relation $[Q,P]=i\hbar I$ cannot hold for finite matrices when $\hbar\neq 0$, because the trace of a commutator is zero. Clock and shift operators provide a consistent finite replacement, with multiplicative relation $UV=qVU$. Taking determinants forces q to be a root of unity, so a rational phase is an exact existence condition, whereas an irrational phase cannot have an exact finite invertible representation. Rational approximation changes the algebraic parameter; it does not remove this obstruction.

The finite models connect naturally to symplectic geometry. Pauli operators modulo scalar phase are labelled by binary vectors, and their commutation sign is an alternating bilinear form. Centralizers of a specified nonzero support become exact symplectic kernels. Sector transition counts are images of another pairing map and therefore require its actual kernel, not a superficially similar quotient. These finite linear maps are a constructive common core of symmetry reduction, stabilizer codes and operator selection rules.

Fourier and Gauss operators add projective phases. A quadratic refinement produces a symmetric polar form, while Pauli commutation uses an alternating form. At odd order these cannot be identified indiscriminately. The finite Weil relation $(ST)^3=\gamma C$ records a Gauss phase γ that must be computed rather than normalized away without explanation. For the order-three example $\gamma=i$. This family therefore keeps three distinct certificates together: trace and determinant obstructions; exact finite operator bases; and central phases in group relations. It does not promote one finite matrix pair into categorical coherence or an infinite irrational rotation algebra.

Definitions

For q of order L, $Ue_j=q^je_j$ and $Ve_j=e_{j+1 \bmod L}$ form a clock–shift pair.

For n qubits, Pauli labels $v=(x,z)\in F_2^{2n}$ have symplectic form $\omega(v,w)=x\cdot z'+z\cdot x'$.

For nonzero Pauli support S of H, its Pauli centralizer labels are $\operatorname{span}(S)^\perp$.

A finite quadratic form Q has polar form $b_Q(x,y)=Q(x+y)-Q(x)-Q(y)$; the associated Gauss sum is $\gamma=|A|^{-1/2}\Sigma_x \exp(2\pi i Q(x))$.

Theorem and hypotheses

Finite invertible matrices U,V satisfying $UV=qVU$ in dimension d force $q^d=1$. If q has order L, the displayed L-dimensional clock–shift pair exists and generates $M_L(C)$. For $H=\Sigma_{s\in S}h_sP(s)$, with distinct labels and nonzero h_s, $P(v)$ commutes with H exactly when $\omega(v,s)=0$ for every $s\in S$.

Proof

Taking determinants in $UV=qVU$ yields $\det U \det V=q^d \det V \det U$, and invertibility permits cancellation. Conversely the stated pair satisfies $UV e_j=q^{j+1}e_{j+1}=qVUe_j$, including wraparound. U has distinct eigenvalues; Lagrange interpolation in U gives every coordinate projection E_{jj}. Multiplying by powers of V then gives all matrix units, proving full generation and scalar commutant. For Pauli operators, $XZ=-ZX$ gives $P(v)P(s)=(-1)^{\omega(v,s)}P(s)P(v)$. Vanishing pairings imply commutation term by term. Conversely, in the commutator the anticommuting terms are nonzero multiples of distinct Pauli basis elements labelled $v+s$. They cannot cancel, so every pairing must vanish. Nondegeneracy of ω then gives kernel dimension $2n-\operatorname{rank}_{F_2}(S)$. This counts Pauli labels, not the full vector-space dimension of the operator commutant.

Worked example

Take $n=2$ and $H=Z_1+Z_2$. Compare the finite pair at order three and the order-three Gauss phase.

For H, the two support equations force $x_1=x_2=0$ while z_1,z_2 remain free. The four Pauli symmetries are I,Z_1,Z_2,Z_1Z_2 modulo phase. The full operator commutant is larger because the zero eigenvalue is degenerate, so the binary dimension two must not be called its complex matrix dimension. For a primitive cube root ζ, the nine products U^aV^b are trace-orthogonal: a nontrivial shift has zero trace and $1+\zeta+\zeta^2=0$. Thus they form a basis of $M_3(C)$. For $Q(x)=x^2/3$ on $Z/3$, $\gamma=(1+2\zeta)/\sqrt{3}=i$, and the finite Fourier/Gauss matrices obey $S^2=C$, $(ST)^3=iC$. The phase is part of the exact relation.

The finite system has a complete algebraic identity and an explicit symplectic label geometry. Projective phases and representation dimensions remain separate invariants.

Continuity across the source years

2025 — Finite canonical-commutator impossibility, Harper matrices and Bogoliubov constraints establish exact operator boundaries. Sources: 2025:Y25-07-015, 2025:Y25-04-017, 2025:Y25-04-026

2026 — Root-of-unity Weyl pairs, Gauss phases and symplectic centralizers give constructive finite replacements and sector calculations. Sources: 2026:Y26-06-001, 2026:Y26-06-002, 2026:Y26-06-003, 2026:Y26-07-006, 2026:Y26-07-007, 2026:Y26-05-009, 2026:Y26-06-004

Connections

D06 — Symplectic forms. Pauli commutation is an alternating finite-field pairing. A quadratic polar form is symmetric; odd-order and characteristic-two conventions differ.

D01 — Cyclotomic phases. Root-of-unity order constrains dimensions and exact arithmetic fields. No irrational finite representation follows from rational approximants.

D03 — Representation and commutants. Generating every matrix unit proves irreducibility and scalar commutant. A Hamiltonian centralizer may be larger than the joint clock–shift commutant.

Capacity and interactive scope

Lab ids: ['L07']

Inputs: Conductor L, invertible clock/shift matrices or a reduced nonzero Pauli support table.

Outputs: Commutation phase, generated basis, symplectic kernel, sector image and explicit central scalar.

Validation: Check root order, remove cancelled support terms and distinguish exact root-of-unity arithmetic from floating phases.

Status: The cyclotomic lab illustrates exact finite fibres; Weyl and Pauli calculations remain documented recipes.

Limits

No finite CCR representation or irrational exact pair is claimed. A Pauli centralizer is not the full operator commutant. Finite Weil operators do not supply arbitrary F/R coherence data.

Next questions

Given the original Hamiltonian support, which claimed sector quotient is the actual kernel of its transition map?

Which scalar rephasing or metaplectic lift turns a specified projective modular action into a genuine representation?

Source anchors: 2025:Y25-07-015, 2025:Y25-04-017, 2026:Y26-06-001, 2026:Y26-06-002, 2026:Y26-06-003, 2026:Y26-07-006, 2026:Y26-07-007, 2026:Y26-05-009

Strömberg, Weil representations associated to finite quadratic modules — https://arxiv.org/abs/1108.0202

D16-F03 · Polynomial Moyal exactness and the complete bounded sextic monomial atlas

Exactly when does a polynomial Moyal bracket equal its Poisson bracket, and when does a vanished cubic correction hide a fifth-order defect?

The Moyal strand has a particularly strong cross-year synthesis because it begins with a formal product and ends with a finite coefficient classification. Fixing the Weyl convention makes every polynomial calculation exact. Antisymmetry removes the even bidifferential orders, while division by $i\hbar$ leaves only even powers $\hbar^{k-1}$. A different power convention cannot be mixed with these coefficients. For monomials, falling factorials turn differentiation into an integer formula; exactness is then the

simultaneous vanishing of every active higher odd coefficient.

Two apparently conflicting source ideas become compatible after the degree bound is respected. The family (qp³,q^{3r+2}p^r) has vanishing cubic correction, and all fifth or higher derivatives of the first input vanish because its total degree is four. It is therefore an infinite exact family. Higher-degree pairs can genuinely hide a fifth-order obstruction. The pair (qp⁵,q⁵p³) gives a nonzero $\hbar^4$ term despite $S_3=0$. The atlas adds a smaller witness entirely inside degree six and an exhaustive bounded monomial classification.

This added sextic result is a present synthesis, not a claim about what an earlier conversation proved. Among the twenty-eight monomials of total degree at most six, 504 of 784 ordered pairs are exact and 280 have a defect; four are invisible to the cubic test alone. Among the seven monomials of exactly degree six, fifteen of forty-nine ordered pairs are exact. These statements are complete for the declared finite monomial universe. Arbitrary linear combinations can cancel coefficients across different monomial pairs, so this enumeration is not a classification of all sextic polynomial pairs. At degree seven and above, S_7 and further terms become possible and must be retained.

Definitions

Set f⋆g=f exp[(iℏ/2)(←∂_q→∂_p−←∂_p→∂_q)]g and [f,g]_M=(f⋆g−g⋆f)/(iℏ).

For f=q^ap^b,g=q^cp^d, define S_k=Σ_{r=0}^k(−1)^r binom(k,r)(a)_{k−r}(b)_r(c)_r(d)_{k−r}, with falling factorials zero beyond the exponent.

The order-k term has coefficient (−1)^{(k−1)/2}ℏ^{k−1}S_k/(2^{k−1}k!) for odd k and output q^{a+c−k}p^{b+d−k}.

A bounded sextic monomial universe means nonnegative exponents with a+b≤6 and c+d≤6; it does not mean arbitrary sextic polynomial coefficients.

Theorem and hypotheses

For monomials of total degrees at most six, Poisson=Moyal if and only if $S_3=S_5=0$. The exhaustive universe has 504 exact and 280 defective ordered pairs; four defective pairs have $S_3=0$. For every r≥1, [qp³,q^{3r+2}p^r]_M=−(8r+6)q^{3r+2}p^{r+2}.

Proof

Expand the exponential and binomially expand its two commuting left-right derivative operators. Differentiating monomials gives the displayed falling-factorial sum. Swapping f and g changes the order-k bidifferential by (−1)^k, so even orders cancel and odd orders double. Division by iℏ gives the stated coefficient. Each input receives exactly k derivatives, so k exceeds neither total degree; with both degrees at most six, only k=1,3,5 survive. Distinct formal $\hbar$ powers force $S_3=S_5=0$ separately. Enumerating all nonnegative a+b≤6 gives 28 inputs and 784 ordered pairs; the attached verifier evaluates both coefficients with exact integers and independently recomputes the antisymmetric star expansion. For the infinite family set c=3r+2. Only two terms of S_3 survive and give 6c(c−1)[3r−(c−2)]=0. Orders k≥5 vanish by degree four. The Poisson coefficient is r−3(3r+2)=−8r−6, proving the family uniformly rather than by sampling.

Worked example

Use the newly stipulated sextic pair f=qp⁵, g=q⁴p², and compare the recovered exact family at r=1.

For a=1, odd coefficients factor as S_k=(b)_{k−1}(c)_{k−1}[kd−(b−k+1)(c−k+1)]. At b=5,c=4,d=2, $S_3=(5)_2(4)_2[6-3\cdot2]=0$. At k=5, $S_5=(5)_4(4)_4[10-1\cdot0]=120\cdot24\cdot10=28800$, giving coefficient 28800/(16·120)=15. The exact bracket is $-18q^4p^6+15\hbar^4p^2$. The four cubic-blind ordered exponent pairs are (1,5;4,2),(2,4;5,1) and their reversals. In contrast (qp³,q⁵p) has bracket −14q⁵p³ with no correction.

The higher-degree source replacement (qp^5, q^5p^3) has $S_5=201600$ and coefficient 105, so it gives $-22q^5p^7+105\hbar^4qp^3$.

Exactness is a finite coefficient statement with a declared universe. The cubic term is sufficient under a genuine degree-four bound, but not throughout degree six.

Continuity across the source years

2023 — Formal polynomial star multiplication establishes the convention and the classical-limit question. Sources: 2023:M086

2025 — The reflection-deformed polynomial representation is a related exact deformation calculation, not evidence that the later Moyal classification was already proved. Sources: 2025:Y25-02-015

2026 — The retrieved coefficient formula and family claim motivate corrected higher-order tests; the bounded sextic enumeration and smaller witness are newly authored atlas calculations. Sources: 2026:Y26-06-018, 2026:Y26-06-019, 2026:Y26-06-020

Connections

D03 — Exact polynomial algebra. Falling factorials and coefficient dictionaries reduce a formal identity to rational arithmetic. A numerical sample does not prove a polynomial identity.

D06 — Poisson geometry. The first bidifferential is the canonical Poisson bracket. Changing coordinates requires the appropriate transformed product; the flat formula cannot be copied blindly.

D14 — Finite certification. An exhaustive degree bound makes a finite enumeration complete. Completeness is lost if the exponent bound or polynomial model class is enlarged.

Capacity and interactive scope

Lab ids: ['L06']

Inputs: Four nonnegative monomial exponents and a declared $\hbar$ convention; optional bounded degree for enumeration.

Outputs: Poisson term, active higher coefficients, exactness verdict and first defect order.

Validation: Exact integer falling factorials and rational normalization; independently compare the direct star expansion; retain all active odd orders.

Status: Implemented bounded monomial worked-model lab, with exact coefficient certificates; not a general quantum computer algebra system.

Limits

The bounded classification covers monomial pairs only. General sextic polynomial pairs require coefficient equations allowing cross-pair cancellation. No complete unrestricted high-degree classification, convergence theorem for arbitrary smooth functions, or new historical attribution is claimed.

Next questions

What are the irreducible components of the coefficient variety of arbitrary homogeneous sextic polynomial pairs satisfying $S_3=S_5=0$?

For degree bounds above six, which families cancel S_3 and S_5 while retaining a first S_7 obstruction?

Source anchors: 2023:M086, 2026:Y26-06-018, 2026:Y26-06-019, 2026:Y26-06-020

Curtright and Zachos, Quantum Mechanics in Phase Space — https://arxiv.org/abs/1104.5269

D16-F04 · Braid representations, fusion rules and complete finite coherence data

Which explicit relations turn braid or fusion numbers into a consistent algebraic or categorical structure?

The braid and fusion discussions form a hierarchy of increasingly structured data. A braid representation requires matrices satisfying the braid relations. A Hecke relation imposes an additional quadratic quotient. A fusion ring records integer multiplication coefficients and a dimension homomorphism. A braided fusion category also requires associators and braidings satisfying pentagon and hexagon equations; a unitary realization requires positivity. A larger object count or a familiar exceptional dimension does not skip any of these stages.

Two finite models organise the hierarchy. The fundamental Hecke braid matrix provides a Laurent-polynomial Yang–Baxter calculation, with a simple rational specialization that is visibly nonunitary. The pointed four-object model supplies complete F,R,S data: labels add in F_2^2, the associator is one and a bilinear sign supplies the braiding. Every coherence equation reduces to a finite bilinearity identity, and character orthogonality gives modular nondegeneracy and Verlinde multiplication. This is a fully specified categorical example, whereas diagonalizing the Fibonacci fusion matrix alone certifies its ring data and dimensions but not every categorical coherence equation.

Temperley–Lieb provides a useful bridge between algebraic relations and positive quantum models. Rank-one projections satisfy the finite TL relations at a declared parameter; a trace then needs a separate positive Gram form. Already at two strands, $|\delta|<1$ produces a negative value on x^*x. Passing this small test does not prove positivity throughout an infinite tower. Similarly, a normalized bracket gives a conventioned trefoil Jones polynomial without becoming a complete knot invariant. The synthesis preserves these genuinely constructive models while making each change of mathematical type explicit.

Definitions

The Hecke braid matrix sends $00\mapsto q00$, $11\mapsto q11$, $01\mapsto 10$, $10\mapsto 01+(q-q^{-1})10$.

A fusion ring has a distinguished simple basis with nonnegative integer structure constants and a unit; it does not specify associators.

For $A=F_2^2$, use fusion $a\otimes b=a+b$, $F=1$, $R(a,b)=(-1)^{a_1b_2}$, twist $q(a)=R(a,a)$, and $S_{ab}=R(a,b)R(b,a)/2$.

The TL_2 algebra has $e^2=\delta e$, $e^*=e$, $\mathrm{tr}1=1$ and $\mathrm{tr}e=1/\delta$ for real nonzero δ.

Theorem and hypotheses

The pointed data above satisfy pentagon and hexagon, and S is real orthogonal with $S^2=I$. Its Verlinde coefficients equal one exactly when $c=a+b$. Separately the TL_2 trace form is positive semidefinite exactly when $|\delta|\geq 1$.

Proof

Since every associator equals one, each pentagon product is one. Each hexagon reduces to $R(a,b+c)=R(a,b)R(a,c)$ or $R(a+b,c)=R(a,c)R(b,c)$, which follows from bilinearity of a_1b_2 over F_2. Double braiding is $B(a,b)=(-1)^{a_1b_2+a_2b_1}$. For nonzero a, choose b supported in the complementary coordinate to get $B(a,b)=-1$, so the pairing is nondegenerate. Orthogonality follows from $\Sigma_x B(a+c,x)=4\delta_{ac}$: if $a+c$ is nonzero, pairing x with $x+x_0$ reverses the sign. Thus $S^2=I$. The Verlinde sum is $(1/4)\Sigma_x B(a+b+c,x)$, giving the group-addition fusion law. For TL_2 write $x=a+be$. Expanding x^*x and using $e^2=\delta e$ gives $\mathrm{tr}(x^*x)=|a|^2+2\mathrm{Re}(\mathrm{conj}(a)b)/\delta+|b|^2$. Its Gram matrix has eigenvalues $1\pm 1/\delta$; both are nonnegative exactly when $|\delta|\geq 1$. These proofs establish different, explicitly stated pieces of structure.

Worked example

Order the pointed labels 00,10,01,11, and compare a TL parameter that passes with one that fails.

The twist values are 1,1,1,−1. The S matrix is half of [[1,1,1,1],[1,1,−1,−1],[1,−1,1,−1],[1,−1,−1,1]]. All dimensions are one and the global squared dimension is four. Every fusion triple can be checked exactly using binary addition. For Fibonacci fusion, $N\tau=[[0,1],[1,1]]$ obeys $N\tau^2=I+N\tau$ and has positive dimension φ; this ring identity is independently complete without claiming a pentagon proof from one orthogonal F block. At $\delta=2$, the TL_2 Gram eigenvalues are 3/2 and 1/2. At $\delta=1/2$, $x=1-e$ has trace norm-square −2, excluding a positive Hilbert-space realization with that trace.

The finite pointed model has complete coherence data; the surrounding examples demonstrate exactly which smaller certificates are insufficient for a larger categorical claim.

Continuity across the source years

2024 — Braid deletion, Hecke matrices, TL quotient relations and a trefoil bracket give explicit algebraic tests. Sources: 2024:Y24-06-010, 2024:Y24-11-012, 2024:Y24-04-015, 2024:Y24-06-011

2025 — Fibonacci rings, pointed modular data and trace positivity separate fusion, coherence and unitarity. Sources: 2025:Y25-03-028, 2025:Y25-04-006, 2025:Y25-08-003, 2025:Y25-10-018, 2025:Y25-04-005, 2025:Y25-04-004, 2025:Y25-04-007, 2025:Y25-04-008

2026 — The conditional compact-group determinant twist supplies central arithmetic consequences while the asserted density step remains unproved without its actual matrices. Sources: 2026:Y26-10-035

Connections

D04 — Monoidal coherence. Associators and braidings must satisfy pentagon and hexagon; universal relations determine factorization through quotients. General higher categorical constructions are outside the finite model.

D05 — Braids and knots. A braid representation and normalized trace can produce knot invariants. Equal polynomials are not generally isotopy certificates.

D03 — Representations and positivity. Hecke matrices and trace Gram forms supply different certificates. An algebraic braid matrix need not be a unitary quantum gate.

Capacity and interactive scope

Lab ids: []

Inputs: Fusion labels, F and R data, trace normalization and explicit braid matrices.

Outputs: Coherence residuals, transparent subgroup, S orthogonality, fusion coefficients and positivity tests.

Validation: Check every finite pentagon/hexagon instance or a symbolic bilinearity proof; check Gram positivity separately.

Status: Complete four-object worked recipe; no arbitrary anyon, large-category or monodromy-density constructor is implemented.

Limits

The four-object model does not construct a proposed rank-384 or E_8 category. Orthogonality of one Fibonacci block is not all pentagon equations. The claimed SU(5) density requires recovered matrices, a unitarizing form and exclusion of proper closed subgroups.

Next questions

For the actual proposed anyon encoding, are all F/R blocks and leakage sectors available for a complete coherence and instrument audit?

Which explicit subgroup argument establishes the closure of a recovered braid representation after its determinant twist?

Source anchors: 2024:Y24-06-010, 2024:Y24-11-012, 2025:Y25-03-028, 2025:Y25-04-006, 2025:Y25-08-003, 2025:Y25-10-018

Etingof–Gelaki–Nikshych–Ostrik, Tensor Categories — https://math.mit.edu/~etingof/egnobookfinal.pdf
Müger, On the structure of modular categories — https://doi.org/10.1112/S0024611503014187

D16-F05 · Integrable models as finite spectral and tau-function bridges

How can an integrability claim be anchored simultaneously in an explicit operator and a verified scalar equation?

The integrable strand contains two complementary constructive bridges. A finite quantum spin chain can be diagonalized directly and compared with its Bethe parameterization. A classical nonlinear wave can be built from a tau-function whose bilinear identity fixes its interaction coefficient. In both cases a normalization connects descriptions that would otherwise remain a slogan. The bridge is useful because each side supplies an independent exact residual.

The four-site XXX chain has a one-magnon sector where the Hamiltonian becomes the cycle Laplacian. Fourier modes diagonalize it, and the finite regular Bethe roots reproduce all nonuniform modes. The uniform mode is present as an infinite-root limit, exposing why a list of finite roots need not be complete. The two-site Gaudin model offers a different spectral-to-differential bridge: a Bethe root supplies a singlet eigenvector and a Miura potential in which the apparent pole cancels exactly. The eigenvalues appear as residues at the physical marked points. This is a complete small calculation, rather than a proof of arbitrary Bethe completeness or a global automorphic correspondence.

The tau-function side similarly distinguishes interaction from superposition. For KdV, two independent one-soliton exponentials need a mixed exponential term, and its coefficient is forced by the Hirota polynomial. Positivity of the coefficients makes the real tau-function nonsingular. The same construction gives phase shifts and mass by asymptotics and boundary values of $\log\tau$. A y-independent solution embeds in the chosen KP equation, but transverse stability and general two-dimensional interactions are additional problems. The unifying capacity is not a universal solver: it is the ability to compare a finite spectral model, a scalar differential operator or a finite exponential identity under explicitly matched conventions.

Definitions

In the periodic XXX one-magnon sector, $H|j\rangle=2|j\rangle-|j-1\rangle-|j+1\rangle$ with indices modulo N.

For the two-site sl_2 Gaudin model at $z_1=0,z_2=1$, $\Omega=h\otimes h/2+e\otimes f+f\otimes e$, $H_1=-\Omega,H_2=\Omega$.

For $k_1>k_2>0$, $\eta_i=k_ix-k_i^3t+\delta_i$ and $\tau=1+e^{\eta_1}+e^{\eta_2}+A\,e^{\eta_1+\eta_2}$, $A=((k_1-k_2)/(k_1+k_2))^2$.

Use $u=2\partial_x^2\log\tau$ and the KdV convention $u_t+6uu_x+u_{xxx}=0$.

Theorem and hypotheses

For N=4, the one-magnon XXX sector has eigenvalues 0,2,2,4, with finite regular Bethe roots 1/2,0,−1/2 for the nonzero modes. The stated KdV tau-function satisfies $(D_xD_t+D_x^4)\tau\cdot\tau=0$; A is forced by the mixed exponential coefficient.

Proof

For a plane wave with $p=2\pi m/N$, substitution into the cycle difference gives $E=2-2\cos p$. With $N=4$ this is $0,2,4,2$. The one-root Bethe relation is $((\lambda+i/2)/(\lambda-i/2))^4=1$, so $\lambda=(1/2)\cot(p/2)$, giving $1/2,0,-1/2$; $p=0$ is the infinite limit. For KdV use $P(k,\omega)=k\omega+k^4$. Each individual exponent obeys $P(k_i,-k_i^3)=0$. The only new mixed condition is $A\cdot P(k_1+k_2,-k_1^3-k_2^3)+P(k_1-k_2,-k_1^3+k_2^3)=0$. The two polynomial values are $3k_1k_2(k_1+k_2)^2$ and $-3k_1k_2(k_1-k_2)^2$, which force A. The remaining exponential coefficients reduce to these equations. Dividing the bilinear identity by τ^2 and differentiating gives KdV. Since all four tau coefficients are positive, $\tau>0$ on real x,t, so this solution has no real tau zeros.

Worked example

Use the integer cycle Hamiltonian at $N=4$ and the two-site Gaudin Bethe root $w=1/2$.

The vectors $(1,1,1,1),(1,0,-1,0),(0,1,0,-1),(1,-1,1,-1)$ have exact eigenvalues $0,2,2,4$ and form a basis. For Gaudin, $w^{-1}|\downarrow\uparrow\rangle+(w-1)^{-1}|\uparrow\downarrow\rangle=2|\downarrow\uparrow\rangle-2|\uparrow\downarrow\rangle$ is a singlet with Ω eigenvalue $-3/2$. Thus H_1,H_2 have values $3/2,-3/2$. With $u(x)=1/(2x)+1/(2(x-1))-1/(x-1/2)$, direct rational simplification gives $u^2-u'=3/[4x^2(x-1)^2]$. The $x=1/2$ pole disappears because the Bethe equation $1/w+1/(w-1)=0$ cancels its residue. This gives a scalar-oper certificate tied to the same eigenvector.

Independent operator and scalar identities are a strong finite bridge. Their scope is the stated sector, solution family and marked-point normalization.

Continuity across the source years

2023 — KdV/KP tau-functions, XXX roots and Gaudin scalar equations form the directly evidenced integrability cluster. Their proofs in the annual source volumes are later expositions of these classical models. Sources: 2023:M011, 2023:M012, 2023:M104, 2023:M105, 2023:M009, 2023:M087

Connections

D08 — Nonlinear differential equations. A finite exponential coefficient identity produces an exact nonlinear PDE solution. Transverse stability and general scattering need further analytic work.

D03 — Representation decomposition. The sl_2 singlet/triplet splitting diagonalizes the two-site invariant interaction. A small decomposition does not establish arbitrary-site Bethe completeness.

D07 — Spectral evolution. Self-adjoint mode decompositions give norm-preserving fractional evolution. Boundary conventions determine which fractional operator is meant.

Capacity and interactive scope

Lab ids: []

Inputs: Finite Hamiltonian and sector, Bethe normalization, marked points or positive tau parameters.

Outputs: Eigenvector residuals, scalar-potential identity and bilinear exponential coefficients.

Validation: Check independent matrix products and polynomial/rational identities; include infinite or singular-root limits explicitly.

Status: Exact worked spectral and tau-function recipes; no general Bethe, KP or scalar-oper solver is implemented.

Limits

No arbitrary-sector Bethe completeness, full Gaudin theorem or all-KP classification is proved. Coincident soliton parameters and singular roots require their own limiting treatment. The source years do not evidence a continuous year-by-year integrability project.

Next questions

Can the next recovered finite magnetization sector be diagonalized with every singular or infinite Bethe state accounted for?

For a specified higher-site Gaudin system, which exact eigenvectors certify all joint eigenvalues and apparent-pole cancellations?

Source anchors: 2023:M011, 2023:M012, 2023:M104, 2023:M105

Faddeev, How Algebraic Bethe Ansatz works for integrable model — https://arxiv.org/abs/hep-th/9605187
Frenkel, Gaudin model and opers — https://arxiv.org/abs/math/0407524
Biondini and Chakravarty, Soliton solutions of the KP II equation — https://arxiv.org/abs/nlin/0511068

D16-F06 · Deformations, obstruction equations and Poisson transport

When does a first-order algebraic change extend to a genuine family, and which structural invariants can still jump?

Deformation is a family of equations, not an adjective attached to a new notation. Its data include a coefficient ring, an object, an equivalence relation and a compatibility equation at every order. In associative algebra, the first-order equation is a Hochschild cocycle condition. In a differential graded Lie model it becomes a Maurer–Cartan equation, with a second-order cohomology class measuring obstruction. Neither a tangent vector nor a long finite truncation proves an all-order extension without a theorem controlling the remaining equations.

Three small models make the distinctions explicit. A characteristic-two character of C_2 has a nontrivial tangent direction that cannot lift one order further. A dual-number multiplication can be deformed to $x^2=t$ as a free rank-two algebra, yet its positive-degree Hochschild cohomology jumps at the special fibre. A reflection-deformed derivative on polynomials supplies an exact represented relation $[D\kappa,X]=I+\kappa s$ for every parameter; its proof uses the monomial action rather than an unsupported universal classification. These are constructive examples of obstruction, flatness with changing derived invariants, and a genuinely defined deformation law.

Poisson wall transport adds a geometric version. A skew lattice pairing gives an automorphism x^m↦x^m f(x^γ)^{B(γ,m)}. Skew symmetry and B(γ,γ)=0 ensure the distinguished monomial is fixed and make the inverse and bracket compatibility work. Replacing the pairing by an arbitrary matrix loses the proof. The same principle links back to Moyal exactness: each higher equation must actually vanish, and its degree or filtration bound determines whether a finite check is complete. The atlas presents finite obstruction calculations with exact coefficient rings and reports the boundary between formal existence, finite truncation, convergence and physical interpretation.

Definitions

A first-order associative deformation is $m_t=m+t\varphi$ modulo t^2, with multiplication unit and equivalence under basis changes specified.

Its cocycle equation is $a\varphi(b,c)-\varphi(ab,c)+\varphi(a,bc)-\varphi(a,b)c=0$.

For a differential graded Lie algebra in characteristic zero, α satisfies $d\alpha+[\alpha,\alpha]/2=0$; coefficients at each order give extension equations.

A flat rank-two family $A_t=k[x]/(x^2-t)$ is free on $1,x$ over its parameter ring; assume characteristic different from two for the stated Hochschild comparison.

Theorem and hypotheses

Over a characteristic-two field, the C_2-character tangent $u=1+\varepsilon$ modulo ε^2 satisfies $u^2=1$ but cannot lift to $u=1+\eta+b\eta^2$ modulo η^3. Over an algebraically closed field of characteristic different from two, $A_t=k[x]/(x^2-t)$ has $HH^n=0$ for $t\neq0$ and $n>0$, while at $t=0$ every positive-degree HH^n has dimension one.

Proof

In characteristic two, $(1+\varepsilon)^2=1$ because $2\varepsilon=\varepsilon^2=0$. For a prospective lift, $(1+\eta+b\eta^2)^2=1+\eta^2$ modulo η^3: doubled terms vanish and the η^2 coefficient cannot be changed by b. Thus the tangent relation has an unavoidable next-order obstruction. For the flat algebra family, division by the monic polynomial leaves a unique representative $a+bx$, proving freeness of rank two. At $t\neq0$, choose $a^2=t$; $x-a$ and $x+a$ are coprime, so Chinese remainders give $A_t\cong k\times k$, a separable algebra with no positive-degree Hochschild cohomology. At $t=0$ the standard two-periodic bimodule resolution has maps $x\otimes1-1\otimes x$ and $x\otimes1+1\otimes x$. Applying Hom over A^e gives alternating maps zero and multiplication by $2x$. In positive degrees the kernels or cokernels have dimension one: $\ker(2x)=kx$ and $A/(2x)=k$. Thus flatness of the underlying module does not force derived-invariant constancy.

Worked example

Compare a finite-order formal solution, a polynomial deformation and its special fibre.

The scalar equation $s^{13}=0$ is satisfied by $s=\varepsilon$ modulo ε^{13}, so it passes all coefficients through order twelve. The same leading direction fails modulo ε^{14} because its ε^{13} coefficient is one; no higher correction changes that leading term. This gives a precise finite-truncation boundary. For the reflection model $D\kappa x^n=[n+(\kappa/2)(1-(-1)^n)]x^{n-1}$. Subtracting $XD\kappa$ from $D\kappa X$ gives coefficient $1+\kappa$ on even n and $1-\kappa$ on odd n, exactly $I+\kappa s$. At the dual-number fibre $x^2=0$, $D(x)=x$ is a nonzero derivation; it represents the HH^1 direction that disappears in the split generic fibre.

One model can be fully verified at each claimed level without carrying unrelated historical quiver or Galois conclusions. Extension and flatness are distinct mathematical outputs.

Continuity across the source years

2024 — A finite character obstruction and first-order associativity introduce the coefficient-ring and cocycle viewpoint. Sources: 2024:Y24-08-005, 2024:Y24-08-010

2025 — The reflection-deformed polynomial representation provides an all-parameter exact relation, while a Maurer–Cartan model specifies curvature. Sources: 2025:Y25-02-015, 2025:Y25-03-018

2026 — Second-order obstruction classes, jumping Hochschild cohomology and Poisson wall maps make extension, flatness and transport separately testable. Sources: 2026:Y26-05-029, 2026:Y26-08-012, 2026:Y26-08-014, 2026:Y26-08-011

Connections

D04 — Derived obstruction classes. The coefficient of t^2 in Maurer–Cartan is a cocycle whose class in H^2 must vanish. An L_∞ structure has further brackets and higher-order conditions.

D02 — Flat algebraic families. A locally free module can have jumping cohomology under specialization. A basis dimension does not compute a quiver Kuranishi space.

D06 — Poisson wall automorphisms. Skew pairings make the monomial transport bracket-preserving and invertible. A non-skew source matrix needs a different definition or proof.

Capacity and interactive scope

Lab ids: ['L06']

Inputs: Multiplication table or graded bracket, coefficient ring, truncation order and specified equivalences.

Outputs: Order-by-order equations, exact obstruction class or failed coefficient, flat basis and jump locus in the worked family.

Validation: Expand coefficients exactly, verify cocycle and boundary conditions, retain characteristic assumptions and prove any all-order closure.

Status: The Moyal lab implements a related finite deformation test; the obstruction and wall-transport examples are exact recipes.

Limits

No recovered order-twelve packet is silently extended to all orders. The dual-number comparison does not compute the actual quiver Hochschild profile. Formal existence does not establish convergence or a physical deformation.

Next questions

For the recovered quiver family, what actual cochain representatives and higher obstruction equations determine its Kuranishi locus?

Which degree, nilpotence or cohomological vanishing theorem turns a proposed finite-order deformation check into an all-order certificate?

Source anchors: 2024:Y24-08-005, 2024:Y24-08-010, 2025:Y25-02-015, 2026:Y26-05-029, 2026:Y26-08-012, 2026:Y26-08-014

Mazur, Deforming Galois Representations — https://bpb-us-e1.wpmucdn.com/sites.harvard.edu/dist/a/189/files/2023/01/Deforming-Galois-Representations.pdf
Ames, Cagliero and Tirao, Hochschild cohomology of truncated quiver algebras — https://arxiv.org/abs/math/0603056

Six cross-domain bridges

Each bridge states what is actually transported between domains. Its scope distinguishes a theorem in a named model from a shared formal pattern or an open programme.

B01 · Quotients, observability and the information a decision actually needs

Proved bridge in a linear model · D03, D14, D15, D11, D12

When can a target be recovered without reconstructing the entire hidden state?

Let X, Y and Z be finite-dimensional vector spaces, with linear observation $H:X\to Y$ and linear target $G:X\to Z$. Exact target recovery means that some map K on the reachable observations satisfies $G=KH$. The necessary and sufficient condition is $\ker H\subseteq\ker G$. Necessity follows immediately by applying KH to an invisible state. For sufficiency, define $K(Hx)=Gx$. If $Hx=Hx'$, then $x-x'$ lies in $\ker H$ and hence in $\ker G$, so the definition is independent of the representative. It is linear on $\operatorname{im} H$ and may be extended linearly to Y. Thus the target descends through the observable quotient $X/\ker H$. There is no requirement that H be injective on X.

This gives a precise relationship between algebraic quotients and experimental design. Rows of H are available measurements; rows of G are the desired quantities. The inclusion of kernels is equivalent to $\operatorname{row}(G)\subseteq\operatorname{row}(H)$, and therefore to $\operatorname{rank}([H;G])=\operatorname{rank}(H)$. A finite candidate-probe catalogue with costs becomes a discrete optimization problem: find a cheapest subset whose row span contains row(G). The lab checks every subset of six probes by exact rational elimination. The distinction between full-state observability and target observability is visible immediately: one probe x_1+x_2 suffices for the sum target even though it leaves a three-dimensional hidden subspace.

The same quotient principle applies to a deterministic machine only after its equivalence relation is specified. Histories u and v may be identified when every future continuation gives the same output; this is a congruence compatible with concatenation. For a linear realization, reachable states and unobservable states lead to the usual reachable-observable reduction. These are rigorous constructions in their respective categories. Calling any two lossy summaries the same quotient would skip the essential work of defining the objects, equivalence relation and compatible operations.

A robust decision can need less information than either absolute value. In $J_A=c+\delta$ and $J_B=c$, the difference is δ. Common uncertainty c disappears under the difference functional. The joint uncertainty set, rather than two marginal intervals, carries the fact that the same c appears twice. With $|c|\le 10$ and $\delta\in[1,2]$, the individual intervals $[-9,12]$ and $[-10,10]$ overlap substantially, while J_A-J_B is strictly positive throughout the joint set. This is an exact cancellation theorem for the stated model. It is not an argument for canceling two unrelated uncertain inputs.

The useful general question is therefore not “How much data can we collect?” but “Which distinctions between states can alter the target?” Its answer specifies an equivalence relation and then a sufficient observation map. In nonlinear problems the linear kernel condition must be replaced by constancy of the target on observation fibres; continuity, stability and computational tractability require additional hypotheses. An algebraic factorization can exist while being numerically unstable or impractical. The atlas keeps those three questions—existence, stability and implementation—separate.

Working models: L01, L08

B02 · Local compatibility, cohomology and transport around a loop

Exact finite chain model; geometric bridge under stated hypotheses · D04, D05, D06, D13

Why can locally consistent data fail to assemble into one global object?

A cochain complex consists of groups or vector spaces C^k and maps $d^k:C^k\to C^{k+1}$ with $d^{k+1}d^k=0$. A compatible datum is a cocycle, an element of $\ker d^k$. A change generated at the preceding level is a coboundary, an element of $\operatorname{im} d^{k-1}$. Cohomology $H^k=\ker d^k/\operatorname{im} d^{k-1}$ records the remaining compatible data modulo those changes. The equation $d^2=0$ makes the quotient legitimate. This formal pattern becomes useful only when the complex is built from the actual restriction, transition or boundary maps of the problem.

For a finite oriented simplicial complex, the boundary of an oriented triangle $[0,1,2]$ is $[1,2]-[0,2]+[0,1]$. Its next boundary is zero because every vertex contribution cancels with the appropriate sign. Over $\mathbb{Q}$, $\beta_1=E-\operatorname{rank}\partial_1-\operatorname{rank}\partial_2$. The graph consisting of three edges around a triangle has $\beta_1=1$. Adding the filled two-simplex introduces a boundary generator and changes β_1 to zero. A drawing of three mutually adjacent vertices does not tell us whether the face is present. The interactive homology model makes this modeling decision explicit, which prevents confusing a graph with its clique complex.

In an additive gluing problem, differences on overlaps form a Čech 1-cocycle. If that cocycle is a coboundary, local corrections can remove it and the local sections glue. If its cohomology class is nonzero, the failure survives all allowed local corrections. This statement requires the relevant sheaf, cover and restriction maps; the abstract slogan “there is an obstruction” supplies none of them. Higher-degree classes arise when compatibility itself must be made compatible at the next level. Integer coefficients can also carry torsion that a rational rank calculation cannot detect.

Transport supplies a related but distinct view. For a local system, moving along paths gives linear maps, and concatenation becomes composition. A closed loop may return a vector transformed by a nonidentity matrix. In the lab, $U=[[1,a],[0,1]]$ and $V=[[1,0],[b,1]]$ give the loop $UVU^{-1}V^{-1}=[[1+ab+a^2b^2,-a^2b],[ab^2,1-ab]]$. Its determinant is one, but the loop can still act nontrivially. With $a=2$ and $b=1$, the result is $[[7,-4],[2,-1]]$. The determinant constraint alone misses the order dependence.

There are important distinctions. Holonomy is the ordered transport around a based loop. A change of fibre coordinates conjugates its matrix, so the conjugacy class is gauge invariant. For a flat connection it depends on the based homotopy class. Curvature concerns infinitesimal transport for a connection; cohomology requires its chosen differential and coefficient system. Flatness permits monodromy on spaces with nontrivial fundamental group, so zero curvature need not force trivial global transport. A finite pair of shears is an exact illustration of noncommuting composition, not a construction of a smooth connection. The bridge worth carrying forward is the explicit passage from local maps to consistency equations, and from those equations to a quotient that measures the failure of a global solution.

Working models: L05, L09

B03 · Spectra, inverse problems and why identifiability is weaker than stability

Proved analytic bridge for diagonal Hilbert-space models · D07, D08, D15, D11

What changes when a target is uniquely determined but small noise is greatly amplified?

Consider a diagonal forward operator $Ae_k=\sigma_k e_k$ on a Hilbert basis. If every σ_k is positive, the noiseless data determine every coordinate x_k. Yet if $\sigma_k\to0$, the formal inverse divides by smaller and smaller numbers and is unbounded on the ambient data space. Uniqueness has therefore been proved without stable inversion. This is the analytic counterpart of the finite quotient criterion: invisible directions can obstruct well-defined target recovery, while nearly invisible directions obstruct conditioning.

Assume the unknown belongs to the ellipsoid $\Sigma_{k\geq1}\, k^{2r}|x_k|^2\leq1$ and the data error has norm at most δ. The spectral cutoff inverse keeps the first N coordinates. The tail obeys $\|x-P_Nx\|\leq(N+1)^{-r}$, because all discarded weights are at least $(N+1)^{2r}$. The propagated noise obeys $\|A_N^\dagger e\|\leq\delta/\sigma_N$ for decreasing σ_k. The triangle inequality gives the fully specified deterministic bound $B(N)=(N+1)^{-r}+\delta/\sigma_N$. This is an upper bound over all admissible signals and noises, not a fitted prediction of the error in one sample.

For mildly decaying singular values $\sigma_k=k^{-p}$, balancing the two terms suggests N of order $\delta^{-1/(p+r)}$ and a rate $\delta^{r/(p+r)}$. Matching lower bounds can be obtained by choosing two signals separated along a coordinate near that cutoff whose forward images differ by no more than the noise scale. Their data neighborhoods overlap, so an estimator cannot distinguish both perfectly. Such a lower-bound argument proves a limit of the information model rather than a defect of one algorithm.

For severe decay $\sigma_k=\exp(-ck)$, write $\delta=\exp(-L)$. A leading estimate $N\approx L/c$ can make the noise term order one. To keep noise on the same polynomial scale as the bias, use a correction of the form $N\approx(L-r\log L)/c$, with integer rounding and bounded-L behavior handled explicitly. The lab displays both terms and their sum so that a visually appealing increase in resolution cannot hide noise amplification. The more general $\exp(-ck^\beta)$ model requires the corresponding β-dependent correction; the implemented lab intentionally fixes $\beta=1$.

Spectral gaps also govern iterative convergence, but their operators must be named. If P and Q are orthogonal projections onto distinct lines meeting at angle θ, then $PQP=\cos^2\theta\,P$. The n-fold product has norm $\cos^{2n}\theta$ on the surviving direction. A small angle gives slow convergence, even though the intersection is trivial and the limiting projection is known. The commutator norm $|\sin 2\theta|/2$ is a different quantity and is small near both alignment and orthogonality, so it cannot substitute for the contraction factor.

The shared capacity is to locate directions that a map suppresses and quantify the cost of recovering them. Inverse regularization, spectral iteration and experimental design can then exchange specific tools: singular values, principal angles, norm bounds and minimax tests. They do not become the same problem merely because an eigenvalue appears in each. The next useful experiment is always tied to a hypothesis: estimate a singular-value tail, perturb a nearly null direction, or compare a proposed cutoff with an explicit uniform error bound.

Working models: L04, L05

B04 · Exact counting, generating functions and the limits of a summary

Exact finite combinatorial bridge · D13, D10, D11, D12, D14

When does a finite summary support a proof, and when can different objects share it?

For a finite configuration set Ω and statistic $T:\Omega\to S$, exact counting produces the fibre sizes $n_s=|T^{-1}(s)|$. Under the uniform distribution, $n_s/|\Omega|$ is the law of T. A generating function packages those same counts as coefficients. If a weight or sampling law is changed, the induced distribution changes accordingly; counting alone does not justify a new probability model. This simple distinction connects combinatorics to inference without assuming that every enumeration is a statistically representative sample.

An optimization problem adds an objective and a feasible set. In a six-item 0–1 knapsack with weights (2,3,4,5,6,7), values (3,4,6,7,9,10), and budget 12, exhaustive enumeration finds 28 feasible subsets and a unique optimum of value 18, selecting items 1,3,5. Because all 64 masks have been checked, existence, optimality and uniqueness are separate certified statements with a transparent common witness set. Ordering feasible masks also gives a ranking and unranking operation. A larger problem may require a recurrence, decision diagram, branch bound or formal certificate instead; the six-item lab does not recover missing large-instance input data.

A summary can be exact and still incomplete. For a graph G and vertex heights h, let $E_G(h)=\Sigma_{uv\in E(G)}(h_u-h_v)^2$. Two explicitly recovered seven-vertex graphs have identical joint counts of Hamming weight and energy for all 128 binary assignments $h\in\{0,1\}^7$. Consequently every invariant that is only a function of that binary census agrees on the pair. This is a rigorous information-loss result. It does not imply that the graphs agree under a richer measurement scheme.

Allow nonnegative integer heights with total mass three. The corresponding energy generating polynomials differ by $q^3(1-q)(1-q^3)^2(1-q^4)^2$. At $q=1/2$ this difference is 11025/262144, so a higher-height measurement distinguishes them. At $q=1$ both polynomials count the same total number of configurations, and the difference vanishes. The choice of evaluation point is therefore part of the probe. The interactive lab exposes the integer coefficient census alongside the floating evaluation, making it possible to see precisely what was counted and which information was discarded.

Cyclotomic phase fibres provide a second exact counting-to-quotient bridge. Reduction of units $U(1680)\to U(30)$ is surjective. Its fibres are cosets of a kernel and hence all have equal size. Since $\varphi(1680)=384$ and $\varphi(30)=8$, each fibre has 48 elements. The 56th powers of primitive 1680th roots correspond to the eight primitive 30th-root phases. This proves a finite phase multiplicity theorem; it does not prove density or closure of a matrix group generated by braiding operators.

A useful research discipline emerges: identify the full configuration space, the statistic, the admissible weights, and the reconstruction goal. Then ask whether the statistic separates the objects relevant to that goal. A collision is constructive evidence of a missing distinction and a guide to designing a richer probe. Exact coefficients are especially valuable because they separate a mathematical collision from a numerical tolerance accident. The source bundles preserve unrecovered-instance boundaries so that a historical claim is not silently converted into a newly verified computation.

Working models: L02, L03, L07

B05 · Poisson structure, Moyal deformation and the first nonzero obstruction

Exact polynomial deformation calculation · D03, D06, D16, D14

How can a classical bracket agree with a deformed bracket at one order and still fail later?

On the canonical plane with coordinates (q,p), the Poisson bracket is $\{f,g\}=\partial_q f\partial_p g-\partial_p f\partial_q g$. Fix the Weyl–Moyal product $f\star g=f\exp[(i\hbar/2)(\overleftarrow{\partial_q}\overrightarrow{\partial_p}-\overleftarrow{\partial_p}\overrightarrow{\partial_q})]g$. Its normalized commutator $(f\star g-g\star f)/(i\hbar)$ has only odd derivative orders, and its leading term is the Poisson bracket. For polynomials the series terminates, so the comparison is an exact finite algebraic calculation rather than a convergence claim about a formal infinite sum.

For $f=q^a p^b$ and $g=q^c p^d$, define $S_k=\Sigma_{j=0}^k(-1)^j \binom{k}{j}(a)_{k-j}(b)_j(c)_j(d)_{k-j}$, with falling factorials zero beyond the degree. The order-k contribution for odd k is $(-1)^{(k-1)/2}\hbar^{k-1}S_k\, q^{a+c-k}p^{b+d-k}/(2^{k-1}k!)$. All coefficients are rational integers divided by an explicit denominator. The lab uses arbitrary-precision integer arithmetic and rational normalization, so no cancellation is decided by a numerical threshold.

For $f=q^3$ and $g=p^3$, the result is $9q^2p^2-(3/2)\hbar^2$. This fixes the sign and normalization convention. More revealing is $f=qp^5$ and $g=q^5p^3$: $S_3=0$, but $S_5=201600$, so the bracket is $-22q^5p^7+105\hbar^4qp^3$. Stopping after the cubic derivative test would certify the wrong conclusion. The first surviving obstruction may occur later than the first possible obstruction.

There are also infinite exact families for structural reasons. For $f=qp^3$ and g=q^(3r+2)p^r with nonnegative integer r, the third-order coefficient cancels, and higher odd orders vanish because the relevant total degree of f is four. In this family the cancellation plus a degree cutoff is sufficient. The missing step in an informal argument is often exactly that cutoff: a vanishing coefficient proves nothing about orders that were not examined unless another theorem excludes them.

This illustrates the proper role of deformation language. A first-order Poisson structure can describe the leading deformation, while associativity and equivalence involve higher compatibility equations. An infinitesimal solution need not integrate to a global family, and a small verified monomial family need not generate a closed algebra under the desired operations. A classification must state its ambient vector space, degree bounds, closure conditions and equivalences. The new bounded sextic monomial enumeration in the algebra chapter is labeled as an atlas computation, not a historical discovery and not a classification of arbitrary sextic polynomial subspaces.

The bridge to exact certification is practical: expand the defining operator with an explicit convention, use structural vanishing to bound the search, and certify every surviving order. The bridge to geometry is conceptual but specific: the Poisson bivector is the operator contracted repeatedly in this canonical constant-coefficient model. Extending the same formula to a curved manifold without charts, connections or a compatible star-product construction would require new mathematics.

Working models: L06

B06 · Arithmetic reduction, orbit completeness and finite certificates of infinite families

Exact unit action with a reduction proof · D01, D02, D09, D14

How can a finite list certify an infinite arithmetic family?

A recurrence can produce infinitely many solutions without proving that it produces all solutions. For $x^2-2y^2=-7$, multiplication of $\alpha=x+y\sqrt{2}$ by the norm-one unit $u=3+2\sqrt{2}$ gives $(x,y)\mapsto(3x+4y,2x+3y)$. Direct expansion proves that x^2-2y^2 is invariant. The inverse unit gives $(x,y)\mapsto(3x-4y,-2x+3y)$, again preserving integrality and the norm. This establishes a group action on the solution set. Completeness is a separate reduction theorem.

For $y>0$ and $x>0$, $\alpha>\sqrt{7}$ and its conjugate is $-7/\alpha$. Multiplying by a suitable integral power of u brings α into the half-open interval $[\sqrt{7},u\sqrt{7})$. Then $x=(\alpha-7/\alpha)/2$ and $y=(\alpha+7/\alpha)/(2\sqrt{2})$ show that the reduced positive representative has $0\le x<8$ and $0<y<6$. Exact checking in that finite rectangle leaves (1,2) and (5,4). The endpoint $\alpha=\sqrt{7}$ would give $x=0$, which cannot solve $2y^2=7$ in integers. Thus every positive solution reduces to one of those two seeds. Overall sign and signed unit powers cover the other sign choices in the corresponding real embeddings.

The numerical interval is merely a way to derive a finite search bound. The proof rests on the norm identity and the fundamental multiplicative interval, and the final candidate test is an exact integer equation. The interactive lab moves in either direction along each orbit with arbitrary-precision integers and checks the norm at every displayed point. Watching many solutions appear is useful intuition; the interval reduction is what makes the list complete.

Congruence conditions can further restrict the orbit. Reducing the recurrence matrix modulo m gives a finite permutation of its admissible residue states. One may determine which powers hit a desired residue class, then combine those periodic conditions with the unit parameter. This is a precise connection between infinite Diophantine families and finite automata. It requires a proof that all solutions have entered the chosen orbit parametrization before a residue sieve can claim completeness.

The same finite-certificate strategy appears elsewhere in arithmetic, with different hypotheses. A rational point on an elliptic curve can be checked by substitution; a claim that a point generates all rational points needs a rank and saturation argument. A finite-field group order can be checked by exhaustive counting for a small field or by a certified algorithm for a large field. A p-adic approximation is an element of a compatible residue system, and lifting a root requires its own derivative or multiplicity hypotheses. None of these certificates can be replaced by the fact that a calculation produced a plausible point.

This is a durable connection across the annual work: turn an unbounded question into finitely many representatives through a proved action, reduction or compactness argument; then verify the finite residue exactly. The bridge from arithmetic to dynamics is the explicit recurrence matrix, not a metaphor about motion. The bridge to computation is a terminating reduction algorithm plus a checkable certificate, not a claim that brute force settles every infinite search.

Working models: L07, L10

Interactive capacity contracts

The interactive companion has two different functions. Its catalogue provides search, domain/year/capability filters and the complete retained text of all 1,051 source treatments. Its ten labs implement small, explicitly specified mathematical models. A capability tag on a chapter means the mathematics supports that kind of operation; it does not mean a general solver for that domain has been implemented. Each lab identifies its inputs, arithmetic, output and limits. Exact rational or integer results are distinguished from floating evaluations, analytic upper bounds and conceptual research directions.

Capacity	Implemented worked-model links
C01 · Calculate exactly	L02, L03, L06, L07, L09, L10
C02 · Enumerate and count	L02, L03, L07
C03 · Certify a result	L01, L02, L03, L06, L09, L10
C04 · Classify and quotient	L01, L03, L07, L10
C05 · Reconstruct a target	L01, L04
C06 · Optimize and decide	L01, L02, L08
C07 · Evolve or simulate	L05, L10
C08 · Approximate with error bounds	L04
C09 · Analyze spectra and transforms	L04, L05
C10 · Transport and glue	L05, L09
C11 · Deform and bifurcate	L06
C12 · Quantify uncertainty	L04, L08
C13 · Inspect geometry	L05, L09
C14 · Design an experiment	L01, L08
C15 · Compose mathematical structures	Conceptual recipes in the chapters; no general solver

L01 · Observable quotient and cheapest probes

Arithmetic

Exact rational row reduction and all 64 candidate subsets

Inputs

Target

Probe subset mask (bits 0–5)

Limits

Four real coordinates and six fixed linear probes. This is a finite design problem, not a general nonlinear observability solver.

Domains: D14, D15, D12 · Capacities: C03, C04, C05, C06, C14

L02 · Finite selection, counting and ranking

Arithmetic

Exhaustive integer enumeration of 64 masks

Inputs

Weight budget

Requested zero-based feasible rank

Mask to rank (−1 result means infeasible)

Limits

Six fixed items; requested rank is clamped to the feasible range. The recovered annual records do not contain the complete Petersen-50 input table. Inverse ranking returns −1 for an infeasible query mask.

Domains: D13, D12, D14 · Capacities: C01, C02, C03, C06

L03 · A graph census collision that higher heights resolve

Arithmetic

Exact integer census; floating evaluation of its polynomial

Inputs

Total nonnegative integer height

Polynomial evaluation q

Limits

Two explicit seven-vertex graphs. Equality of the binary census is verified for this pair only; it is not a universal graph reconstruction theorem.

Domains: D13, D09, D11 · Capacities: C01, C02, C03, C04

L04 · Spectral cutoff: bias, noise and stable reconstruction

Arithmetic

Analytic upper bounds evaluated in floating arithmetic

Inputs

Singular-value decay

Smoothness r

Decay p or c

Noise exponent L, $\delta=\exp(-L)$

Selected cutoff N

Limits

Diagonal Hilbert-space model with a Sobolev ellipsoid and deterministic noise. The severe model fixes $\beta=1$. Bounds are not empirical error estimates.

Domains: D15, D07, D08 · Capacities: C05, C08, C09, C12

L05 · Transport order and alternating projections

Arithmetic

Integer shears; floating trigonometric projection model

Inputs

Model

Shear a

Shear b

Line angle in degrees

Limits

Two explicit 2×2 constructions. Noncommutativity, curvature and convergence are related through specified operators; they are not interchangeable quantities.

Domains: D06, D05, D07, D15 · Capacities: C07, C09, C10, C13

L06 · Exact polynomial Moyal bracket

Arithmetic

Exact BigInt falling factorials and rational coefficients

Inputs

Exponent a in $f=q^a p^b$

Exponent b

Exponent c in $g=q^c p^d$

Exponent d

Limits

One canonical pair (q,p), polynomial monomials, and the stated Weyl–Moyal convention. It does not classify arbitrary polynomial subspaces, nonlinear star products or all deformations.

Domains: D16, D03, D06 · Capacities: C01, C03, C11

L07 · Cyclotomic phase fibres

Arithmetic

Enumeration of all 384 units modulo 1680

Inputs

Target modulus m dividing 1680

Limits

Exact phase counting. Projective braid-image closure requires the actual generating matrices and cannot be inferred from phase counts alone.

Domains: D01, D03, D16 · Capacities: C01, C02, C04

L08 · Shared uncertainty and robust comparison

Arithmetic

Exact interval formulas evaluated at decimal inputs

Inputs

Common uncertainty radius C

Lower differential effect

Upper differential effect

Limits

A specified jointly coupled additive model. Cancellation requires the same common variable in both alternatives; independent uncertainties would give a different set.

Domains: D10, D11, D12 · Capacities: C06, C12, C14

L09 · Homology changes when a face is filled

Arithmetic

Exact rational boundary ranks

Inputs

Finite complex

Fill the triangle face

Limits

Homology over Q of these small finite complexes. It does not compute integral torsion, sheaf cohomology or arbitrary persistent homology.

Domains: D04, D05, D13 · Capacities: C01, C03, C10, C13

L10 · Complete Pell orbits, exact integer navigation

Arithmetic

Exact integer unit recurrence and norm identity

Inputs

Reduced seed

Signed power of $3+2\sqrt{2}$

Limits

The norm equation $x^2-2y^2=-7$ with its two reduced orbit seeds. The general Pell theory and other quadratic fields need their own reduction and local analysis.

Domains: D01, D09, D14 · Capacities: C01, C03, C04, C07

Complete treatment index

Every retained treatment is indexed below by stable year-prefixed identifier and revised domain. The interactive companion contains the complete definitions, theorem, proof, example, limits, extension, original question and references of each treatment. Titles are revised annual-volume titles, not verbatim historical conversation titles. Source evidence codes remain as originally recorded and are not reinterpreted as confidence scores.

Source identifier / date	Revised treatment	Domain
2023:M001 2023-01-23	Separate the meanings of Q	D14
2023:M002 2023-01-23	p-adic completion with certified digits	D01
2023:M003 2023-01-23	Affine dynamics on Z_p	D01
2023:M004 2023-01-23	A valid probability model and identifiable parameters	D11
2023:M005 2023-01-23	Specify a hard problem before using arithmetic complexity	D14
2023:M006 2023-02-02	Zeno through a convergent series and a time law	D09
2023:M007 2023-02-20	Octonions with a declared multiplication convention	D03
2023:M008 2023-02-20	The sharp octonion regrouping bound	D03
2023:M009 2023-03-11	Hyperbolic identities that actually verify a wave	D08
2023:M010 2023-03-11	Gudermannian coordinates with the measure retained	D07
2023:M011 2023-03-11	KdV and KP with matched conventions	D16
2023:M012 2023-03-11	A complete two-soliton construction in a fixed KdV convention	D16
2023:M013 2023-03-11	Radial conservation and stereographic geometry	D06
2023:M014 2023-03-11	q-differentiation of the Gudermannian	D07
2023:M015 2023-03-11	Quantitative Jacobian bounds for deep compositions	D15
2023:M016 2023-03-11	Minimal universal recurrences on an observable quotient	D15
2023:M017 2023-03-11	The reachable and observable parts of a realization	D15
2023:M018 2023-03-11	Exact finite closure and a polynomial obstruction	D07

Source identifier / date	Revised treatment	Domain
2023:M019 2023-04-23	Hardy–BMO duality from atoms	D07
2023:M020 2023-04-23	A complete heat-signature collision	D13
2023:M021 2023-04-24	Exact modularity phases on a four-cycle	D12
2023:M022 2023-04-24	Ensemble assumptions determine the spectral baseline	D10
2023:M023 2023-04-25	A reversible q-dependent chain with an exact gap	D10
2023:M024 2023-04-25	Continuous q-Hermite polynomials through a Jacobi operator	D07
2023:M025 2023-04-24	A graph game with explicit equilibria	D12
2023:M026 2023-04-29	Relative entropy along a finite Markov chain	D11
2023:M027 2023-05-11	Neat groups and a smooth modular-curve example	D02
2023:M028 2023-05-11	Minimal and toroidal compactifications in an explicit chart	D02
2023:M029 2023-05-11	General type and log-general type on modular curves	D02
2023:M030 2023-05-11	A projective embedding from a degree bound	D02
2023:M031 2023-05-13	Adjunction fixes the quintic dimension	D02
2023:M032 2023-05-13	Formal inversion has a public coefficient algorithm	D14
2023:M033 2023-05-13	The A_2 Weyl denominator expanded exactly	D03
2023:M034 2023-05-14	An affine A_1 identity with its normalization proved	D07
2023:M035 2023-05-14	Young tableaux counted by recursion and hooks	D13
2023:M036 2023-05-13	A nonconstant Kazhdan–Lusztig calculation in S_4	D03
2023:M037 2023-05	An independently replayable BCH certificate	D14
2023:M038 2023-05-16	A finite Peter–Weyl calculation with an exact truncation error	D03
2023:M039 2023-05-16	Rephasing invariance and a sharp three-flavor CP bound	D16
2023:M040 2023-05-22	A complete small persistence calculation	D05

Source identifier / date	Revised treatment	Domain
2023:M041 2023-05-22	A two-level Berry holonomy with a gauge check	D06
2023:M042 2023-05-22	Crossing symmetry and an exactly opened spectral gap	D07
2023:M043 2023-05-22	A Moore spectrum as a precise chromatic example	D05
2023:M044 2023-05-30	A conservative positive diffusion stencil with a sharp local condition	D08
2023:M045 2023-05-30	Reflection equivariance as an operator identity	D03
2023:M046 2023-06-05	Affine coordinate changes and time evolution are different operations	D06
2023:M047 2023-06-08	Mutual information detects dependence rather than a causal channel	D11
2023:M048 2023-06-08	One measurement can miss coherence that another detects	D16
2023:M049 2023-06-10	An observable timescale that saturates the energy–time bound	D16
2023:M050 2023-06-12	Modularity from lattice reindexing and a diagnostic zero	D07
2023:M051 2023-06-12	A classical abelian Chern–Simons calculation with its global boundary	D06
2023:M052 2023-06-12	Explicit genus-one modular data with the anomaly retained	D16
2023:M053 2023-06-12	Two-oscillator locking with stable and unstable branches	D09
2023:M054 2023-06-12	Invariant gates and an exact identifiability obstruction in conductance ODEs	D09
2023:M055 2023-07-18	Decision thresholds and genuine strategic equilibria	D12
2023:M056 2023-07-18	Infinite utilities through controlled finite limits	D12
2023:M057 2023-07-28	A Cauchy model has no ordinary mean or variance	D10
2023:M058 2023-07-28	Feedback instability is a mechanism to specify	D09
2023:M059 2023-08-29	Frobenius energy across disjoint blocks	D03
2023:M060 2023-08-29	Population weights versus amplitude weights	D10
2023:M061 2023-08-29	Covariance inequalities with finite moments	D11
2023:M062 2023-08-29	A quadratic program with singular cases included	D12

Source identifier / date	Revised treatment	Domain
2023:M063 2023-08-16	Open-system dynamics must preserve physical states	D16
2023:M064 2023-08-10	Population growth and the actual logarithmic time	D09
2023:M065 2023-08-23	A unified equation needs shared variables and invariances	D09
2023:M066 2023-09-01	Bessel versus Parseval	D07
2023:M067 2023-09-01	Hölder and Young with equality cases	D07
2023:M068 2023-09-01	Matrix norms and the quantity being bounded	D03
2023:M069 2023-09-03	Lyapunov conditions in the right direction	D15
2023:M070 2023-09-09	Units of the Einstein equation	D09
2023:M071 2023-09-09	Why the coefficient is 8π	D09
2023:M072 2023-09-09	Sixteen entries but ten independent components	D06
2023:M073 2023-09-09	Covariant conservation is a structural condition	D06
2023:M074 2023-09-30	Grothendieck universes and their logical cost	D04
2023:M075 2023-09-30	Stirling partitions do not supply torsional energy	D13
2023:M076 2023-09-23	Vector motion and rotation conventions	D09
2023:M077 2023-09-23	A proposed analogy needs a dimensionless correspondence	D09
2023:M078 2023-10-04	Residual energy and exact Gaussian calibration	D11
2023:M079 2023-10-05	The chain rule as a statement about local linear maps	D07
2023:M080 2023-10-05	Graph heat flow and the stability of a discrete gradient step	D12
2023:M081 2023-10-07	Exact exponent arithmetic before approximation	D01
2023:M082 2023-10-07	Competing response laws and identifiable parameters	D15
2023:M083 2023-10-07	Tensor order and matrix rank are different invariants	D03
2023:M084 2023-10-07	Directional cancellation as a limited shearlet illustration	D07

Source identifier / date	Revised treatment	Domain
2023:M085 2023-10-07	A fully normalized spin one-half Wigner representation	D16
2023:M086 2023-10-07	Exact polynomial Moyal products	D16
2023:M087 2023-10-07	A spectral definition of fractional quantum evolution	D07
2023:M088 2023-10-07	Lagrange identity and its equality geometry	D03
2023:M089 2023-10-12	Winding number and Gauss–Bonnet with boundaries explicit	D05
2023:M090 2023-10-12	An explicit marked-torus deformation	D06
2023:M091 2023-10-16	Certified first logistic-map period doublings	D09
2023:M092 2023-10-16	Parameter loci, orbit closures and dissipative flows	D09
2023:M093 2023-10-21	Convex conjugates with their actual domains	D12
2023:M094 2023-10-21	The CHSH bound and a finite-sample test	D16
2023:M095 2023-10-26	An exact free resolution and a separate geometric blowup	D04
2023:M096 2023-10-26	A complete two-dimensional toric subdivision	D02
2023:M097 2023-10-26	Hodge numbers constrained by geometry	D02
2023:M098 2023-10-26	A rational parametrization and a nonexistence certificate	D02
2023:M099 2023-10-09	A typed glossary that prevents invalid compositions	D14
2023:M100 2023-11-01	Bernstein–Sato certificates and a minimality proof	D02
2023:M101 2023-11-21	Conic polarity as a typed point-to-line map	D06
2023:M102 2023-11-21	Cross-ratio with a fixed ordering and homogeneous coordinates	D02
2023:M103 2023-11-21	A Heisenberg field and its verified local algebra	D03
2023:M104 2023-11-22	Four-site XXX spectrum and the regular one-magnon roots	D16
2023:M105 2023-11-22	A complete two-site Gaudin to scalar-oper calculation	D16
2023:M106 2023-11-06	Joint reliability is a dependence problem	D10

Source identifier / date	Revised treatment	Domain
2023:M107 2023-11-24	Nonlinearity does not imply chaos or identifiability	D09
2023:M108 2023-12-02	The Klein–Gordon dispersion relation and its massless limit	D08
2023:M109 2023-12-02	A controlled formal route from Dirac to Pauli	D08
2023:M110 2023-12-02	Maxwell equations as field equations and identities	D06
2023:M111 2023-12-02	Ward–Takahashi constrains a vertex without determining it	D16
2023:M112 2023-12-02	A complete finite-dimensional perturbation calculation	D10
2023:M113 2023-12-02	An exact RG toy flow and an invariant local exponent	D08
2023:M114 2023-12-16	Moving level sets and the singularity of threshold coordinates	D07
2024:Y24-02-001 2024-02-13T06:01:20Z	Concentration times area does not determine an amount	D09
2024:Y24-02-003 2024-02-19T01:19:06Z	Stored energy needs a state function and a power balance	D09
2024:Y24-02-002 2024-02-25T03:31:43Z	An area ratio is not a geometric packing theorem	D13
2024:Y24-03-001 2024-03-30	Picard classes and the Néron–Severi quotient	D02
2024:Y24-03-002 2024-03-30	Replace Picard intransigence by a defined fixed-space calculation	D02
2024:Y24-03-003 2024-03-31	Independent variations give separate Euler–Lagrange equations	D09
2024:Y24-03-004 2024-03-31	Normalize the density before differentiating entropy	D11
2024:Y24-03-005 2024-03-31	A numerical optimizer does not certify an equation	D08
2024:Y24-03-006 2024-03-31	Quantum evolution needs an explicit state space and norm	D16
2024:Y24-03-007 2024-03-31	A complexity score requires units, weights and a stated inference	D11
2024:Y24-03-020 2024-03-31T02:28:25Z	Laplace transforms turn an initial-value equation into algebra	D08
2024:Y24-03-021 2024-03-31T02:28:25Z	Legendre polynomials from a derivative formula to orthogonal projection	D07
2024:Y24-03-022 2024-03-31T02:28:25Z	Bessel functions are the radial modes of the Helmholtz equation	D08
2024:Y24-03-023 2024-03-31T02:28:25Z	The Schwarzian chain rule detects projective rather than affine structure	D06

Source identifier / date	Revised treatment	Domain
2024:Y24-03-024 2024-03-31T02:28:25Z	Euclidean certificates and a complete Pell example	D01
2024:Y24-03-025 2024-03-31T02:28:25Z	Brouwer and hairy-ball obstructions require the right domain	D05
2024:Y24-03-026 2024-03-31T02:28:25Z	Cauchy’s integral formula is coefficient extraction with hypotheses	D07
2024:Y24-03-027 2024-03-31T02:28:25Z	Maxwell–Boltzmann speed is a radial Gaussian distribution	D10
2024:Y24-03-028 2024-03-31T02:28:25Z	A catenary is fixed by force balance and an explicit length constraint	D09
2024:Y24-03-029 2024-03-31T02:28:25Z	Navier plate modes and the Rayleigh quotient share a spectral calculation	D08
2024:Y24-03-030 2024-03-31T02:28:25Z	Laplace tidal equations yield a local rotating-wave certificate	D08
2024:Y24-03-031 2024-03-31T02:28:25Z	Hamilton, Hamilton–Jacobi and Riccati equations meet at a caustic	D06
2024:Y24-03-032 2024-03-31T02:28:25Z	A cubic Routh–Hurwitz test with an exact stability boundary	D09
2024:Y24-03-033 2024-03-31T02:28:25Z	A linear chirp has quadratic phase, with a precise sampling ambiguity	D11
2024:Y24-03-034 2024-03-31T02:28:25Z	Stokes drag comes from a stress integral, not dimensional analysis alone	D09
2024:Y24-04-001 2024-04-04	Specify the clustering objective before adding weights	D12
2024:Y24-04-002 2024-04-04	Spatial coherence gives an exact cut-versus-fit threshold	D12
2024:Y24-04-003 2024-04-04	Three regularizers produce three different shrinkage rules	D12
2024:Y24-04-004 2024-04-04	An online centroid is a mean only for the right step sizes	D12
2024:Y24-04-005 2024-04-04	A similarity becomes a kernel only after a positivity check	D11
2024:Y24-04-006 2024-04-04	Bayesian uncertainty and data fusion depend on covariance	D11
2024:Y24-04-007 2024-04-04	Transfer learning needs a support and invariance assumption	D11
2024:Y24-04-012 2024-04-04	PCA reconstruction rank is not intrinsic dimension	D11
2024:Y24-04-008 2024-04-07	Ratios, percentages and lottery events require a denominator	D10
2024:Y24-04-009 2024-04-20	Perimeter–area measures need units and an edge-count convention	D06
2024:Y24-04-010 2024-04-20	Replace a single-patch log ratio by a dimension definition	D06

Source identifier / date	Revised treatment	Domain
2024:Y24-04-011 2024-04-20	Landscape division is a collision probability with exact merge laws	D10
2024:Y24-04-013 2024-04-20	An E^2 page is not already the homology of the total space	D04
2024:Y24-04-014 2024-04-20	Sheaf notation needs a gluing rule and cohomology needs a complex	D05
2024:Y24-04-016 2024-04-20	A scalar stress law is not the full elasticity tensor	D09
2024:Y24-04-017 2024-04-20	Conservation and first integrals distinguish two population models	D09
2024:Y24-04-018 2024-04-20	Fixed-point stability separates maps from planar oscillators	D09
2024:Y24-04-019 2024-04-20	Lorenz and Rössler labels do not constitute a chaos certificate	D09
2024:Y24-04-020 2024-04-20	Reaction stability and diffusion-driven instability are different tests	D08
2024:Y24-04-021 2024-04-20	Gradient flows separate mass conservation from energy decay	D08
2024:Y24-04-022 2024-04-20	Kuramoto coherence has a Lyapunov function under precise symmetry	D09
2024:Y24-04-023 2024-04-30	A finite lattice gauge action needs a local transformation law	D06
2024:Y24-04-015 Date unavailable	A Brunnian pure braid has an exact deletion certificate	D16
2024:Y24-05-001 2024-05-06	Finite mixtures and typed kernel pairings	D12
2024:Y24-05-002 2024-05-06	Winding number from a lifted angle	D05
2024:Y24-05-003 2024-05-06	An Euler–Lagrange equation with a coercive minimization certificate	D12
2024:Y24-05-004 2024-05-24	A consistent gl(1|1) replacement for unspecified graded brackets	D03
2024:Y24-05-005 2024-05-24	Point-counting zeta functions and one elliptic local factor	D02
2024:Y24-05-006 2024-05-24	Schrödinger continuity and guidance on a nonvanishing region	D16
2024:Y24-05-007 2024-05-25	A nonlinear wave equation derived from its energy	D08
2024:Y24-05-008 2024-05-25	Finite Gaussian and Grassmann integrals before formal path integrals	D10
2024:Y24-05-009 2024-05-25	Source derivatives and dimensional claims in a proposed unification	D11
2024:Y24-05-010 2024-05-27	Intertwining maps define a category of dynamical systems	D04

Source identifier / date	Revised treatment	Domain
2024:Y24-05-011 2024-05-27	Coupled equilibria with a complete two-variable stability test	D09
2024:Y24-05-012 2024-05-27	A specified noisy dynamics and its stationary covariance	D10
2024:Y24-05-013 2024-05-27	Finite probability kernels compose but do not determine topology	D10
2024:Y24-06-001 2024-06-02	A number-system catalogue organized by actual algebraic properties	D03
2024:Y24-06-002 2024-06-05	Lie-bracket closure does not imply cyclic time evolution	D03
2024:Y24-06-003 2024-06-05	Hamiltonian conservation is not the Casimir property	D06
2024:Y24-06-004 2024-06-05	Rational coordinates do not select an ambient dimension	D02
2024:Y24-06-005 2024-06-05	A real constant and a geometric shape require different definitions	D06
2024:Y24-06-006 2024-06-05	The circle is irreducible and its radius scaling has a contravariant map	D02
2024:Y24-06-007 2024-06-05	The recovered quartic has Klein-four rather than dihedral Galois group	D01
2024:Y24-06-008 2024-06-05	The weight-dependent Hecke formula and an exact Eisenstein eigencheck	D01
2024:Y24-06-009 2024-06-05	A nonzero H^1 from two arcs without violating sheaf gluing	D05
2024:Y24-06-010 2024-06-05	A braid R-matrix with a verified Hecke relation	D16
2024:Y24-06-011 2024-06-05	A conventioned trefoil Jones polynomial from the bracket algebra	D16
2024:Y24-06-012 2024-06-05	Hyperbolic curvature and geodesics in a declared metric	D06
2024:Y24-06-013 2024-06-05	Witt brackets and the Virasoro central term	D03
2024:Y24-06-014 2024-06-06	Fundamental groups distinguish boundaries, fillings and quotients	D05
2024:Y24-06-015 2024-06-06	Wedge products, Hodge duality and a genuine closed-form obstruction	D05
2024:Y24-06-016 2024-06-06	Affine density integrates exactly on a simplex	D07
2024:Y24-06-017 2024-06-06	The SPD conditions behind conjugate gradients and multigrid	D15
2024:Y24-06-019 2024-06-07	A renormalization map with a proved Gaussian fixed point	D10
2024:Y24-06-021 2024-06-07	A complete two-spin Gibbs and Metropolis model	D10

Source identifier / date	Revised treatment	Domain
2024:Y24-06-020 2024-06-08	Antisymmetry does not imply a magnetic monopole	D06
2024:Y24-06-022 2024-06-30	A qubit rotation has an exact gradient and a testable learning rule	D16
2024:Y24-06-023 2024-06-30	Entanglement, dephasing and observability in a two-qubit calculation	D16
2024:Y24-06-024 2024-06-30	An additive correction ansatz does not identify a new mechanism	D09
2024:Y24-06-018 Date unavailable	A toroidal toy Tonnetz separates graph cycles from surface homology	D05
2024:Y24-07-001 2024-07-02	An undefined Ψ measure needs an actual transport law	D09
2024:Y24-07-002 2024-07-02	Ψ mutual information and capacity need a channel	D11
2024:Y24-07-003 2024-07-02	Functorial relabeling does not automatically preserve code distance	D14
2024:Y24-07-004 2024-07-03	A purchasing-power ratio is a stipulated unit conversion	D09
2024:Y24-07-005 2024-07-14	PID gains require a plant and a stability calculation	D15
2024:Y24-07-006 2024-07-14	A logarithmic threshold needs a reference and a squared-amplitude convention	D11
2024:Y24-07-007 2024-07-20	Ideal power identities as a typed algebraic system	D09
2024:Y24-07-008 2024-07-20	Euler’s product yields an exact partition recurrence	D13
2024:Y24-07-009 2024-07-21	An eleven-dimensional space does not determine an operator	D06
2024:Y24-07-010 2024-07-21	A coupled diffusion model needs an energy balance	D08
2024:Y24-07-011 2024-07-21	A product of motivic measure and entropy needs a scalar realization	D02
2024:Y24-07-012 2024-07-21	A flat connection is not automatically a mixed Hodge module	D04
2024:Y24-07-013 2024-07-21	Reduced homology and intersection cohomology remove different phenomena	D05
2024:Y24-07-014 2024-07-21	Legendrian structure begins with a contact form and a tangent condition	D06
2024:Y24-07-015 2024-07-21	Spatial averages can erase the diffusion parameter	D15
2024:Y24-08-001 2024-08-04	Diagram distances require diagonal matching and a stated filtration	D05
2024:Y24-08-002 2024-08-04	A graph Dirac gap is a concrete operator statement	D13

Source identifier / date	Revised treatment	Domain
2024:Y24-08-004 2024-08-04	Photon number density times photon energy is energy density	D09
2024:Y24-08-003 2024-08-05	The stipulated return and covariance example has two arithmetic corrections	D11
2024:Y24-08-005 2024-08-17	A tangent deformation can exist while its next lift is obstructed	D16
2024:Y24-08-006 2024-08-17	A modular-curve quotient has a concrete finite-index calculation	D02
2024:Y24-08-007 2024-08-19	The Poincaré group is a noncommutative semidirect product	D03
2024:Y24-08-008 2024-08-20	Finite zero checks and an arbitrary spectral gap cannot prove a Riemann hypothesis	D07
2024:Y24-08-009 2024-08-20	A finite-field zeta example gives a genuine cohomological counting bridge	D02
2024:Y24-08-010 2024-08-22	Operadic corrections need actual compositions and an associativity constraint	D16
2024:Y24-08-011 2024-08-22	Formal, rigid and Berkovich disks are related constructions with different data	D02
2024:Y24-08-012 2024-08-25	A BV operator generates a concrete odd bracket	D03
2024:Y24-08-014 2024-08-26	Two power laws are assumptions with testable ratios, not a derived physical theory	D09
2024:Y24-08-013 Date unavailable	Controlled irregularity needs a norm and a perturbation estimate	D07
2024:Y24-09-012 2024-09-08T20:07:59Z	Frobenius traces for 37.a1 can be counted exactly	D02
2024:Y24-09-013 2024-09-08T20:07:59Z	A rational point of infinite order without assuming BSD	D02
2024:Y24-09-014 2024-09-08T20:07:59Z	Sato–Tate is a distribution of genuine normalized traces	D02
2024:Y24-09-015 2024-09-10T10:34:37Z	A Benford law needs logarithmic equidistribution	D10
2024:Y24-09-001 2024-09-20T17:56:10Z	What can actually determine the deformation parameters α and β?	D11
2024:Y24-09-021 2024-09-20T17:58:47Z	A Toeplitz eigenvalue law with a complete elementary proof	D07
2024:Y24-09-022 2024-09-20T17:58:47Z	A Toeplitz log determinant needs positivity	D07
2024:Y24-09-023 2024-09-20T18:05:37Z	The atomic Hardy-space inequality requires cancellation	D07
2024:Y24-09-024 2024-09-20T18:05:37Z	A6 automorphisms must preserve element orders and class structure	D03
2024:Y24-09-025 2024-09-20T18:05:37Z	Weight38 modular forms: specify the group, then compute the space	D01

Source identifier / date	Revised treatment	Domain
2024:Y24-09-002 2024-09-20T18:19:30Z	A sublinear signed deformation is a homeomorphism, with a cusp at zero	D07
2024:Y24-09-005 2024-09-20T18:19:30Z	An irrational angular frequency does not define a closed polar curve	D07
2024:Y24-09-006 2024-09-20T18:19:30Z	Regularity of the recovered lacunary angular series	D07
2024:Y24-09-007 2024-09-20T18:19:30Z	Homology follows from a specified space, not a deformation name	D05
2024:Y24-09-008 2024-09-20T18:19:30Z	What persistence stability can certify for a deformation	D05
2024:Y24-09-009 2024-09-20T18:19:30Z	A complete small sheaf calculation before spectral-sequence claims	D05
2024:Y24-09-010 2024-09-20T18:19:30Z	Hochschild homology and K-theory detect different data	D04
2024:Y24-09-026 2024-09-20T23:48:20Z	The magic square has a defined Lie bracket and integer dimensions	D03
2024:Y24-09-003 2024-09-20T23:49:35Z	A genuine deformed algebra requires transported operations	D03
2024:Y24-09-004 2024-09-20T23:49:35Z	Constant width is a condition on the support function	D06
2024:Y24-09-027 2024-09-20T23:49:35Z	A quiver dimension count is valid on a free regular quotient	D06
2024:Y24-09-028 2024-09-20T23:49:35Z	The logistic universal deformation is not injective on the whole real line	D07
2024:Y24-09-011 2024-09-20T23:54:44Z	Fractional running becomes a memory equation after choosing a derivative	D08
2024:Y24-09-019 2024-09-22T20:50:03Z	The Bell expression has a local bound2 and a quantum value2√2	D16
2024:Y24-09-020 2024-09-22T20:50:03Z	Unitary evolution does not make neighboring states orthogonal	D16
2024:Y24-09-016 2024-09-27T23:45:40Z	A residual certifies spectral proximity only with operator hypotheses	D07
2024:Y24-09-017 2024-09-27T23:45:40Z	The GUE Wigner surmise has exact normalization, but is a model	D11
2024:Y24-09-018 2024-09-29T14:26:48Z	Euler products require a convergence domain and controlled tails	D07
2024:Y24-10-002 2024-10-05T12:33:58Z	A Weyl-law obstruction for a bounded-interval RH model	D07
2024:Y24-10-003 2024-10-05T12:33:58Z	A determinant identity needs a proved arithmetic correspondence	D07
2024:Y24-10-001 2024-10-07T03:51:27Z	A formal Lakeland operator becomes an operator only after its domain is fixed	D07
2024:Y24-10-028 2024-10-10T13:05:52Z	Exponent towers require a parse tree	D01

Source identifier / date	Revised treatment	Domain
2024:Y24-10-029 2024-10-10T13:05:52Z	The uncertainty inequality is a Cauchy–Schwarz theorem with domain conditions	D07
2024:Y24-10-030 2024-10-10T13:05:52Z	Least action reduces to a first variation with fixed endpoints	D09
2024:Y24-10-031 2024-10-10T13:05:52Z	Entropy is a concave function with an exact uniform maximum	D11
2024:Y24-10-032 2024-10-10T13:05:52Z	The differential identities behind Maxwell-type constraints are exact complexes	D13
2024:Y24-10-007 2024-10-11T13:27:29Z	Adelic data require a common restricted-product space	D07
2024:Y24-10-008 2024-10-11T13:27:29Z	Adding time as an operator changes the spectral problem	D07
2024:Y24-10-004 2024-10-13T04:34:36Z	A heat trace and partition function have precise convexity identities	D11
2024:Y24-10-005 2024-10-13T04:34:36Z	A nonlinear eigenproblem is a constrained variational equation	D07
2024:Y24-10-006 2024-10-13T04:34:36Z	Spectral rigidity is a fitted statistic with a declared unfolding	D11
2024:Y24-10-026 2024-10-13T19:38:24Z	Propositional connectives can be specified by complete truth tables	D14
2024:Y24-10-027 2024-10-13T19:42:21Z	Quantifier order and variable binding change the theorem	D14
2024:Y24-10-014 2024-10-18T12:28:50Z	Haar analysis of prime gaps has an exact finite energy identity	D07
2024:Y24-10-009 2024-10-20T15:59:22Z	The pointwise product $\mu\Lambda$ has the wrong sign for the recovered prime sum	D01
2024:Y24-10-010 2024-10-20T15:59:22Z	The convolution identity $\mu * \log = \Lambda$ explains the Euler logarithmic derivative	D01
2024:Y24-10-011 2024-10-20T15:59:22Z	Prime ideals in Q(i) give an exact local-factor model	D01
2024:Y24-10-012 2024-10-22T00:10:37Z	A two-state transfer gap proves mixing, not the prime number theorem	D09
2024:Y24-10-013 2024-10-22T00:10:37Z	The prime-counting main term cannot be truncated at x/logx for a square-root error	D01
2024:Y24-10-015 2024-10-25T19:49:16Z	A modified discrete logarithm may be only a group relabeling	D03
2024:Y24-10-016 2024-10-25T19:49:16Z	A homomorphic encoding must prove an operation-preserving identity	D03
2024:Y24-10-018 2024-10-25T19:49:16Z	Class groups and Galois groups are different invariants	D01
2024:Y24-10-019 2024-10-25T19:49:16Z	The feature map $x \mapsto (x^2, x^3)$ is injective but singular at zero	D02
2024:Y24-10-020 2024-10-25T19:49:16Z	A deformed graph Laplacian is controlled by its edge weights	D13

Source identifier / date	Revised treatment	Domain
2024:Y24-10-017 2024-10-26T20:12:18Z	An invertible LWE coordinate change preserves the exact sample equation	D14
2024:Y24-10-023 2024-10-27T14:58:14Z	A concrete perfectoid field requires completeness and Frobenius surjectivity	D01
2024:Y24-10-024 2024-10-27T14:58:14Z	A prism is a structured pair, not a new label for a height weight	D02
2024:Y24-10-025 2024-10-27T14:58:14Z	A cohomology comparison must survive a coefficient check	D04
2024:Y24-10-021 2024-10-28T01:43:15Z	The height sum is meaningful only with normalized local absolute values	D02
2024:Y24-10-022 2024-10-28T01:43:15Z	A categorical decomposition needs a chain complex and an actual invariant	D04
2024:Y24-11-001 2024-11-01	Compound growth, resource totals, and consistent denominators	D09
2024:Y24-11-004 2024-11-01	Euler products: convergence, nonvanishing, and sensitivity	D01
2024:Y24-11-002 2024-11-02	Frequency filtering with an actual stability certificate	D09
2024:Y24-11-003 2024-11-03	A positive Schrödinger discretization and the limits of zero matching	D08
2024:Y24-11-005 2024-11-04	A typed unification test: supersymmetry and finite Fourier duality	D03
2024:Y24-11-006 2024-11-04	BSD ranks and the logical burden of a spectral realization	D02
2024:Y24-11-007 2024-11-04	Finite Gibbs states, the KMS boundary condition, and a Laplace resolvent	D07
2024:Y24-11-008 2024-11-12	Wave, photon, charge, and simulation formulas with consistent units	D09
2024:Y24-11-009 2024-11-15	Maxima, constructible images, and the danger of missing compactness	D02
2024:Y24-11-012 2024-11-15	Temperley–Lieb quotients and why the affine relation must be specified	D16
2024:Y24-11-013 2024-11-15	Homoclinic sequences and a precise horseshoe model	D09
2024:Y24-11-014 2024-11-15	Cylindrical Gaussian data versus a Radon Gaussian on Hilbert space	D10
2024:Y24-11-015 2024-11-15	Bratteli diagrams as exact branching and path-count data	D03
2024:Y24-11-011 2024-11-16	Higher-dimensional slices, projections, and visible singularities	D06
2024:Y24-11-016 2024-11-16	Geometric complexity needs an input model and a certificate	D14
2024:Y24-11-010 2024-11-19	Invariant averaging as a legitimate symmetry reduction	D03

Source identifier / date	Revised treatment	Domain
2024:Y24-11-017 2024-11-19	Brauer height zero: precise theorem and a quaternion audit	D03
2024:Y24-11-020 2024-11-19	Flow scaling and a dissipative energy identity	D09
2024:Y24-11-018 2024-11-20	Harper operators: norm bounds, flux symmetries, and touching bands	D16
2024:Y24-11-019 2024-11-20	IDS gap labels and a rational Diophantine calculation	D16
2024:Y24-11-021 2024-11-30	Frenet frames, Darboux rotation, and covariant differentiation	D06
2024:Y24-11-022 2024-11-30	Characteristic forms and why a topological density is not a bulk field equation	D05
2024:Y24-11-023 2024-11-30	Twisted boundary conditions and a self-adjoint circle spectrum	D07
2024:Y24-11-024 2024-11-30	Schmidt decomposition, superselection, and a limited error-correction claim	D16
2024:Y24-11-025 2024-11-30	Lax–Milgram coercivity as a usable existence theorem	D08
2024:Y24-11-026 2024-11-30	Renormalization-flow formulas as ordinary differential equations	D09
2024:Y24-12-001 2024-12-01	Sphere eversion and a correctly normalized Willmore bound	D06
2024:Y24-12-002 2024-12-01	The two transverse-traceless polarizations	D06
2024:Y24-12-003 2024-12-01	A trace anomaly as a scale derivative with stated conventions	D08
2024:Y24-12-004 2024-12-01	Triangulation quiddities and continuants with a corrected example	D13
2024:Y24-12-005 2024-12-01	Jordan chains, nullity increments, and symmetry that really commutes	D03
2024:Y24-12-037 2024-12-01	A finite Stinespring dilation with a complete channel calculation	D16
2024:Y24-12-006 2024-12-03	Bundle curvature, parallel sections, and a determinant identity that needs evidence	D06
2024:Y24-12-007 2024-12-03	Projectors, stable generators, and nonnormal transient growth	D08
2024:Y24-12-008 2024-12-05	Graded characters, mock-series domains, and the limits of formal measures	D16
2024:Y24-12-009 2024-12-05	An exact energy-rate conversion and a controlled barrier model	D09
2024:Y24-12-010 2024-12-12	A normalized Laguerre–spherical basis with the correct measure	D07
2024:Y24-12-011 2024-12-12	Gram energy and why orthogonal modes have no phase gradient	D07

Source identifier / date	Revised treatment	Domain
2024:Y24-12-012 2024-12-13	Traveling waves, Laurent symbols, and a well-typed nonlinear equation	D08
2024:Y24-12-013 2024-12-13	Interference logic and gradients through a linear modal layer	D15
2024:Y24-12-014 2024-12-14	Heat traces and spectral zeta functions on a compact circle	D07
2024:Y24-12-015 2024-12-14	A Lie superalgebra and a differential that really squares to zero	D03
2024:Y24-12-016 2024-12-15	Spectral fitting is not prediction: six-point statistics and prescribed masses	D11
2024:Y24-12-017 2024-12-16	SU(2) rotations, area asymptotics, and a metric counterexample	D06
2024:Y24-12-018 2024-12-16	Galois data and a small local-to-global arithmetic example	D01
2024:Y24-12-019 2024-12-17	S_4 through cycle classes, its character table, and the resolvent quotient	D03
2024:Y24-12-020 2024-12-17	E_2, the number 24, and the exact modular value j(i)	D01
2024:Y24-12-021 2024-12-17	Musical ratios and an explicit frequency-to-pitch convention	D07
2024:Y24-12-022 2024-12-17	Crystallographic restrictions and finite root systems without a mineral numerology	D03
2024:Y24-12-023 2024-12-17	A simplex is not a cavity: a flag complex with genuine H_5	D05
2024:Y24-12-036 2024-12-17	Running-coupling factors and the logarithm convention	D08
2024:Y24-12-024 2024-12-18	The P30 point fails exactly: a polynomial and an interval certificate	D02
2024:Y24-12-025 2024-12-18	Mordell–Weil torsion and height lattices with their actual meaning	D02
2024:Y24-12-027 2024-12-18	Landau symmetry and a two-band Bloch model	D16
2024:Y24-12-026 2024-12-19	A precise version of algebrogeometric width	D02
2024:Y24-12-028 2024-12-26	The enhanced Hamiltonian: Hermiticity and a two-spin diagonalization	D16
2024:Y24-12-029 2024-12-26	GHZ dephasing and the precise scope of the quantum Zeno limit	D16
2024:Y24-12-030 2024-12-26	A_2 cluster mutation and an exact pentagon of Laurent expressions	D03
2024:Y24-12-031 2024-12-26	A quiver Grassmannian whose Euler characteristic can be counted	D03
2024:Y24-12-032 2024-12-26	F_1 heuristics, Krasner addition, and the two points of a projective line	D02

Source identifier / date	Revised treatment	Domain
2024:Y24-12-033 2024-12-26	A flag incidence equation and a changing matroid support	D02
2024:Y24-12-034 2024-12-31	Braiding, Reidemeister moves, and an exact toric modular datum	D16
2024:Y24-12-035 2024-12-31	Toric-code stabilizers and the exact limits of local protection	D16
2025:Y25-01-025 2025-01-01	The6.02dB identity refers to an amplitude ratio	D11
2025:Y25-01-024 2025-01-02	A controlled gate is unitary but protection needs an error model	D16
2025:Y25-01-021 2025-01-10	The oscillator connects a connection, a star product and an eigenstate	D16
2025:Y25-01-022 2025-01-10	Equivariant cohomology retains the stabilizer of a fixed point	D05
2025:Y25-01-001 2025-01-11	A nilpotent monodromy has a computable weight filtration	D02
2025:Y25-01-002 2025-01-11	A kernel transform is an equivalence only when it has an inverse	D04
2025:Y25-01-003 2025-01-15	Noncommutative observables still have identities and symmetries	D16
2025:Y25-01-004 2025-01-15	Closing a curve is distinct from choosing a polygon or a winding	D05
2025:Y25-01-005 2025-01-15	Boundary conditions are part of a spectral problem	D08
2025:Y25-01-006 2025-01-15	A positive metric can repair a matrix inner product	D03
2025:Y25-01-023 2025-01-15	A modular Hamiltonian is defined by a state	D11
2025:Y25-01-007 2025-01-17	Moonshine coefficients are graded dimensions and traces	D03
2025:Y25-01-008 2025-01-17	Modular invariance has a concrete weight and coefficient test	D01
2025:Y25-01-009 2025-01-18	Projective phases require a genuine two-cocycle	D05
2025:Y25-01-010 2025-01-21	Six pentads give an exhaustive outer automorphism certificate	D03
2025:Y25-01-012 2025-01-24	Platonic boundaries share homology but have different symmetry groups	D05
2025:Y25-01-013 2025-01-24	A scalar wave potential has critical points, not a square Jacobian	D06
2025:Y25-01-014 2025-01-24	Intersection homology retains cycles that an ordinary cone contracts	D05
2025:Y25-01-015 2025-01-26	The derived tangent complex comes from the cotangent complex	D04

Source identifier / date	Revised treatment	Domain
2025:Y25-01-016 2025-01-26	Perverse degree is fixed by support and cosupport	D04
2025:Y25-01-017 2025-01-26	A persistence module retains finite bars that its colimit forgets	D05
2025:Y25-01-018 2025-01-26	A weight filtration need not split canonically	D04
2025:Y25-01-019 2025-01-26	Localization gives a valid motivic and cohomological subtraction	D02
2025:Y25-01-011 2025-01-28	An E_8 projection has exact fibers and changes root lengths	D03
2025:Y25-01-020 2025-01-28	A Jacobiator detects nonassociativity only after multiplication is defined	D03
2025:Y25-02-016 2025-02-01	Berry curvature and a Chern number require a state bundle	D06
2025:Y25-02-002 2025-02-05	A network multiplier is an inverse matrix with a convergence condition	D15
2025:Y25-02-003 2025-02-05	Weighted scores and capacity quotas need different constraints	D12
2025:Y25-02-004 2025-02-05	Debt-to-output dynamics include the primary flow	D09
2025:Y25-02-001 2025-02-07	A cumulative wealth equation is a stock–flow identity	D09
2025:Y25-02-006 2025-02-16	A simplified metric RHS has an exact solution but no relativity certificate	D09
2025:Y25-02-007 2025-02-16	CFL stability and halo exchange have separate exact tests	D08
2025:Y25-02-008 2025-02-16	A spline element has exact partition and stiffness certificates	D07
2025:Y25-02-009 2025-02-17	Weighted atoms must cancel in the same measure as their pairing	D07
2025:Y25-02-010 2025-02-17	Hyperbolic volume growth changes global analysis	D07
2025:Y25-02-011 2025-02-17	Frame bounds, coordinate changes and sampling errors are separate	D07
2025:Y25-02-005 2025-02-18	A Gini calculation and a growth decomposition measure different objects	D11
2025:Y25-02-025 2025-02-18	A hyperbolic quotient needs an action and can have a noncompact cusp	D06
2025:Y25-02-012 2025-02-19	Searching many strategies needs an independent validation contract	D11
2025:Y25-02-022 2025-02-19	Creation operators and Boolean cumulants have distinct finite tests	D16
2025:Y25-02-013 2025-02-22	A categorical equivalence has explicit units and counits	D04

Source identifier / date	Revised treatment	Domain
2025:Y25-02-014 2025-02-22	Monodromy-free means identity, not eigenvalues equal to one	D08
2025:Y25-02-015 2025-02-24	A reflection deformation has an explicit polynomial representation	D16
2025:Y25-02-017 2025-02-25	Heat dimension and summability are definitions with calculable limits	D07
2025:Y25-02-018 2025-02-25	A logarithmic L-derivative has a pole at a zero	D07
2025:Y25-02-019 2025-02-25	A repetition code corrects specified flips but has quantum distance one	D16
2025:Y25-02-020 2025-02-27	Descent and factorization need the actual diagram	D04
2025:Y25-02-021 2025-02-27	A CPTP update gives a quantitative mixing law	D16
2025:Y25-02-024 2025-02-27	Pentagon consistency is a path identity with a norm error bound	D04
2025:Y25-02-023 Date unavailable	Parity observations alone do not certify quantum coherence	D16
2025:Y25-03-001 2025-03-01T15:38:22Z	A low-rank approximation has an exact error certificate	D15
2025:Y25-03-003 2025-03-01T15:38:22Z	Persistent topology starts with a chain complex whose boundary squares to zero	D05
2025:Y25-03-004 2025-03-01T15:44:25Z	A QUBO and a one-edge QAOA instance can be solved exactly	D12
2025:Y25-03-005 2025-03-01T15:44:25Z	A Poisson response model gives an identifiable rate, not a latency theorem	D10
2025:Y25-03-002 2025-03-01T17:37:28Z	Tensor compression uses isometries, not an inverse of a rectangular map	D14
2025:Y25-03-029 2025-03-02T01:53:39Z	A fractal probability measure has a quantitative weak-convergence proof	D10
2025:Y25-03-027 2025-03-03T01:13:00Z	A stabilizer code needs commuting checks and an error model	D16
2025:Y25-03-032 2025-03-03T21:07:25Z	A spin-foam face cannot integrate a four-form as its surface action	D06
2025:Y25-03-034 2025-03-03T21:07:25Z	Circle factorization has an explicit Hochschild chain model	D04
2025:Y25-03-028 2025-03-04T15:53:49Z	Fibonacci fusion has a complete two-dimensional algebraic certificate	D16
2025:Y25-03-030 2025-03-04T15:53:49Z	A homotopy fiber retains paths that an ordinary preimage discards	D05
2025:Y25-03-031 2025-03-04T15:56:30Z	A projection loss controls only the directions that its matrix sees	D15
2025:Y25-03-033 2025-03-07T00:40:57Z	The eta factors carry a calculable modular phase	D01

Source identifier / date	Revised treatment	Domain
2025:Y25-03-035 2025-03-07T00:42:53Z	A square-root eigenvalue law determines a spectral dimension	D07
2025:Y25-03-006 2025-03-09T18:48:28Z	The recovered renormalization series has a conditional convergence theorem	D08
2025:Y25-03-007 2025-03-10T01:55:28Z	An automorphic transformation law comes from reindexing a lattice sum	D07
2025:Y25-03-010 2025-03-13T07:19:49Z	Connes distance can be computed exactly on two points	D16
2025:Y25-03-011 2025-03-13T07:19:49Z	Coherence is an identity of operations, and a spectral page still needs a differential	D04
2025:Y25-03-008 2025-03-14T14:27:39Z	Spectral thresholds and curvature must survive changes of scale	D06
2025:Y25-03-009 2025-03-14T16:49:48Z	A categorical surface theory begins with a Frobenius algebra	D04
2025:Y25-03-014 2025-03-17T19:01:02Z	A modular cocycle obeys a twisted composition law	D03
2025:Y25-03-016 2025-03-17T19:01:02Z	A zero-dimensional Gaussian realizes an exact Wetterich flow	D09
2025:Y25-03-017 2025-03-17T19:01:02Z	Spectral flow counts oriented crossings; odd-dimensional integrals need odd forms	D05
2025:Y25-03-012 2025-03-18T01:23:25Z	Finite modular flow distinguishes the state Hamiltonian from the standard-form generator	D03
2025:Y25-03-013 2025-03-18T01:23:25Z	The entropy first law is an exact first derivative at a faithful state	D11
2025:Y25-03-015 2025-03-18T01:23:25Z	Wilson-loop words need a connection or a well-defined group assignment	D06
2025:Y25-03-019 2025-03-21T06:13:18Z	A semidirect product requires an action, even when its factors are familiar	D03
2025:Y25-03-018 2025-03-21T06:14:55Z	A Maurer–Cartan equation needs graded brackets and a curvature calculation	D16
2025:Y25-03-020 2025-03-21T06:14:55Z	A projective multiplier gives a concrete H^2 cocycle	D05
2025:Y25-03-021 2025-03-21T06:14:55Z	Cohomology-preserving blocking requires a chain homotopy certificate	D04
2025:Y25-03-026 2025-03-24T06:01:49Z	An energy bound exposes incompatible power and thrust scalings	D09
2025:Y25-03-025 2025-03-28T19:55:14Z	Second-order correlations are factorial moments with distinguishable models	D10
2025:Y25-03-022 2025-03-28T19:56:25Z	A power-law phase-noise spectrum needs cutoffs to have finite variance	D10
2025:Y25-03-023 2025-03-28T19:56:25Z	Eliminating a hidden variable derives a memory kernel and a source term	D09
2025:Y25-03-024 2025-03-28T19:56:25Z	The recovered self-bilinear Hirota equation loses its time derivative	D08

Source identifier / date	Revised treatment	Domain
2025:Y25-04-001 2025-04-01T15:59:45Z	The number384 is a cyclotomic degree, with a computable Galois group	D01
2025:Y25-04-002 2025-04-01T15:59:47Z	A finite characteristic polynomial cannot equal a zeta entire function	D07
2025:Y25-04-024 2025-04-01T20:19:02Z	Dimensionful constants cannot be numerical predictions of unit-free integers alone	D09
2025:Y25-04-003 2025-04-10T00:59:30Z	The sine matrix at π/840 is orthogonal only with the full839 indices	D16
2025:Y25-04-004 2025-04-10T00:59:30Z	Quantum-dimension probabilities must be normalized by their actual sum	D16
2025:Y25-04-009 2025-04-18T17:31:12Z	Milnor numbers come from a finite Jacobian algebra	D02
2025:Y25-04-010 2025-04-18T17:31:12Z	A Brieskorn link can have torsion invisible to its Betti numbers	D05
2025:Y25-04-017 2025-04-18T17:31:12Z	A rational Harper operator is a finite clock–shift problem	D16
2025:Y25-04-018 2025-04-18T17:31:12Z	A gap-label Diophantine equation determines a residue class, not an open gap	D16
2025:Y25-04-014 2025-04-18T17:35:51Z	Hyperbolic volume growth prevents a global Euclidean doubling argument	D06
2025:Y25-04-015 2025-04-18T17:35:51Z	Local hyperbolic Hardy atoms pair with the matching BMO norm	D07
2025:Y25-04-016 2025-04-18T17:35:51Z	A hyperbolic spectral lower bound follows from a completed square	D06
2025:Y25-04-012 2025-04-19T20:21:13Z	A logarithmic derivative is local analytic data, not an undefined regulator identity	D07
2025:Y25-04-013 2025-04-19T20:21:13Z	Golden-ratio dilogarithms have exact functional-identity evaluations	D07
2025:Y25-04-019 2025-04-20T00:41:00Z	Golden-ratio contraction with two overlapping branches fills an interval	D09
2025:Y25-04-020 2025-04-20T00:41:00Z	A multiplicative attractor update is a linear recurrence with a sharp stability test	D09
2025:Y25-04-021 2025-04-20T00:41:00Z	Wavelet coherence needs smoothing to avoid an identically-one ratio	D07
2025:Y25-04-011 2025-04-20T20:11:26Z	The recovered top-form quotient would vanish if all exact forms are removed	D04
2025:Y25-04-022 2025-04-20T20:11:26Z	A homotopy pushout needs a common category and actual arrows	D04
2025:Y25-04-028 2025-04-20T20:11:26Z	The E8 dimension248 follows from an explicit root count	D03
2025:Y25-04-025 2025-04-23T20:06:04Z	A two-level gap produces an exact doubled-frequency observable	D16
2025:Y25-04-026 2025-04-23T20:06:04Z	A Bogoliubov transformation must preserve the canonical commutator	D16

Source identifier / date	Revised treatment	Domain
2025:Y25-04-027 2025-04-23T20:06:04Z	A switched oscillator gives exact parametric instability without a resonance slogan	D09
2025:Y25-04-006 2025-04-23T20:13:27Z	Müger nondegeneracy can be checked through an explicit bicharacter	D16
2025:Y25-04-007 2025-04-23T20:13:27Z	The quantum double of Z2 has distinct flux and charge labels	D16
2025:Y25-04-005 2025-04-23T20:22:00Z	Fusion coefficients are constrained by a recurrence and parity	D16
2025:Y25-04-008 2025-04-23T20:22:00Z	A Frobenius algebra and a Lagrangian dimension test require actual multiplication	D16
2025:Y25-04-023 2025-04-27T04:14:14Z	Chromatic height is visible in a formal-group p-series	D02
2025:Y25-05-001 2025-05-04	The Mexican-hat Fourier pair needs a declared normalization	D07
2025:Y25-05-002 2025-05-04	Curl flux is a boundary quantity before it is a topological invariant	D05
2025:Y25-05-003 2025-05-09T19:22:59Z	A primitive fourth-power counterexample defeats the proposed exclusion	D01
2025:Y25-05-004 2025-05-14T16:27:05Z	The periodic prime-grid Laplacian has an exact Fourier spectrum	D07
2025:Y25-05-005 2025-05-14T16:28:34Z	Galerkin projection explains what a coarse correction can guarantee	D15
2025:Y25-05-006 2025-05-14T16:29:47Z	Functional-calculus and Newton error bounds need explicit domains	D08
2025:Y25-05-007 2025-05-18T02:28:11Z	Tripartite information vanishes for every pure three-party state	D16
2025:Y25-05-008 2025-05-18T02:28:11Z	A filtration changes homology when simplices enter, not when labels sound singular	D05
2025:Y25-05-011 2025-05-21T01:13:59Z	Derived tensor products record information an ordinary product erases	D04
2025:Y25-05-009 2025-05-21T03:29:31Z	Tropicalization retains valuation constraints and forgets unit data	D02
2025:Y25-05-010 2025-05-21T03:29:31Z	A p-adic integral can be evaluated exactly by valuation shells	D01
2025:Y25-05-012 2025-05-22T03:48:58Z	Energy ratios require a shared boundary and a complete denominator	D09
2025:Y25-05-013 2025-05-22T04:04:37Z	The stated gravity-wave dispersion has perpendicular phase and group directions	D08
2025:Y25-05-014 2025-05-23T23:36:13Z	A tolerance relation cannot serve as field equality	D03
2025:Y25-05-015 2025-05-23T23:36:13Z	Uncertainty propagation needs covariance and does not create a field	D11
2025:Y25-05-016 2025-05-23T23:52:46Z	Contextual arithmetic needs restriction maps before sheaf claims	D05

Source identifier / date	Revised treatment	Domain
2025:Y25-05-017 2025-05-24T00:26:43Z	Totient-weighted variation is a hybrid optimization problem	D09
2025:Y25-05-018 2025-05-24T00:26:43Z	The recovered gcd coupling defines a valid graph, not an automatic rigidity law	D13
2025:Y25-05-019 2025-05-24T04:27:42Z	Finite spectral zeta and heat transforms do not imply a universal density law	D07
2025:Y25-05-021 2025-05-24T04:27:42Z	A PAC–Bayes guarantee needs a sampling law, prior and bounded loss	D11
2025:Y25-05-020 2025-05-24T04:44:21Z	Multiplicative convolution has a precise divisor algebra	D01
2025:Y25-05-022 2025-05-30T02:40:47Z	A thermal pulse bound reverses the source area-to-volume factor	D09
2025:Y25-05-024 2025-05-31T03:21:40Z	A bonded-layer stress calculation needs a compatibility model	D09
2025:Y25-05-023 2025-05-31T17:23:55Z	Effective conductivity depends on geometry and interface resistance	D08
2025:Y25-06-001 2025-06-01T03:08:28Z	ABC quality starts with an exact triple audit	D01
2025:Y25-06-002 2025-06-01T05:43:50Z	A smoothed radical has computable derivatives but no automatic curvature	D01
2025:Y25-06-003 2025-06-03T18:11:03Z	Power-residue filters prove exclusions only for their declared sequence	D01
2025:Y25-06-005 2025-06-03T20:46:33Z	The tanh fixed-point model has a provable threshold at$\lambda=1$	D09
2025:Y25-06-004 2025-06-04T12:43:38Z	The factorial gcd ratio is an exact valuation calculation	D01
2025:Y25-06-006 2025-06-08T04:10:12Z	A prime-delta operator needs a quadratic form and jump conditions	D08
2025:Y25-06-007 2025-06-12T05:44:12Z	A fibre functor reconstructs a finite group only with its whole category	D04
2025:Y25-06-013 2025-06-12T05:50:33Z	Étale and crystalline comparisons require different base data	D02
2025:Y25-06-008 2025-06-12T13:25:59Z	A modal reflector can be exact while failing tensor preservation	D04
2025:Y25-06-009 2025-06-12T13:55:56Z	Projective motivic cohomology retains the base field in every summand	D02
2025:Y25-06-010 2025-06-12T14:38:37Z	Motivic realization is not the claimed universal Betti isomorphism	D02
2025:Y25-06-011 2025-06-12T15:00:47Z	Künneth signs and Tor corrections survive a tiny exact example	D04
2025:Y25-06-012 2025-06-12T15:00:47Z	Gysin localization and a torus example force codimension shifts	D04
2025:Y25-06-015 2025-06-13T15:09:39Z	Hamiltonian tuning dynamics need an energy and damping law	D09

Source identifier / date	Revised treatment	Domain
2025:Y25-06-014 2025-06-13T15:42:40Z	Rational musical intervals have an exact logarithmic lattice map	D07
2025:Y25-06-018 2025-06-15T11:40:21Z	Tripod overlap and sensor aliasing have exact finite certificates	D15
2025:Y25-06-019 2025-06-15T12:22:14Z	Resonance and regenerative accounting are separate mathematical claims	D09
2025:Y25-06-021 2025-06-16T15:20:01Z	An invented coefficient cannot turn unlike quantities into a validated scalar	D09
2025:Y25-06-020 2025-06-17T17:13:49Z	A strong correlation is not a causal policy proof	D11
2025:Y25-06-016 2025-06-22T16:55:09Z	A nonlinear dyad has a rank-one Jacobian, not a bilinear tensor law	D03
2025:Y25-06-017 2025-06-22T17:17:35Z	Nonlocal dyad learning needs an adjoint in its variational update	D12
2025:Y25-06-022 2025-06-24T05:21:32Z	Area, fourth-power temperature and efficiency ratios need the same balance	D09
2025:Y25-06-024 2025-06-28T00:08:43Z	Exhaustive finite tests prove a polynomial identity only with a degree bound	D14
2025:Y25-06-023 2025-06-28T00:09:50Z	The mod-three B_4 action has five exact orbits	D03
2025:Y25-07-030 2025-07-02	A SAT avoidance complex does not make all 2-SAT instances contractible	D05
2025:Y25-07-031 2025-07-02	A decision reduction does not automatically inject homology	D14
2025:Y25-07-033 2025-07-02	A singleton gives an exact Frankl injection	D13
2025:Y25-07-032 2025-07-03	Ollivier curvature is a transport calculation, not a hardness oracle	D06
2025:Y25-07-035 2025-07-04	A Mackey decomposition can be proved by orbit splitting	D03
2025:Y25-07-034 2025-07-06	Bernoulli691 belongs to a precise coefficient identity	D01
2025:Y25-07-001 2025-07-08	Modular symbols begin with a boundary relation	D01
2025:Y25-07-002 2025-07-08	Local Euler factors distinguish three coefficient sequences	D01
2025:Y25-07-003 2025-07-08	Weierstrass invariants are a first step toward Tate data	D02
2025:Y25-07-004 2025-07-08	Selmer bounds require actual local conditions	D01
2025:Y25-07-005 2025-07-08	Height pairings turn independence into a Gram certificate	D02
2025:Y25-07-011 2025-07-10	Normalized conformal growth preserves area, not every proposed entropy	D06

Source identifier / date	Revised treatment	Domain
2025:Y25-07-012 2025-07-10	A fixed-geometry heat model has a complete energy law	D08
2025:Y25-07-013 2025-07-12	The dimension of Ext is not the splitting class of one extension	D04
2025:Y25-07-006 2025-07-13	Quaternion invariants obey a global reciprocity constraint	D01
2025:Y25-07-007 2025-07-13	Exact certificates can cross an untrusted solver boundary	D14
2025:Y25-07-008 2025-07-15	Vector clocks describe a partial order with a join	D14
2025:Y25-07-009 2025-07-15	Bayesian prediction differs from plug-in independence	D10
2025:Y25-07-010 2025-07-15	A scalar LQR controller has an exact value certificate	D15
2025:Y25-07-014 2025-07-16	The arithmetic-ribbon ODE is an exact scalar connection	D08
2025:Y25-07-015 2025-07-16	Finite matrices cannot satisfy an exact canonical commutator	D16
2025:Y25-07-016 2025-07-17	The numbers 24 and 240 arise from different exact constructions	D03
2025:Y25-07-017 2025-07-17	A multiple-zeta product has an exact combinatorial decomposition	D07
2025:Y25-07-018 2025-07-17	A two-point spectral triple computes a genuine distance	D03
2025:Y25-07-019 2025-07-17	An sl_2 crystal records weights without inventing a quantum model	D03
2025:Y25-07-020 2025-07-23	A genuine 7-adic obstruction explains the ternary exclusions	D01
2025:Y25-07-021 2025-07-23	Cyclic decimals and the tuple identities require separate proofs	D01
2025:Y25-07-022 2025-07-25	The finite ray class group of $Q(\sqrt{10})$ can be computed exactly	D01
2025:Y25-07-023 2025-07-25	The claimed Hilbert class field has the wrong ramification	D01
2025:Y25-07-024 2025-07-25	A real quartic is not its D_4 splitting field	D01
2025:Y25-07-025 2025-07-27	The quadratic sequence has a rational generating function	D13
2025:Y25-07-026 2025-07-27	The proposed mock-modular shadow has incompatible weight and domain	D07
2025:Y25-07-027 2025-07-27	A_2 and D_4 lattice data precede any Jacobi completion	D03
2025:Y25-07-028 2025-07-28	A fitted density line and three beta values do not define a theory	D11

Source identifier / date	Revised treatment	Domain
2025:Y25-07-029 2025-07-28	A quartic splitting field cannot have Mathieu Galois group	D01
2025:Y25-08-909 2025-08-01T16:13:11Z	A calibrated search has a computable multiplicity cost	D11
2025:Y25-08-905 2025-08-02T00:45:27Z	An Eisenstein correction has an exact modular transformation law	D07
2025:Y25-08-901 2025-08-02T05:10:45Z	An exact KP line soliton from a traveling-wave reduction	D16
2025:Y25-08-001 2025-08-03	Spirograph closure is a commensurability theorem	D06
2025:Y25-08-002 2025-08-03	SO(3) rotations have a precise infinitesimal commutator	D03
2025:Y25-08-908 2025-08-03T03:56:33Z	The classical Heegner hypothesis is a local test	D01
2025:Y25-08-907 2025-08-04T02:41:44Z	Filtered Frobenius membership needs a precision certificate	D01
2025:Y25-08-903 2025-08-04T15:22:05Z	A nonprincipal ideal gives an actual Picard obstruction	D01
2025:Y25-08-902 2025-08-04T15:37:10Z	Finite-field Hilbert90 as an exact descent calculation	D01
2025:Y25-08-904 2025-08-05T06:22:34Z	Clutching produces a computable cohomological obstruction	D05
2025:Y25-08-910 2025-08-05T15:20:07Z	Compounding distinguishes price growth from purchasing-power decay	D09
2025:Y25-08-906 2025-08-08T23:35:01Z	A Mellin split gives a reproducible central L-value formula	D07
2025:Y25-08-003 2025-08-09	A Temperley–Lieb trace needs a positivity check	D16
2025:Y25-08-004 2025-08-09	Perron data define a Markov measure and a pressure value	D09
2025:Y25-08-007 2025-08-09	Cubic KKT equations do not certify a global maximum	D12
2025:Y25-08-008 2025-08-09	Consistent Gaussian coordinates need not live in the Hilbert space	D10
2025:Y25-08-005 2025-08-10	The Perron projection must carry the Perron eigenvalue	D03
2025:Y25-08-006 2025-08-10	Chebyshev acceleration needs an interval and a norm	D07
2025:Y25-08-020 2025-08-10	BFGS positivity requires curvature, not Armijo alone	D12
2025:Y25-08-021 2025-08-10	ISTA and conjugate gradients need different mathematical contracts	D12
2025:Y25-08-009 2025-08-12	Čech obstruction transfer requires a chain equivalence	D04

Source identifier / date	Revised treatment	Domain
2025:Y25-08-016 2025-08-13	Gauge transformations and Picard–Fuchs coordinates must be normalized	D08
2025:Y25-08-010 2025-08-15	Translated cellular-automaton recurrence is a relative periodic orbit	D09
2025:Y25-08-028 2025-08-15	Complete-graph labels must obey the edge-count bound	D13
2025:Y25-08-018 2025-08-16	The denominator5760 follows from a characteristic power series	D05
2025:Y25-08-019 2025-08-17	A B_5 symmetry label does not force empirical rank five	D03
2025:Y25-08-011 2025-08-20	Measurability and L^p convergence need different hypotheses	D07
2025:Y25-08-012 2025-08-20	A torus delta integral requires a coarea factor	D07
2025:Y25-08-013 2025-08-20	Hardy atoms bound BMO pairings with an explicit constant	D07
2025:Y25-08-014 2025-08-20	A Landau quartic has a precise minimizer set	D09
2025:Y25-08-015 2025-08-20	Hall coefficients count subobjects even when Ext^1 vanishes	D03
2025:Y25-08-017 2025-08-20	Phase consistency is a cycle condition, not a missing matrix certificate	D05
2025:Y25-08-023 2025-08-23	The Heawood spectrum follows from Fano-plane incidence	D13
2025:Y25-08-024 2025-08-23	Paley Laplacian moments have closed rational values	D13
2025:Y25-08-025 2025-08-24	CRT and finite entropy do not prove decimal normality	D11
2025:Y25-08-027 2025-08-25	The Riccati flow and its wall times are exact within the model	D08
2025:Y25-08-026 2025-08-26	The lattice symbol has a proved dispersion and anisotropy bound	D08
2025:Y25-08-029 2025-08-31	A rank-one CRO matrix has an exact spectrum and inverse	D07
2025:Y25-08-030 2025-08-31	A residual for one vector cannot certify vanishing of H^1	D05
2025:Y25-08-022 Date unavailable	Optimizer hypergradients must follow the entire state recursion	D12
2025:Y25-09-001 2025-09-01	A compositional contraction certificate with an exact fixed point	D15
2025:Y25-09-019 2025-09-02	A modular minor certifies a rational rank lower bound	D03
2025:Y25-09-020 2025-09-06	Adjacent exact inequalities prove full monotonicity of a finite clock	D14

Source identifier / date	Revised treatment	Domain
2025:Y25-09-021 2025-09-08	A genuine KdV soliton requires a PDE identity, not a checkpoint metaphor	D08
2025:Y25-09-005 2025-09-09	A trace invariant follows from local rewrite equations by induction	D14
2025:Y25-09-018 2025-09-10	A geometric checkpoint clock estimates a decay rate without being a soliton	D11
2025:Y25-09-015 2025-09-14	The square-root smoothing map has dynamics but is not a linear Hecke operator	D09
2025:Y25-09-007 2025-09-15	The symmetric-square factor needs all three eigenvalue products	D01
2025:Y25-09-008 2025-09-15	FFT derivatives are exact for an interpolant, not for an unknown function	D07
2025:Y25-09-009 2025-09-15	A contour count can certify two zeros without certifying a double zero	D07
2025:Y25-09-010 2025-09-15	A μ-invariant is a valuation, not a real-valued twist statistic	D01
2025:Y25-09-016 2025-09-15	Integer micro-arithmetic needs a proved rounding envelope	D14
2025:Y25-09-006 2025-09-17	Genuine local arithmetic replaces a synthetic Hecke trace	D02
2025:Y25-09-013 2025-09-19	Fractional ownership is a partial algebra, not duplicable permission	D14
2025:Y25-09-014 2025-09-19	Bisimulation preserves observable traces under a stated transition relation	D14
2025:Y25-09-017 2025-09-19	Chebyshev polynomials give exact normalized Hecke recurrences	D07
2025:Y25-09-003 2025-09-20	Transport defects satisfy a cocycle law and a sharp norm bound	D05
2025:Y25-09-004 2025-09-20	Cokernel certificates detect torsion that real residuals can miss	D03
2025:Y25-09-011 2025-09-20	A cross-ratio produces an auditable map into elliptic-curve moduli	D02
2025:Y25-09-012 2025-09-21	A Legendre-symbol certificate has a precise and limited arithmetic meaning	D14
2025:Y25-09-002 2025-09-29	The cosine iteration has a rigorous tail certificate	D09
2025:Y25-10-001 2025-10-01	The RG leaf has an exact first integral, and its two-loop extension introduces a logarithm	D08
2025:Y25-10-002 2025-10-01	Whitening and signed-permutation invariance impose different equations	D03
2025:Y25-10-003 2025-10-01	Newton lifting supplies an exact p-adic certificate rather than a digest comparison	D01
2025:Y25-10-004 2025-10-03	The mapping cone turns an echo equivalence into an acyclicity test	D04

Source identifier / date	Revised treatment	Domain
2025:Y25-10-005 2025-10-03	Transition functions satisfy a cocycle equation, and gauge changes preserve it	D05
2025:Y25-10-006 2025-10-03	A pushout glues compatible maps by a universal property	D04
2025:Y25-10-007 2025-10-05	A rank-two Mordell–Weil group forces a4-Selmer group of at least sixteen elements	D02
2025:Y25-10-008 2025-10-05	A regulator is a Gram determinant tied to a generator basis	D02
2025:Y25-10-009 2025-10-09	A real quadratic L-derivative is exactly the logarithm of a unit	D01
2025:Y25-10-011 2025-10-09	A heat supertrace equals an Euler index because nonzero modes pair	D04
2025:Y25-10-013 2025-10-09	A theta heat equation and an elliptic transformation can be proved term by term	D07
2025:Y25-10-014 2025-10-10	An index does not determine Yukawa singular values	D05
2025:Y25-10-015 2025-10-10	A Tor group requires modules and a resolution before it can model an obstruction	D04
2025:Y25-10-012 2025-10-12	A Picard–Fuchs basis comparison is determined by values and derivatives at one regular point	D08
2025:Y25-10-016 2025-10-14	The E_8 roots, its Lie dimension and the totient384 are different exact counts	D03
2025:Y25-10-017 2025-10-14	Semisimplicity kills ordinary extension classes	D04
2025:Y25-10-018 2025-10-14	A four-object pointed model has fully checkable F, R and S data	D16
2025:Y25-10-019 2025-10-14	A Pareto frontier contains points that no linear selector can choose	D12
2025:Y25-10-010 2025-10-16	Heat residues follow from a Mellin transform and need not be integer indices	D08
2025:Y25-10-029 2025-10-19	Fiber integration commutes with d on closed product fibers in a fixed convention	D06
2025:Y25-10-030 2025-10-19	The Bloch–Wigner correction has coefficient one, and a scalar residue leaves a level curve	D07
2025:Y25-10-020 2025-10-21	Character orthogonality gives an exact frame bound before a large-sieve claim	D07
2025:Y25-10-021 2025-10-21	Projecting the updated state has a descent proof; clipping the gradient is a different operation	D12
2025:Y25-10-027 2025-10-21	Trace class is contained in Hilbert–Schmidt, and strong convergence alone preserves neither	D07
2025:Y25-10-036 2025-10-22	Product-variable reduction gives exact zeta integrals and a controlled fractional-part transform	D07
2025:Y25-10-022 2025-10-23	A finite prime oscillatory sum has exact calculus but no automatic zero correspondence	D01

Source identifier / date	Revised treatment	Domain
2025:Y25-10-033 2025-10-23	Prime-log tones have exact beat differences but no common nonzero period	D09
2025:Y25-10-034 2025-10-27	Second and fourth mean values expose multiplicative resonances exactly	D07
2025:Y25-10-035 2025-10-27	The asset recurrence has an exact stability threshold independent of its interpretation	D09
2025:Y25-10-028 2025-10-29	Cahn–Hilliard flow conserves mass and dissipates a declared energy	D08
2025:Y25-10-031 2025-10-30	Finite-group symmetry permits frequency ratios but does not usually force one	D03
2025:Y25-10-032 2025-10-30	Damping ratio, decay and Q must use the same oscillator convention	D09
2025:Y25-10-023 2025-10-31	The sine quotient has an exact parity integral and a finite quadrature certificate	D07
2025:Y25-10-024 2025-10-31	The logarithmic sine integral separates into an elementary term and Catalan’s constant	D07
2025:Y25-10-025 2025-10-31	The recovered fourth-power logarithmic integral has an exact elementary evaluation	D07
2025:Y25-10-026 2025-10-31	Cyclotomic trace orthogonality fixes the coefficient p, not p+1	D01
2025:Y25-11-001 2025-11-01	The cosine–quartic integral has an exact closed form	D07
2025:Y25-11-006 2025-11-01	Exact forms have zero periods; winding and entropy need different data	D05
2025:Y25-11-003 2025-11-02	A positive kernel is not necessarily positive definite	D07
2025:Y25-11-004 2025-11-02	A bridge functional with an exact variance	D10
2025:Y25-11-005 2025-11-02	Boundary conditions determine a beam operator and its stability	D08
2025:Y25-11-002 2025-11-05	Three integral checks from the integer-breath atlas	D07
2025:Y25-11-007 2025-11-05	The proposed resonator spectrum grows factorially, not geometrically	D07
2025:Y25-11-011 2025-11-05	A radial heat kernel and a convergent Volterra expansion	D08
2025:Y25-11-012 2025-11-06	A Hecke relation is not an arbitrary prime-index power ratio	D01
2025:Y25-11-010 2025-11-07	Wallis recurrences and what a hypergeometric operator really proves	D07
2025:Y25-11-008 2025-11-10	Lamé’s first finite-gap example from Jacobi identities	D08
2025:Y25-11-009 2025-11-10	The product locus is Humbert discriminant one, not four	D02

Source identifier / date	Revised treatment	Domain
2025:Y25-11-017 2025-11-10	An inflection ridge is not automatically a phase transition	D09
2025:Y25-11-016 2025-11-11	Kronecker flow is densely recurrent and has zero entropy	D09
2025:Y25-11-013 2025-11-12	The scalar Laplacian variation under Ricci flow	D06
2025:Y25-11-014 2025-11-12	Commutator traces, matrix words, and the missing regularization	D03
2025:Y25-11-015 2025-11-13	Circular concentration measures evidence, not a motivic sector	D11
2025:Y25-11-018 2025-11-15	Tomita–Takesaki theory in a faithful finite matrix model	D03
2025:Y25-11-019 2025-11-15	Spectral multiplicity defeats the sealed-commutant claim	D07
2025:Y25-11-027 2025-11-16	IMP projection equals an iteration limit only under spectral hypotheses	D15
2025:Y25-11-029 2025-11-18	Low-rank computation has a measurable error, not a universal speedup	D15
2025:Y25-11-031 2025-11-18	Poisson-consistent smoothing follows from commuting convolution	D08
2025:Y25-11-030 2025-11-19	A Riccati equation with a complete optimal-control proof	D15
2025:Y25-11-028 2025-11-20	The recovered curvature block is unstable and not the standard convex stencil	D08
2025:Y25-11-021 2025-11-21	Levi adjacency is controlled by singular values	D13
2025:Y25-11-020 2025-11-24	Catmull–Clark Fourier blocks derived from the actual stencil	D07
2025:Y25-11-024 2025-11-24	The recovered S_6 has square-zero Jordan mixing	D03
2025:Y25-11-025 2025-11-24	The plastic constant is a genuine Pisot recurrence eigenvalue	D01
2025:Y25-11-022 2025-11-25	Discrete Hodge splitting and the correct cohomology transport group	D04
2025:Y25-11-023 2025-11-25	Forced block dynamics and a genuine Schur-complement correction	D09
2025:Y25-11-026 2025-11-29	Cancelling quadratic drift requires an expansion, not a benchmark label	D08
2025:Y25-12-024 2025-12-02	Weighted threshold rank needs a tensorial transformation law	D15
2025:Y25-12-025 2025-12-03	A spherical portal can cancel only specified spectral components	D07
2025:Y25-12-026 2025-12-03	Greedy spectral deflation converges to zero in trace norm	D07

Source identifier / date	Revised treatment	Domain
2025:Y25-12-008 2025-12-04	Freidlin–Wentzell action as an exact linear control problem	D15
2025:Y25-12-007 2025-12-06	Euler–Maruyama and an exact Ornstein–Uhlenbeck comparison	D10
2025:Y25-12-010 2025-12-07	A constant rectangular metric cannot make a detour shorter	D06
2025:Y25-12-012 2025-12-09	Abel–Liouville controls volume, not every mode	D09
2025:Y25-12-001 2025-12-10	The Stirling–Bernoulli matrix is exact and singular	D03
2025:Y25-12-002 2025-12-10	Bernoulli inversion and the corrected Blissard product	D07
2025:Y25-12-003 2025-12-10	Weighted ℓ^1 bounds use weighted columns	D07
2025:Y25-12-004 2025-12-10	Finite differences are genuine FIR annihilators	D15
2025:Y25-12-005 2025-12-10	Composition by $x+x^d$ is triangular but not uniformly banded	D03
2025:Y25-12-009 2025-12-10	Polyhedral Fenchel duality and the drift-action distinction	D12
2025:Y25-12-011 2025-12-10	Escape probability inversion needs a rate reference scale	D10
2025:Y25-12-027 2025-12-10	Cumulant evolution is triangular for an actual OU process	D10
2025:Y25-12-006 2025-12-12	The first logistic period doublings can be derived exactly	D09
2025:Y25-12-013 2025-12-12	Decimal remainders and the exact Koopman period	D01
2025:Y25-12-016 2025-12-14	Quantization error and Layer Norm require a positive denominator	D14
2025:Y25-12-015 2025-12-15	Mahalanobis proximal steps and what a tube projector preserves	D12
2025:Y25-12-017 2025-12-20	A noisy pitchfork has an exact barrier and a conditional Kramers law	D09
2025:Y25-12-014 2025-12-22	Softmax Fisher geometry and a uniform fixed-point diagnostic	D11
2025:Y25-12-018 2025-12-22	The recovered oscillator has minimal continuous state dimension eight	D15
2025:Y25-12-019 2025-12-22	Exact windings of the recovered oscillator	D05
2025:Y25-12-021 2025-12-24	A rigorous two-point-to-gradient inequality	D07
2025:Y25-12-023 2025-12-24	A conditional proof chain is not a verified closure	D14

Source identifier / date	Revised treatment	Domain
2025:Y25-12-022 2025-12-25	Misalignment is nonnegative but cannot universally control gradient energy	D08
2025:Y25-12-020 2025-12-31	A capped-addition monoid with an exact trace ideal	D03
2026:Y26-01-001 2026-01-01	Saturation products preserve kernels for a precise algebraic reason	D03
2026:Y26-01-002 2026-01-01	A lattice encoding of a discrete logarithm still needs an efficient decoder	D14
2026:Y26-01-025 2026-01-04	Cumulative counts require a time-dependent rate and exposure denominator	D11
2026:Y26-01-003 2026-01-05	A semantic quotient is determined by all future observations	D14
2026:Y26-01-004 2026-01-05	Quadratic-form invariants require one consistent cross-term convention	D03
2026:Y26-01-034 2026-01-05	Radical layers and extensions distinguish finite algebras beyond state counts	D03
2026:Y26-01-005 2026-01-06	The reciprocal-maximum kernel has an exact covariance factorization	D07
2026:Y26-01-006 2026-01-06	Divisor weights give positive arithmetic kernels and a sieve formula	D01
2026:Y26-01-007 2026-01-08	A stable Jordan block can transiently amplify a trajectory	D09
2026:Y26-01-008 2026-01-08	Trace-estimator variance depends on the probe distribution	D11
2026:Y26-01-009 2026-01-11	A cubic family makes discriminants, caustics and Airy scaling explicit	D08
2026:Y26-01-010 2026-01-13	The vorticity logarithm has two different diffusion contributions	D08
2026:Y26-01-011 2026-01-13	Polynomial misalignment measures angle but cannot control all dissipation	D09
2026:Y26-01-012 2026-01-13	Gradient-flow curvature is a projected Hessian quantity	D12
2026:Y26-01-013 2026-01-13	A clothoid is the minimizer of a specified curvature-rate functional	D06
2026:Y26-01-014 2026-01-15	ADE graph spectra are exact and do not certify fitted RG intercepts	D13
2026:Y26-01-015 2026-01-15	A finite Gibbs state satisfies an exact KMS identity	D03
2026:Y26-01-026 2026-01-16	Partition derivatives are expectations, while mixtures need an algebra	D11
2026:Y26-01-016 2026-01-17	Cyclotomic degree and prime-density products give exact arithmetic checks	D01
2026:Y26-01-027 2026-01-18	Multiplicative random products give a testable CLT and convergence threshold	D10

Source identifier / date	Revised treatment	Domain
2026:Y26-01-029 2026-01-20	Log folding is a monoid action and recovery is an idempotent projection	D14
2026:Y26-01-017 2026-01-21	Turning-point time and Jacobi distance have different exponents	D09
2026:Y26-01-018 2026-01-21	Thermodynamic contact geometry derives a Maxwell relation	D06
2026:Y26-01-019 2026-01-21	Affine RG coordinates linearize approach to a finite blowup boundary	D08
2026:Y26-01-020 2026-01-21	Fokker–Planck entropy decay follows from an explicit gradient structure	D11
2026:Y26-01-028 2026-01-24	Angular diffusion loses each Fourier mode at its own exact rate	D08
2026:Y26-01-022 2026-01-25	Dimensional consistency changes a thin-plate threshold exponent	D09
2026:Y26-01-023 2026-01-26	A computational monotone needs a specified coarse-graining map	D11
2026:Y26-01-024 2026-01-28	Coalescing Frobenius exponents generate logarithms by a divided difference	D08
2026:Y26-01-901 2026-01-28	Itô correction and a Feynman–Kac model with an exact solution	D10
2026:Y26-01-902 2026-01-28	The semicircle law has exact Catalan moments before any random-matrix limit	D10
2026:Y26-01-903 2026-01-28	A frozen tangent kernel gives an exact learning dynamics with a visible nullspace	D11
2026:Y26-01-904 2026-01-28	A two-point spectral distance can be computed from the commutator norm	D06
2026:Y26-01-905 2026-01-28	An index counts a kernel–cokernel defect in an explicit infinite model	D07
2026:Y26-01-906 2026-01-28	Two-dimensional sewing has a concrete Frobenius-algebra model	D04
2026:Y26-01-907 2026-01-28	The recovered Picard–Fuchs operator has a checkable Frobenius recurrence	D08
2026:Y26-01-908 2026-01-28	A certified arctangent acceleration follows from a geometric expansion	D07
2026:Y26-01-909 2026-01-29	A factorial series has a fully visible Borel singularity and Stokes jump	D08
2026:Y26-01-910 2026-01-29	A triangular regular-singular connection encodes a dilogarithm period	D08
2026:Y26-01-030 2026-01-30	An exact Čech calculation distinguishes local primitives from global gluing	D05
2026:Y26-01-031 2026-01-30	A Selmer bound separates visible rank from hidden torsors	D02
2026:Y26-01-032 2026-01-30	A CM automorphism explains an exact family of local Frobenius traces	D02

Source identifier / date	Revised treatment	Domain
2026:Y26-01-033 2026-01-30	A normalized maximum residual gives a robust acceptance margin	D12
2026:Y26-01-021 2026-01-31	A causal graph separates conservation laws, cycle space and mixing	D05
2026:Y26-02-025 2026-02-02	Linear covariance transport can vanish while nonlinear uncertainty remains	D11
2026:Y26-02-026 2026-02-02	Quasi-static elimination is a resolvent step with a stability condition	D12
2026:Y26-02-001 2026-02-03	Hypergeometric face survival gives an exact expected Euler characteristic	D05
2026:Y26-02-002 2026-02-03	Random birth times connect order statistics to an exact barcode law	D05
2026:Y26-02-003 2026-02-03	Weighted sampling replaces hypergeometric ratios by symmetric polynomials	D10
2026:Y26-02-017 2026-02-03	Empirical Fisher, Gauss–Newton and Hessian are different matrices	D12
2026:Y26-02-028 2026-02-04	Genus-two point counts determine a Frobenius polynomial	D02
2026:Y26-02-029 2026-02-04	Swan conductor measures ramification rather than vector-space rank	D01
2026:Y26-02-032 2026-02-04	A fund target is an annuity recurrence with explicit assumptions	D09
2026:Y26-02-004 2026-02-06	Compatibility and gluing can be tested by a linear obstruction	D05
2026:Y26-02-027 2026-02-07	A prime-supported quadratic energy has an elementary constrained minimizer	D12
2026:Y26-02-010 2026-02-08	Public depth recursions certify algebraic consistency, not sequential work	D14
2026:Y26-02-024 2026-02-08	Spectral flatness is an AM–GM statistic, not a false-acceptance exponent	D11
2026:Y26-02-031 2026-02-10	A rectified accumulator is path-dependent through suffix deficits	D09
2026:Y26-02-033 2026-02-10	Changing the scalar domain changes solvability	D01
2026:Y26-02-012 2026-02-11	A power spectrum can fail to identify a finite sequence	D07
2026:Y26-02-013 2026-02-12	An anisotropic cone has a declared homothety and a link eigenvalue equation	D06
2026:Y26-02-030 2026-02-12	Fibonacci transfer matrices give exact word growth and spectral data	D09
2026:Y26-02-014 2026-02-13	A secretless feature verifier factors through its observation fibers	D15
2026:Y26-02-011 2026-02-14	Hash chains preserve order while holonomy uses a different algebra	D14

Source identifier / date	Revised treatment	Domain
2026:Y26-02-009 2026-02-16	Exponential connections produce periods without proving their independence	D07
2026:Y26-02-007 2026-02-17	An elliptic curve has nontrivial self-extensions even for a stable line bundle	D04
2026:Y26-02-008 2026-02-17	Independent shifts classify a semisimple model only under its own hypotheses	D04
2026:Y26-02-019 2026-02-19	A damped mode gives a Floquet multiplier and an exact memory recursion	D08
2026:Y26-02-005 2026-02-20	Prefix geometry has an exact Hausdorff dimension and a carry obstruction	D06
2026:Y26-02-018 2026-02-20	A Smith certificate computes a quotient only after compatibility is imposed	D05
2026:Y26-02-020 2026-02-20	Eliminating a PT-style coupler yields Hill form with extra harmonics	D08
2026:Y26-02-021 2026-02-20	The Mathieu discriminant has a negative q^2 correction near $a=0$	D08
2026:Y26-02-022 2026-02-20	Path induction proves a contractible based-path space, not conservativity	D14
2026:Y26-02-034 2026-02-21	Prefix entropy grows by conditional information, not by metric contraction alone	D11
2026:Y26-02-006 2026-02-23	Retaining a rounding residual repairs a concrete cancellation failure	D14
2026:Y26-02-023 2026-02-24	Fractional analog states still have a finite resolution capacity	D11
2026:Y26-02-015 2026-02-26	A derived inverse-limit obstruction can be computed without a geometric analogy	D04
2026:Y26-02-035 2026-02-26	A quadratic Wasserstein proximal step has an exact pushforward solution	D12
2026:Y26-02-016 2026-02-28	Contraction yields finite-resolution stabilization only with a compatible observation	D15
2026:Y26-03-001 2026-03-01T02:25:36Z	A semigroup contraction is exactly a generator inequality in the same metric	D15
2026:Y26-03-005 2026-03-01T20:47:09Z	A vanishing gap produces inverse time scales only for the modes it controls	D07
2026:Y26-03-002 2026-03-03T02:18:28Z	The Lyapunov Gramian certifies stability and exposes nonnormal amplification	D09
2026:Y26-03-003 2026-03-03T02:23:49Z	A resolvent measures gain; it does not determine arbitrary parameter curvature	D07
2026:Y26-03-004 2026-03-03T02:23:49Z	Discrete stability has its own Gramian and a unit-circle resolvent	D09
2026:Y26-03-022 2026-03-04T07:18:50Z	An alignment regression needs a declared sample and a nondegenerate denominator	D11
2026:Y26-03-006 2026-03-06T20:42:57Z	Hypergraph matching bounds must distinguish disjointness from induced tensor restrictions	D13

Source identifier / date	Revised treatment	Domain
2026:Y26-03-007 2026-03-06T20:42:57Z	A proposed concentration inequality yields a necessary numerical threshold	D12
2026:Y26-03-026 2026-03-08T03:49:27Z	A reflected complex map is an involution; a damped Fourier recurrence has a separate stability test	D09
2026:Y26-03-027 2026-03-08T03:49:27Z	Fibonacci binomial sums and Pisot contraction have exact algebraic certificates	D13
2026:Y26-03-008 2026-03-08T22:57:42Z	The logarithmic-cosine integral has an exact evaluation on its convergence strip	D07
2026:Y26-03-009 2026-03-08T23:01:30Z	A two-parameter exponent collapses along explicit level curves	D15
2026:Y26-03-024 2026-03-09T03:56:19Z	An exponential kernel family has an explicit transcendental flip candidate	D09
2026:Y26-03-025 2026-03-12T03:23:50Z	A born two-cycle has square-root branching while the fixed point may remain single-valued	D09
2026:Y26-03-023 2026-03-14T09:19:00Z	Gradient descent reaches a flip boundary at twice the inverse top curvature	D09
2026:Y26-03-014 2026-03-14T20:03:09Z	An exponential–rational master equation has a conditional transcendence obstruction	D01
2026:Y26-03-015 2026-03-15T00:58:50Z	A Lambert-W rewriting needs a branch condition and adds no missing equation	D07
2026:Y26-03-011 2026-03-17T01:23:03Z	A normalized analytic germ has coordinate-dependent jets and invariantly stated derivatives	D07
2026:Y26-03-012 2026-03-17T22:30:35Z	A short recurrence is testable at the first coefficient it did not fit	D14
2026:Y26-03-013 2026-03-17T22:37:28Z	A bounded-degree differential relation becomes a finite linear algebra problem	D07
2026:Y26-03-021 2026-03-19T11:52:05Z	Four reported invariants need an injectivity theorem before they identify one singularity	D03
2026:Y26-03-016 2026-03-26T01:31:43Z	Commuting projections give an exact Boolean block decomposition	D03
2026:Y26-03-018 2026-03-27T23:07:13Z	A coupled residual energy is uniquely minimized when its Hessian is uniformly positive	D12
2026:Y26-03-017 2026-03-28T01:46:08Z	Local nonsingularity does not imply domain-global uniqueness	D15
2026:Y26-03-010 2026-03-28T17:48:44Z	The fixed-point flip reduces to one explicit scalar equation	D09
2026:Y26-03-019 2026-03-29T20:33:43Z	Singular-value anisotropy depends on a declared input and output metric	D03
2026:Y26-03-020 2026-03-29T20:33:43Z	A saturating scale kernel has an exact logarithmic area	D07
2026:Y26-04-024 2026-04-01T20:11:49Z	A nested fit improves training R^2; a residual runs test asks a different question	D11
2026:Y26-04-025 2026-04-02T07:43:38Z	A quadratic compatibility surface has a conditional minimum with a declared objective	D12

Source identifier / date	Revised treatment	Domain
2026:Y26-04-011 2026-04-03T21:13:13Z	A cyclic difference design has one exact gauge mode	D15
2026:Y26-04-012 2026-04-03T21:13:13Z	Family-constant targets vanish under a symmetric difference design	D15
2026:Y26-04-013 2026-04-06T22:00:15Z	Nested supports do not by themselves force a low-rank sensitivity matrix	D11
2026:Y26-04-014 2026-04-09T11:49:09Z	Participation ratio quantifies spectral concentration without replacing rank	D11
2026:Y26-04-015 2026-04-09T11:49:09Z	Temporal spectral color needs a specified stationary or integrated process	D10
2026:Y26-04-009 2026-04-12T20:40:02Z	A triangular map admits an exact spectral-submanifold graph	D09
2026:Y26-04-010 2026-04-12T23:10:45Z	A spectral gap alone does not select the smoothest invariant graph	D09
2026:Y26-04-005 2026-04-14T15:49:49Z	Noncommuting shears have an exact loop commutator	D06
2026:Y26-04-026 2026-04-14T20:38:43Z	A zeta spiral is a sampling layout with a conjugation symmetry	D01
2026:Y26-04-006 2026-04-16T07:52:11Z	A return-map jet at multiplier minus one has a computable second iterate	D09
2026:Y26-04-007 2026-04-16T08:45:46Z	A finite transfer operator has a genuine pressure and an implicit root derivative	D09
2026:Y26-04-028 2026-04-16T13:39:16Z	A Čech-type cocycle is removable exactly when its loop periods vanish	D05
2026:Y26-04-004 2026-04-17T01:26:30Z	A secondary transport defect requires distinguishing coboundary from differential	D05
2026:Y26-04-008 2026-04-17T01:26:30Z	A two-term coboundary complex has exact kernels and ranges in a finite Markov example	D04
2026:Y26-04-032 2026-04-17T03:32:11Z	A finite-volume derivative has an exact product-rule correction	D07
2026:Y26-04-016 2026-04-19T03:48:18Z	A determinant measures volume but does not determine a directional product field	D06
2026:Y26-04-017 2026-04-19T04:01:36Z	Spherical product moments explain 105 and 945 Gaussian pairings	D10
2026:Y26-04-029 2026-04-21T15:09:42Z	The silver-mean substitution has exact eigenvalues and normalized letter frequencies	D09
2026:Y26-04-033 2026-04-21T15:09:42Z	A quotient has a universal factorization only after its equivalence relation is specified	D04
2026:Y26-04-031 2026-04-22T02:53:03Z	A block resolvent controls weak coupling only relative to a spectral separation	D07
2026:Y26-04-030 2026-04-23T04:25:37Z	The roots of x^4-2 have a dihedral Galois action, not a physical trajectory	D01
2026:Y26-04-018 2026-04-23T06:55:19Z	Trace-range realization is explicit for cyclic and dyadic groups	D03

Source identifier / date	Revised treatment	Domain
2026:Y26-04-019 2026-04-23T07:04:52Z	Gap-projection invariants factor through K_0 under an actual spectral gap	D16
2026:Y26-04-020 2026-04-23T07:04:52Z	Morita trace scaling is an arithmetic lattice identity after normalization	D16
2026:Y26-04-023 2026-04-23T13:57:36Z	Weighted projection is unique on a convex set but can have several competing branches	D12
2026:Y26-04-001 2026-04-26T02:32:14Z	The circulation space of $K_{3,3}$ has an explicit four-cycle basis	D13
2026:Y26-04-002 2026-04-26T02:32:14Z	A Gaussian quartic probe extracts one exact multilinear coefficient	D10
2026:Y26-04-003 2026-04-26T05:15:15Z	Cycle-monomial Gram norms contain one squared coefficient, not a power indexed by length	D03
2026:Y26-04-027 2026-04-28T17:45:36Z	Path differences lie in a circulation kernel, with a useful converse on a finite DAG	D13
2026:Y26-04-022 2026-04-29T20:58:53Z	Pooling standardized domains can create a correlation absent from each domain	D11
2026:Y26-04-021 2026-04-29T21:10:45Z	AUROC is an exact pairwise statistic with a specified tie convention	D11
2026:Y26-05-008 2026-05-02	Z_2 gauge classes are counted by cycle rank and retain a flux detectable in the spectrum	D13
2026:Y26-05-009 2026-05-02	Schwinger bosons realize the spin algebra on each fixed-particle sector	D16
2026:Y26-05-010 2026-05-04	Observation geometry has exact invisible subspaces and a noise-to-state bound	D15
2026:Y26-05-024 2026-05-05	Two-qubit Pauli locality is tautological, while mutual information need not define a metric	D16
2026:Y26-05-025 2026-05-05	Incidence rank computes cycle homology and remains separate from a drawing	D13
2026:Y26-05-004 2026-05-06	A quotient obstruction energy has a computable Hessian and an exact Fourier spectrum	D12
2026:Y26-05-006 2026-05-06	Mechanical words give a constructive Sturmian balance certificate	D09
2026:Y26-05-005 2026-05-07	Fibonacci matrices connect units, geodesic length and an exact fourth-power decay	D01
2026:Y26-05-007 2026-05-08	A finite maximal-ideal algorithm separates constructive witnesses from unrestricted choice	D03
2026:Y26-05-013 2026-05-11	Pressure distinguishes a deterministic Bowen root from an annealed random statistic	D09
2026:Y26-05-014 2026-05-11	The Poincaré disk distance and hyperbolic generator gate have exact formulas	D06
2026:Y26-05-015 2026-05-11	A heat-trace asymptotic is different from a trace-class statement at a fixed time	D07
2026:Y26-05-016 2026-05-15	Hamiltonian conservation follows from the symplectic convention and has an exact pendulum separatrix	D09

Source identifier / date	Revised treatment	Domain
2026:Y26-05-017 2026-05-15	Symplectic reduction removes a cyclic coordinate only after fixing momentum	D06
2026:Y26-05-011 2026-05-16	A Lyapunov metric certifies coercivity even when the Euclidean symmetric part fails	D09
2026:Y26-05-012 2026-05-16	An SDP primal-dual pair gives an exact certificate and a traceable gap	D12
2026:Y26-05-026 2026-05-16	Hybrid switching consumes a logarithmic metric budget at each jump	D09
2026:Y26-05-033 2026-05-17	Prime-factor closure is a Boolean lattice, while pairwise compatibility need not imply joint consistency	D13
2026:Y26-05-001 2026-05-19	Authority composition is a meet calculation with an explicit proof dependency graph	D14
2026:Y26-05-002 2026-05-19	A chain contraction proves homology equivalence and survives a controlled elementary perturbation	D04
2026:Y26-05-003 2026-05-19	The RG characteristic-function map has explicit stable fixed points and a finite-variance Gaussian obstruction	D10
2026:Y26-05-029 2026-05-19	A second-order Maurer–Cartan obstruction is a cohomology class, not a numerical residual	D04
2026:Y26-05-030 2026-05-19	A Bézout identity certifies inconsistency, while Buchberger replay certifies the final ideal basis	D14
2026:Y26-05-027 2026-05-20	Finite-field construction needs an irreducible polynomial and an explicit generator image	D01
2026:Y26-05-028 2026-05-20	A binary Hamming code follows from a rank-three parity map	D14
2026:Y26-05-018 2026-05-23	Stoichiometric recovery is a dimensionally exact mass balance with an explicit yield parameter	D09
2026:Y26-05-031 2026-05-24	The cusp parametrization, local intersection and blowup are compatible exact calculations	D02
2026:Y26-05-032 2026-05-25	A spectral-sequence page does not determine an integral ring or its extension data	D04
2026:Y26-05-021 2026-05-27	Species counting is an exact weighted partition problem with a convergent Euler product	D13
2026:Y26-05-022 2026-05-27	The species exponential scale follows from a tilted-measure argument under explicit prime asymptotics	D13
2026:Y26-05-019 2026-05-30	An exponent species has a unique least integer representative	D01
2026:Y26-05-020 2026-05-30	The divisor lattice is a product of chains with a symmetric rank polynomial	D01
2026:Y26-05-023 2026-05-30	A coercivity debit ledger and a finite gluing obstruction have independent exact meanings	D05
2026:Y26-06-021 2026-06-01	Breath-stream capture arithmetic is a transparent volume-to-mass balance	D09
2026:Y26-06-002 2026-06-03	Quadratic polarization is symmetric and cannot equal an alternating form at odd order	D16

Source identifier / date	Revised treatment	Domain
2026:Y26-06-006 2026-06-06	Sequential quantum instruments preserve probability only when every branch is included	D16
2026:Y26-06-007 2026-06-06	Two-sector failure has a correlation parameter and a rigorous confidence envelope	D10
2026:Y26-06-003 2026-06-08	Finite Fourier and Gauss operators satisfy a projective modular relation with a computed phase	D16
2026:Y26-06-023 2026-06-08	A torus mirror cocycle can carry exactly one binary obstruction	D05
2026:Y26-06-024 2026-06-08	The totient Dirichlet series has the source zeta majorant on its correct half-plane	D01
2026:Y26-06-008 2026-06-10	A weighted path majorant proves an exponential tail under an explicit row-sum bound	D13
2026:Y26-06-022 2026-06-10	The SU_2 sine transform proves orthogonality and its global dimension formula	D16
2026:Y26-06-011 2026-06-11	Bloch–Wigner values give a convergent regular-shape computation with an explicit tail	D07
2026:Y26-06-001 2026-06-15	Finite Weyl pairs exist exactly at root-of-unity phases	D16
2026:Y26-06-004 2026-06-15	CRT factorization and idempotent splitting give constructive finite closure	D01
2026:Y26-06-009 2026-06-15	The discriminant-minus-twenty-three cubic has S_3 closure and cannot lie in an abelian cyclotomic field	D01
2026:Y26-06-010 2026-06-15	A singular corner equation is a birational form of the cubic and has a computable divisor	D02
2026:Y26-06-012 2026-06-15	Branch periods detect monodromy that a local curl test misses	D07
2026:Y26-06-014 2026-06-15	A lattice parallelotope tiles space and basis changes preserve covolume	D13
2026:Y26-06-015 2026-06-15	Planar zonotope area is a determinant sum, not a standalone runtime ratio	D13
2026:Y26-06-025 2026-06-15	The recovered Habiro matrix has a free direction that must be removed before Gaussian integration	D16
2026:Y26-06-005 2026-06-16	SELECT plus state preparation gives an exact linear-combination block	D16
2026:Y26-06-013 2026-06-18	A cubic fold produces the three-halves action law before any thimble claim	D08
2026:Y26-06-032 2026-06-19	The recovered seven-crossing quintic admits an exact field and Hessian inverse certificate	D02
2026:Y26-06-031 2026-06-21	Finite symplectic group order gives the reported projective denominator, not a magic-state proof	D03
2026:Y26-06-028 2026-06-24	Lossless reconstruction forces injectivity and gives a counting lower bound on compression	D11
2026:Y26-06-026 2026-06-25	Alternating projections have an exact contraction factor and a distinct commutator norm	D09

Source identifier / date	Revised treatment	Domain
2026:Y26-06-027 2026-06-25	Finite closure iteration terminates by rank, but noncommuting closures can hide new elements	D14
2026:Y26-06-030 2026-06-26	Tangram areas and chirality give exact invariants but do not classify all assemblies	D06
2026:Y26-06-029 2026-06-27	Read-write footprint independence gives a sufficient commutation certificate	D14
2026:Y26-06-016 Date unavailable	The quartic discriminant−283 is exact, but its claimed global uniqueness is not established	D01
2026:Y26-06-017 Date unavailable	Corner pins generate exceptional units and a finite anharmonic orbit	D01
2026:Y26-06-018 Date unavailable	The Moyal bracket has a finite falling-factorial formula with even powers of $\hbar$	D16
2026:Y26-06-019 Date unavailable	The proposed cubic-cancellation counterexample is actually an infinite exact Moyal family	D16
2026:Y26-06-020 Date unavailable	A genuine fifth-order Moyal obstruction appears in a higher-degree family	D16
2026:Y26-07-001 2026-07-01	Finite spectral rigidity needs annihilation and support witnesses	D03
2026:Y26-07-002 2026-07-03	Quadratic jets need their cubic closure to become finite	D03
2026:Y26-07-003 2026-07-03	The Hilbert–Burch quotient has type two despite its symmetric Hilbert function	D04
2026:Y26-07-004 2026-07-03	Tensoring by a dual-number factor gives Betti numbers(1,4,5,2)	D04
2026:Y26-07-005 2026-07-03	Two-prime agreement is not a rational membership proof	D14
2026:Y26-07-006 2026-07-04	Pauli centralizers are exact symplectic kernels	D16
2026:Y26-07-007 2026-07-04	Sector transition counts use the symplectic-kernel quotient	D16
2026:Y26-07-008 2026-07-04	Sector Schmidt ranks add only across orthogonal blocks	D16
2026:Y26-07-009 2026-07-05	The discriminant−23 cubic and the plastic unit are exactly related	D01
2026:Y26-07-010 2026-07-05	The eta product has exact coefficients and a Fricke law	D07
2026:Y26-07-011 2026-07-05	The five-vertex spectrum reconstructs a graph exactly	D13
2026:Y26-07-012 2026-07-05	Numerical volume closure needs a geometric certificate	D06
2026:Y26-07-013 2026-07-06	A parametrix isolates a singularity and a smooth remainder	D08
2026:Y26-07-014 2026-07-06	The120+128 decomposition is a dimension and weight calculation	D03

Source identifier / date	Revised treatment	Domain
2026:Y26-07-015 2026-07-06	A derivative residual can be an identity rather than a prediction	D09
2026:Y26-07-017 2026-07-13	Monotone phase gives interlacing roots and positive local masses	D07
2026:Y26-07-016 2026-07-14	Finite positive atoms produce a Herglotz function, but differences need a new proof	D07
2026:Y26-07-018 2026-07-16	A positivity atlas needs a covering bound, not a dense scatterplot	D07
2026:Y26-07-019 2026-07-18	The small-alpha frontier is a critical-point normal form	D07
2026:Y26-07-020 2026-07-20	Moment inversion must preserve the root count and its conditioning	D15
2026:Y26-07-021 2026-07-25	Integral homology retains torsion that rational ranks erase	D05
2026:Y26-07-030 2026-07-25	Discrete Gauss–Bonnet is an exact angle-counting identity	D06
2026:Y26-07-022 2026-07-26	A rank-one cellular sheaf on a circle is controlled by holonomy	D05
2026:Y26-07-023 2026-07-26	Total-complex signs and mapping cones are structural data	D04
2026:Y26-07-024 2026-07-26	Tensor–Hom adjunction separates currying from dualizability	D04
2026:Y26-07-025 2026-07-29	The elliptic curve of discriminant37 admits exact group-law and rank-lower-bound certificates	D02
2026:Y26-07-026 2026-07-31	Riemann–Roch proves a code bound, not an unexhibited minimum-weight word	D14
2026:Y26-07-027 2026-07-31	Base twelve changes representation while preserving exact arithmetic	D01
2026:Y26-07-028 2026-07-31	A root atlas combines Rouché counts with explicit root squaring	D07
2026:Y26-07-029 2026-07-31	A contraction certificate gives a genuinely guaranteed Newton enclosure	D15
2026:Y26-07-031 2026-07-31	A p-adic completion changes convergence while keeping rational identities	D01
2026:Y26-08-901 2026-08-01	A cutoff theorem with the quantifiers made explicit	D15
2026:Y26-08-902 2026-08-01	Mild ill-posedness: an optimal rate from a concrete ellipsoid	D15
2026:Y26-08-903 2026-08-01	Severe ill-posedness needs a logarithmic safety margin	D15
2026:Y26-08-904 2026-08-01	Tikhonov filters: qualification and explicit constants	D15
2026:Y26-08-905 2026-08-01	Landweber iteration: regularization by stopping	D15

Source identifier / date	Revised treatment	Domain
2026:Y26-08-906 2026-08-01	The ignition constant: a proved integral behind an unverified front law	D08
2026:Y26-08-001 2026-08-02	A singular descriptor matrix can encode a regular constrained system	D15
2026:Y26-08-004 2026-08-04	Rank-one Rouché bounds preserve structure and leave a separate remainder obligation	D07
2026:Y26-08-008 2026-08-04	A common storage function proves hybrid passivity across modes	D09
2026:Y26-08-015 2026-08-10	The Plücker shell has an essential parity lattice	D02
2026:Y26-08-016 2026-08-10	Correlated validation needs a joint enclosure rather than independent error bars	D15
2026:Y26-08-014 2026-08-14	A skew pairing makes wall transport invertible and Poisson	D16
2026:Y26-08-017 2026-08-15	A small harmonic residual does not make a cycle topologically trivial	D05
2026:Y26-08-018 2026-08-15	Holonomy on a loop and curvature on a surface are different invariants	D06
2026:Y26-08-012 2026-08-16	Flat algebras can have jumping Hochschild cohomology	D16
2026:Y26-08-013 2026-08-16	The regulator fixed-point equation has an exact pitchfork threshold	D09
2026:Y26-08-009 2026-08-17	An improper complex Gaussian is an ordinary two-dimensional covariance problem	D11
2026:Y26-08-010 2026-08-17	Accretivity bounds a passive resolvent in the correct norm	D07
2026:Y26-08-023 2026-08-17	Cyclic symmetry diagonalizes a modal map but does not guarantee full rank	D07
2026:Y26-08-024 2026-08-17	Phase-error fidelity depends on the relative phase convention	D16
2026:Y26-08-025 2026-08-17	Decibels and transfer matrices require power and amplitude conventions	D09
2026:Y26-08-005 2026-08-18	A quadratic Lyapunov certificate survives nonnormal transient growth	D09
2026:Y26-08-006 2026-08-18	Finite-horizon Gramians certify controllability and observability	D15
2026:Y26-08-007 2026-08-19	A danger-to-information quotient is finite only after its null directions are checked	D12
2026:Y26-08-019 2026-08-22	The reported corner count identifies the correct ambient product space	D13
2026:Y26-08-020 2026-08-22	A weighted row is a proof only when its support and derivation are supplied	D14
2026:Y26-08-021 2026-08-23	Rational span does not determine an integer toric lattice	D03

Source identifier / date	Revised treatment	Domain
2026:Y26-08-801 2026-08-28T00:38:04Z	Semantic fibers enlarge the feasible optimum only when deployment constraints are preserved	D12
2026:Y26-08-022 2026-08-30	A local vacancy potential can certify a global resource bound	D14
2026:Y26-08-802 2026-08-30T16:42:29Z	Transformer gauge identities must respect rotations, nonlinear gates and sharing partitions	D03
2026:Y26-08-803 2026-08-30T16:42:29Z	Bellman-safe merging compares complete continuation behavior, not one current optimum	D14
2026:Y26-08-804 2026-08-31T03:50:37Z	Channel erasure has exactly256 states only when forgotten lane identities are irrelevant	D13
2026:Y26-08-805 2026-08-31T23:09:26Z	A frontier state supports exact min-plus transfer when it separates past from future	D14
2026:Y26-08-806 2026-08-31T23:32:35Z	Connectivity compression4140→128 is an exact rank theorem over the field with two elements	D13
2026:Y26-08-002 Date unavailable	Polynomial nullspaces expose minimal indices of singular pencils	D15
2026:Y26-08-003 Date unavailable	Descriptor eigenvalue and damping sensitivities need a nonzero normalization	D15
2026:Y26-08-011 Date unavailable	The projective quiver pencil really has constant dimension thirteen	D03
2026:Y26-09-001 2026-09-01	A rotating residue has an exact period and a complete orbit decomposition	D09
2026:Y26-09-002 2026-09-01	A hidden port variable can prevent a deterministic observation from closing	D15
2026:Y26-09-003 2026-09-01	A period-matched difference separates a linear wave cost from a recurring offset	D11
2026:Y26-09-004 2026-09-01	A paired sign-randomization test has a finite exact null certificate	D11
2026:Y26-09-005 2026-09-02	Typed min-plus matrix multiplication is exact dynamic programming	D14
2026:Y26-09-006 2026-09-02	Future equivalence is a right congruence when continuation costs are retained	D14
2026:Y26-09-007 2026-09-02	A two-by-two tropical minor can rule out a one-register linear factorization	D14
2026:Y26-09-008 2026-09-02	An equivariant derivative intertwines stabilizers but does not remove hidden coordinates	D03
2026:Y26-09-013 2026-09-16	The recovered quintic has directly verifiable simple roots and Hensel lifts	D01
2026:Y26-09-014 2026-09-16	A Fitting ideal and its contraction certify global prime support in a declared quotient	D01
2026:Y26-09-015 2026-09-17	An action barrier is a residual pullback only under stated quotient hypotheses	D03
2026:Y26-09-016 2026-09-17	An abelian kernel can have a perfect and centerless effective actor extension	D03

Source identifier / date	Revised treatment	Domain
2026:Y26-09-033 2026-09-17	A Green-coordinate identity gives an exact Hilbert–Schmidt model and explains $\det_2$	D07
2026:Y26-09-034 2026-09-17	The imaginary-order Bessel boundary equations arise from a confining half-line operator	D07
2026:Y26-09-035 2026-09-17	A Ky Fan edge remainder is a weighted curvature integral	D12
2026:Y26-09-009 2026-09-18	Intertwiner spaces are kernels of explicit integer matrices	D03
2026:Y26-09-010 2026-09-21	Hankel rank measures reachable observable state, not the whole generated algebra	D14
2026:Y26-09-011 2026-09-21	An incidence model needs a dual lower-bound witness before a call count is a theorem	D13
2026:Y26-09-012 2026-09-22	Two exceptional hyperplanes have a common core and an exact repair criterion	D03
2026:Y26-09-017 2026-09-22	Local carrier quotients form a functor but can miss globally generated relations	D04
2026:Y26-09-024 2026-09-23	A local space-time affine velocity has fifteen coefficients but needs a full-rank observation design	D15
2026:Y26-09-021 2026-09-24	Exact SDE propagation can make a driver measurement more informative than a direct present-target measurement	D10
2026:Y26-09-022 2026-09-24	A hidden-target conditional variance is a genuine noise-free error floor	D11
2026:Y26-09-023 2026-09-24	More present information and a longer prediction horizon need separate monotonicity statements	D11
2026:Y26-09-026 2026-09-24	The gradient-wind quadratic has two branches and a singular sensitivity boundary	D09
2026:Y26-09-027 2026-09-24	Positive averaging preserves stress positivity, while signed correction need not	D11
2026:Y26-09-019 2026-09-26	A zero directional derivative does not make a nonlinear target constant on a hidden fiber	D15
2026:Y26-09-020 2026-09-26	A correlated confidence ellipsoid has exact target extrema and an intersection test	D15
2026:Y26-09-025 2026-09-26	Hodge potentials recover divergence and curl but leave a harmonic mean	D15
2026:Y26-09-018 2026-09-28	A target is observable exactly when it factors through the observation map	D15
2026:Y26-09-028 2026-09-29	The linear perfect-power branch has an exact valuation formula and never becomes finite through underflow	D01
2026:Y26-09-029 2026-09-30	A complete Pell orbit has exact congruence filters and a logarithmic population	D01
2026:Y26-09-030 2026-09-30	The recovered Thue form has an exact norm identity, a rank-two lattice restriction and finite unit residue classes	D01
2026:Y26-09-031 2026-09-30	A unimodular substitution preserves a Thue solution set; a nonunimodular one needs a lattice condition	D01

Source identifier / date	Revised treatment	Domain
2026:Y26-09-032 2026-09-30	A finite SMT replacement needs a completeness theorem, while a single exact witness already proves satisfiability	D14
2026:Y26-10-002 2026-10-02	Irreducibility certificates do not by themselves classify rational points	D01
2026:Y26-10-009 2026-10-03	Cubic norms certify unit transport but not an unreplayed global search	D01
2026:Y26-10-005 2026-10-04	Integer lifting turns monomial equations into finite exponent lattices	D01
2026:Y26-10-007 2026-10-04	GCD matrices factor through divisor incidence and discrete Green kernels	D01
2026:Y26-10-008 2026-10-04	Superelliptic genus follows from finite and infinite ramification	D02
2026:Y26-10-004 2026-10-05	Perfect-power images admit complete valuation parametrizations	D01
2026:Y26-10-026 2026-10-05	A distance-regular intersection array fixes a conditional exact spectrum	D13
2026:Y26-10-011 2026-10-06	Squarefree quotient algebras split exceptional and regular branches by gcd	D03
2026:Y26-10-012 2026-10-06	The wall fluctuation operator has exactly two bound-state levels	D07
2026:Y26-10-037 2026-10-06	Elliptic multiplication fibres are kernel cosets, with halving checked by a quartic	D02
2026:Y26-10-038 2026-10-06	The elliptic logarithmic constant log4 follows from an exact integral split	D07
2026:Y26-10-039 2026-10-07	A sufficiently divisible small integer determinant must vanish	D01
2026:Y26-10-001 2026-10-08	Repeated polynomial roots require multiplicities and a squarefree isolation certificate	D03
2026:Y26-10-006 2026-10-08	The negative Pell equation splits into two complete unit orbits	D01
2026:Y26-10-003 2026-10-09	Prime-stripped factorials have an exact recursive block algorithm	D01
2026:Y26-10-010 2026-10-09	Quadratic elimination needs conjugate and zero-denominator branch checks	D01
2026:Y26-10-013 2026-10-09	Darboux cofactors give a certificate for rational first integrals	D09
2026:Y26-10-021 2026-10-09	Permitted actions determine the correct finite observation quotient	D14
2026:Y26-10-036 2026-10-09	The integral Mordell equation $y^2=x^3-2$ has a complete elementary proof	D01
2026:Y26-10-018 2026-10-10	A Hankel witness proves the minimum observable recurrence order	D14
2026:Y26-10-015 2026-10-12	Threshold completion counts certify a population and its optimum	D13

Source identifier / date	Revised treatment	Domain
2026:Y26-10-016 2026-10-12	Completion counts make ranking, inverse ranking and sampling exact	D14
2026:Y26-10-017 2026-10-12	Factor elimination preserves counts only when auxiliary fibres are controlled	D13
2026:Y26-10-014 2026-10-14	Schröder jets are solved degree by degree until a resonance occurs	D09
2026:Y26-10-019 2026-10-16	A target is recoverable exactly when it factors through the observation	D15
2026:Y26-10-025 2026-10-16	Cancellation controls the condition number of an exact sum	D07
2026:Y26-10-030 2026-10-16	Conductor 1680 collapses to eight order30 phases with multiplicity48	D01
2026:Y26-10-031 2026-10-16	A 264-fold root filter differs from a degree240 field extension	D01
2026:Y26-10-032 2026-10-16	The inverse-central-binomial series is an arcsine square	D07
2026:Y26-10-033 2026-10-16	The order30 cyclotomic kernel has an exact Mellin zero and logarithmic companion	D07
2026:Y26-10-040 2026-10-16	Expected surplus and probability of profit require different dependence assumptions	D10
2026:Y26-10-027 2026-10-17	Binary graph censuses can agree while integer-height series separate	D13
2026:Y26-10-029 2026-10-18	Finite observation cutoffs need a stated algebraic or linear closure theorem	D14
2026:Y26-10-020 2026-10-19	A moving observation obeys a closed equation exactly when its transport identity holds	D15
2026:Y26-10-022 2026-10-19	Worst-case experiment design has an exact finite Bellman proof	D14
2026:Y26-10-028 2026-10-19	The centered graph Green matrix gives a sharp constrained energy cost	D13
2026:Y26-10-024 2026-10-20	Raw generator observability is an injective coding problem	D14
2026:Y26-10-034 2026-10-20	The five-phase logarithmic integral equals $955\pi^3/867$ exactly	D07
2026:Y26-10-035 2026-10-20	Determinant twists explain the claimed compact group form but do not prove density	D16
2026:Y26-10-023 2026-10-21	Shared uncertainty should be optimized jointly in a decision difference	D12
2026:Y26-10-901 2026-10-21	Five raw observations determine the entire 107-bit generator state	D14
2026:Y26-10-902 2026-10-21	Seven and nine raw observations have optimal one- and two-error correction	D14

Additional 2023 annual syntheses

Fourteen additional synthesis treatments from the 2023 volume are preserved separately from the 1,051 main-treatment count. Their complete original revised text is available in the interactive companion and source bundle.

Identifier	Synthesis
2023:S001	Minimal recurrences connect observations to finite realizations
2023:S002	A reversible observation transform preserves Fisher information
2023:S003	Finite observations generate an algebra of distinguishable classes
2023:S004	Heat observations separate exact recovery from stable recovery
2023:S005	Positivity conservation entropy and spectra share one graph update
2023:S006	Whitening turns sensor questions into projection geometry
2023:S007	Exact finite polynomial closure for triangular maps and its resonance strata
2023:S008	An exact robust Gaussian sensor panel with a strict greedy gap and a blind direction
2023:S009	Exact tree synchronization certificates and observable phase information
2023:S010	The complete one-magnon Gaudin family and its scalar equations
2023:S011	Minimum induced subgraph order and the sharp heat observation margin
2023:S012	Singular quadratic duality requires an observation quotient
2023:S013	An exact exceptional point and its two-sheeted spectral cover
2023:S014	Complete complex one-magnon Jordan classification

Verification and reproducibility

The four original annual verification suites were replayed successfully from fresh copies. They contain 61 verifier programs and 1,173 scoped records; records are heterogeneous and include a proof-only record, so these counts are not counts of formally proved theorems. New domain scripts check selected worked examples, and the lab engine is tested against independent exact fixtures and boundary cases. A passing finite check supports its named model and does not automatically certify an entire exposition or an unrecovered historical instance.

The portable bundle includes normalized data, full immutable source payloads, domain verifier scripts, the exact lab engine, independent fixtures, interface tests, builders and the four annual source ZIPs. Verification results name their scope. Rebuilding the PDF needs ReportLab and DejaVu fonts; the mathematical domain checks use the Python standard library. Running the HTML companion needs only a modern browser and no network connection.

The catalogue preserves original annual treatment payloads byte-for-byte at the JSON value level. Classification, grouping, navigation and newly authored bridges are additional atlas data. Source hashes and a portable package manifest make accidental changes detectable. Historical gaps—such as unrecovered large-instance tables or actual braid generators—remain gaps, even when a related finite model can be demonstrated.

New computations and conceptual refinements belong to this atlas edition. They are not backdated to the conversation dates. The report does not claim that numerical spot checks constitute formal verification of all prose proofs.